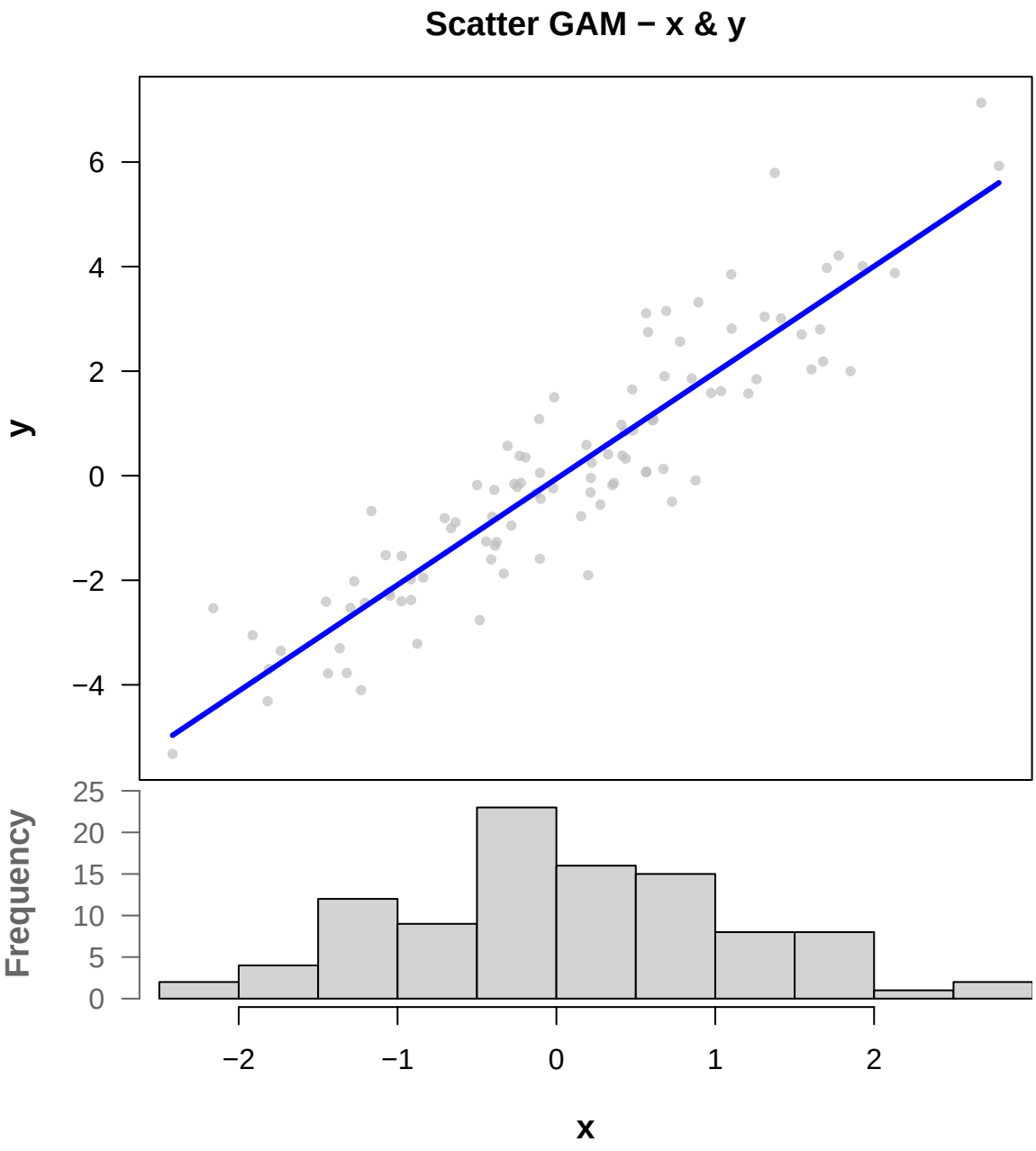
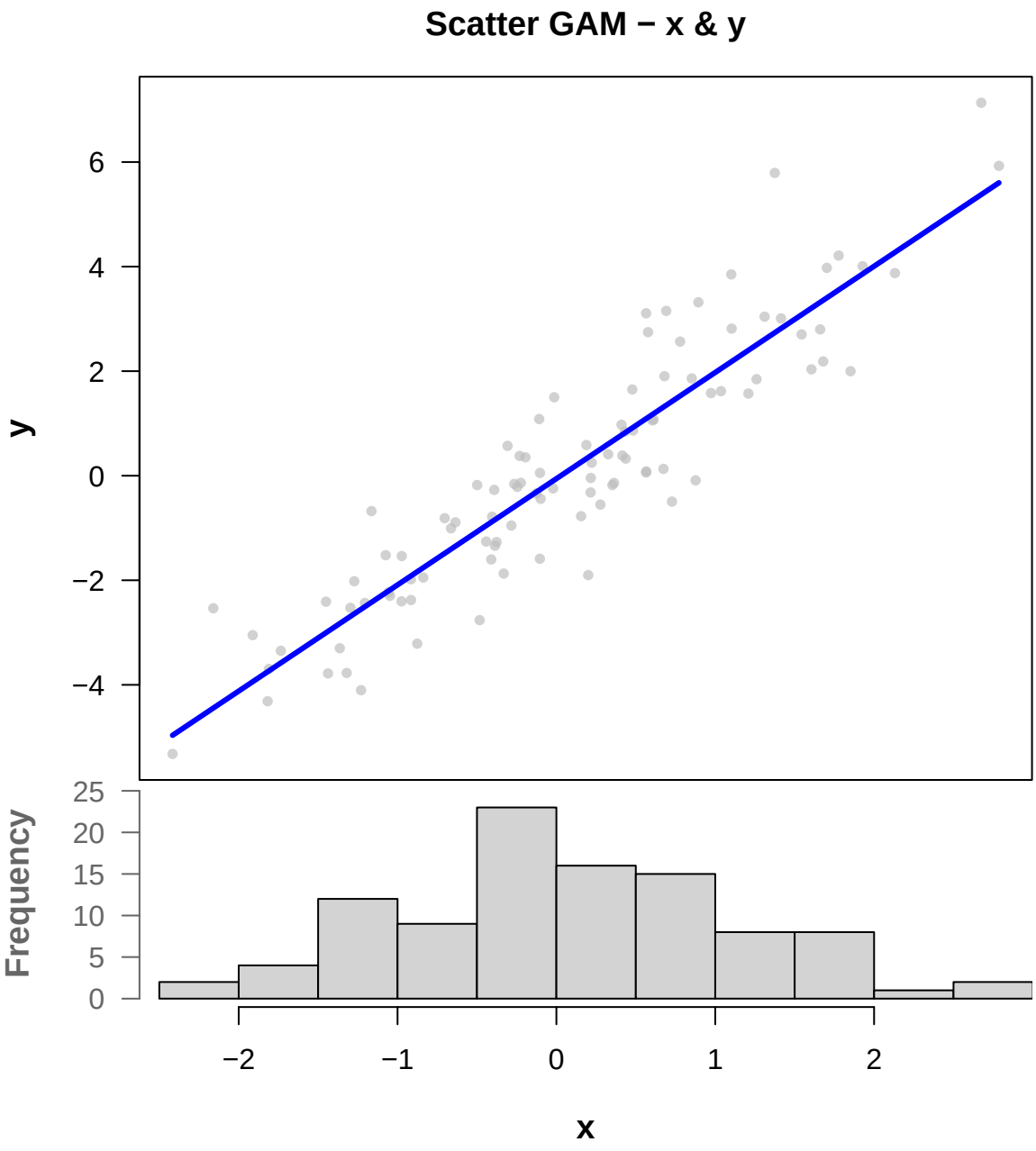
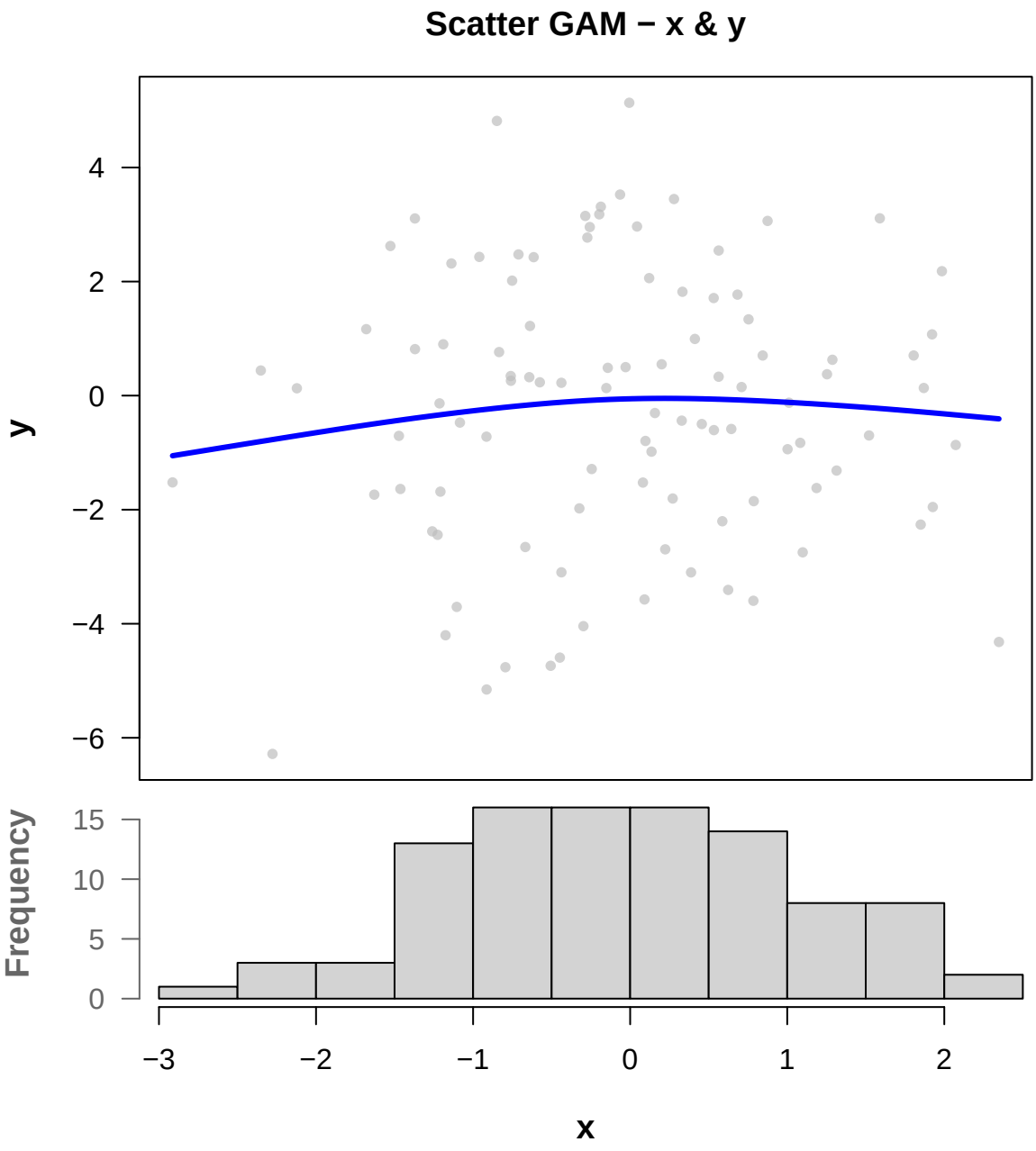
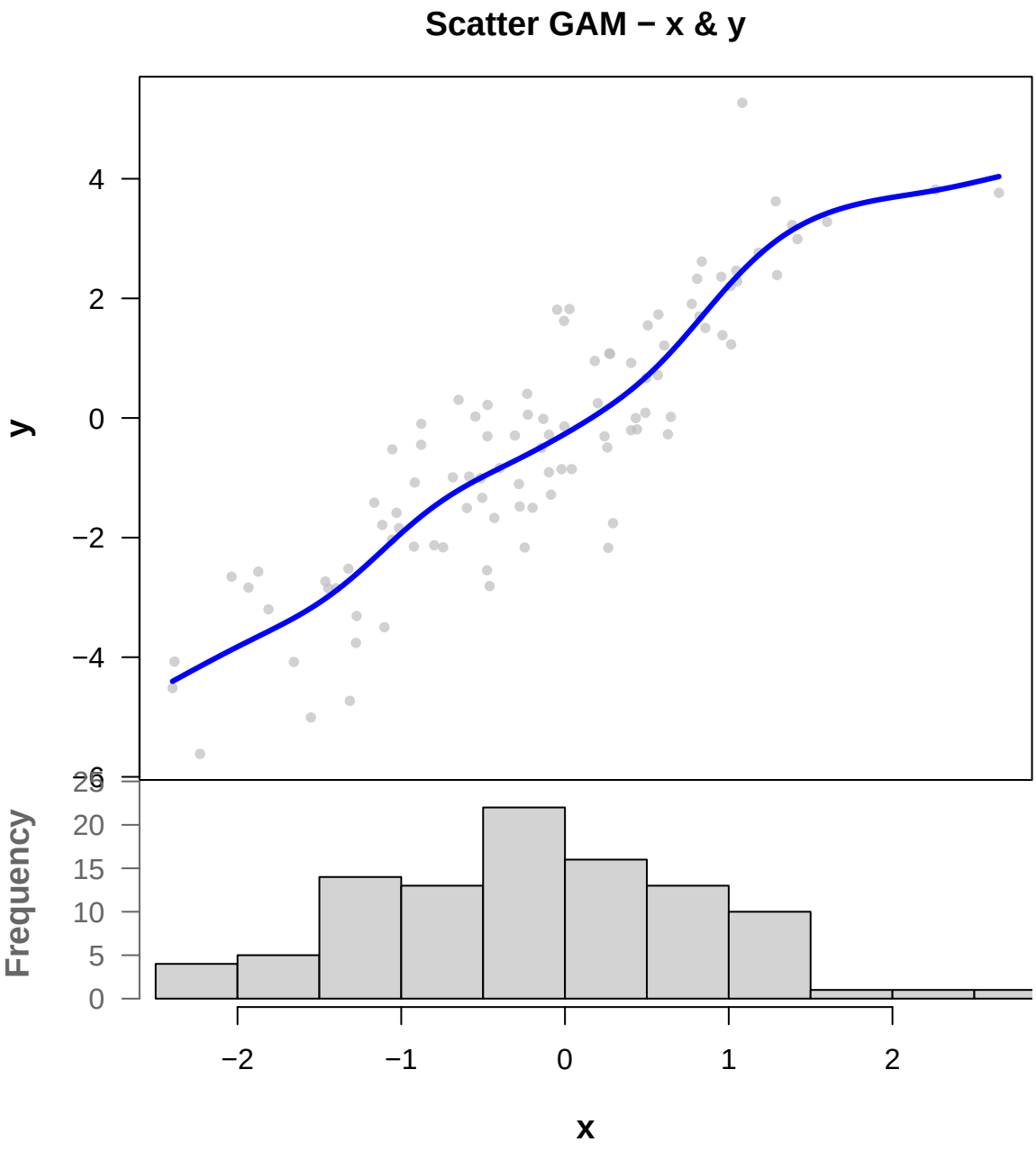
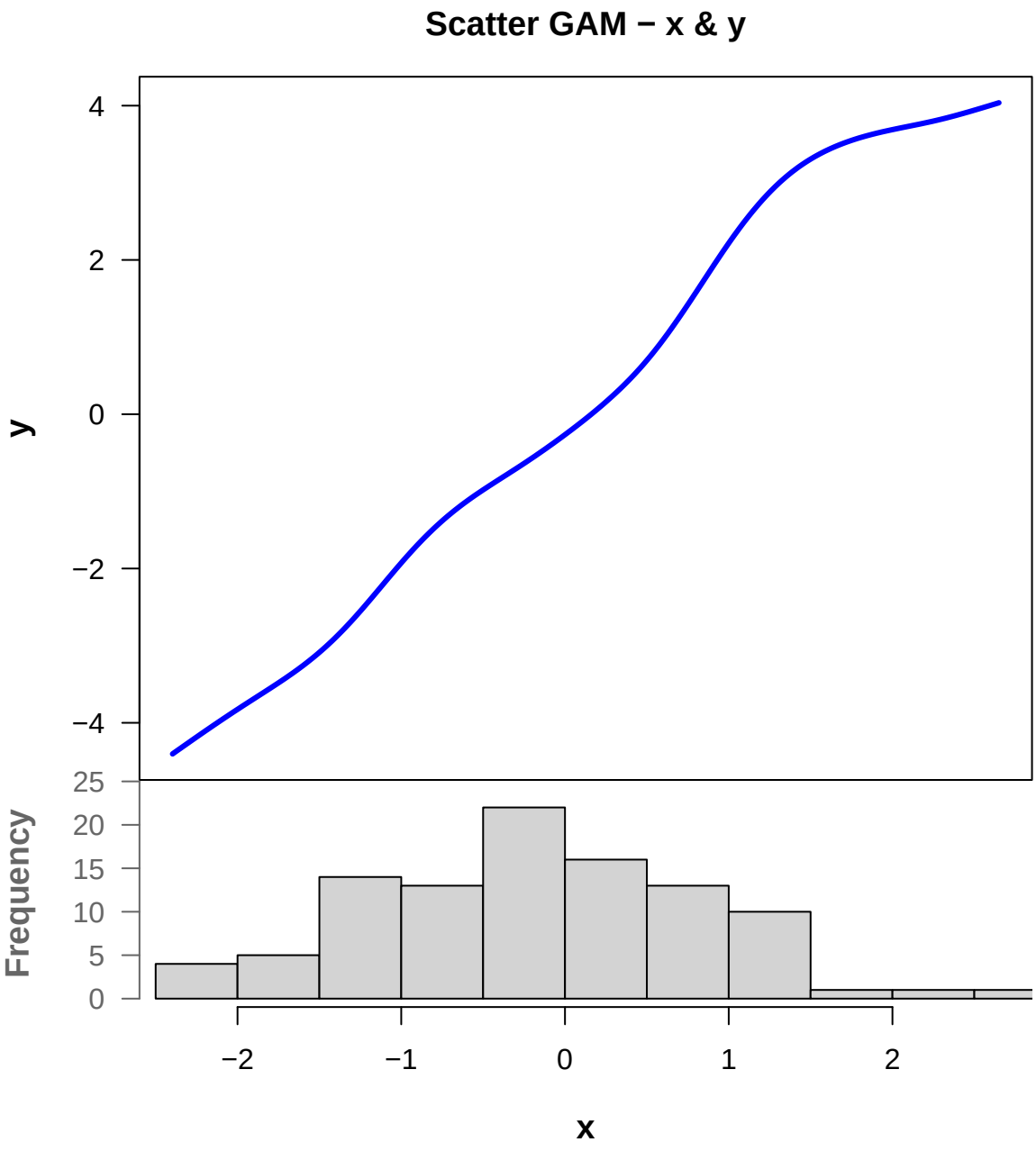
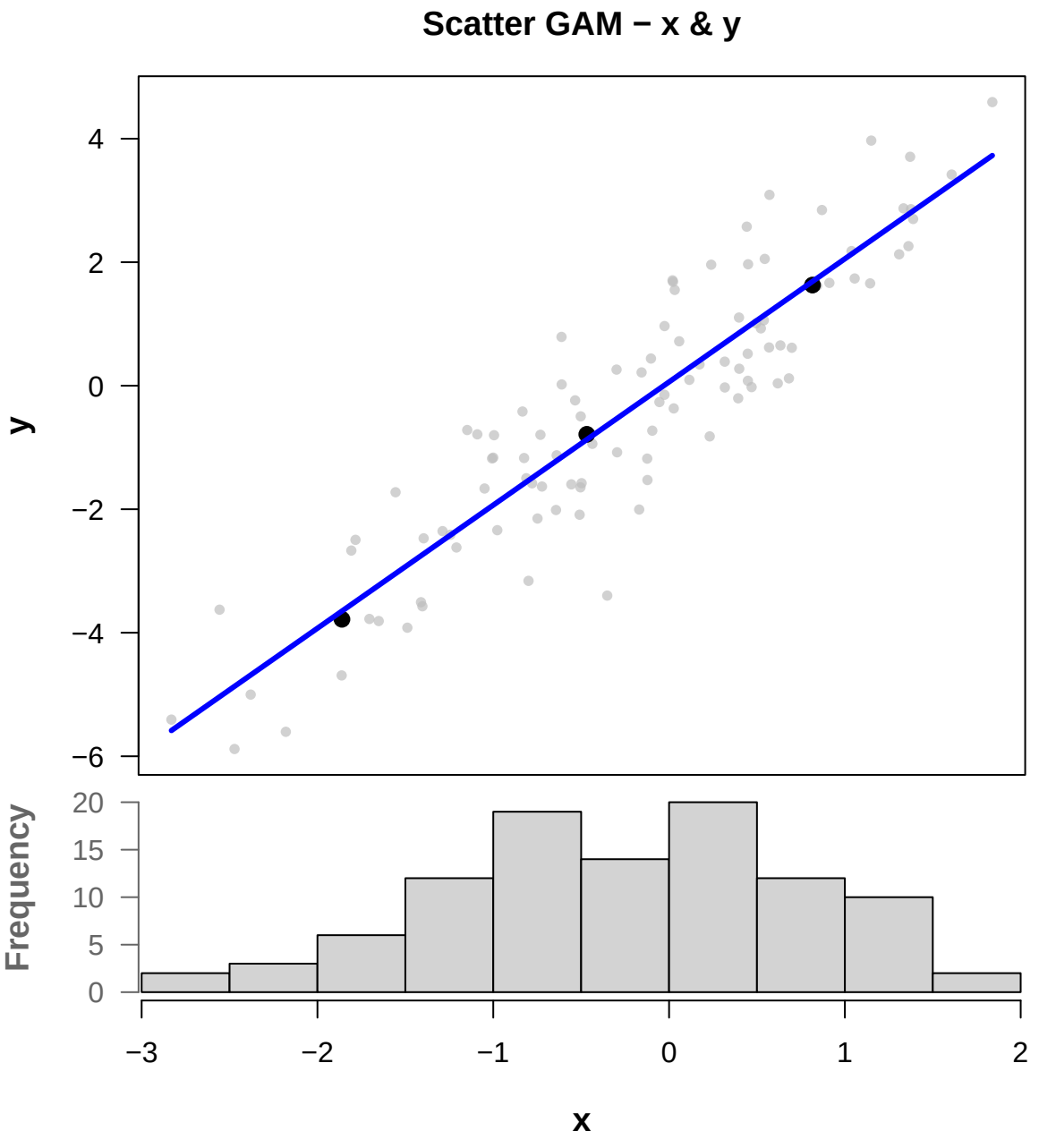

Scatter GAM − x & y
y
6
4
2
0
−2
−4
Frequency
25
20
15
10
5
0
−2
−1
0
1
2
x

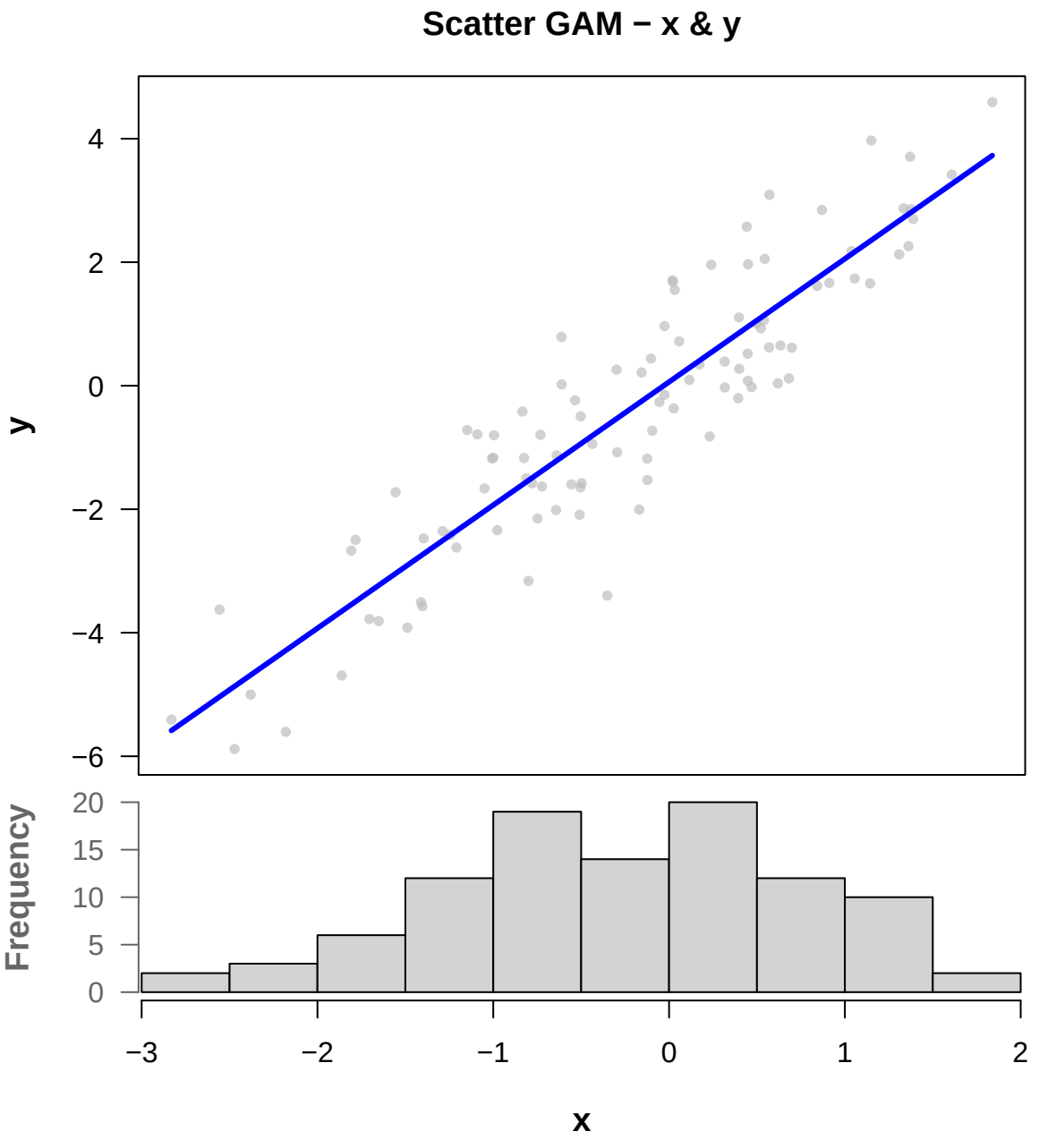

Scatter GAM − x & y
y
6
4
2
0
−2
−4
Frequency
25
20
15
10
5
0
−2
−1
0
1
2
x

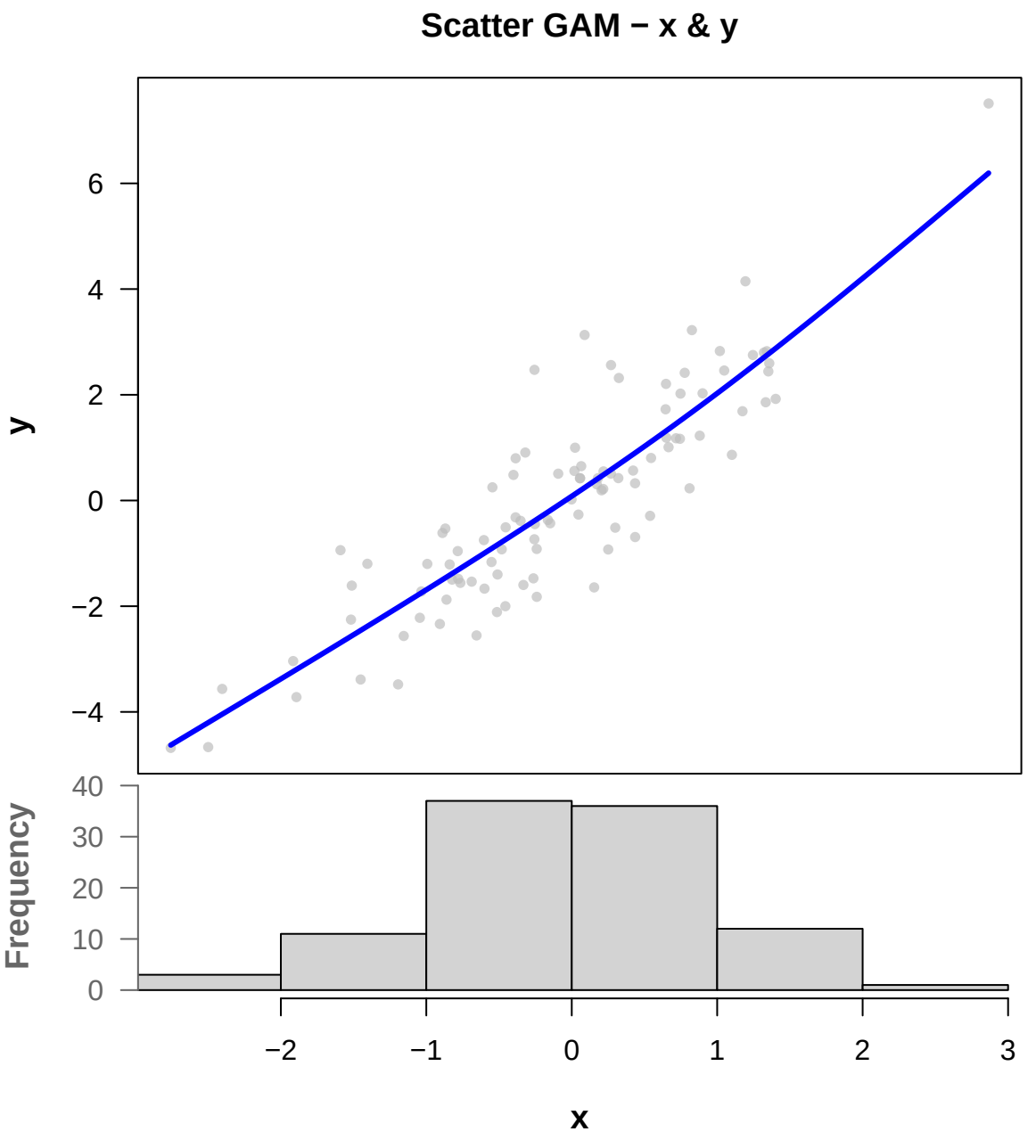

Scatter GAM − x & y
y
4
2
0
−2
−4
−6
Frequency
15
10
5
0
−3
−2
−1
0
1
2
x

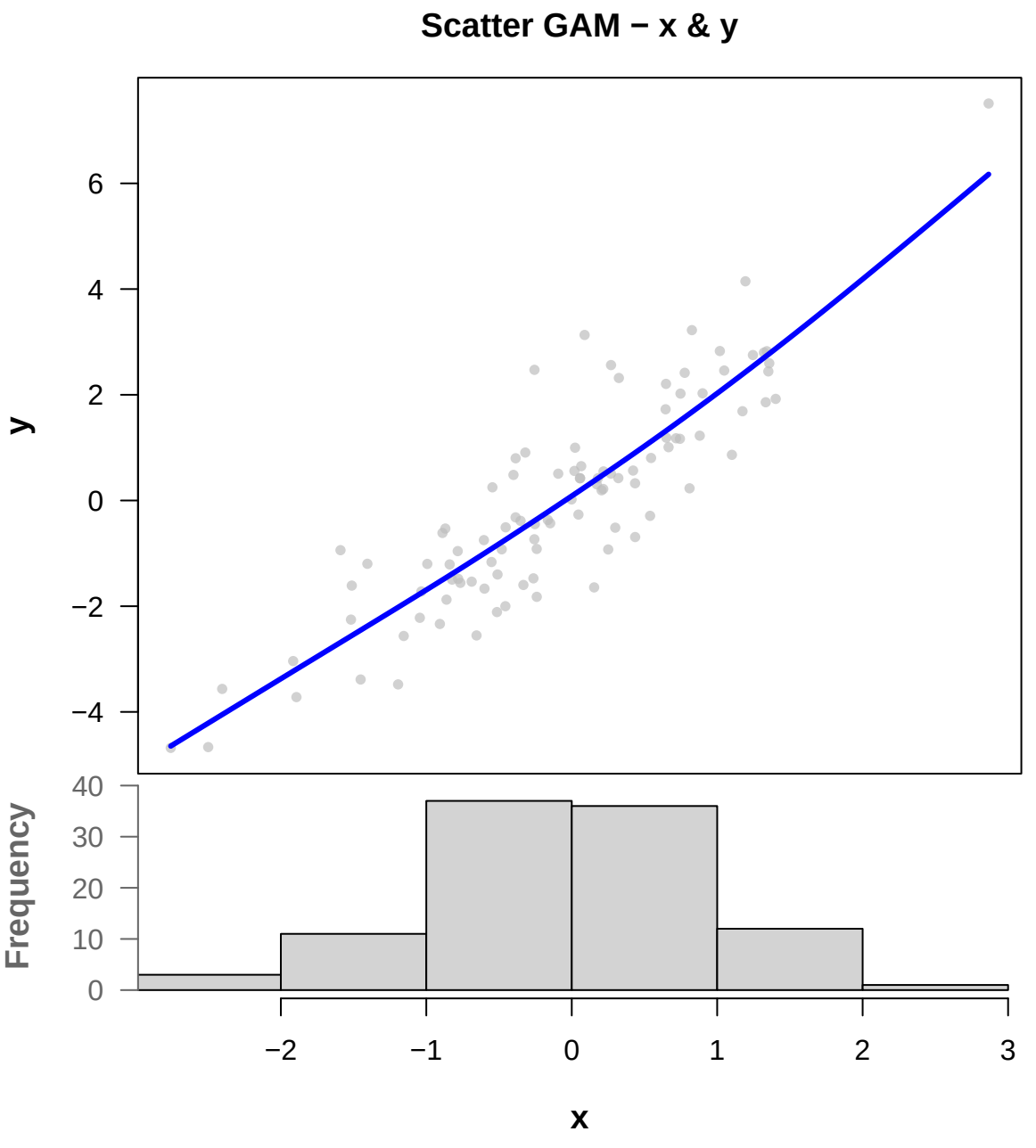

Scatter GAM − x & y
y
4
2
0
−2
−4
−6
Frequency
25
20
15
10
5
0
−2
−1
0
1
2
x

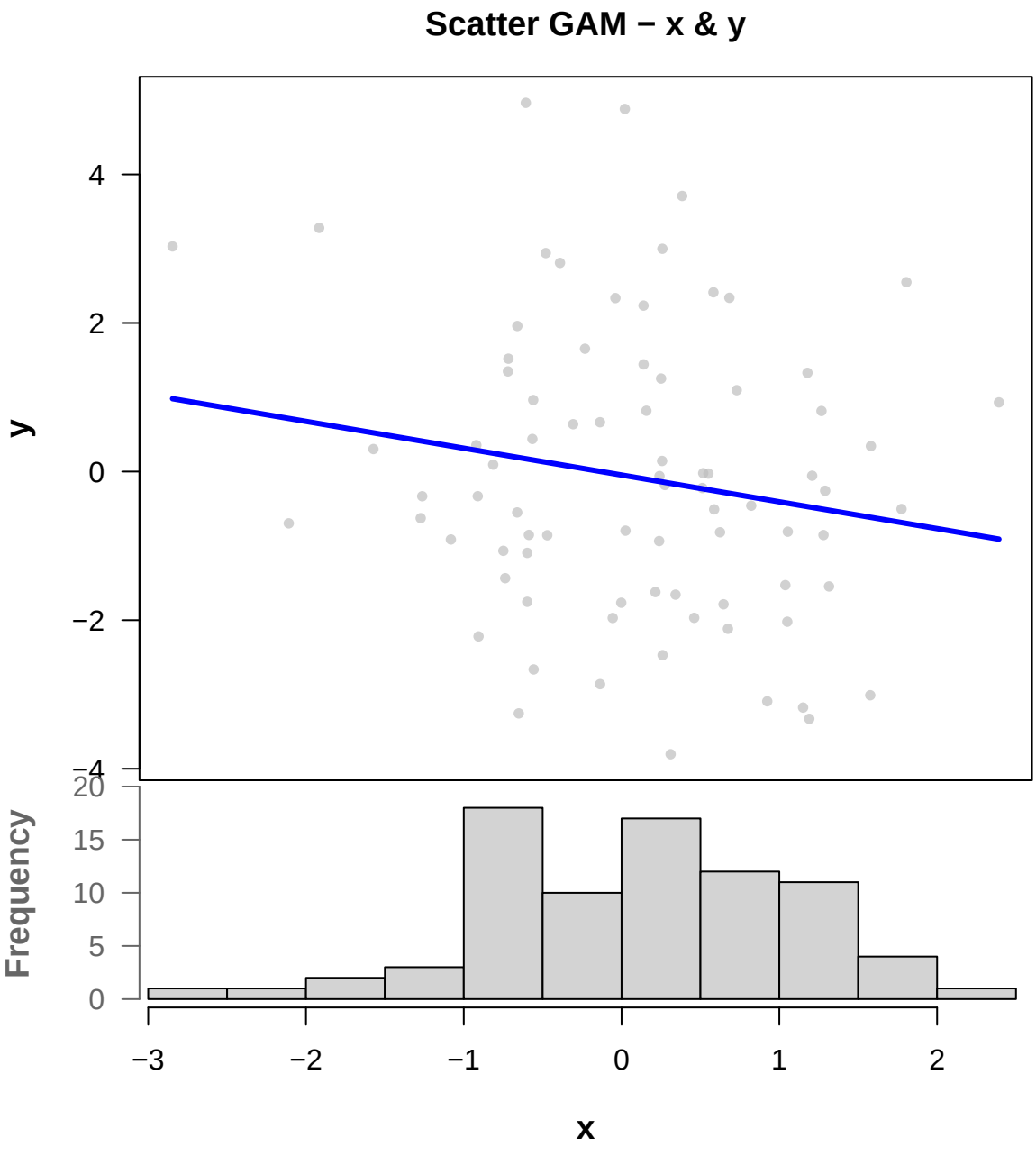

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
25
20
15
10
5
0
−2
−1
0
1
2
x

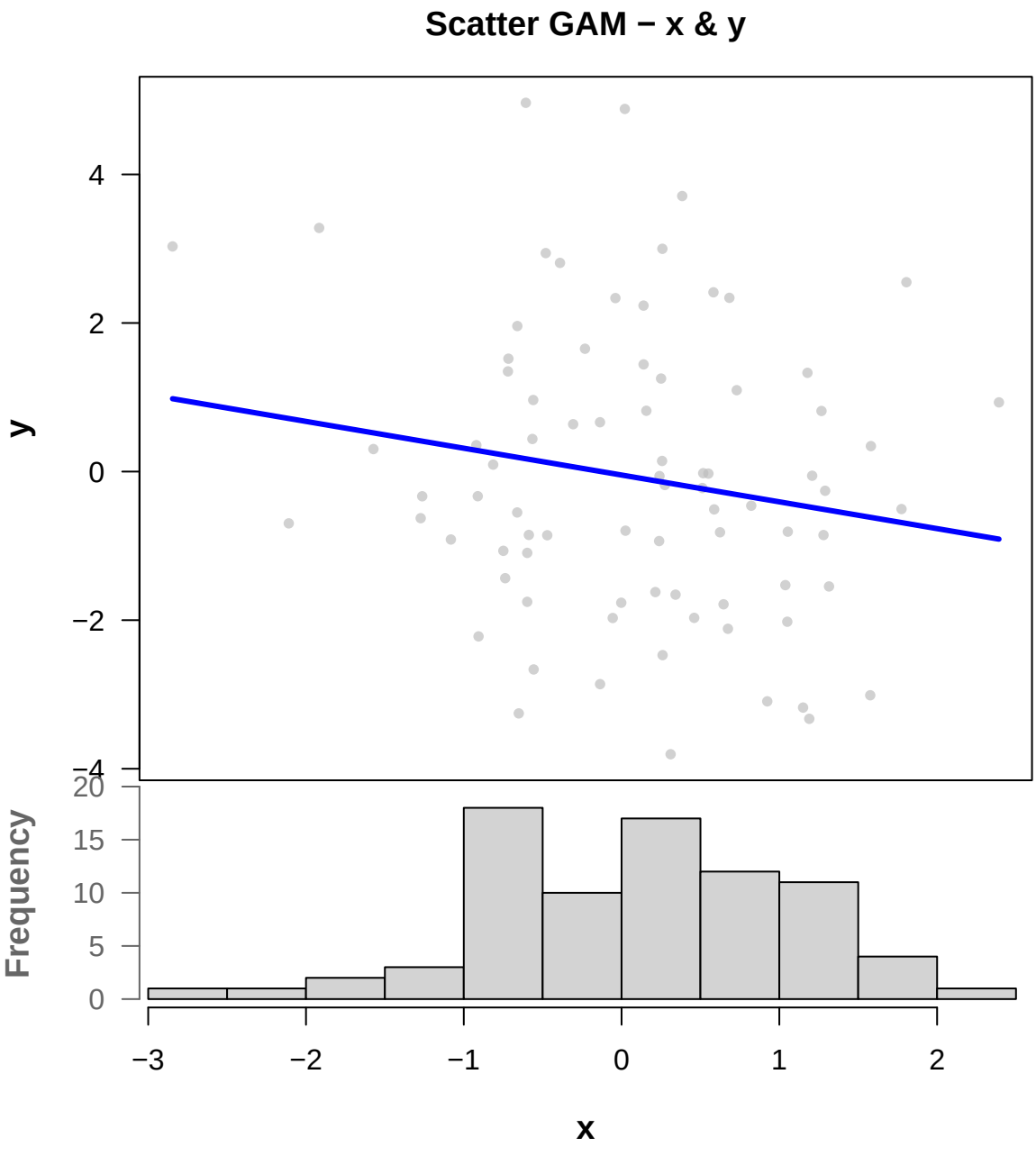

Scatter GAM − x & y
y
4
2
0
−2
−4
−6
Frequency
20
15
10
5
0
−3
−2
−1
0
1
2
x

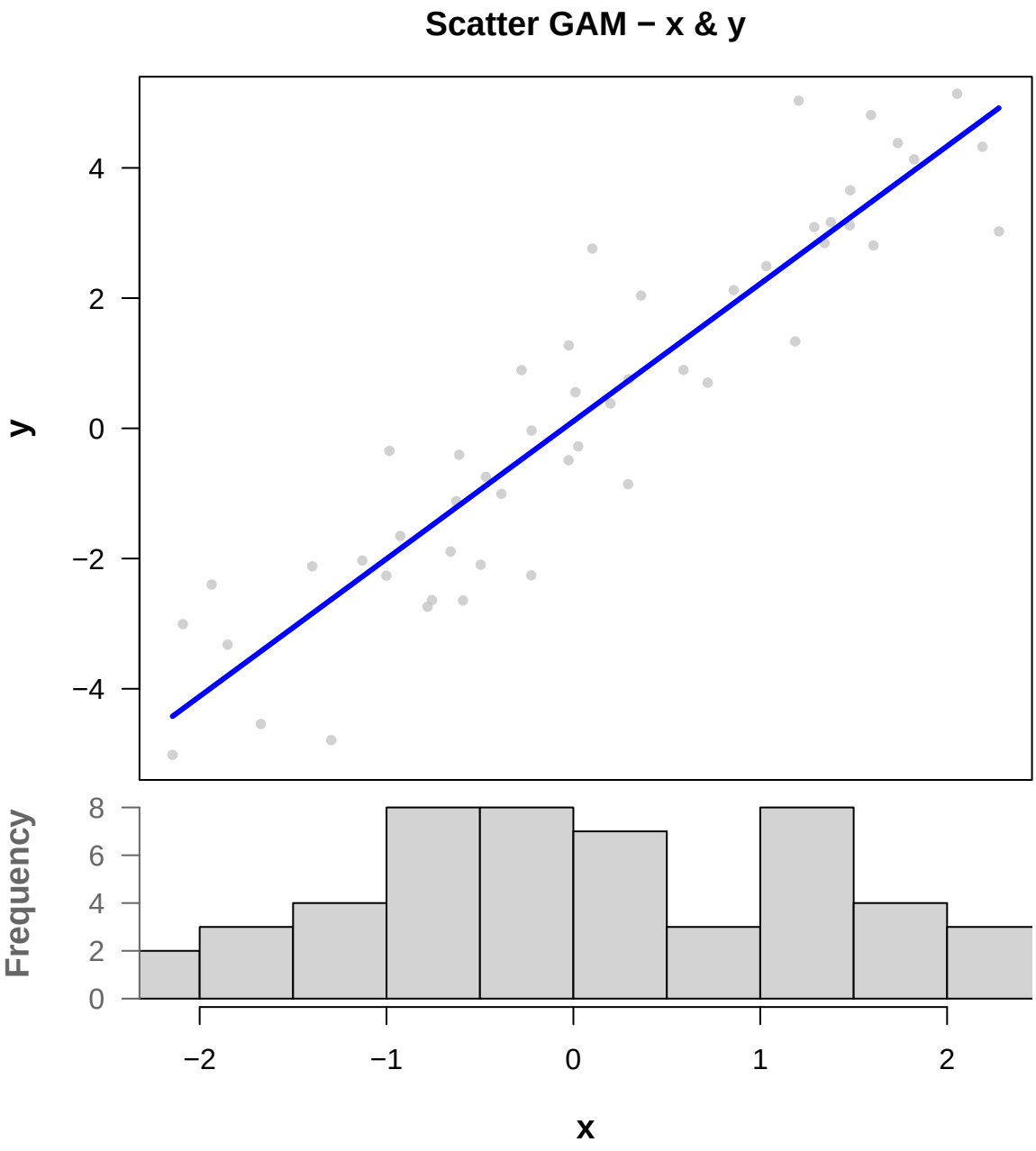

Scatter GAM − x & y
y
4
2
0
−2
−4
−6
Frequency
20
15
10
5
0
−3
−2
−1
0
1
2
x

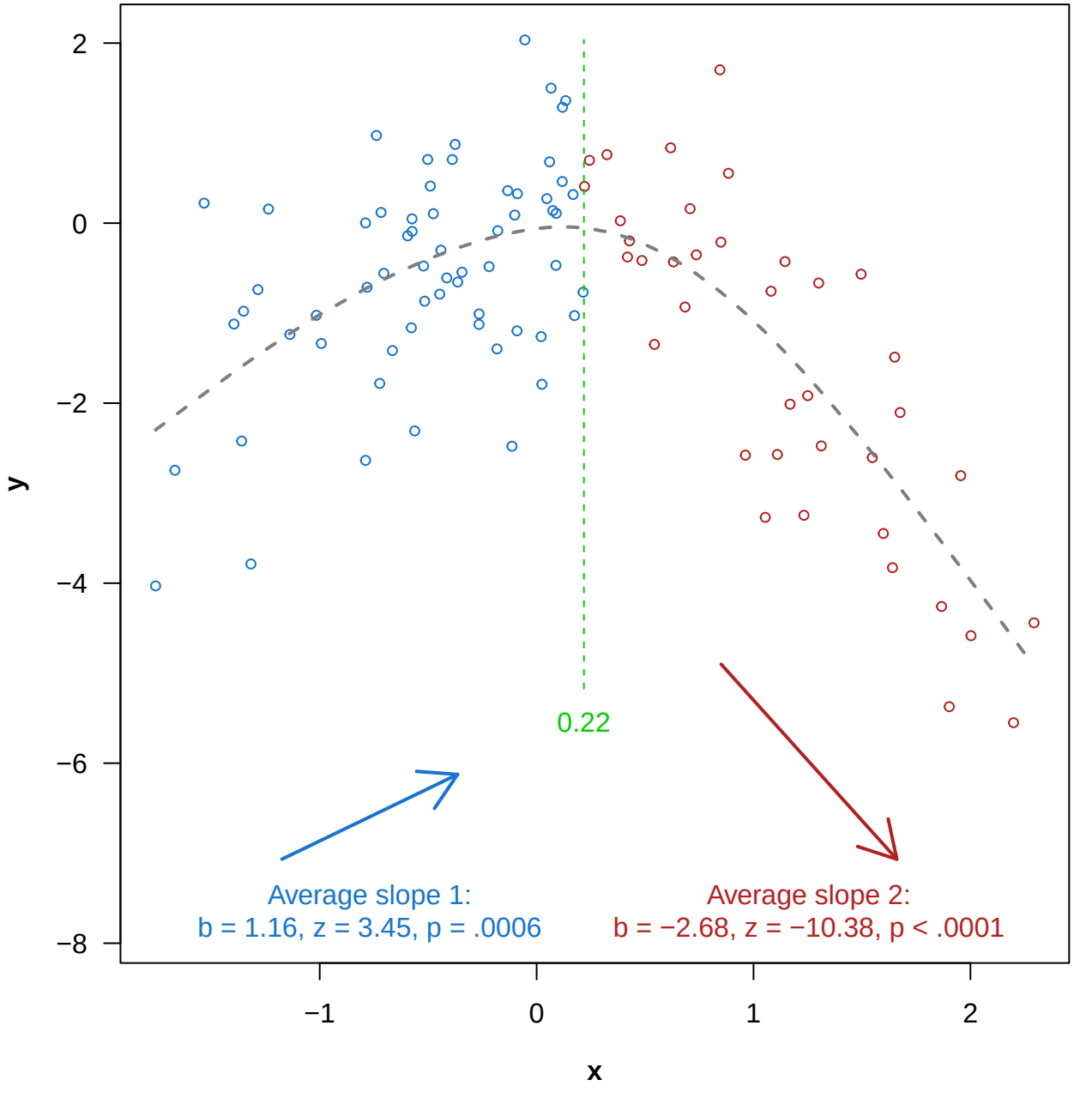

Scatter GAM − x & y
y
6
4
2
0
−2
−4
Frequency
40
30
20
10
0
−2
−1
0
1
2
3
x

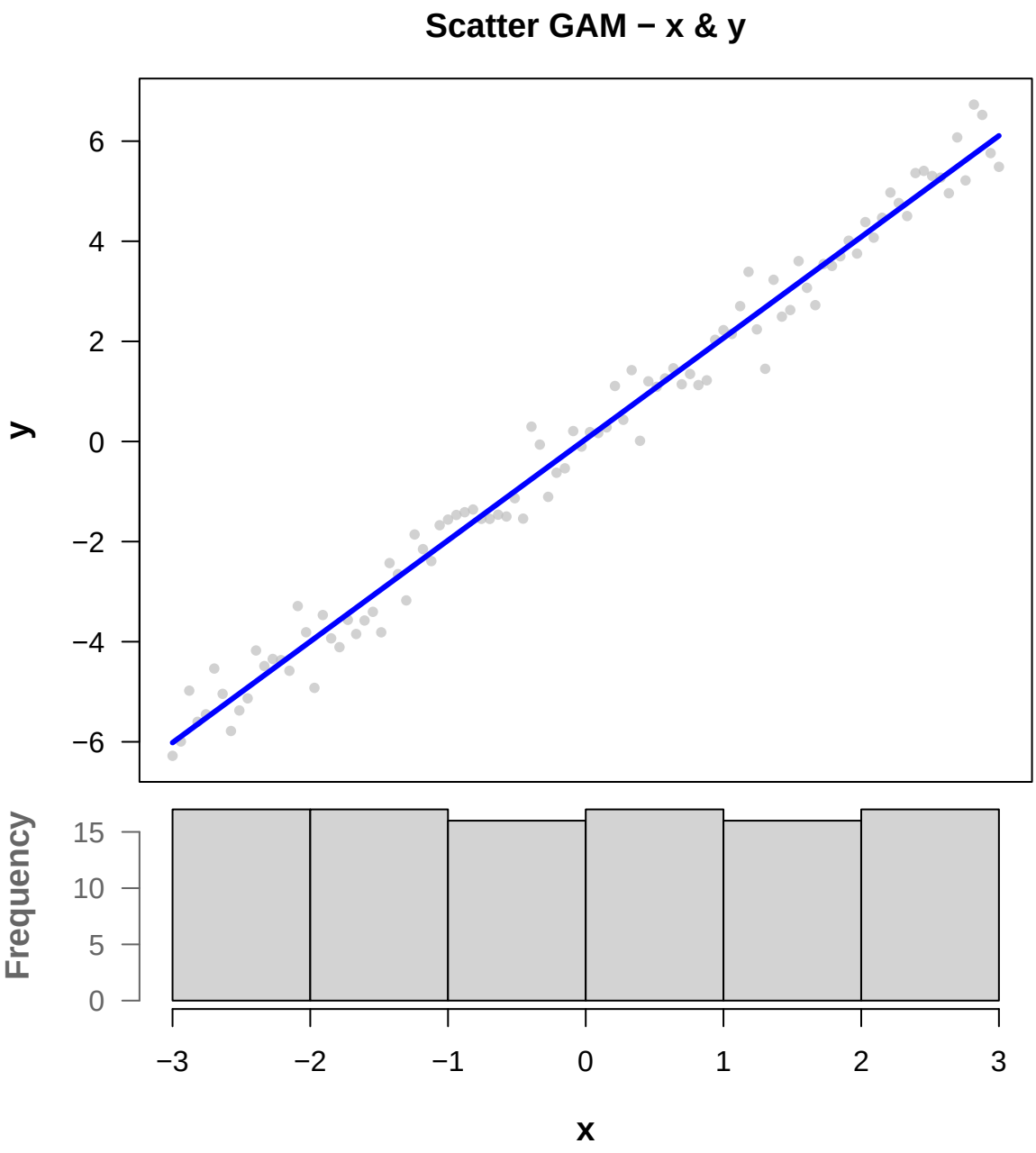

Scatter GAM − x & y
y
6
4
2
0
−2
−4
Frequency
40
30
20
10
0
−2
−1
0
1
2
3
x

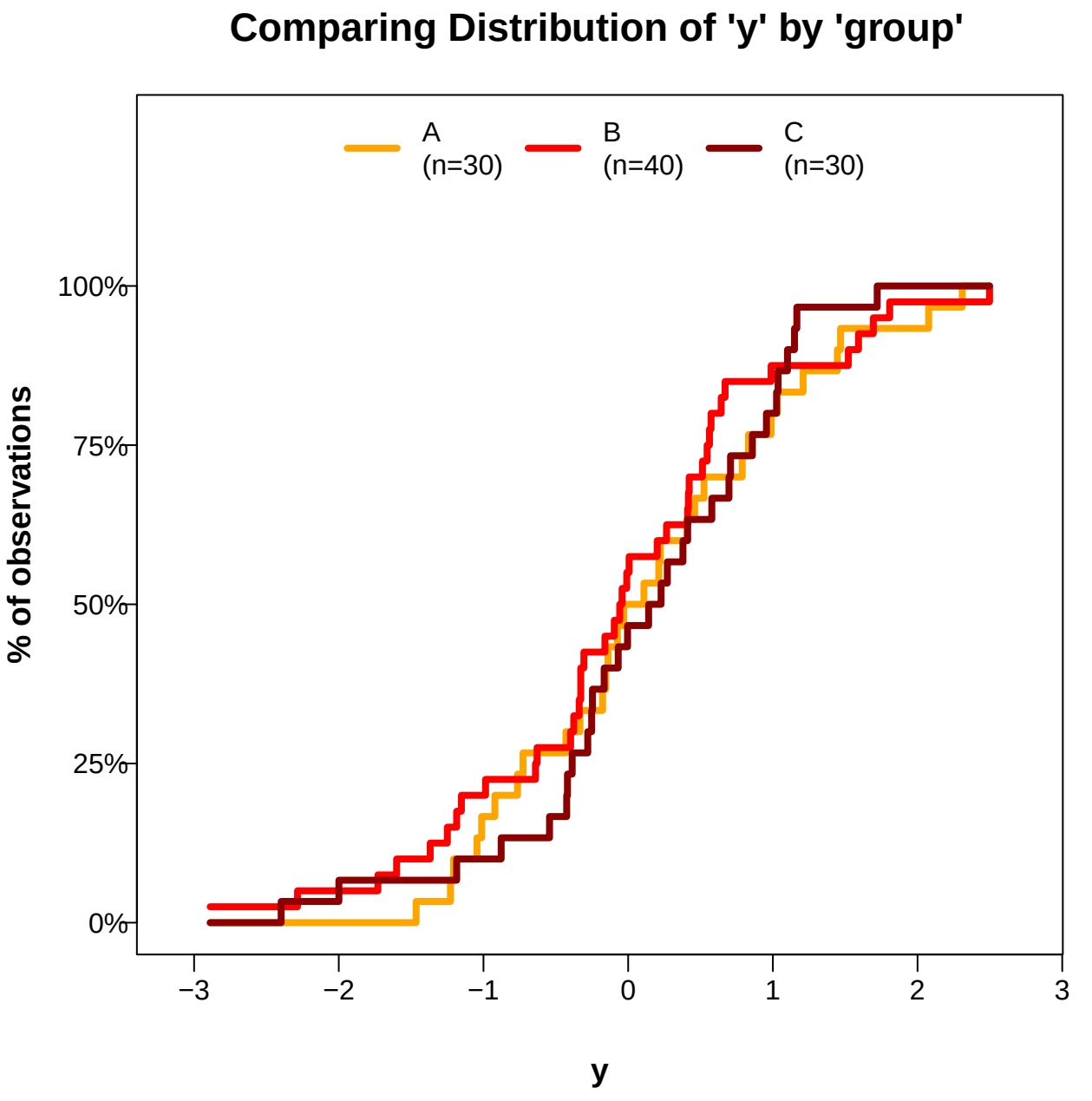

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
20
15
10
5
0
−3
−2
−1
0
1
2
x

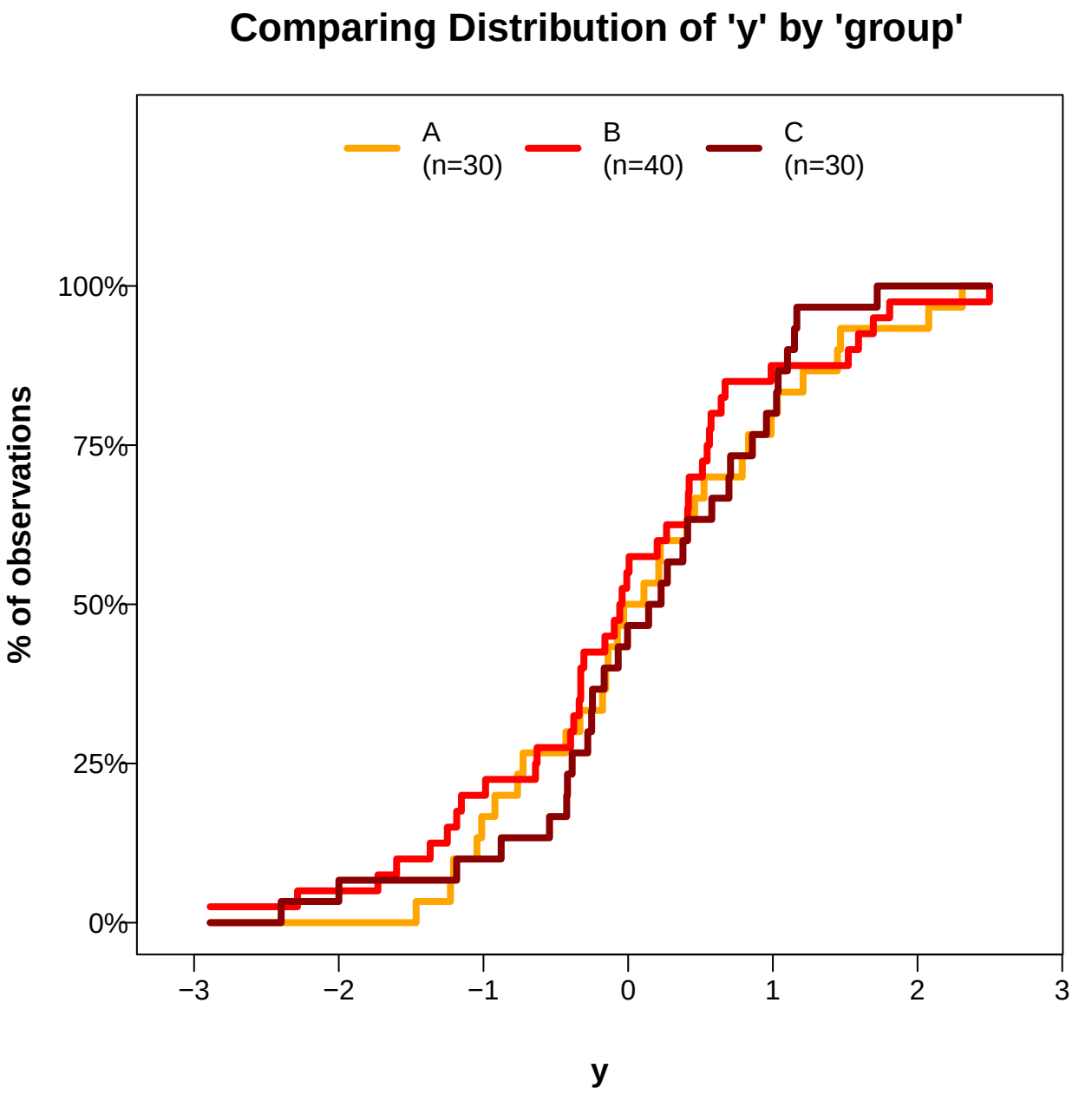

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
20
15
10
5
0
−3
−2
−1
0
1
2
x

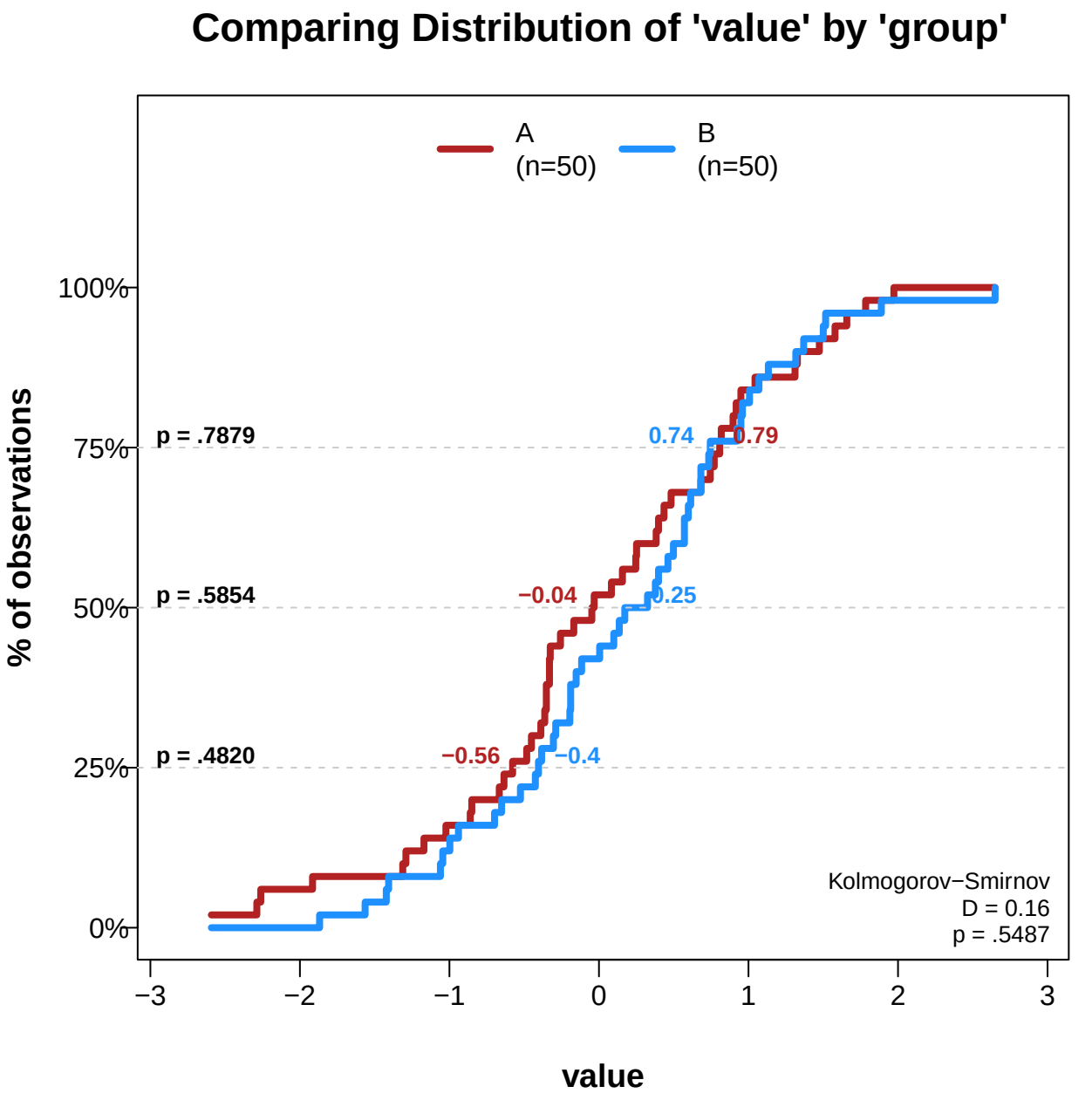

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
8
6
4
2
0
−2
−1
0
1
2
x

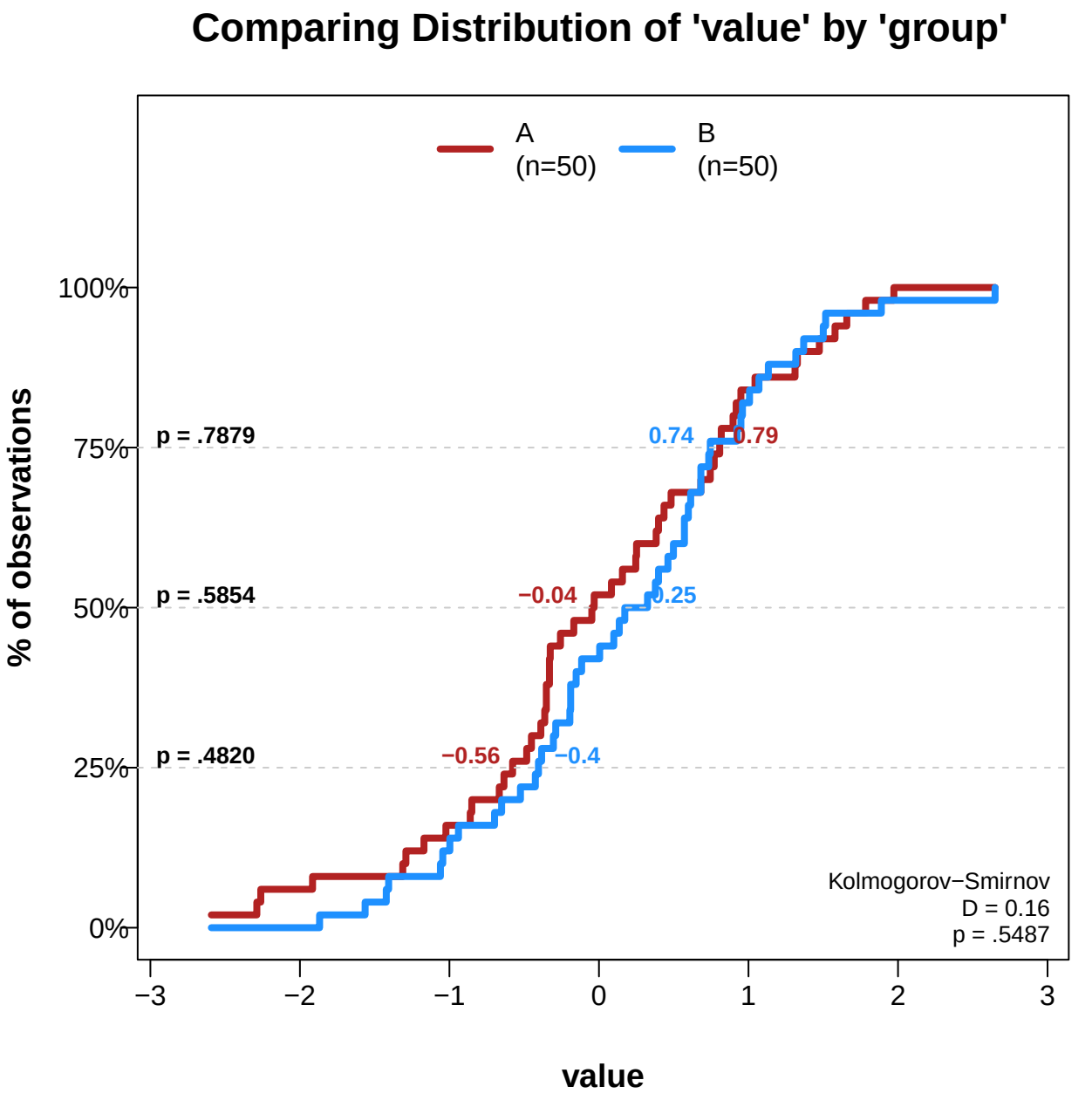

2
0
−2
−4
−6
−8
y
−1
0
1
2
x
0.22
Average slope 1:
b = 1.16, z = 3.45, p = .0006
Average slope 2:
b = −2.68, z = −10.38, p < .0001

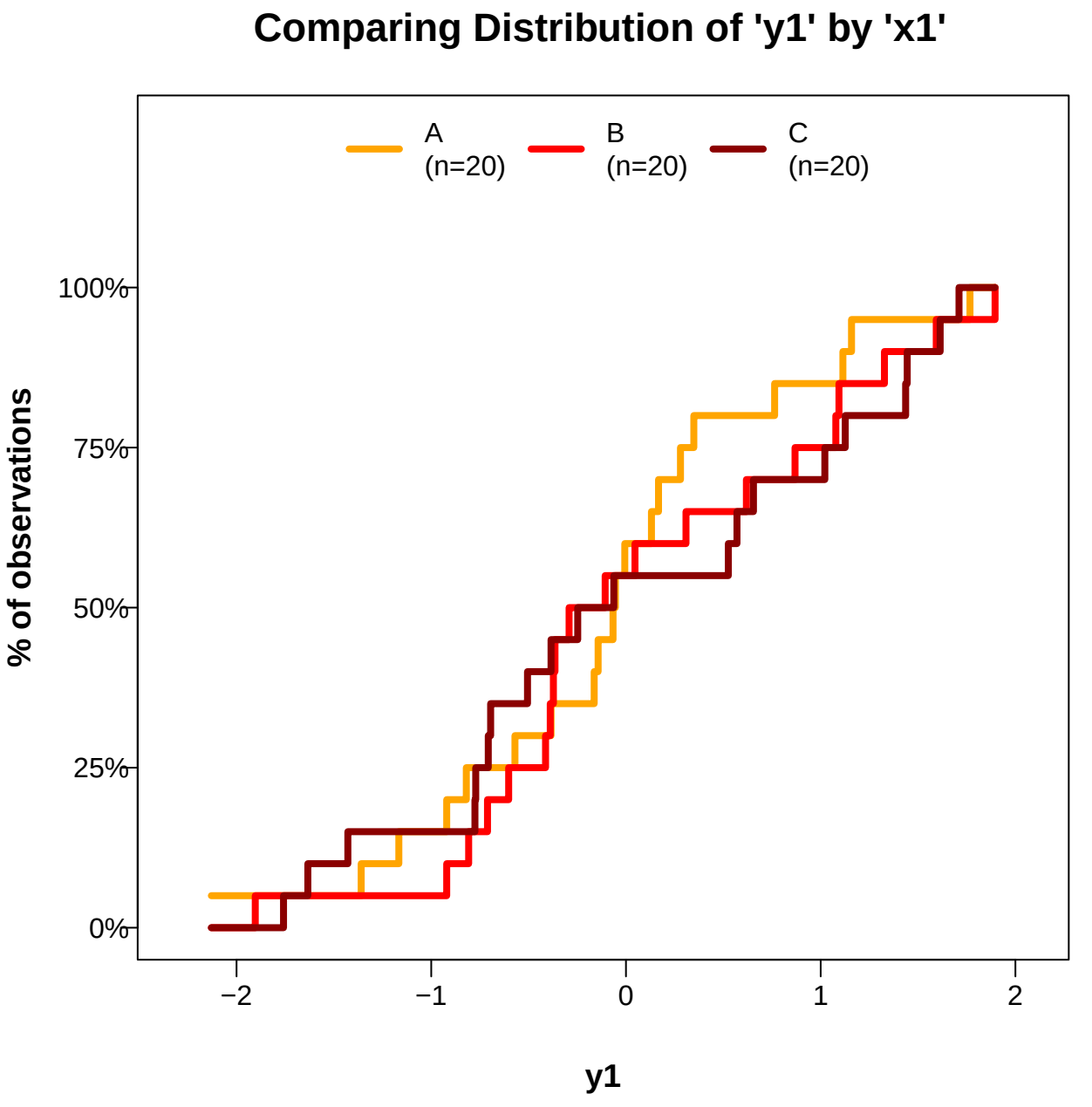

Scatter GAM − x & y
y
6
4
2
0
−2
−4
−6
Frequency
15
10
5
0
−3
−2
−1
0
1
2
3
x

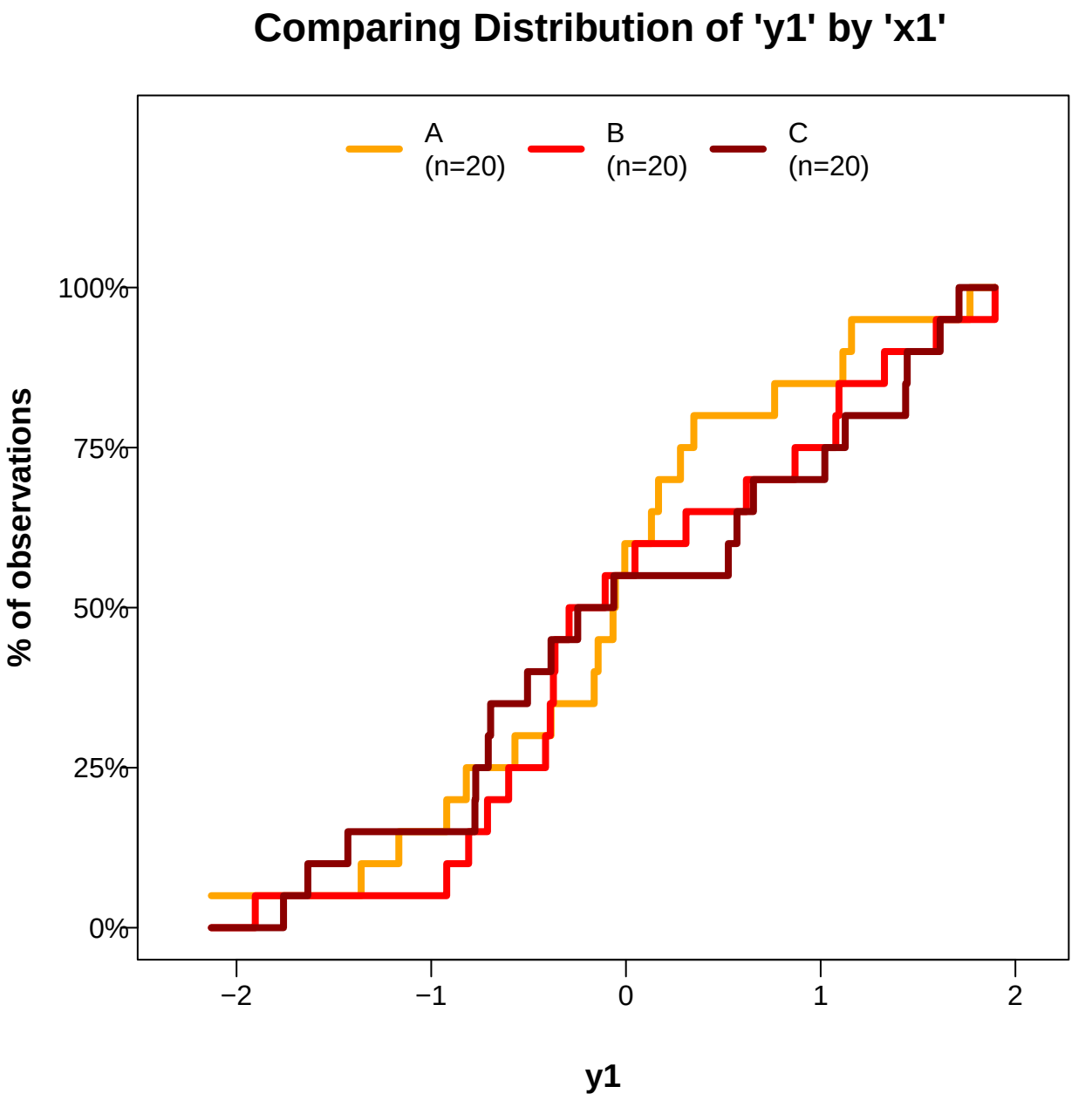

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−3
−2
−1
0
1
2
3
y

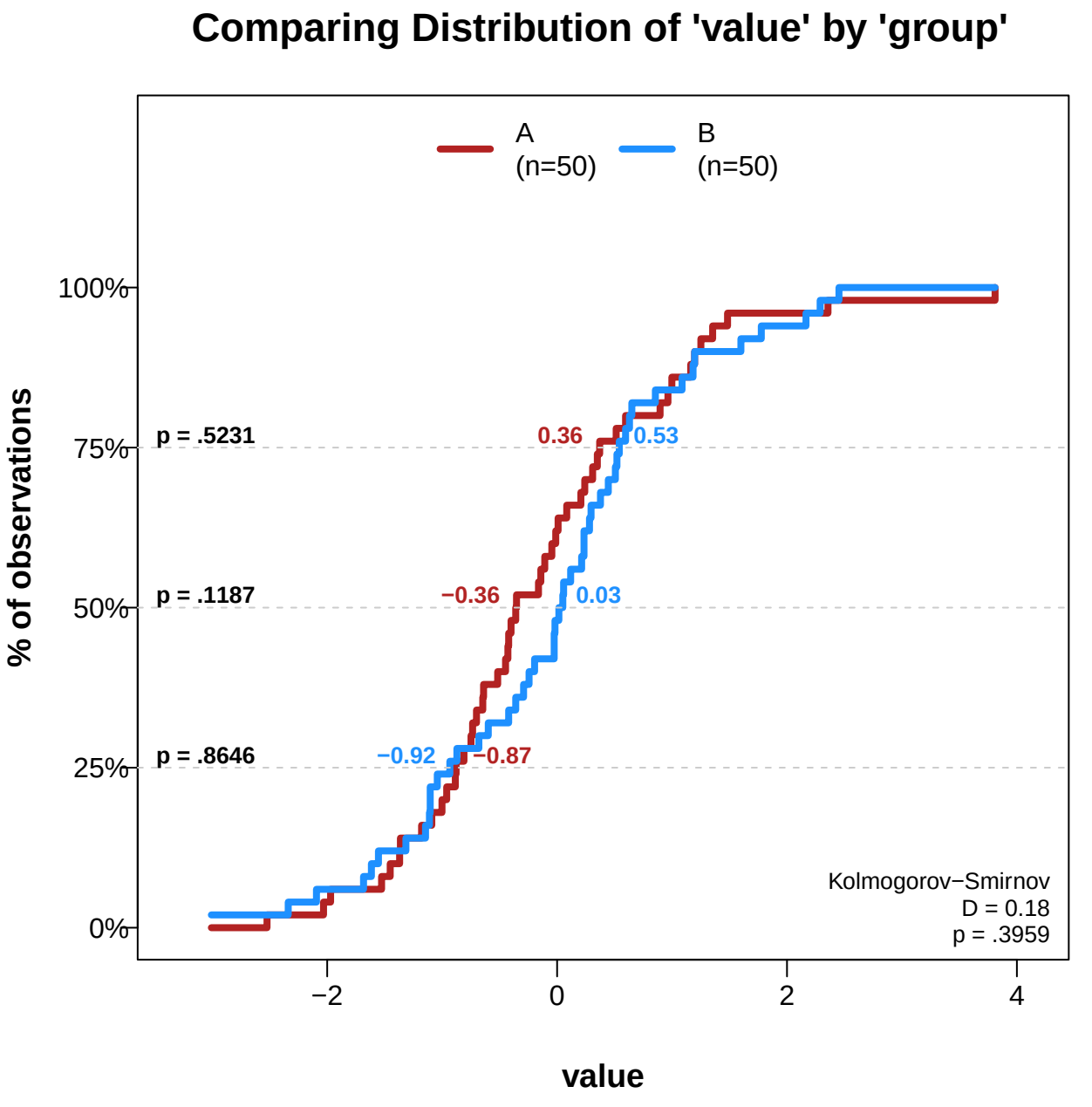

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−3
−2
−1
0
1
2
3
y

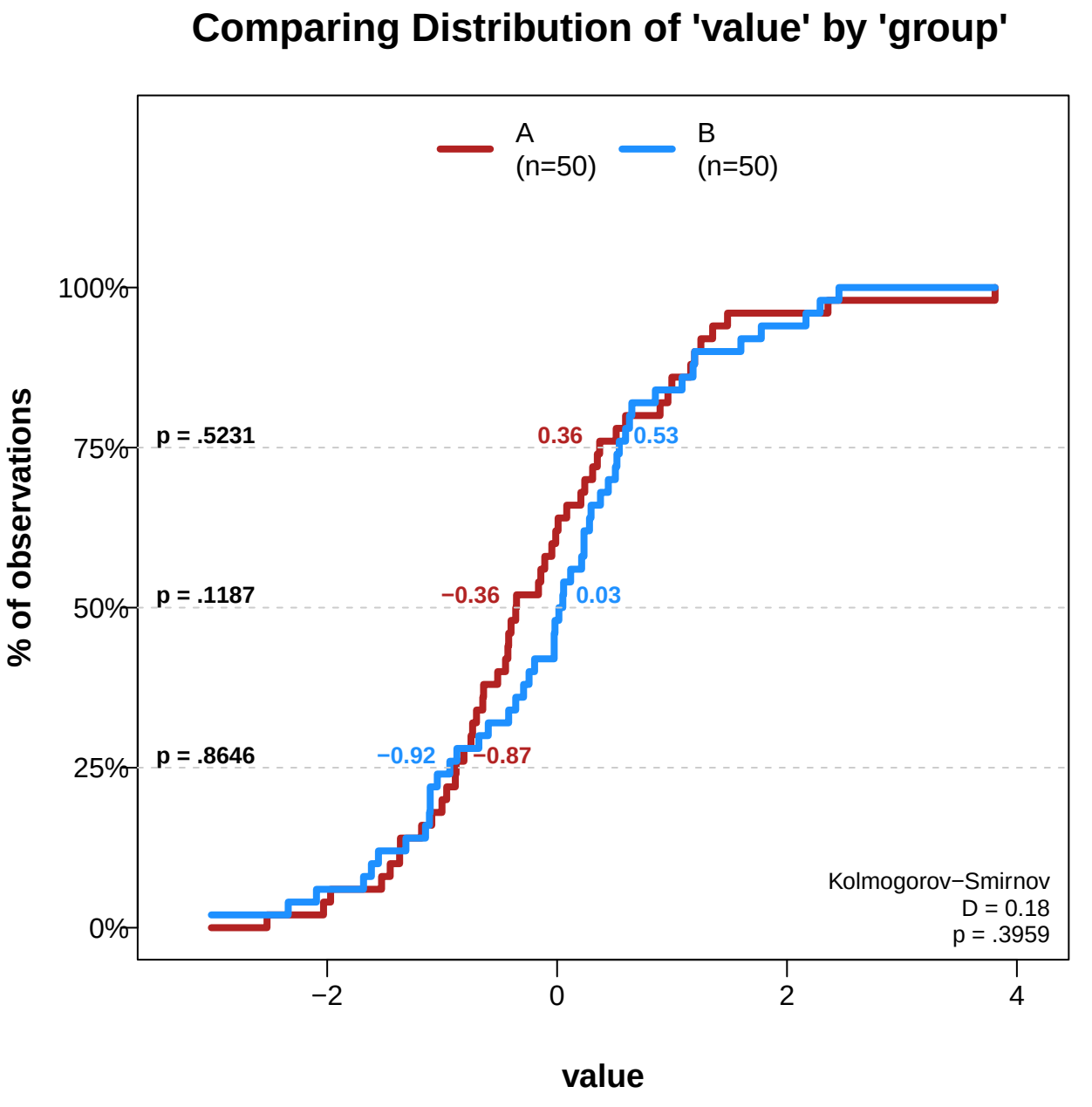

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .7879
p = .5854
p = .4820
0.74
0.79
−0.04
0.25
−0.56
−0.4
Kolmogorov−Smirnov
D = 0.16
p = .5487
−3
−2
−1
0
1
2
3
value

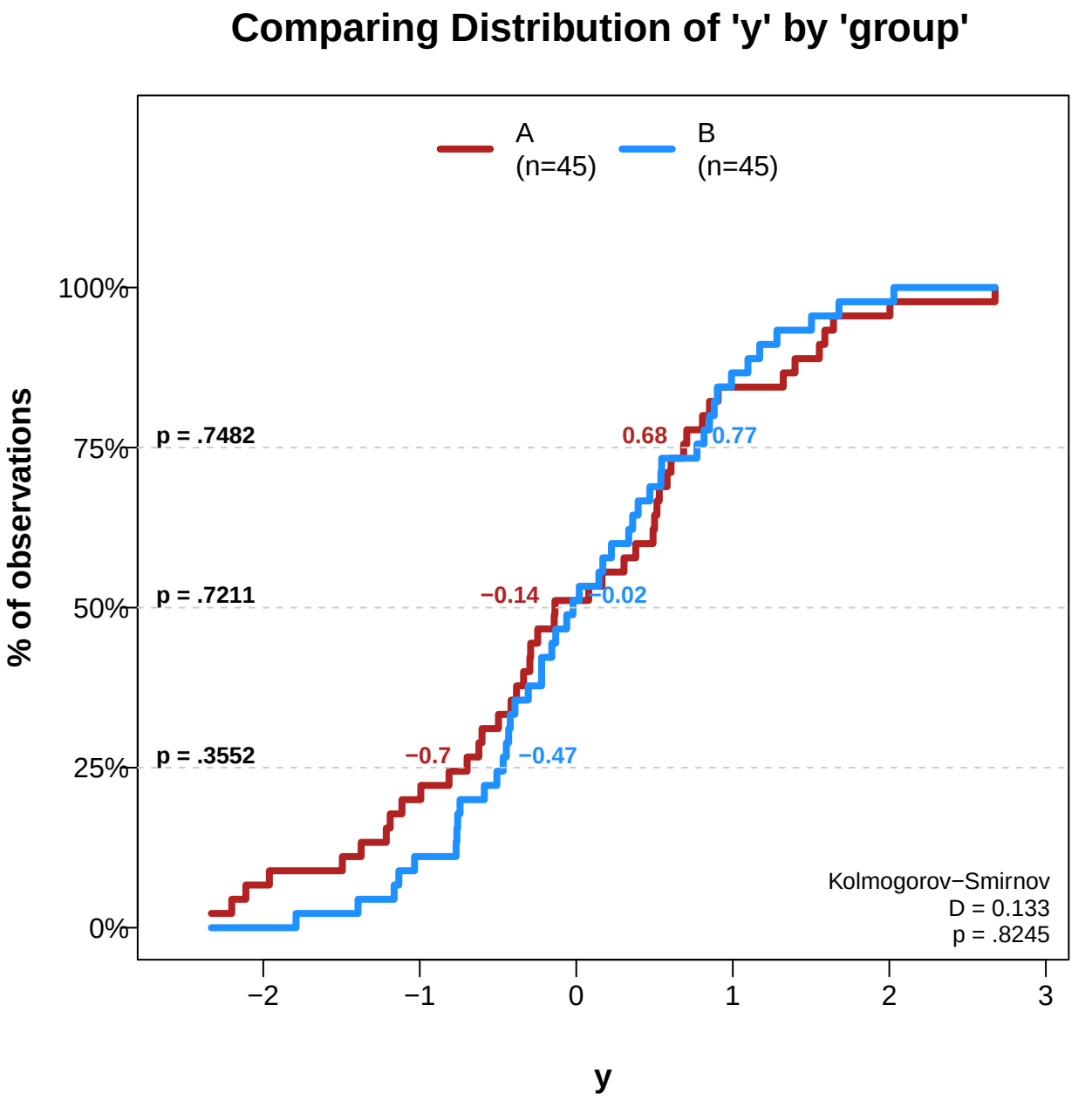

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .7879
p = .5854
p = .4820
0.74
0.79
−0.04
0.25
−0.56
−0.4
Kolmogorov−Smirnov
D = 0.16
p = .5487
−3
−2
−1
0
1
2
3
value

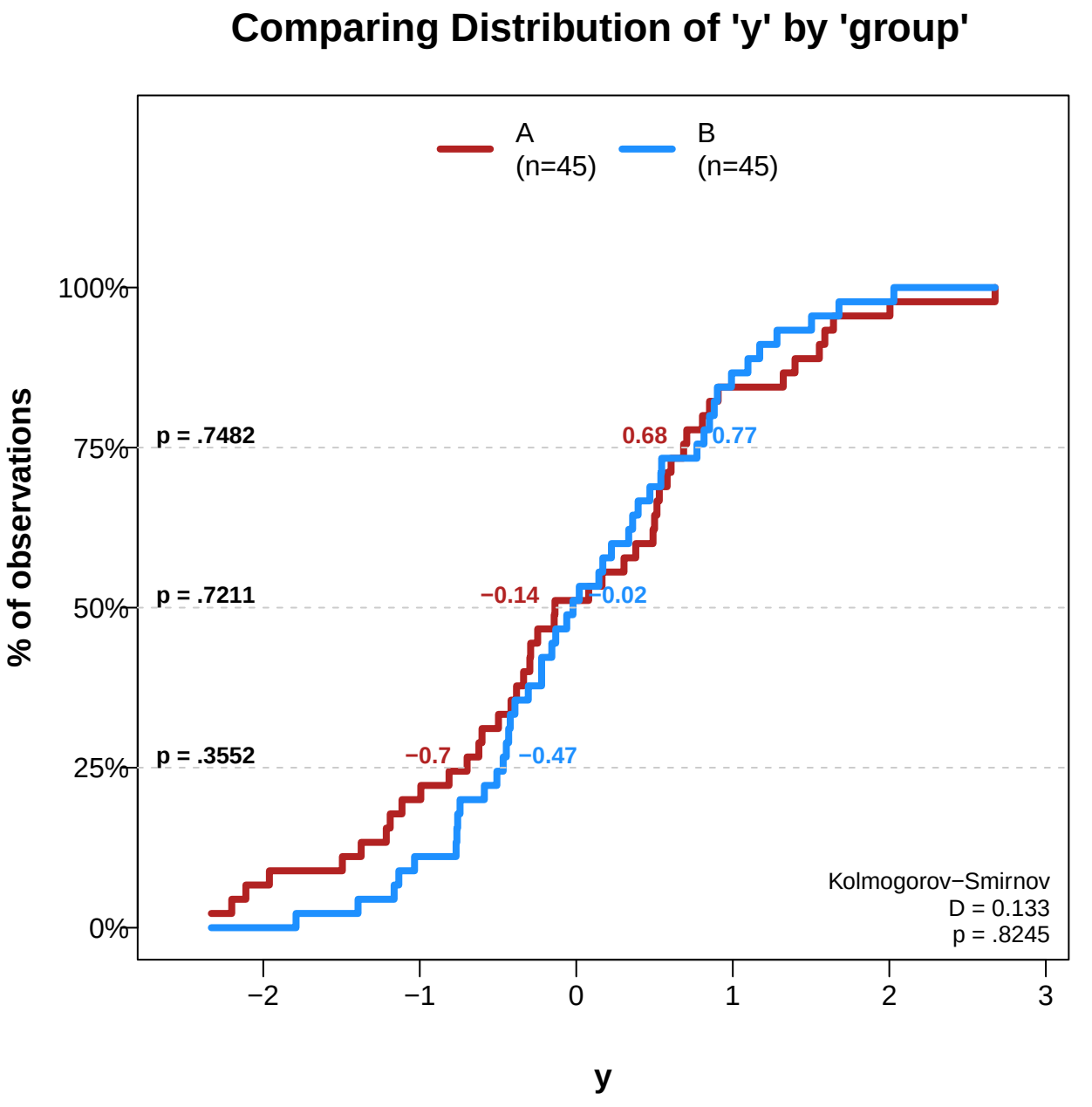

Comparing Distribution of 'y1' by 'x1'
A
(n=20)
B
(n=20)
C
(n=20)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y1

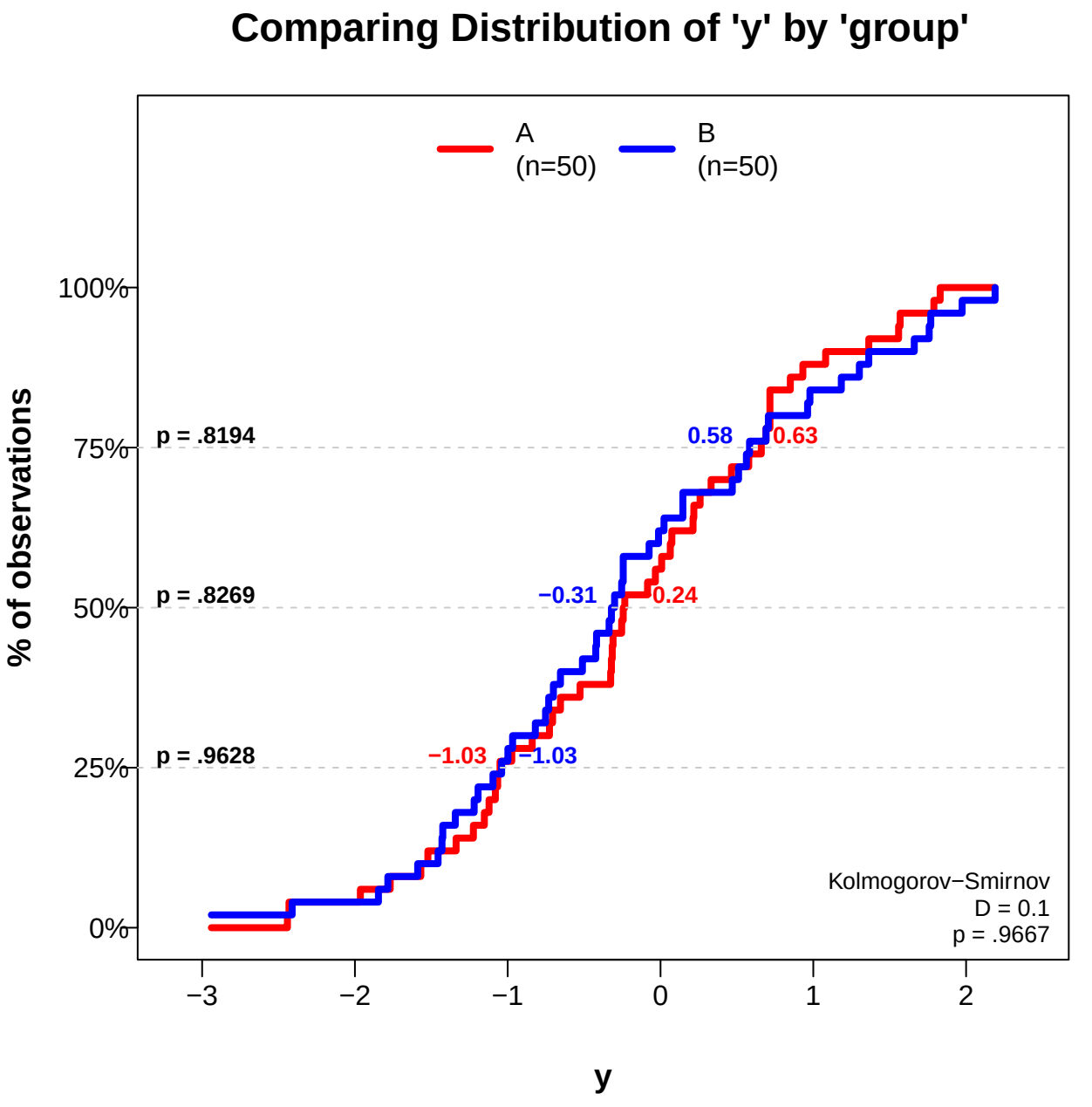

Comparing Distribution of 'y1' by 'x1'
A
(n=20)
B
(n=20)
C
(n=20)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y1

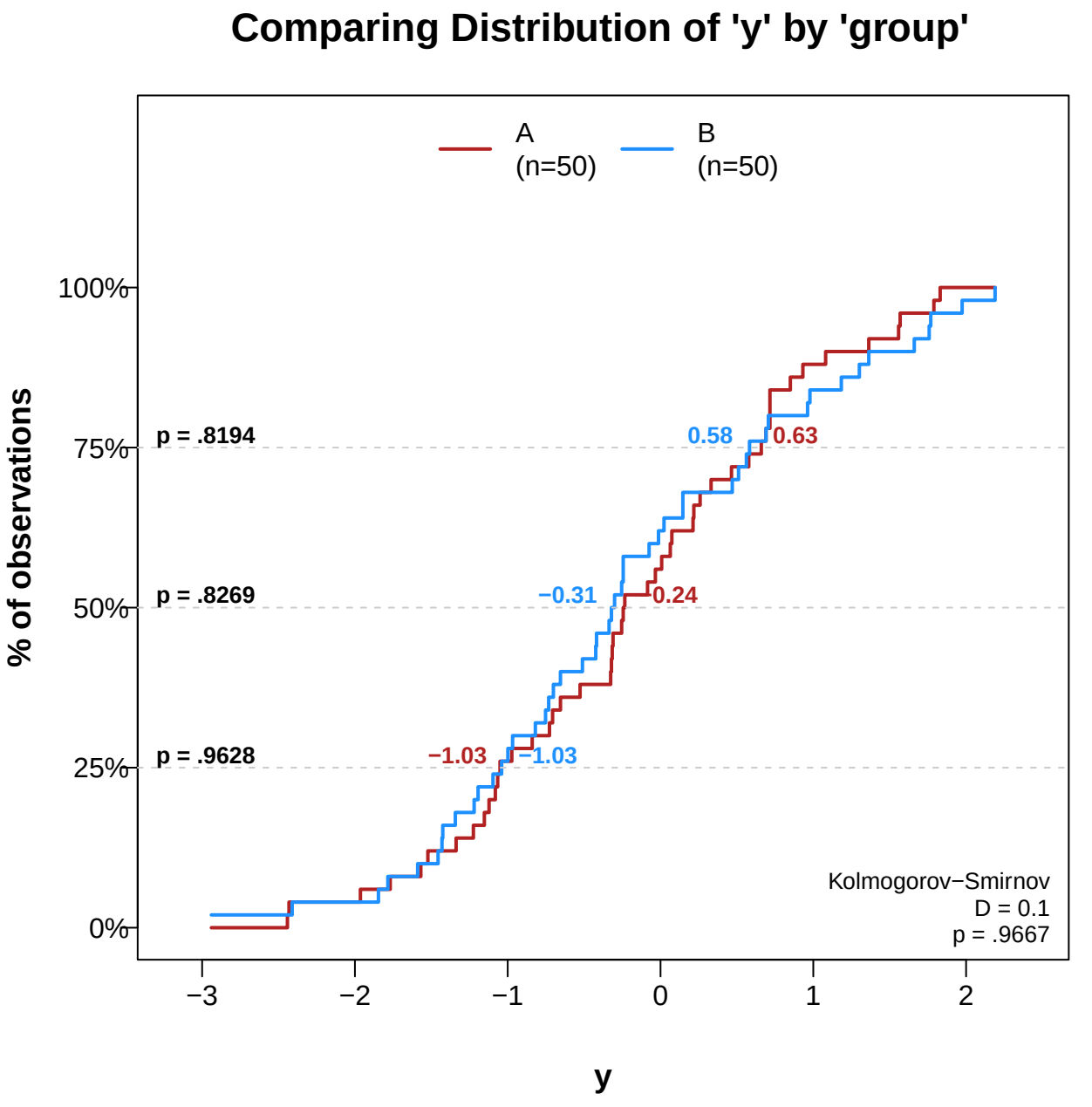

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .5231
p = .1187
p = .8646
0.36
0.53
−0.36
0.03
−0.92
−0.87
Kolmogorov−Smirnov
D = 0.18
p = .3959
−2
0
2
4
value

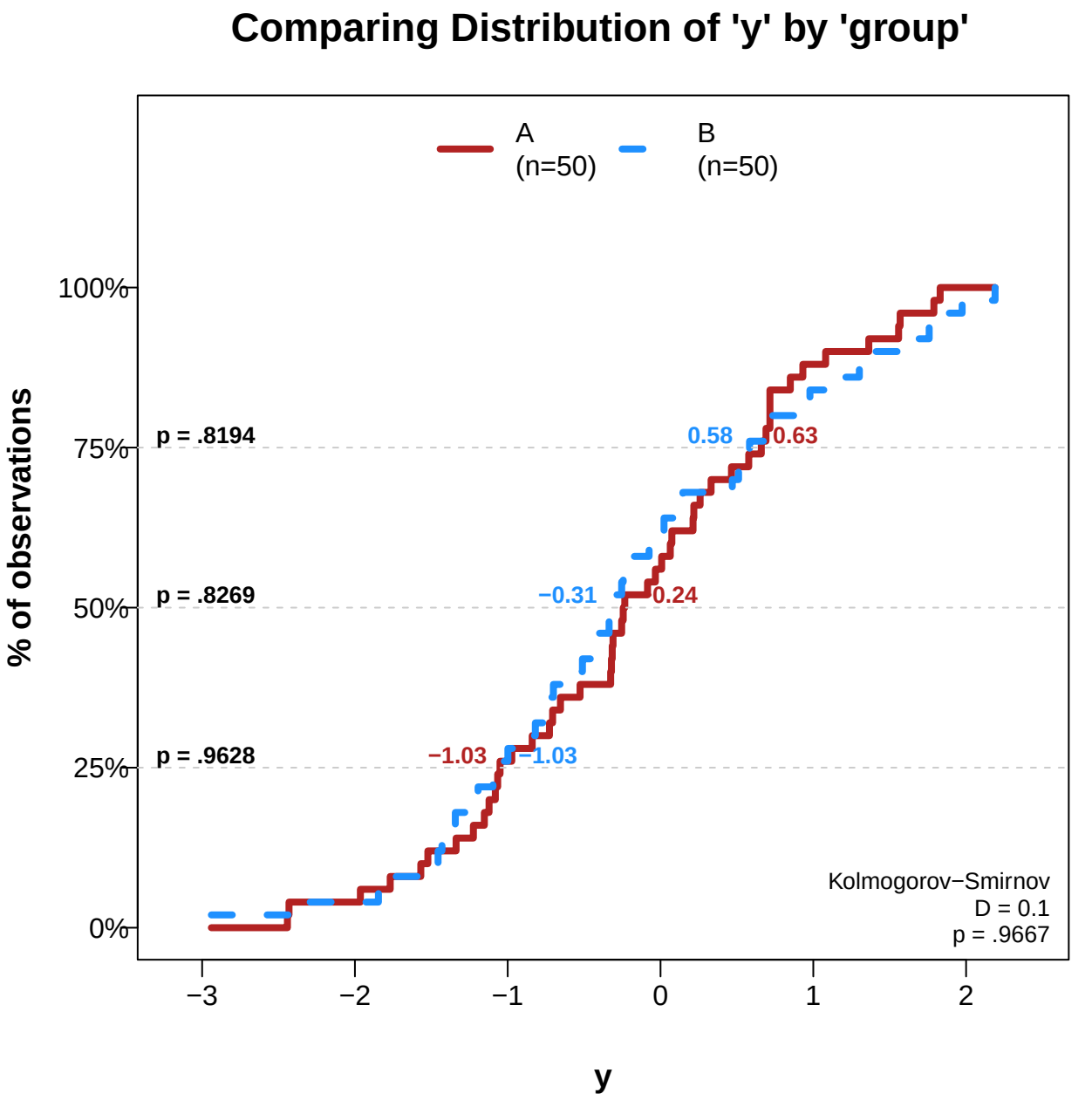

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .5231
p = .1187
p = .8646
0.36
0.53
−0.36
0.03
−0.92
−0.87
Kolmogorov−Smirnov
D = 0.18
p = .3959
−2
0
2
4
value

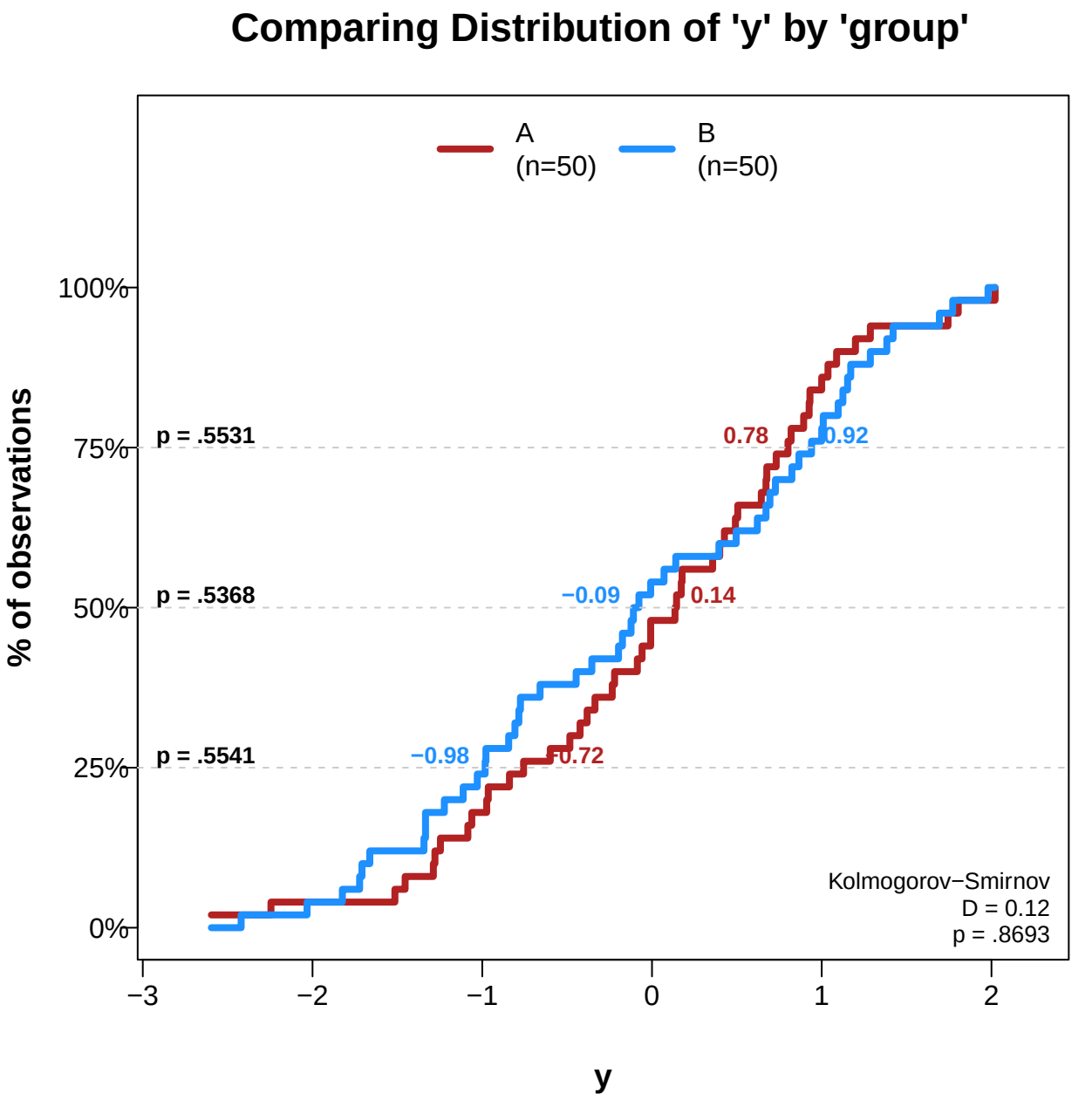

Comparing Distribution of 'y' by 'group'
A
(n=45)
B
(n=45)
100%
75%
50%
25%
0%
% of observations
p = .7482
p = .7211
p = .3552
0.68
0.77
−0.14
−0.02
−0.7
−0.47
Kolmogorov−Smirnov
D = 0.133
p = .8245
−2
−1
0
1
2
3
y

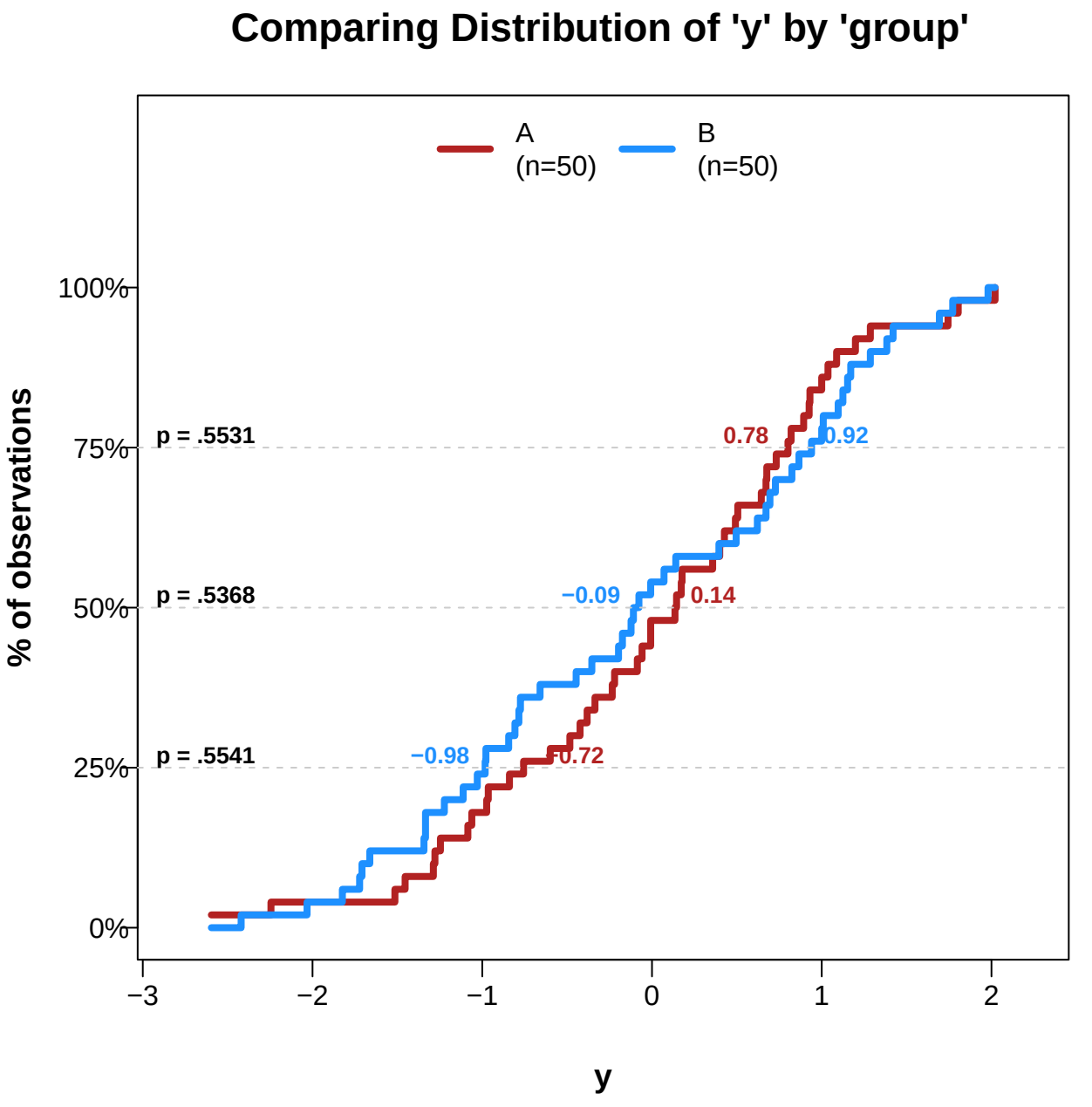

Comparing Distribution of 'y' by 'group'
A
(n=45)
B
(n=45)
100%
75%
50%
25%
0%
% of observations
p = .7482
p = .7211
p = .3552
0.68
0.77
−0.14
−0.02
−0.7
−0.47
Kolmogorov−Smirnov
D = 0.133
p = .8245
−2
−1
0
1
2
3
y

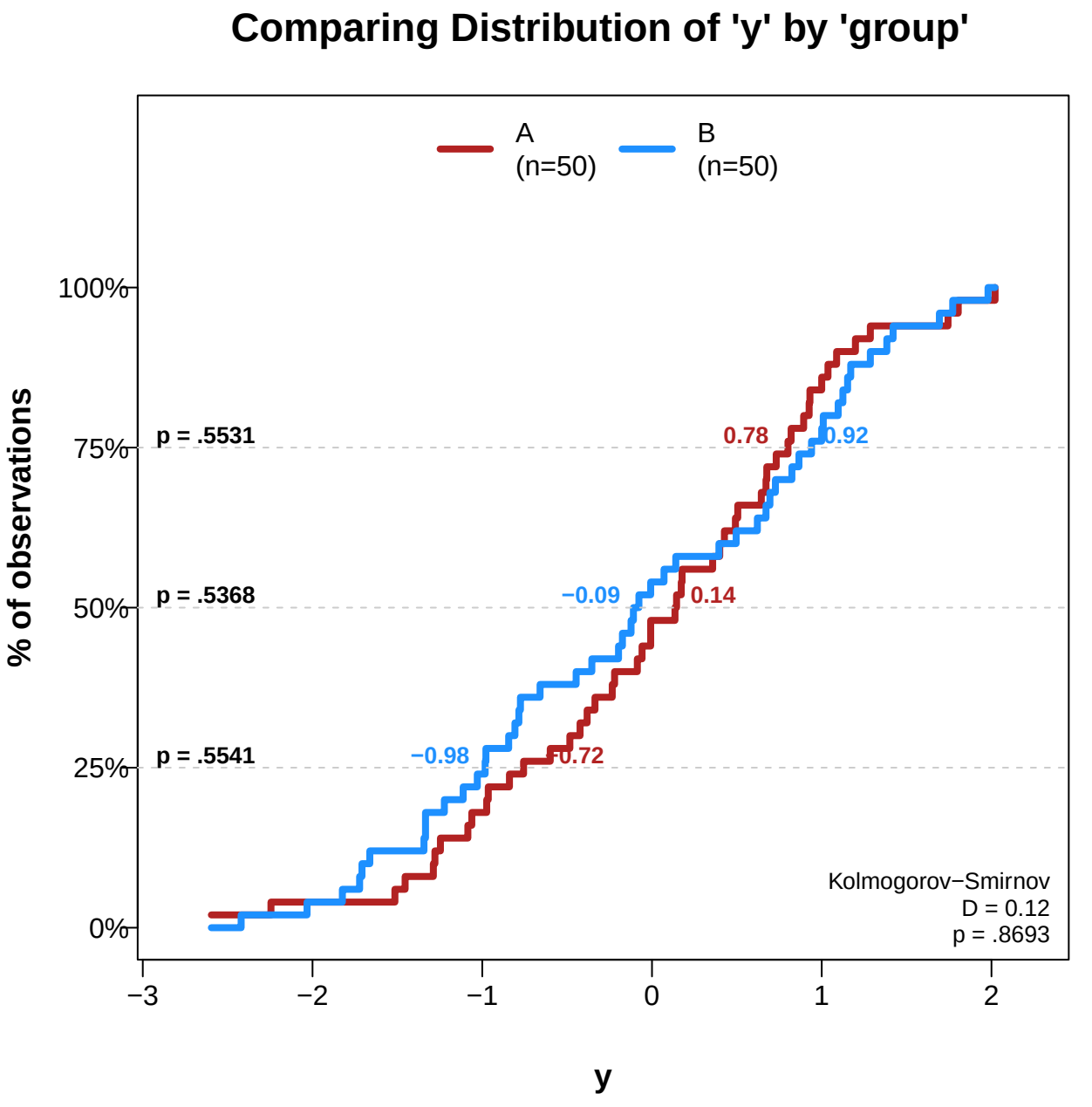

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .8194
p = .8269
p = .9628
0.58
0.63
−0.31
0.24
−1.03
−1.03
Kolmogorov−Smirnov
D = 0.1
p = .9667
−3
−2
−1
0
1
2
y

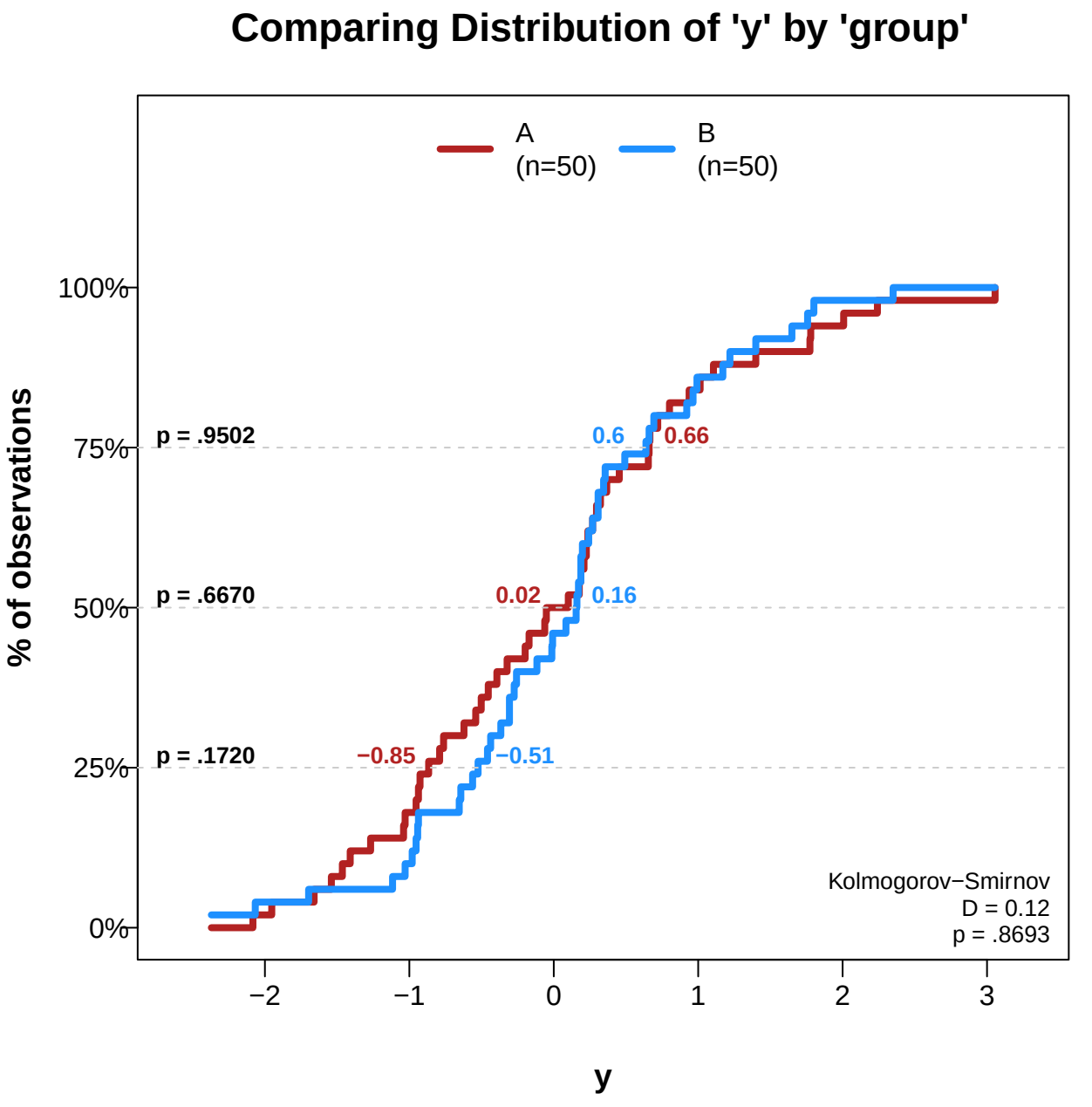

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .8194
p = .8269
p = .9628
0.58
0.63
−0.31
0.24
−1.03
−1.03
Kolmogorov−Smirnov
D = 0.1
p = .9667
−3
−2
−1
0
1
2
y

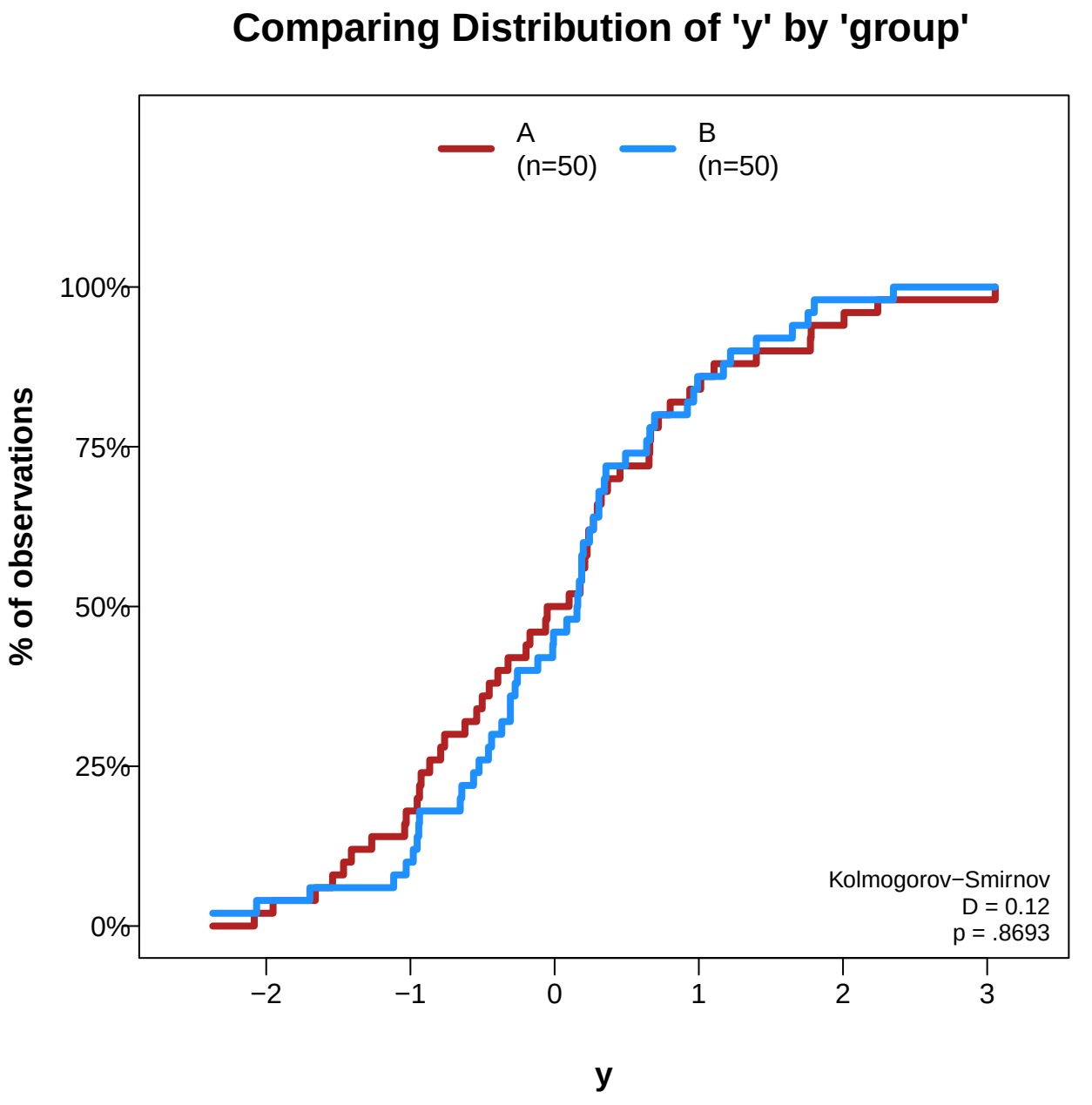

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .8194
p = .8269
p = .9628
0.58
0.63
−0.31
0.24
−1.03
−1.03
Kolmogorov−Smirnov
D = 0.1
p = .9667
−3
−2
−1
0
1
2
y

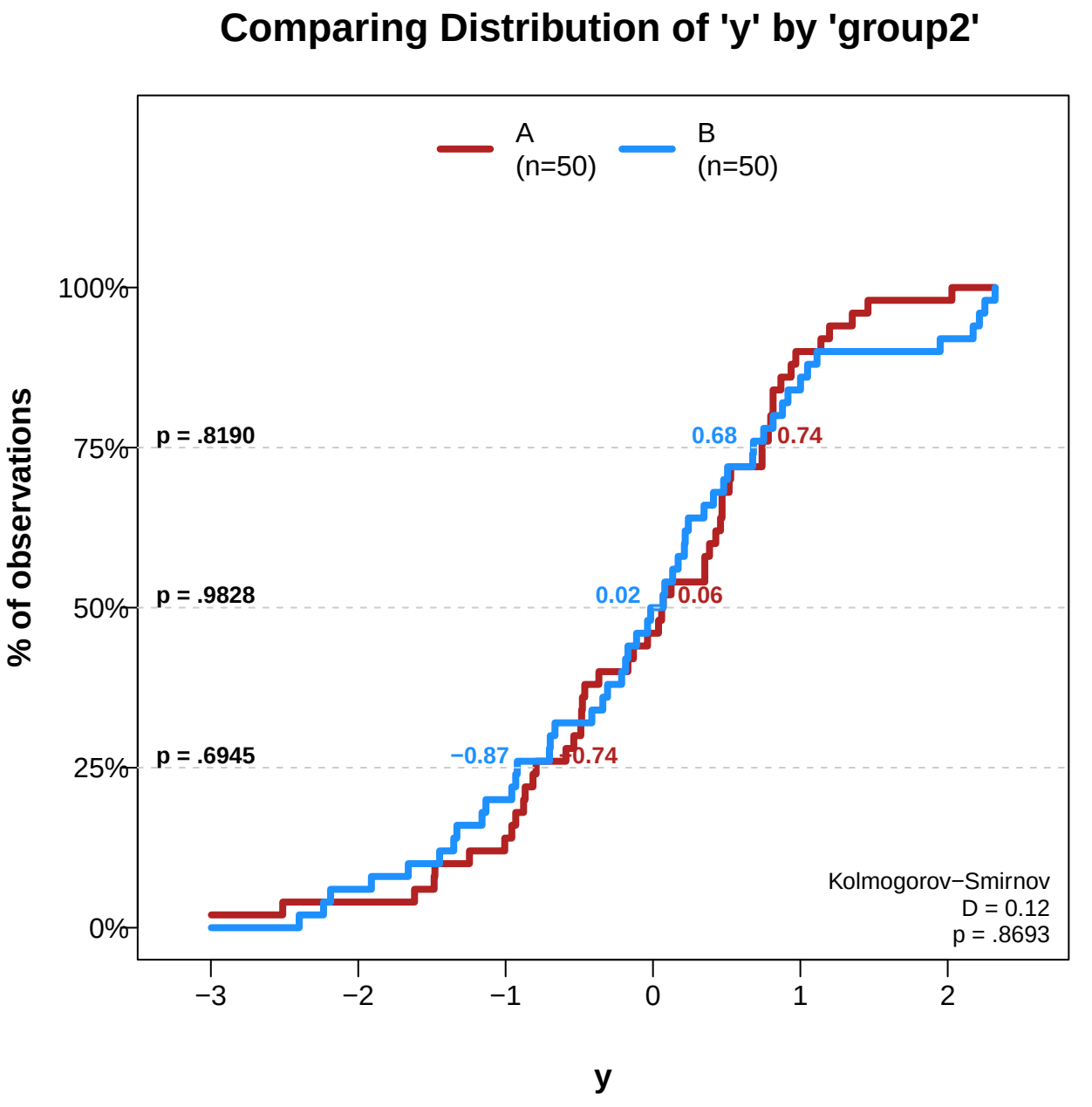

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .5531
p = .5368
p = .5541
0.78
0.92
−0.09
0.14
−0.98
−0.72
Kolmogorov−Smirnov
D = 0.12
p = .8693
−3
−2
−1
0
1
2
y

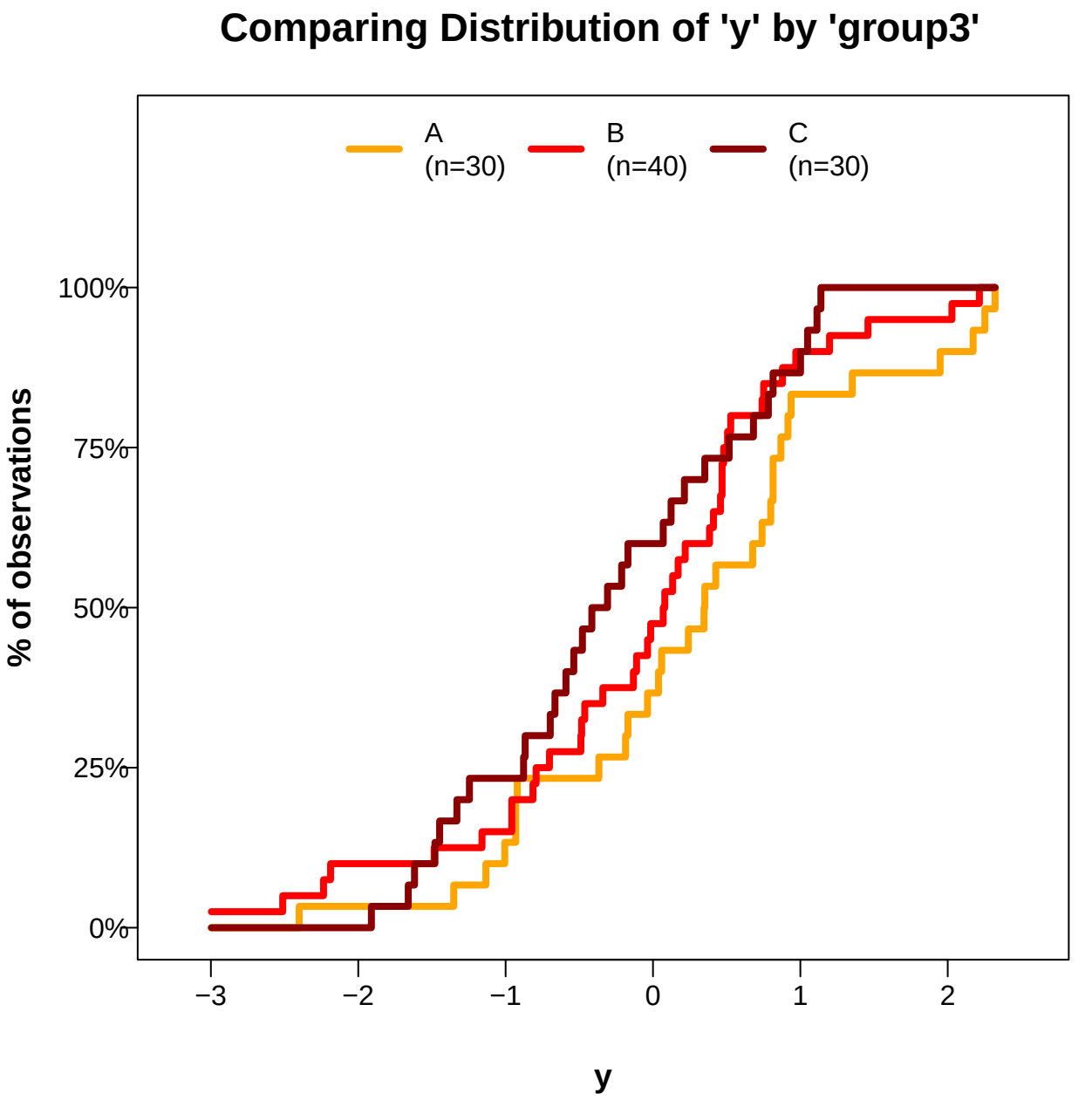

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .5531
p = .5368
p = .5541
0.78
0.92
−0.09
0.14
−0.98
0.72
−3
−2
−1
0
1
2
y

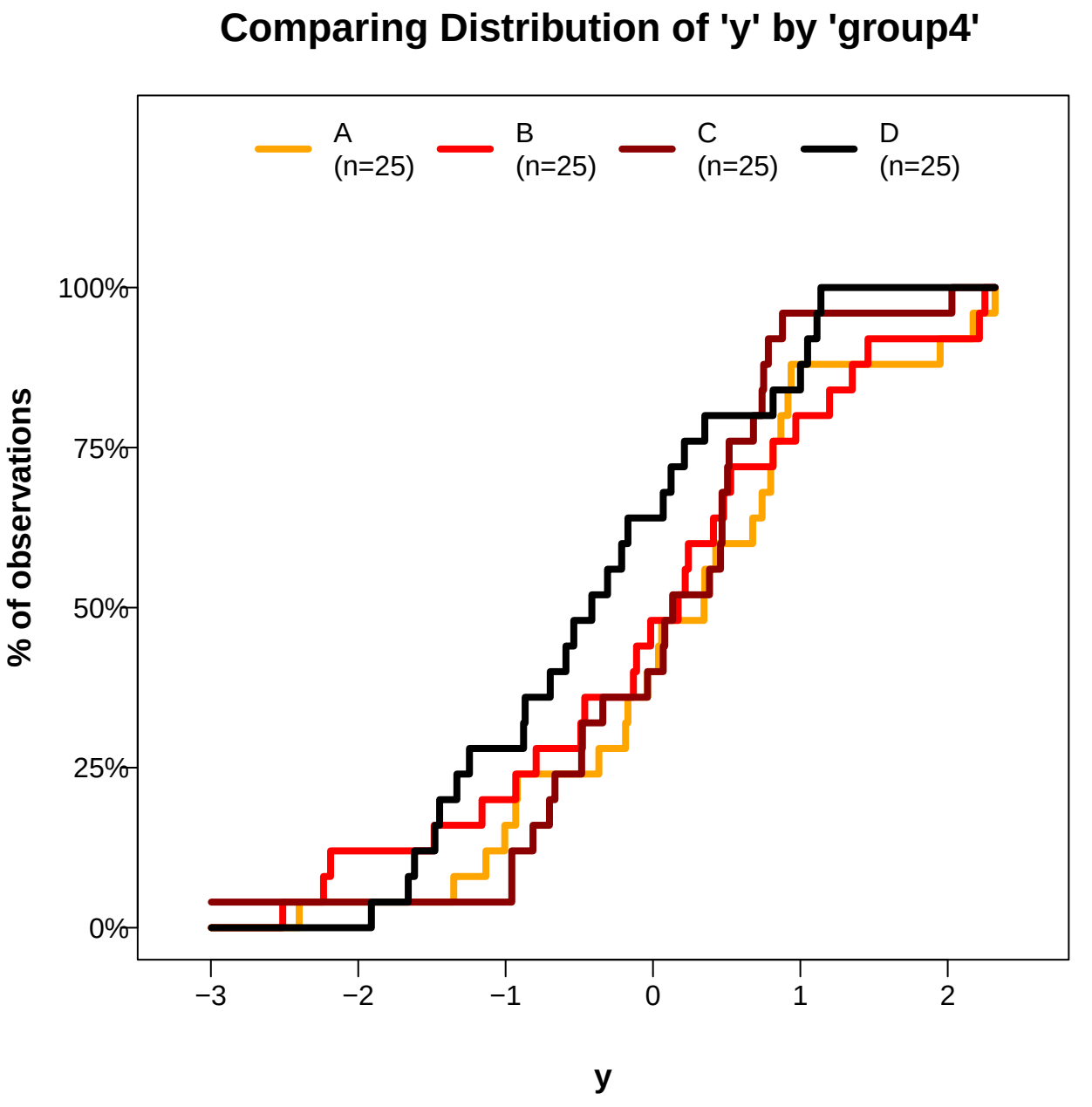

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .5531
p = .5368
p = .5541
0.78
0.92
−0.09
0.14
−0.98
−0.72
Kolmogorov−Smirnov
D = 0.12
p = .8693
−3
−2
−1
0
1
2
y

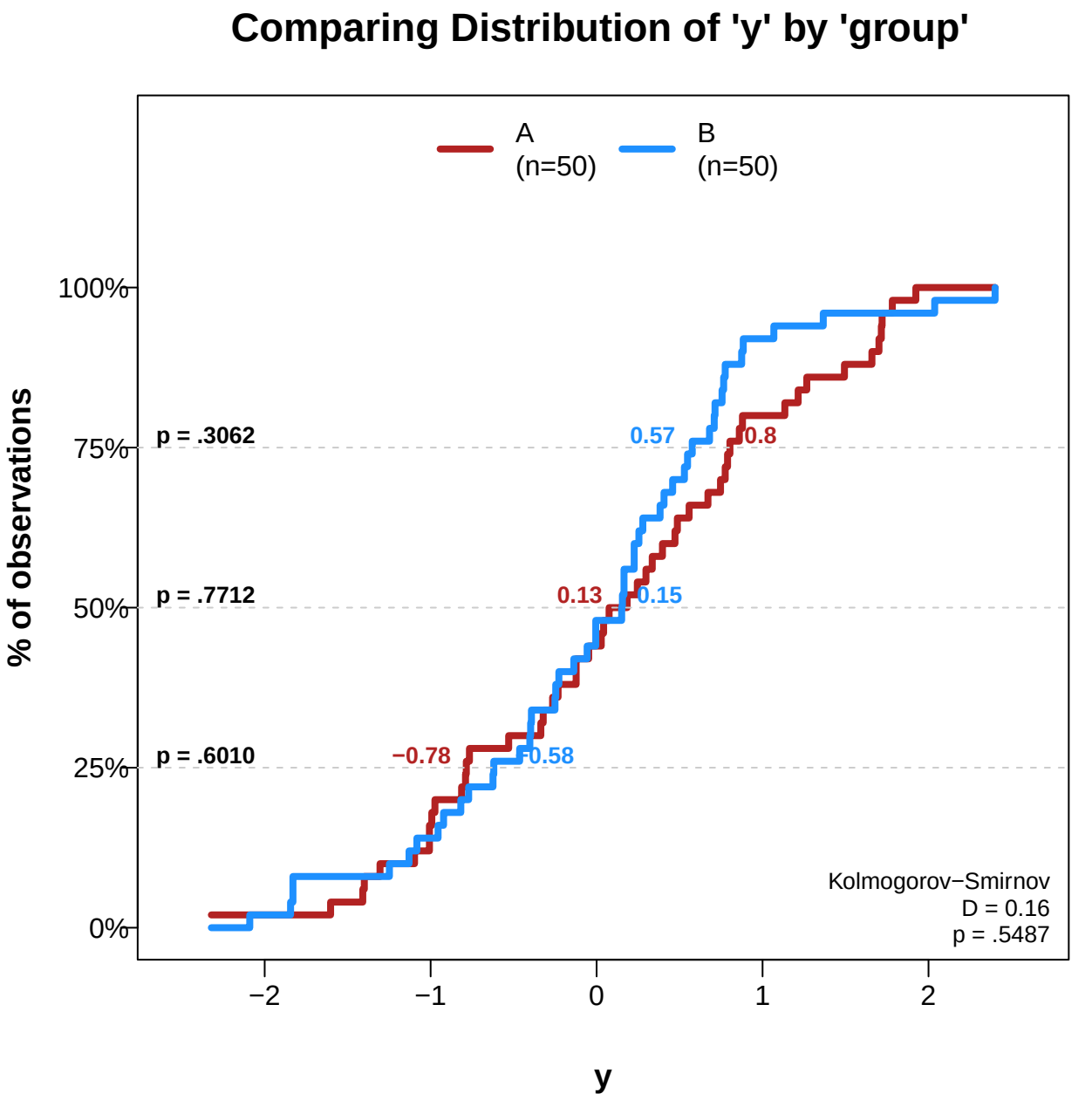

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .9502
p = .6670
p = .1720
0.6
0.66
0.02
0.16
−0.85
−0.51
Kolmogorov−Smirnov
D = 0.12
p = .8693
−2
−1
0
1
2
3
y

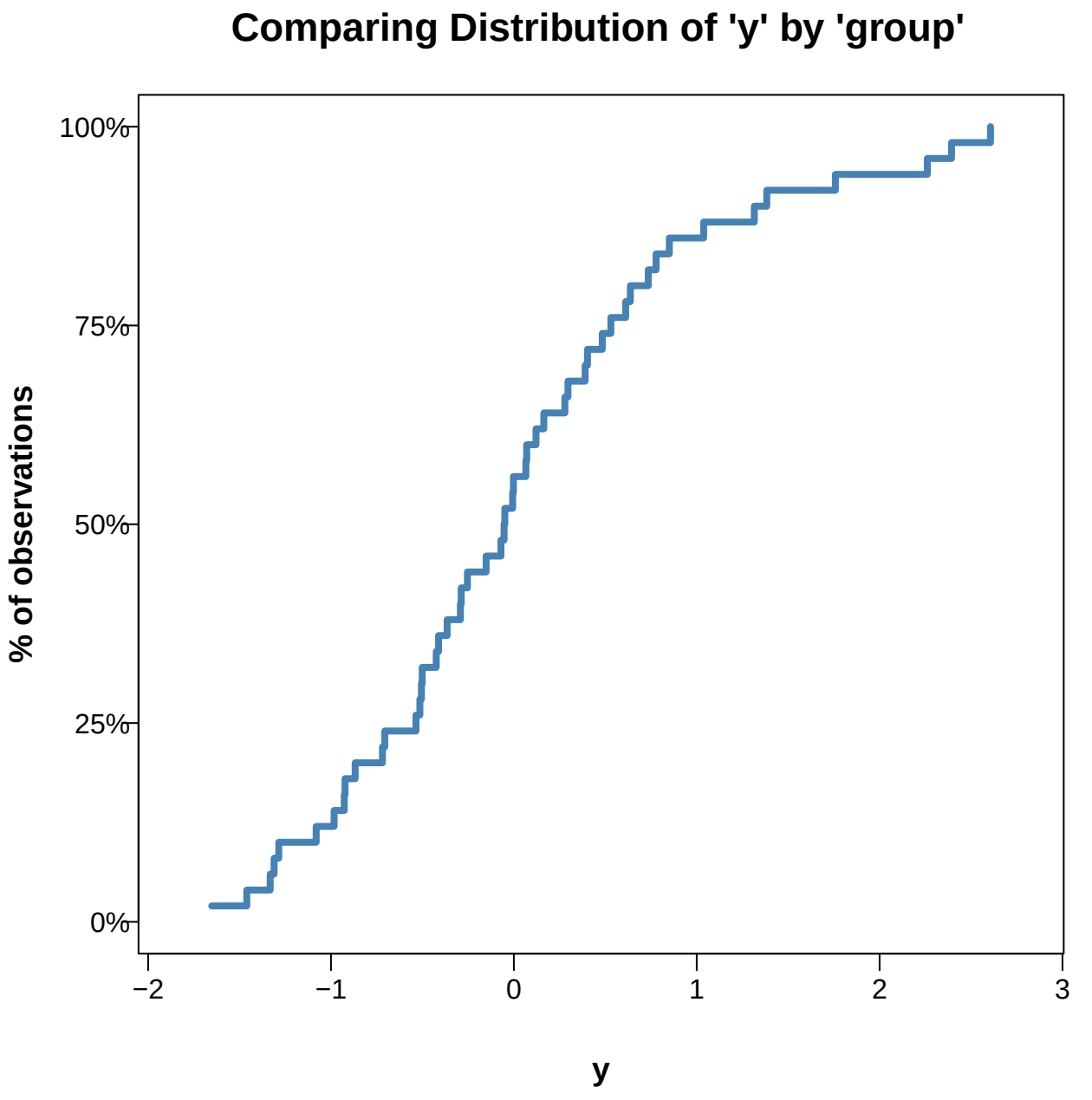

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
3
y
Kolmogorov−Smirnov
D = 0.12
p = .8693

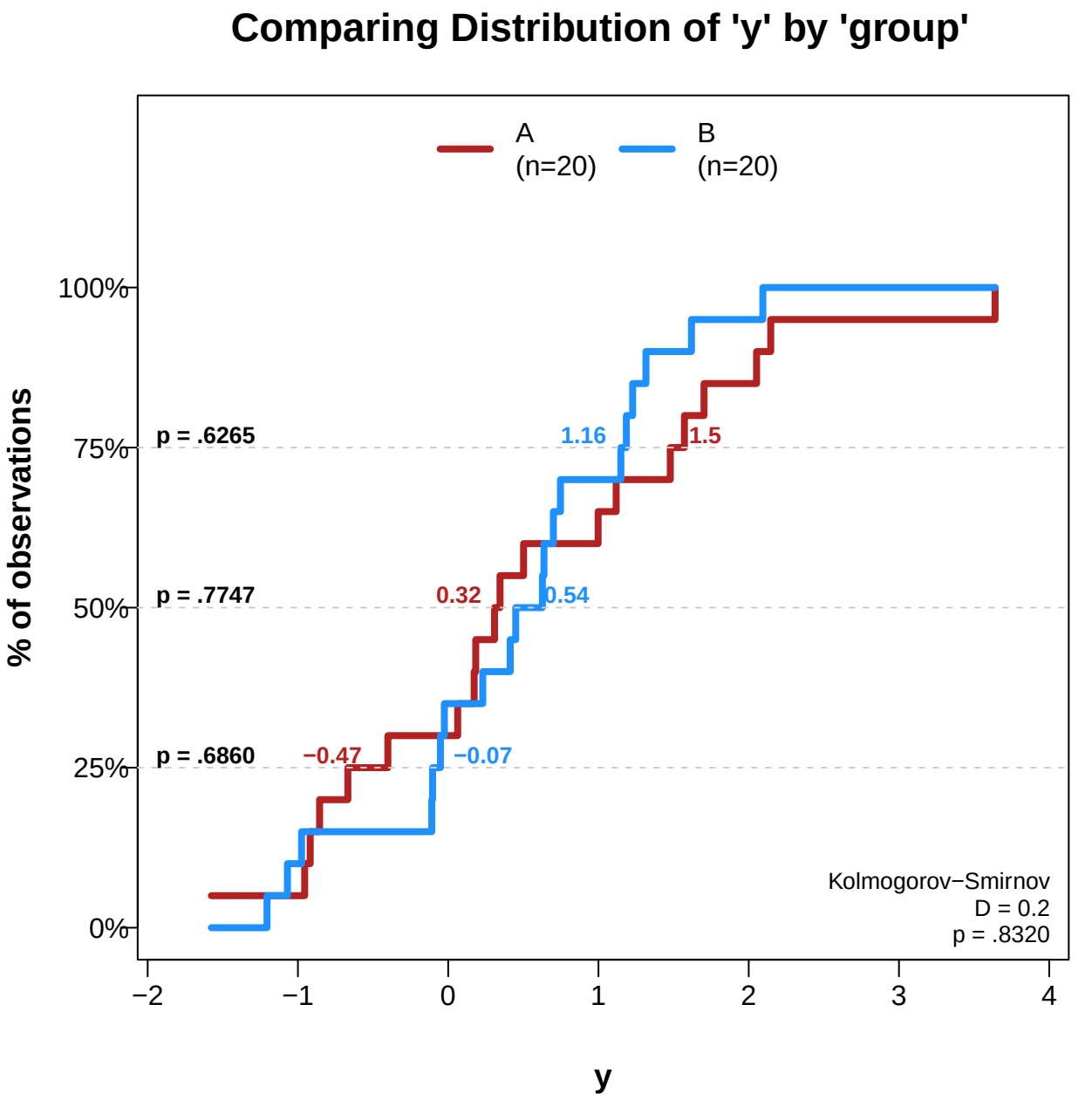

Comparing Distribution of 'y' by 'group2'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .8190
p = .9828
p = .6945
0.68
0.74
0.02
0.06
−0.87
0.74
Kolmogorov−Smirnov
D = 0.12
p = .8693
−3
−2
−1
0
1
2
y

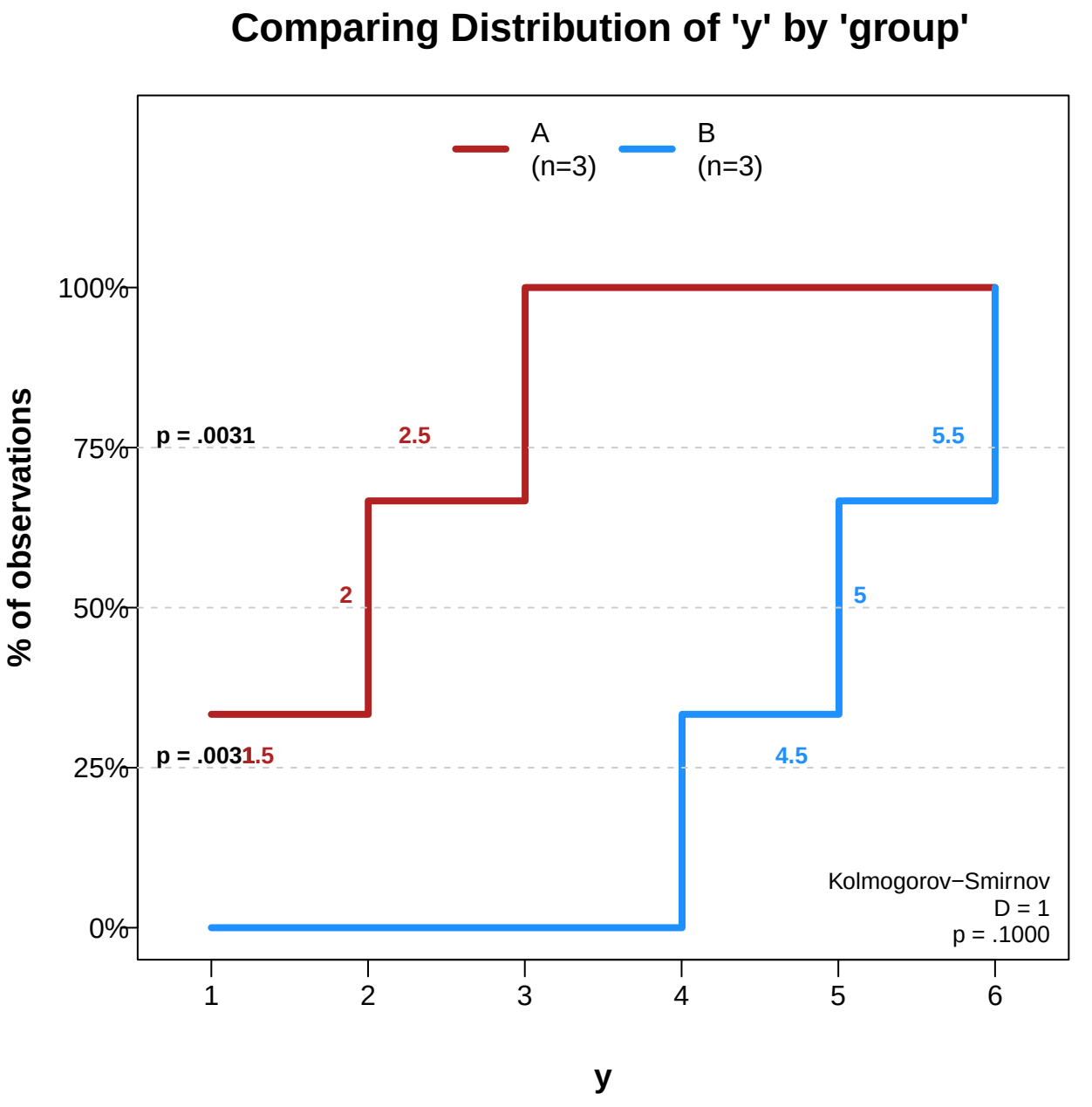

Comparing Distribution of 'y' by 'group3'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−3
−2
−1
0
1
2
y

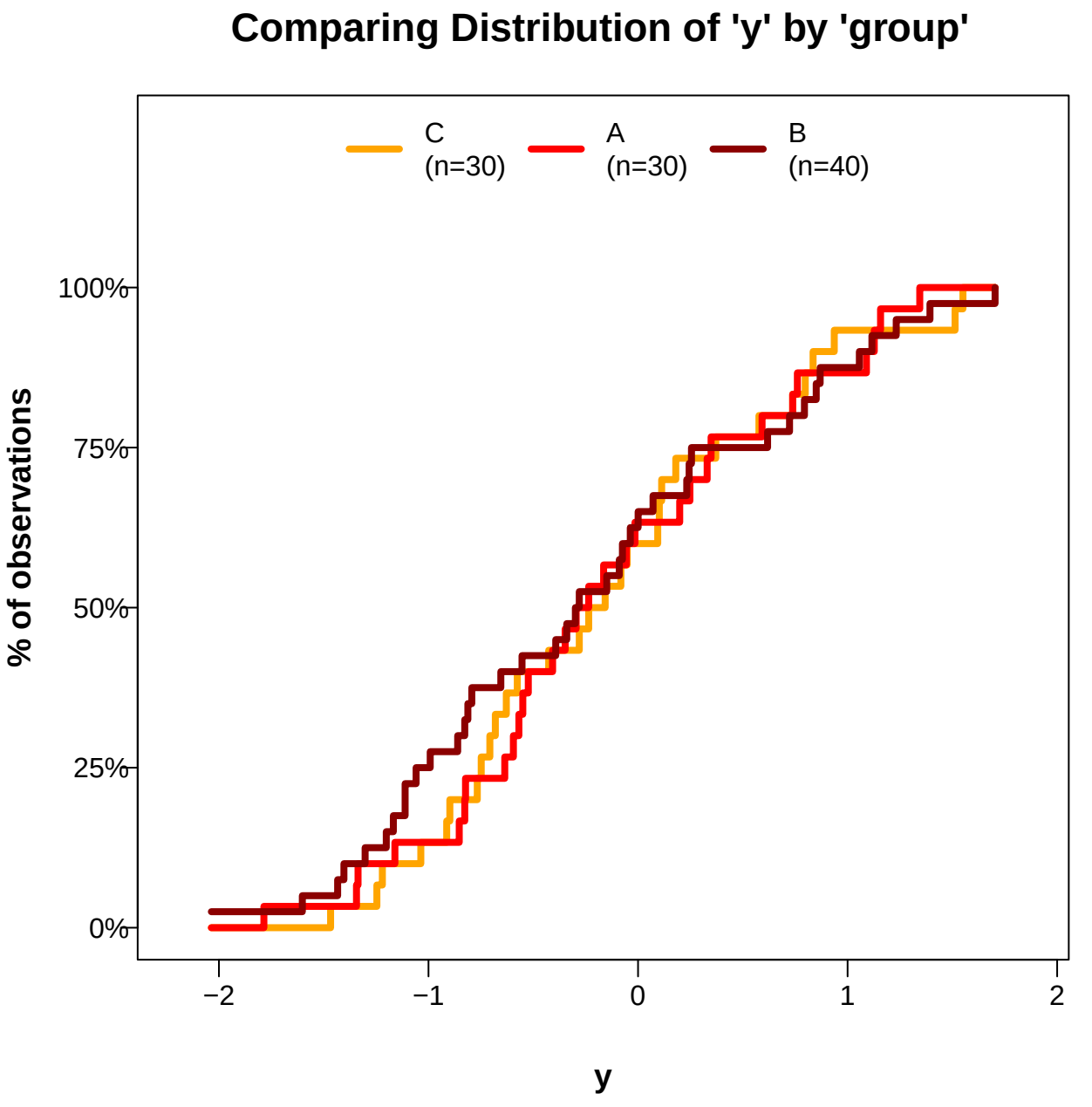

Comparing Distribution of 'y' by 'group4'
A
(n=25)
B
(n=25)
C
(n=25)
D
(n=25)
100%
75%
50%
25%
0%
% of observations
−3
−2
−1
0
1
2
y

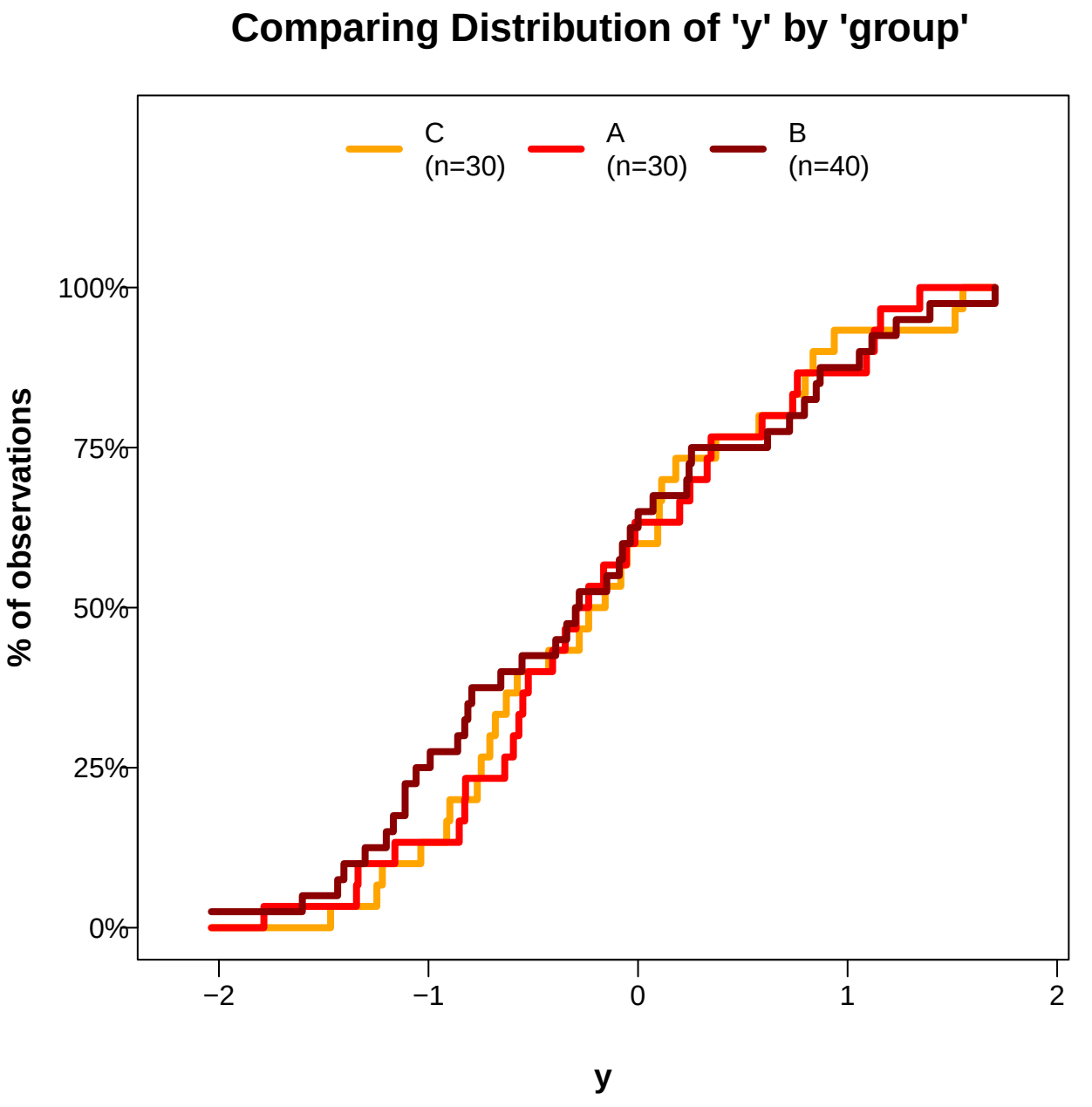

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .3062
p = .7712
p = .6010
0.57
0.8
0.13
0.15
−0.78
−0.58
Kolmogorov−Smirnov
D = 0.16
p = .5487
−2
−1
0
1
2
y

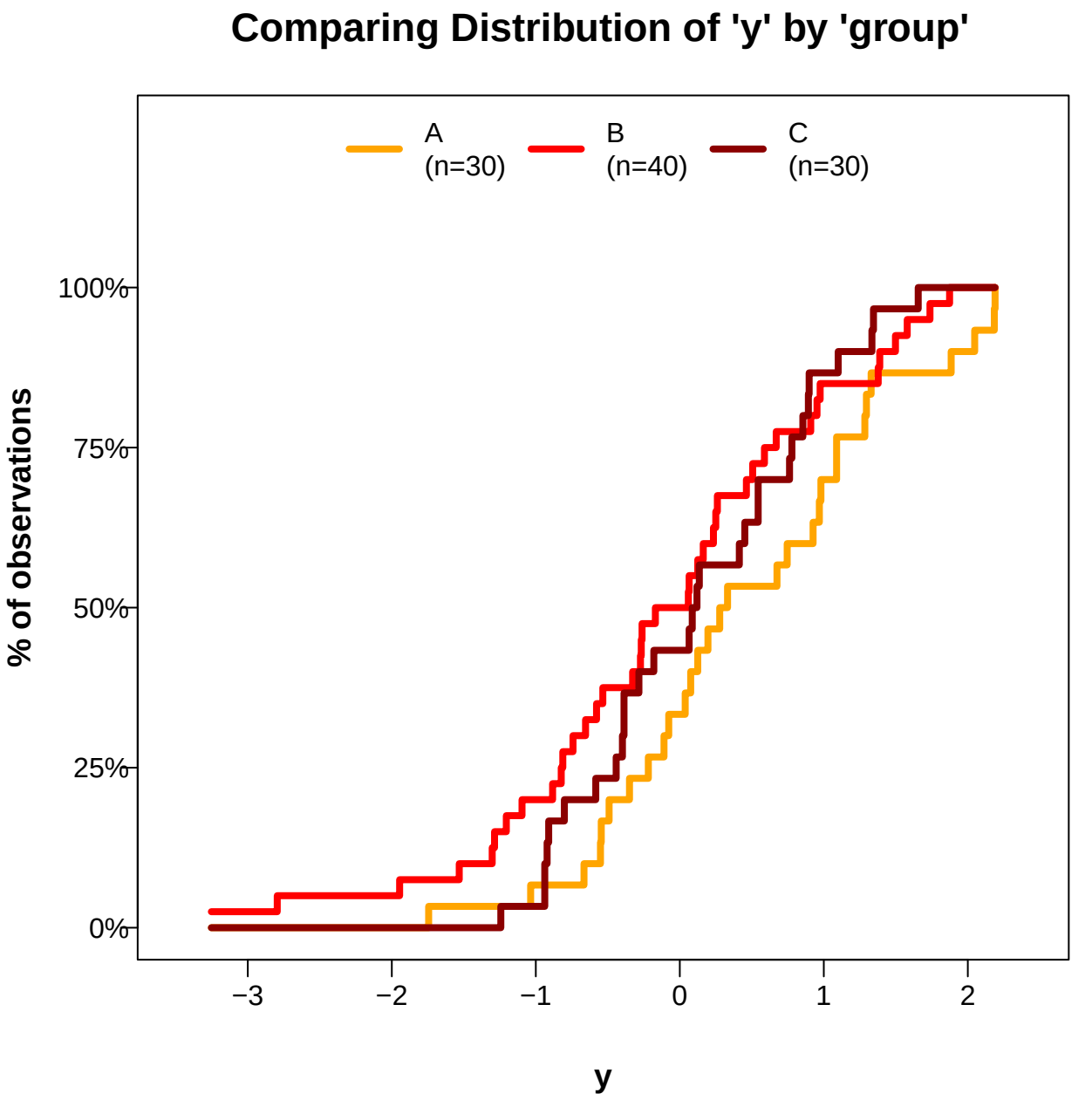

Comparing Distribution of 'y' by 'group'
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
3
y

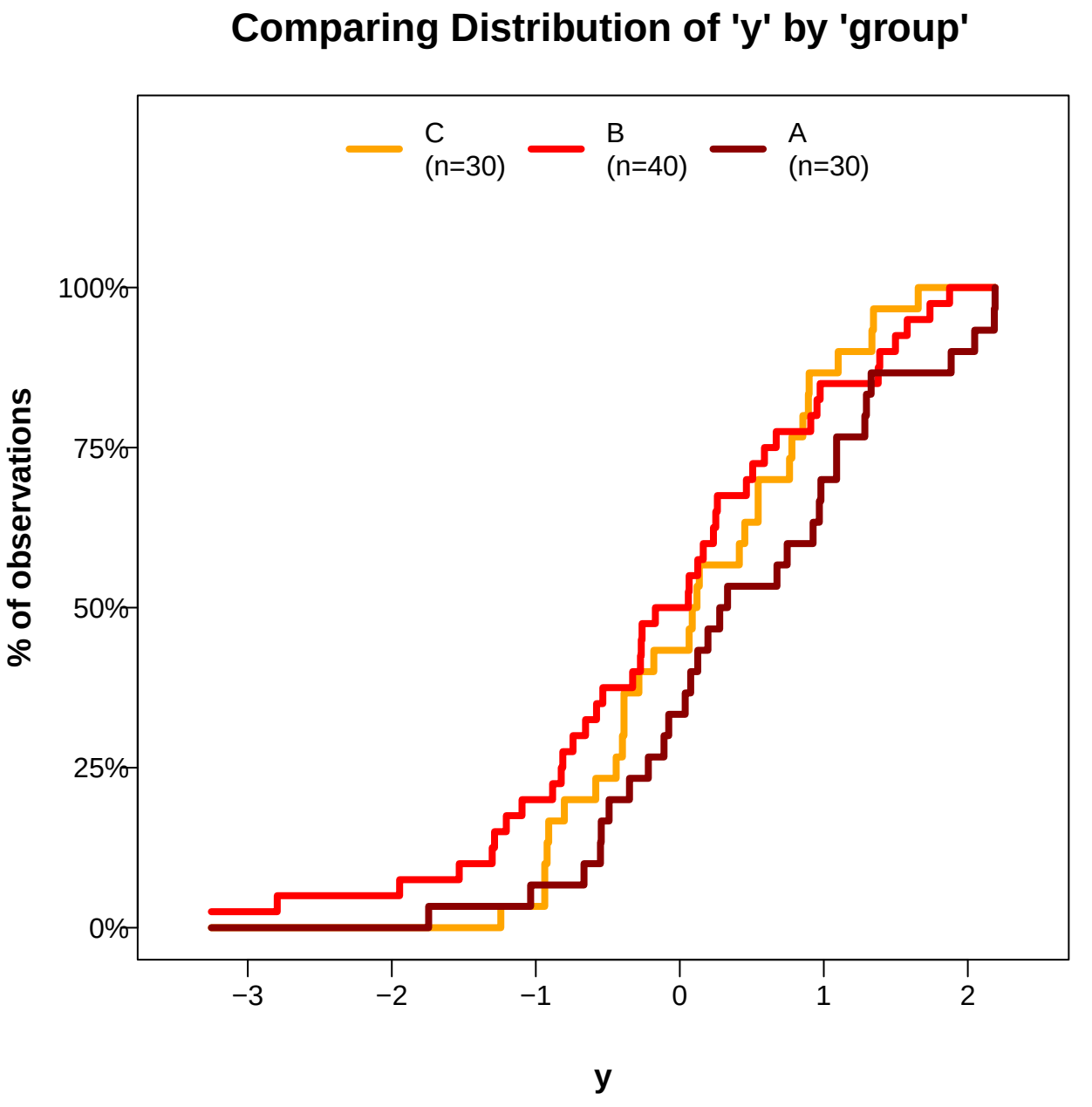

Comparing Distribution of 'y' by 'group'
A
(n=20)
B
(n=20)
100%
75%
50%
25%
0%
% of observations
p = .6265
p = .7747
p = .6860
1.16
1.5
0.32
0.54
−0.47
−0.07
Kolmogorov−Smirnov
D = 0.2
p = .8320
−2
−1
0
1
2
3
4
y

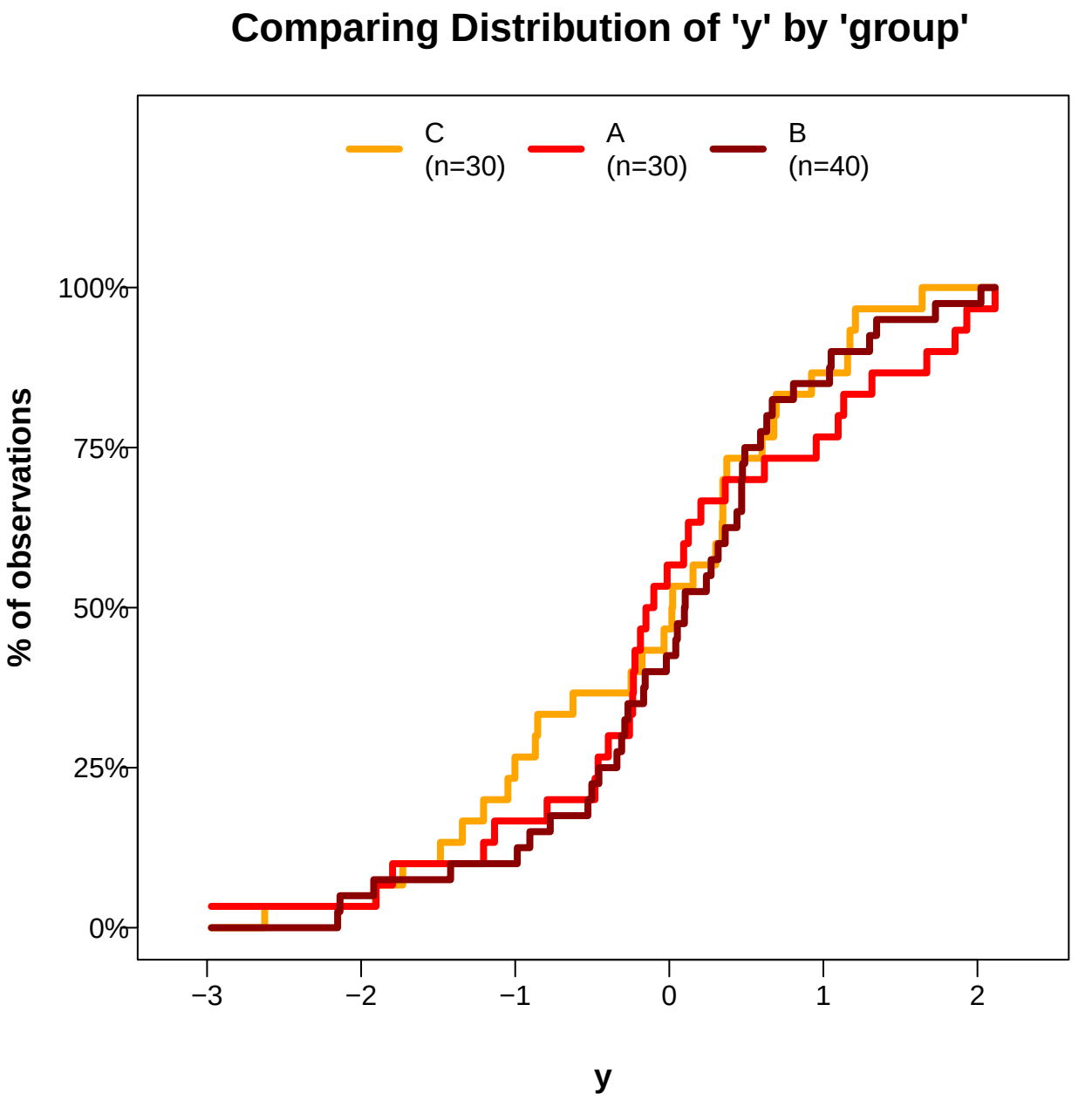

Comparing Distribution of 'y' by 'group'
A
(n=3)
B
(n=3)
100%
75%
50%
25%
0%
% of observations
p = .0031
2.5
5.5
2
5
p = .0031.5
4.5
Kolmogorov−Smirnov
D = 1
p = .1000
1
2
3
4
5
6
y

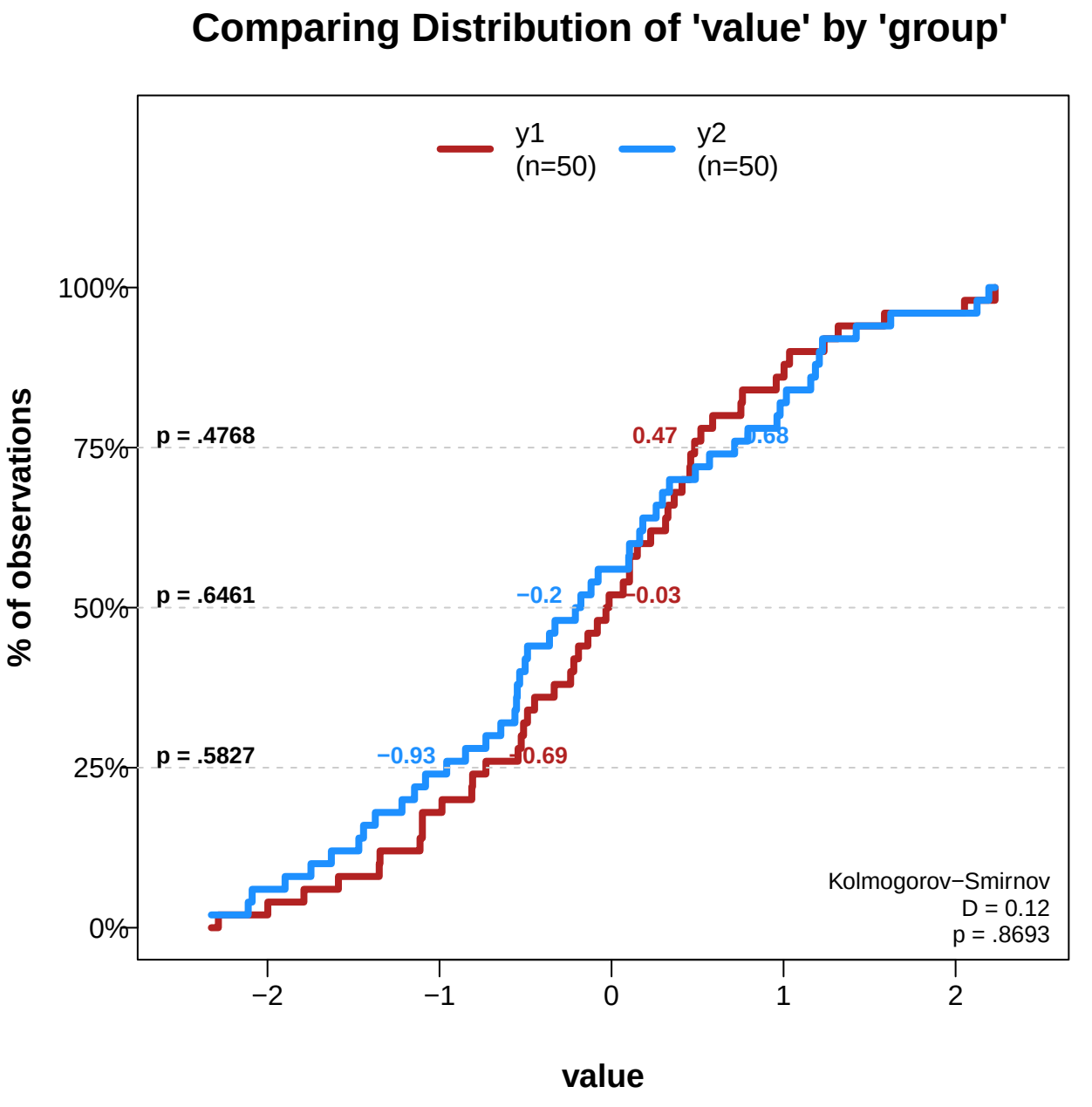

Comparing Distribution of 'y' by 'group'
C
(n=30)
A
(n=30)
B
(n=40)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y

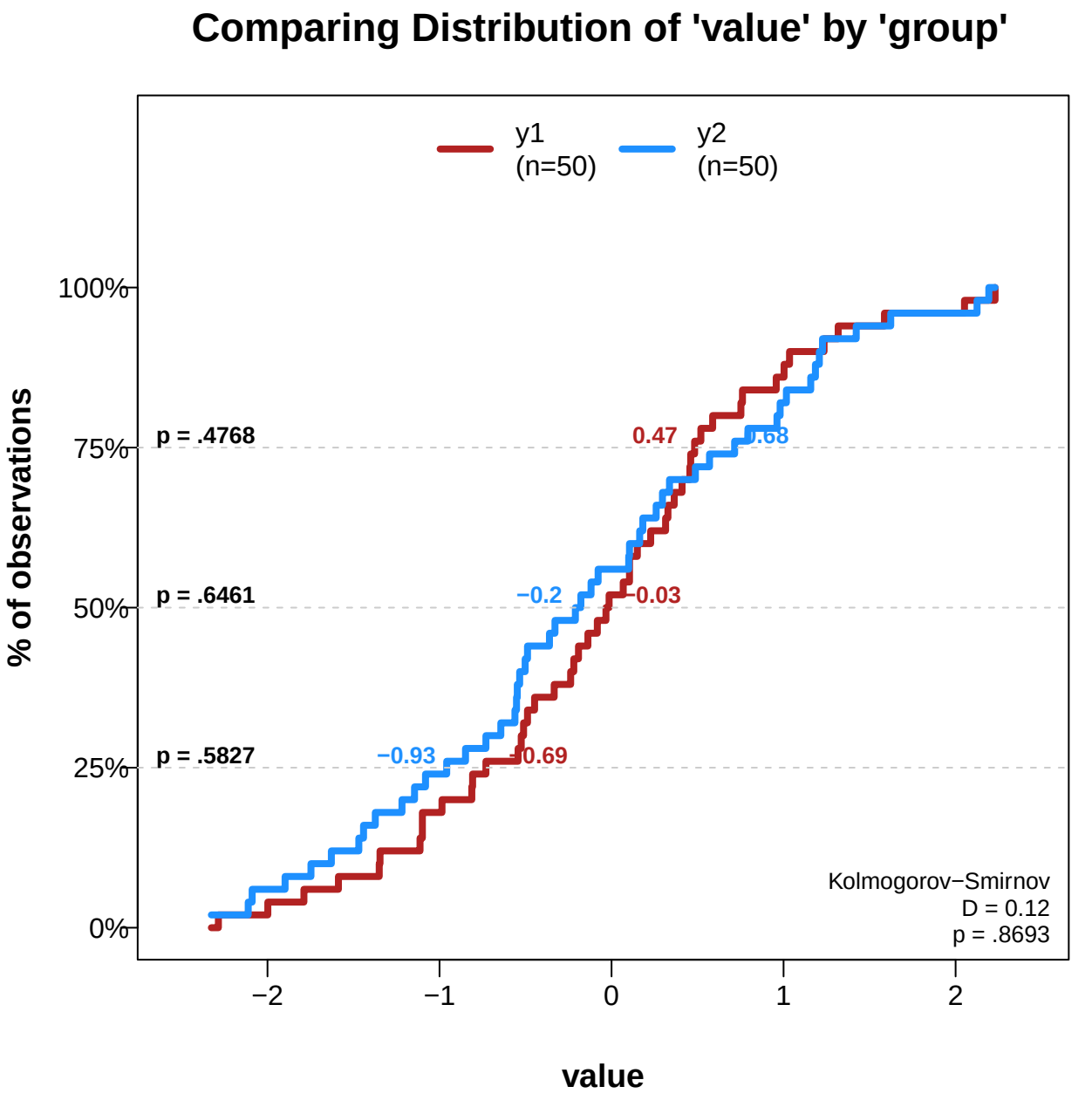

Comparing Distribution of 'y' by 'group'
C
(n=30)
A
(n=30)
B
(n=40)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y

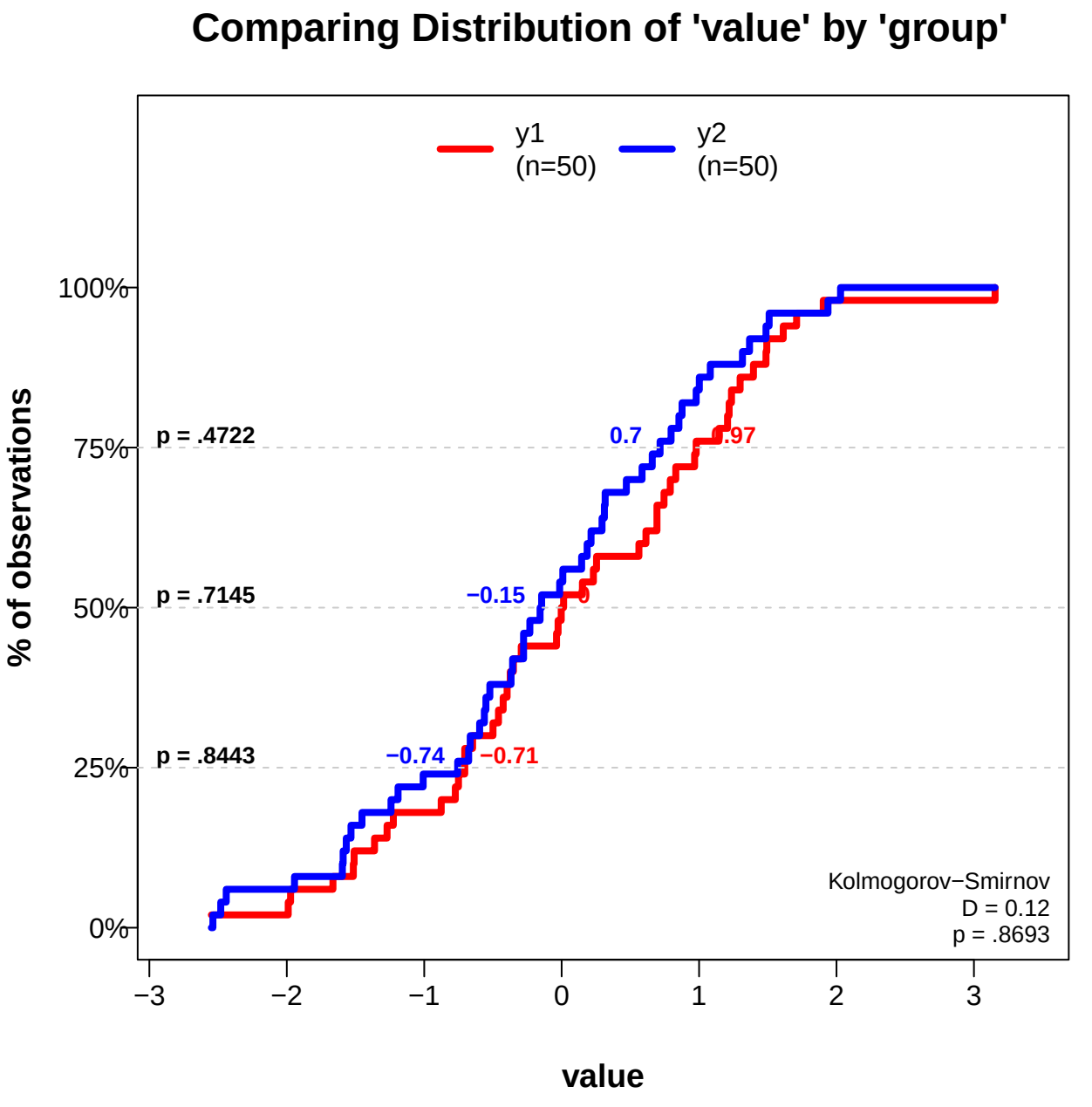

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−3
−2
−1
0
1
2
y

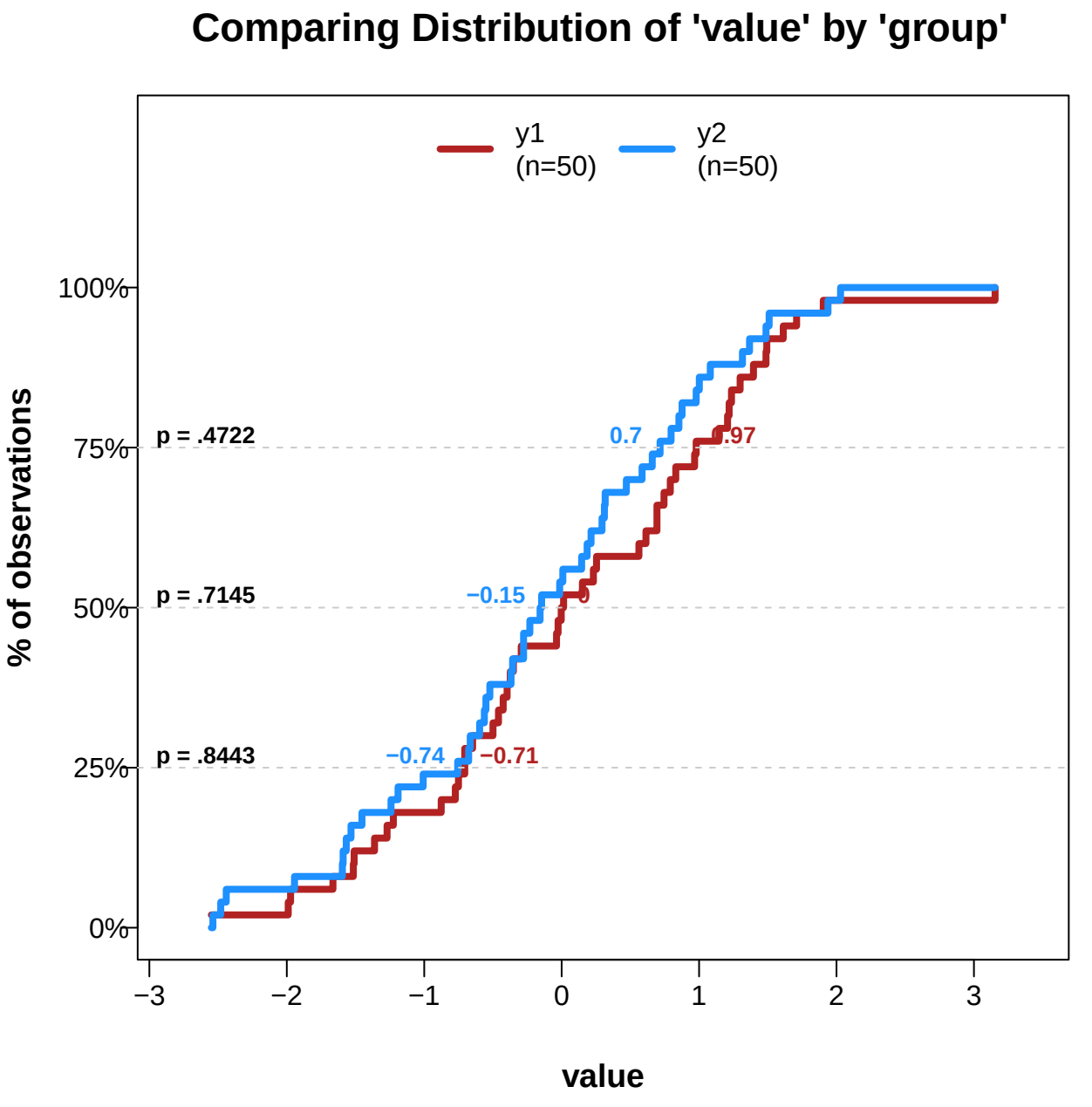

Comparing Distribution of 'y' by 'group'
C
(n=30)
B
(n=40)
A
(n=30)
100%
75%
50%
25%
0%
% of observations
−3
−2
−1
0
1
2
y

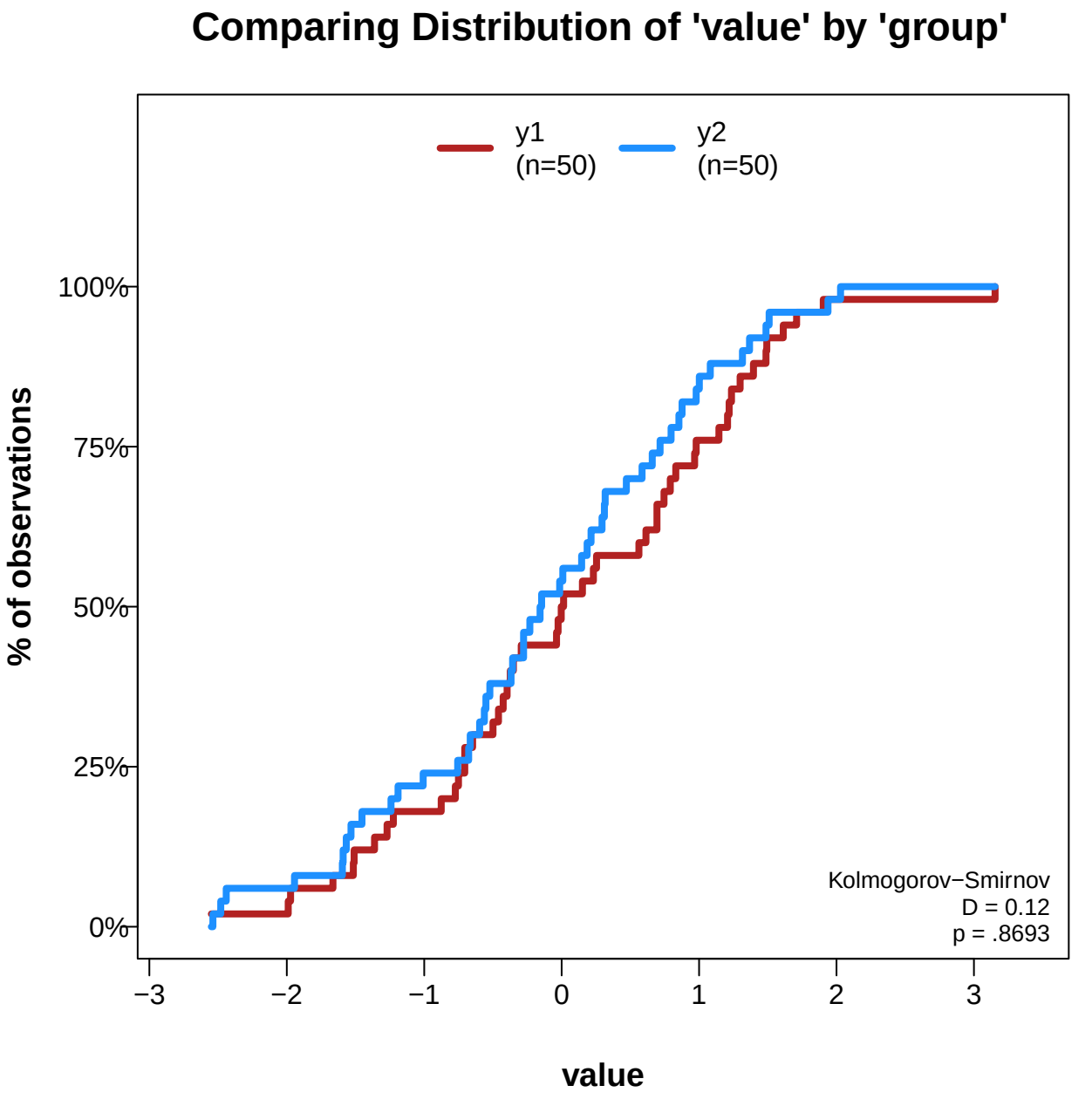

Comparing Distribution of 'y' by 'group'
C
(n=30)
A
(n=30)
B
(n=40)
100%
75%
50%
25%
0%
% of observations
−3
−2
−1
0
1
2
y

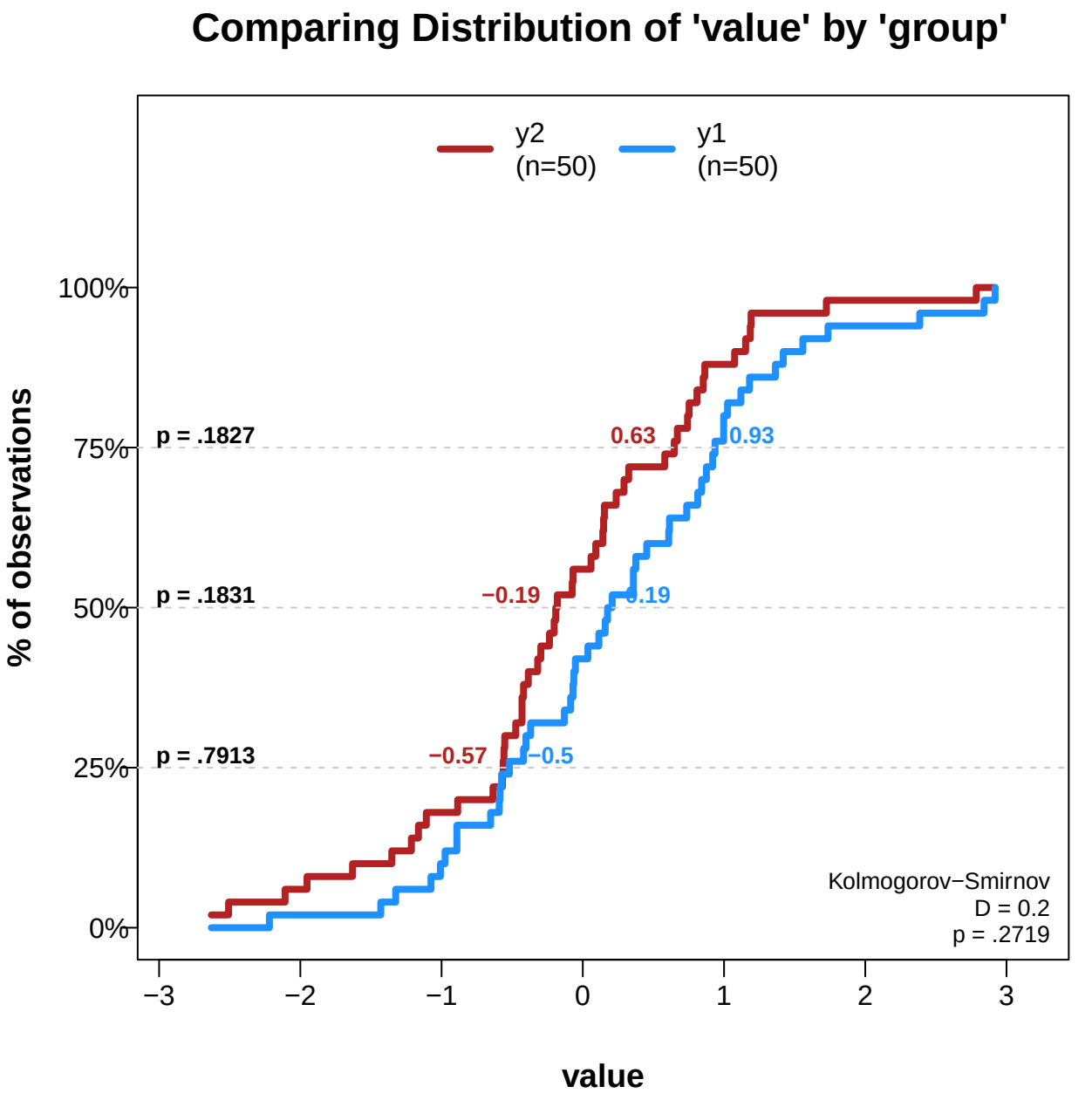

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .4768
p = .6461
p = .5827
0.47
0.68
−0.2
−0.03
−0.93
−0.69
Kolmogorov−Smirnov
D = 0.12
p = .8693
−2
−1
0
1
2
value

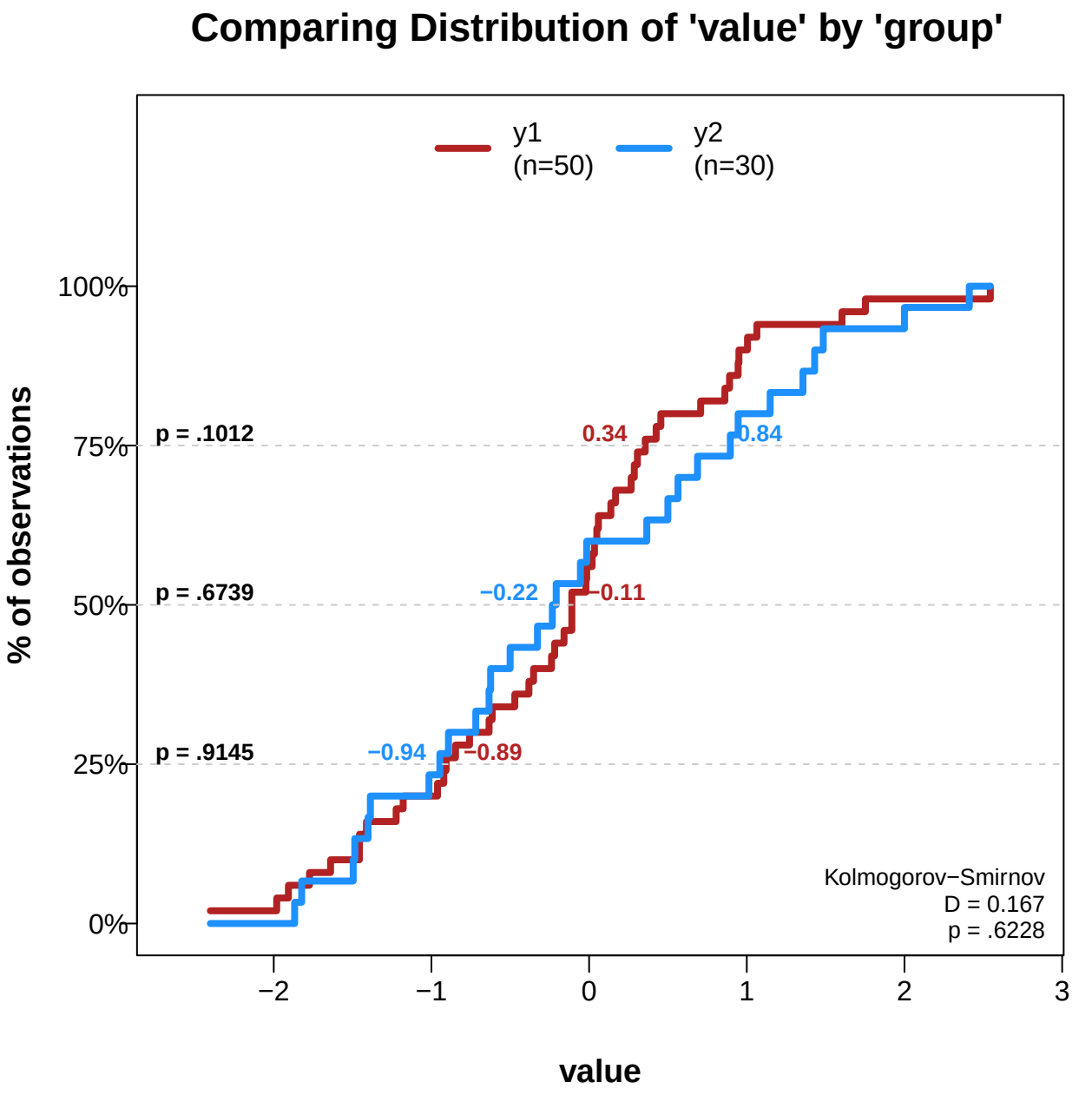

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .4768
p = .6461
p = .5827
0.47
0.68
−0.2
−0.03
−0.93
−0.69
Kolmogorov−Smirnov
D = 0.12
p = .8693
−2
−1
0
1
2
value

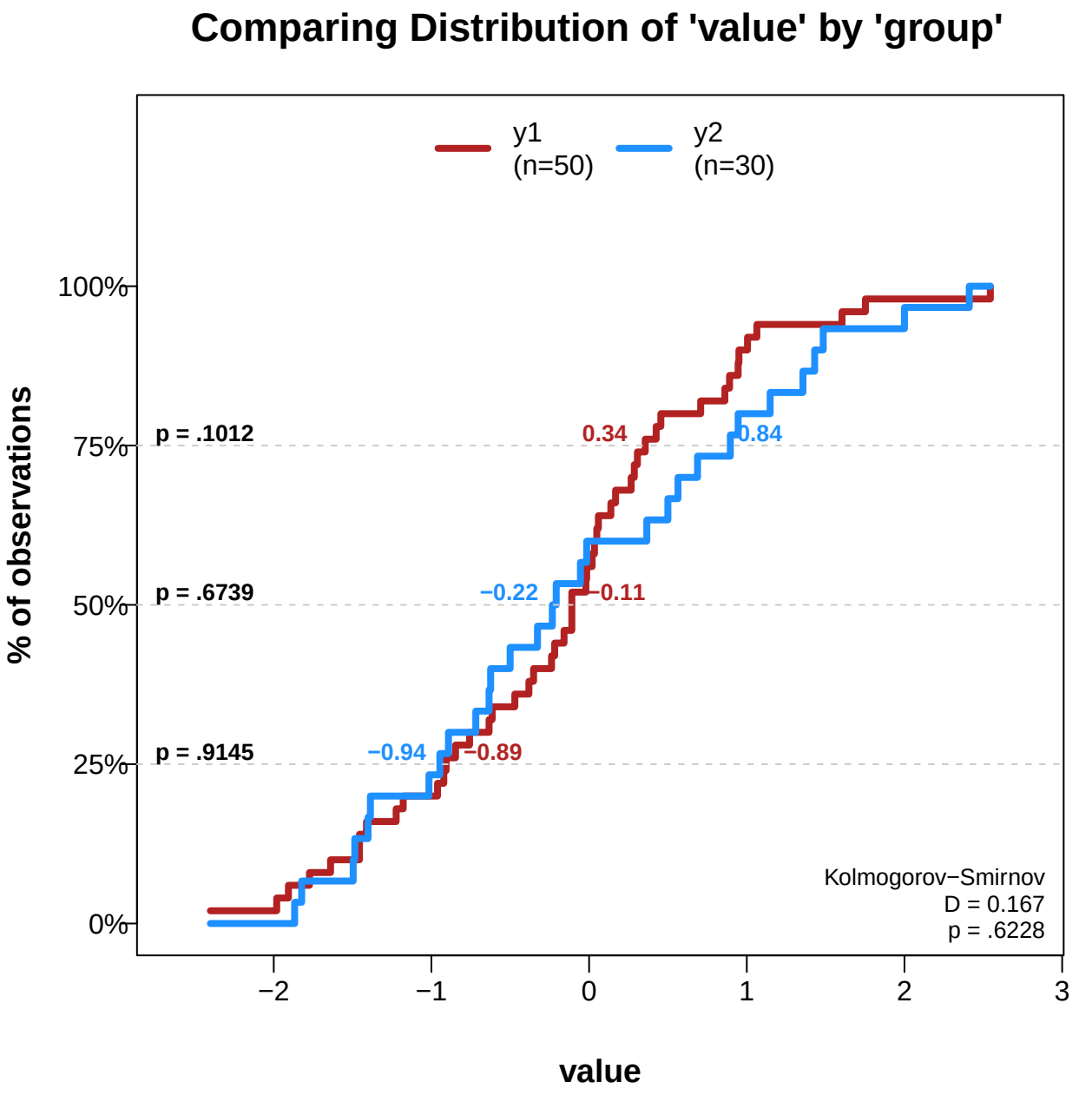

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .4722
p = .7145
p = .8443
0.7
.97
−0.15
0
−0.74
−0.71
Kolmogorov−Smirnov
D = 0.12
p = .8693
−3
−2
−1
0
1
2
3
value

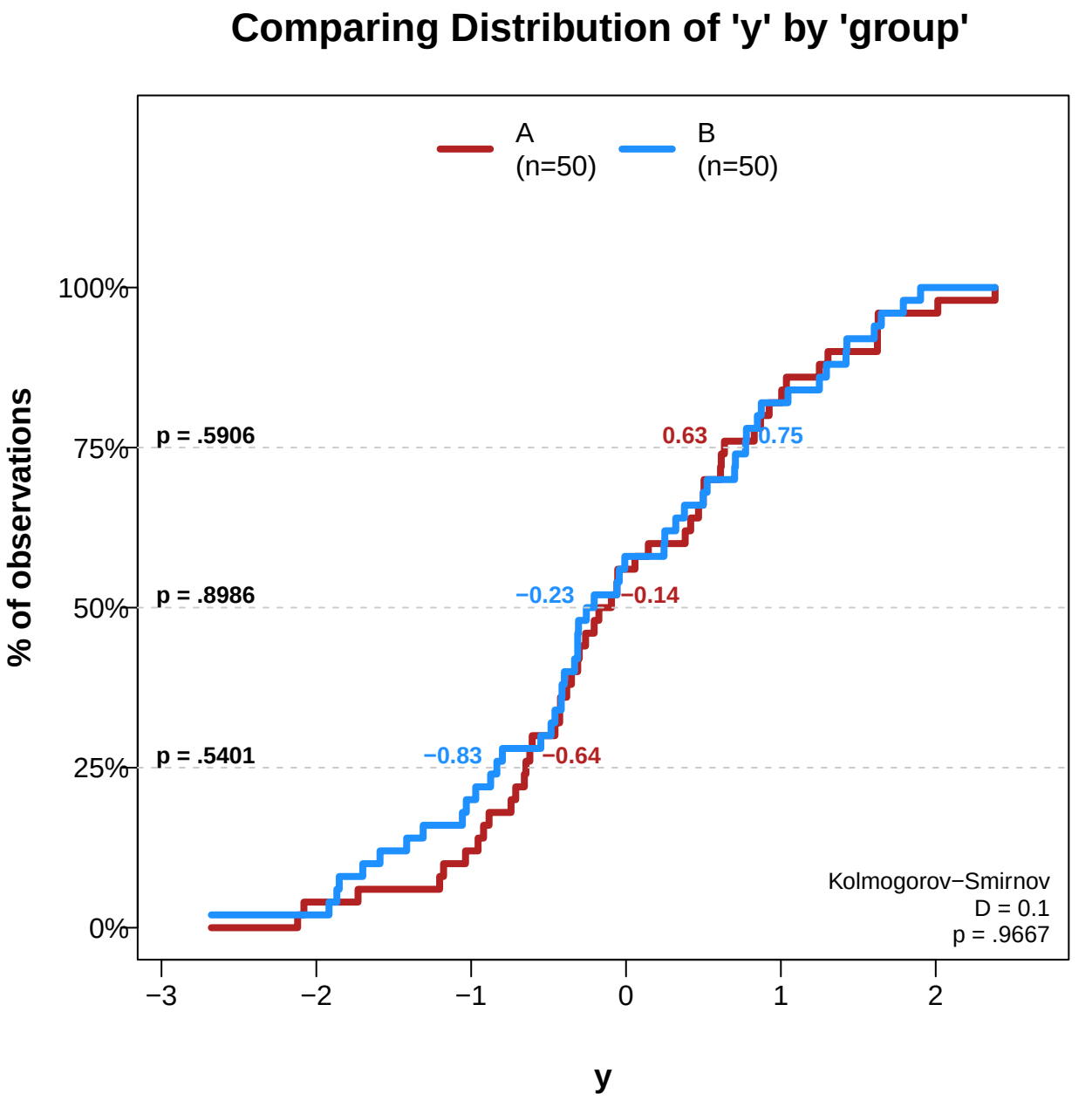

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .4722
p = .7145
p = .8443
0.7
.97
−0.15
−0.74
−0.71
−3
−2
−1
0
1
2
3
value

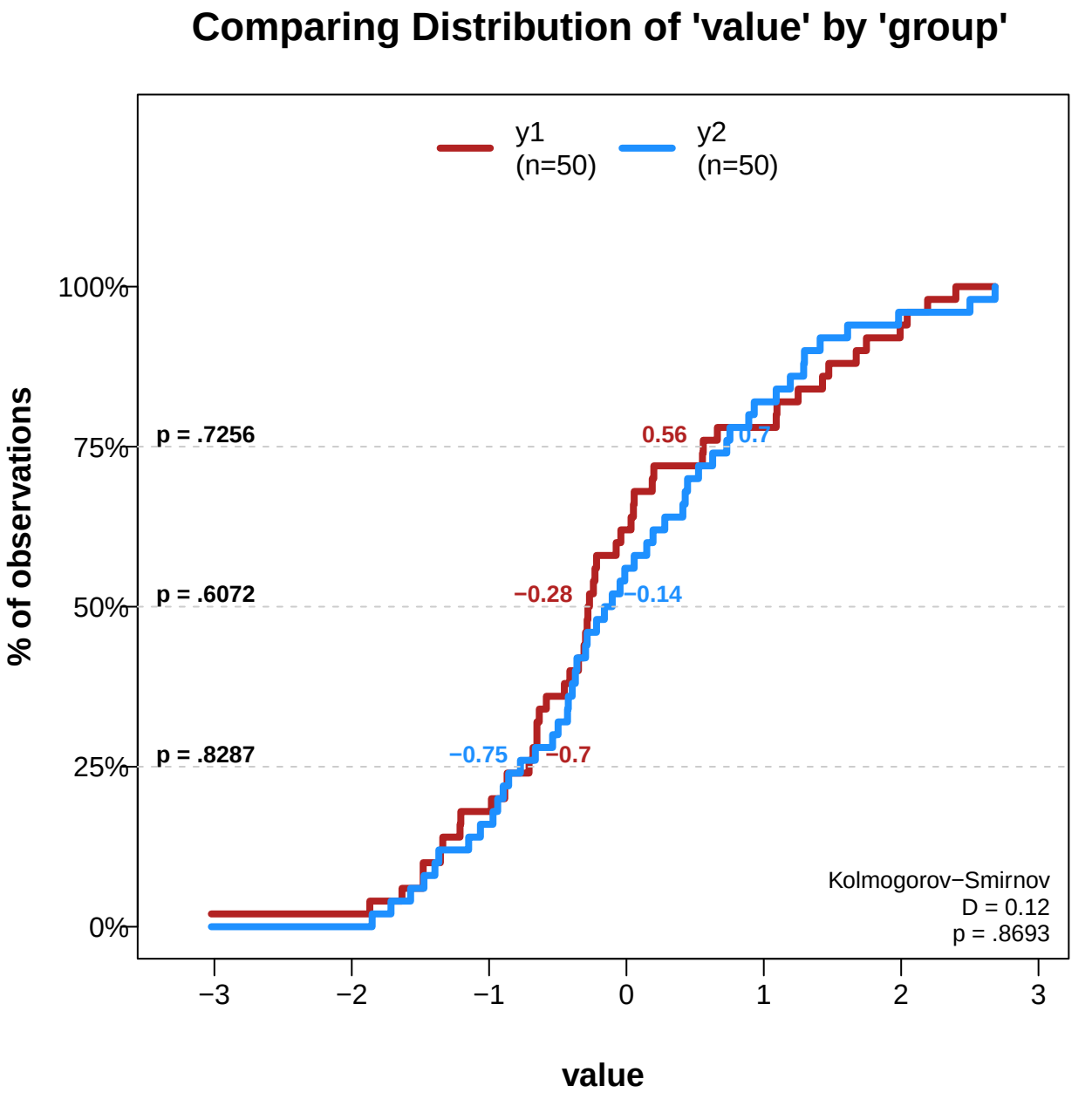

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=50)
100%
75%
50%
25%
0%
% of observations
Kolmogorov−Smirnov
D = 0.12
p = .8693
−3
−2
−1
0
1
2
3
value

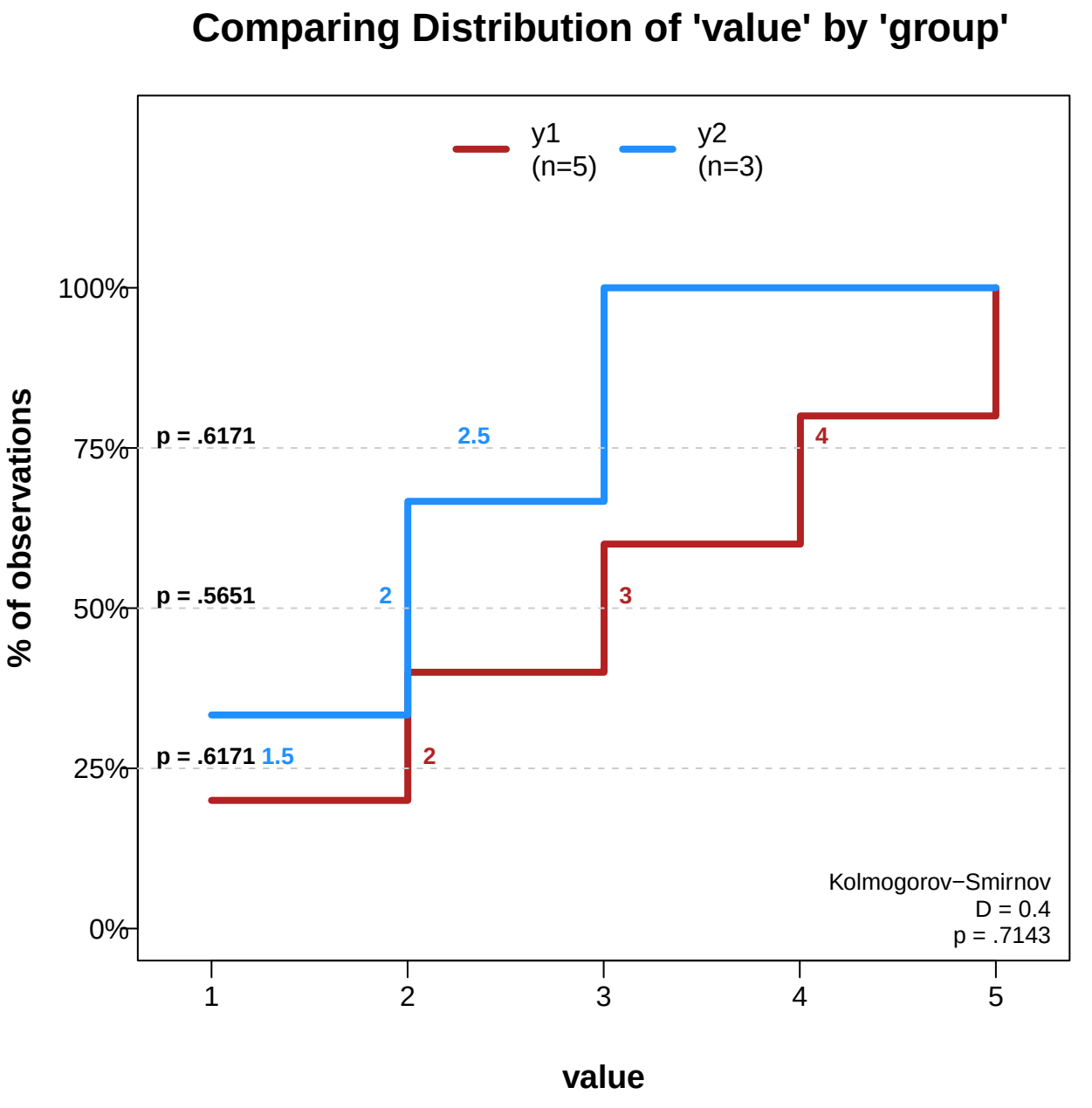

Comparing Distribution of 'value' by 'group'
y2
(n=50)
y1
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .1827
p = .1831
p = .7913
0.63
0.93
−0.19
0.19
−0.57
−0.5
Kolmogorov−Smirnov
D = 0.2
p = .2719
−3
−2
−1
0
1
2
3
value

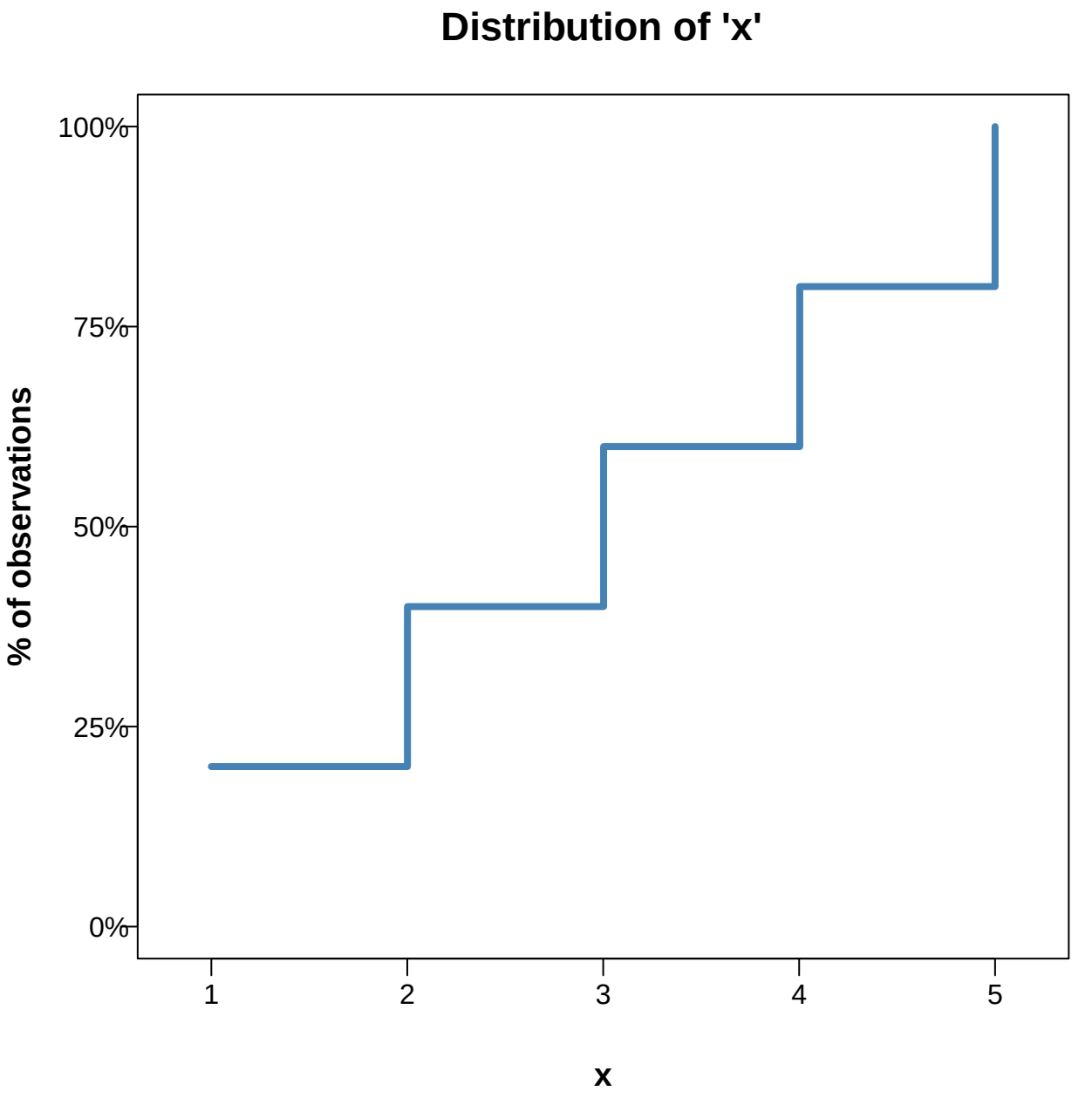

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=30)
100%
75%
50%
25%
0%
% of observations
p = .1012
p = .6739
p = .9145
0.34
0.84
−0.22
−0.11
−0.94
−0.89
Kolmogorov−Smirnov
D = 0.167
p = .6228
−2
−1
0
1
2
3
value

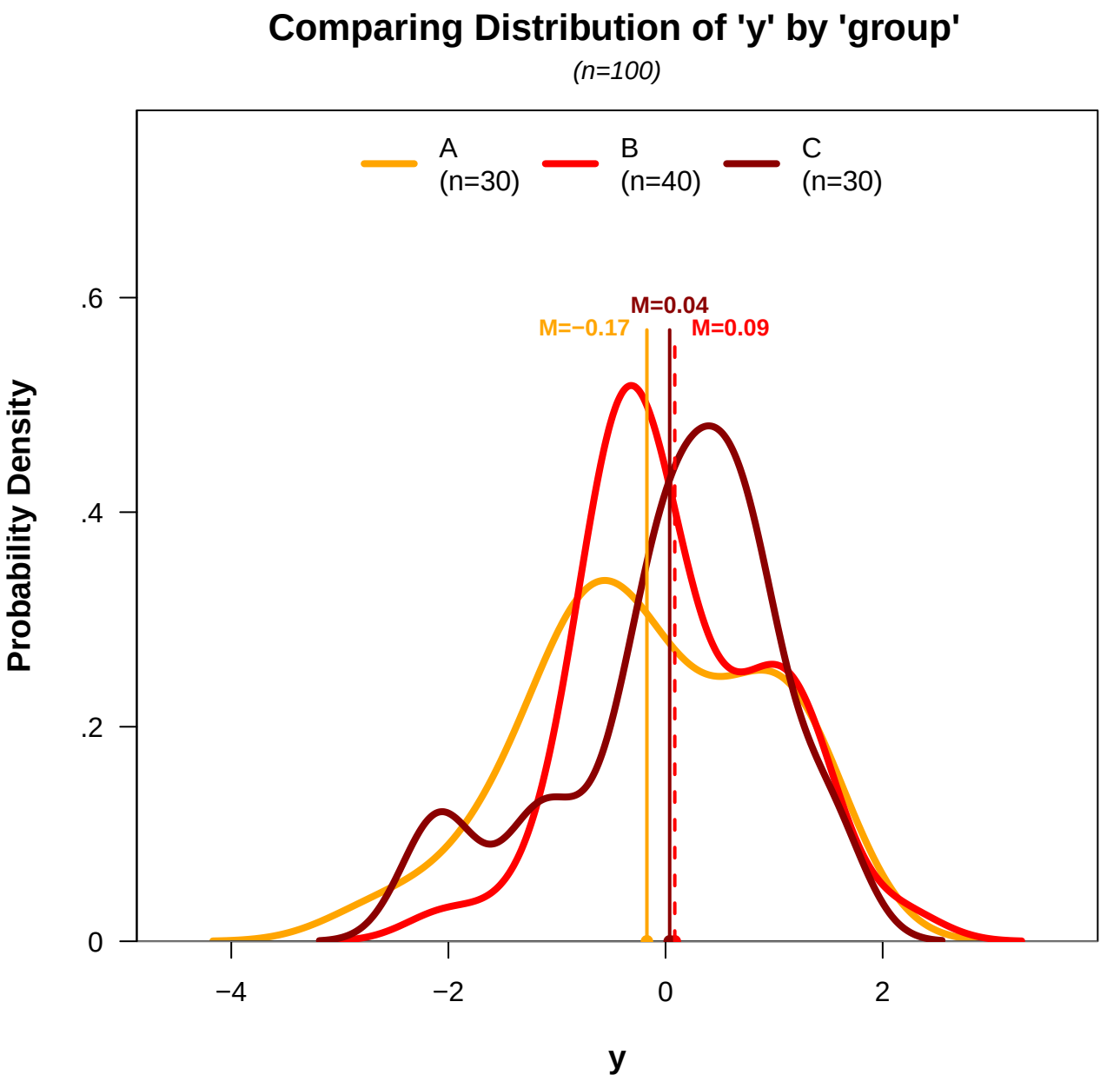

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=30)
100%
75%
50%
25%
0%
% of observations
p = .1012
p = .6739
p = .9145
0.34
0.84
−0.22
−0.11
−0.94
−0.89
Kolmogorov−Smirnov
D = 0.167
p = .6228
−2
−1
0
1
2
3
value

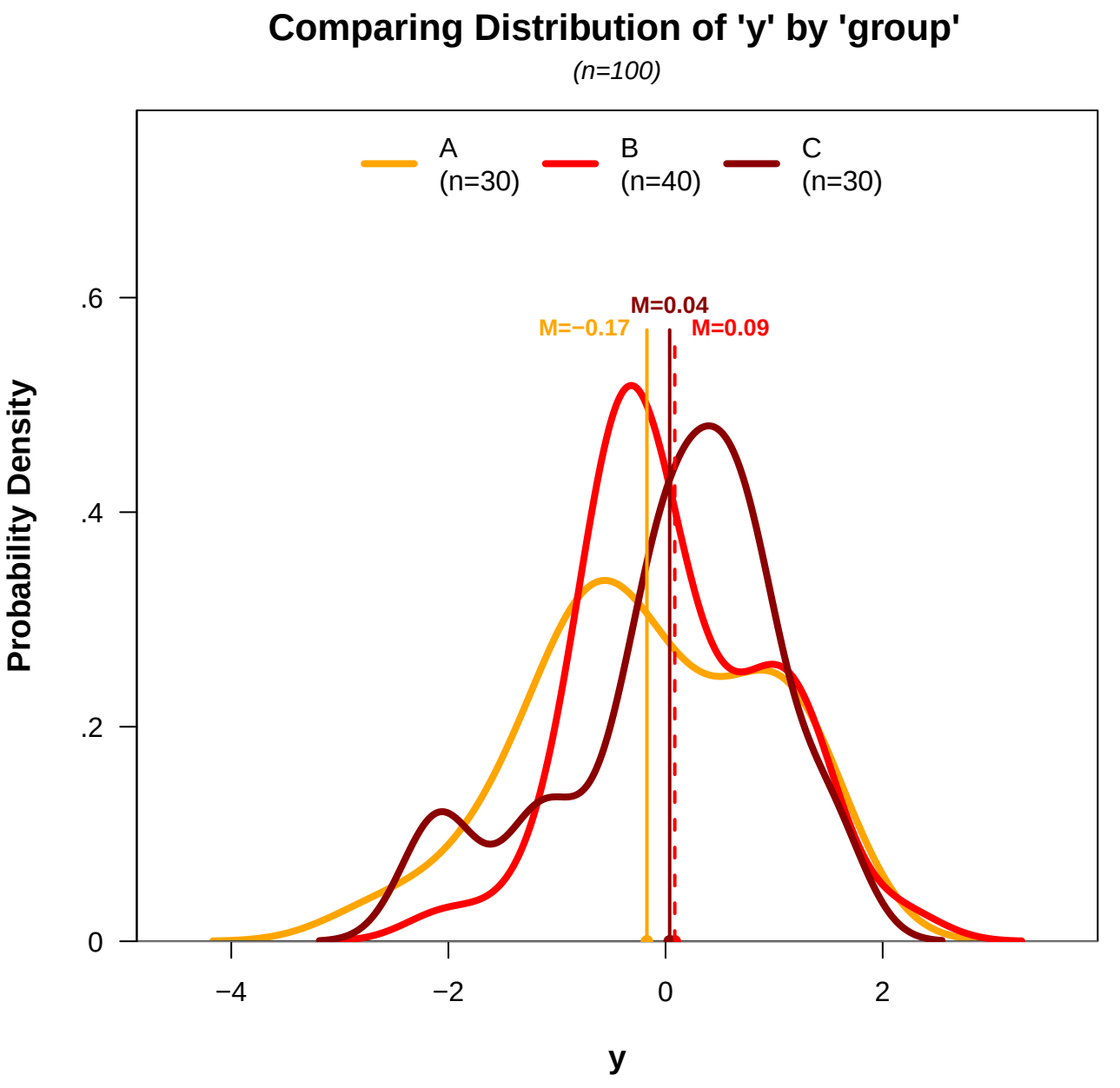

Comparing Distribution of 'y' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .5906
p = .8986
p = .5401
0.63
0.75
−0.23
−0.14
−0.83
−0.64
Kolmogorov−Smirnov
D = 0.1
p = .9667
−3
−2
−1
0
1
2
y

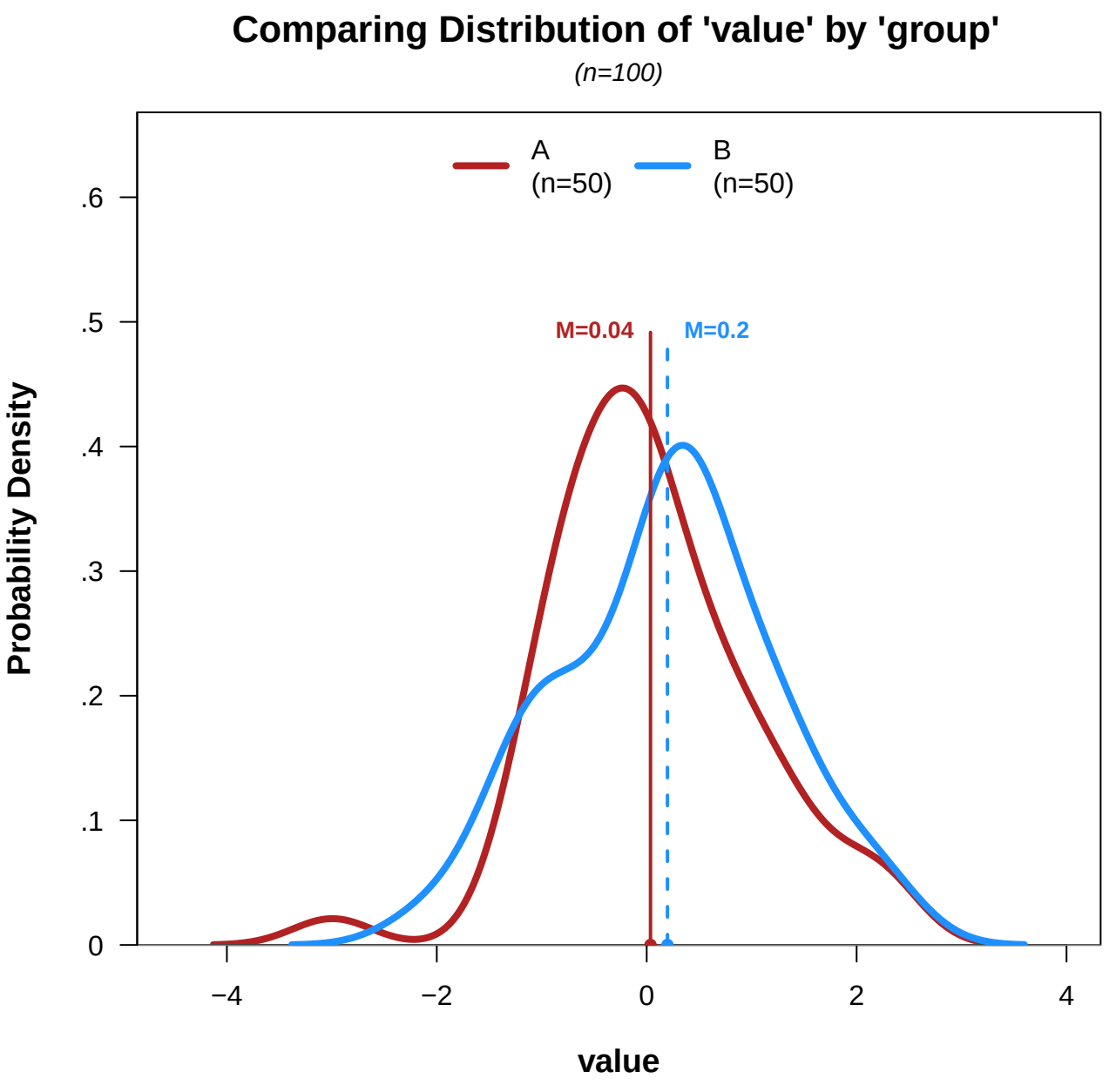

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .7256
p = .6072
p = .8287
0.56
0.7
−0.28
−0.14
−0.75
−0.7
Kolmogorov−Smirnov
D = 0.12
p = .8693
−3
−2
−1
0
1
2
3
value

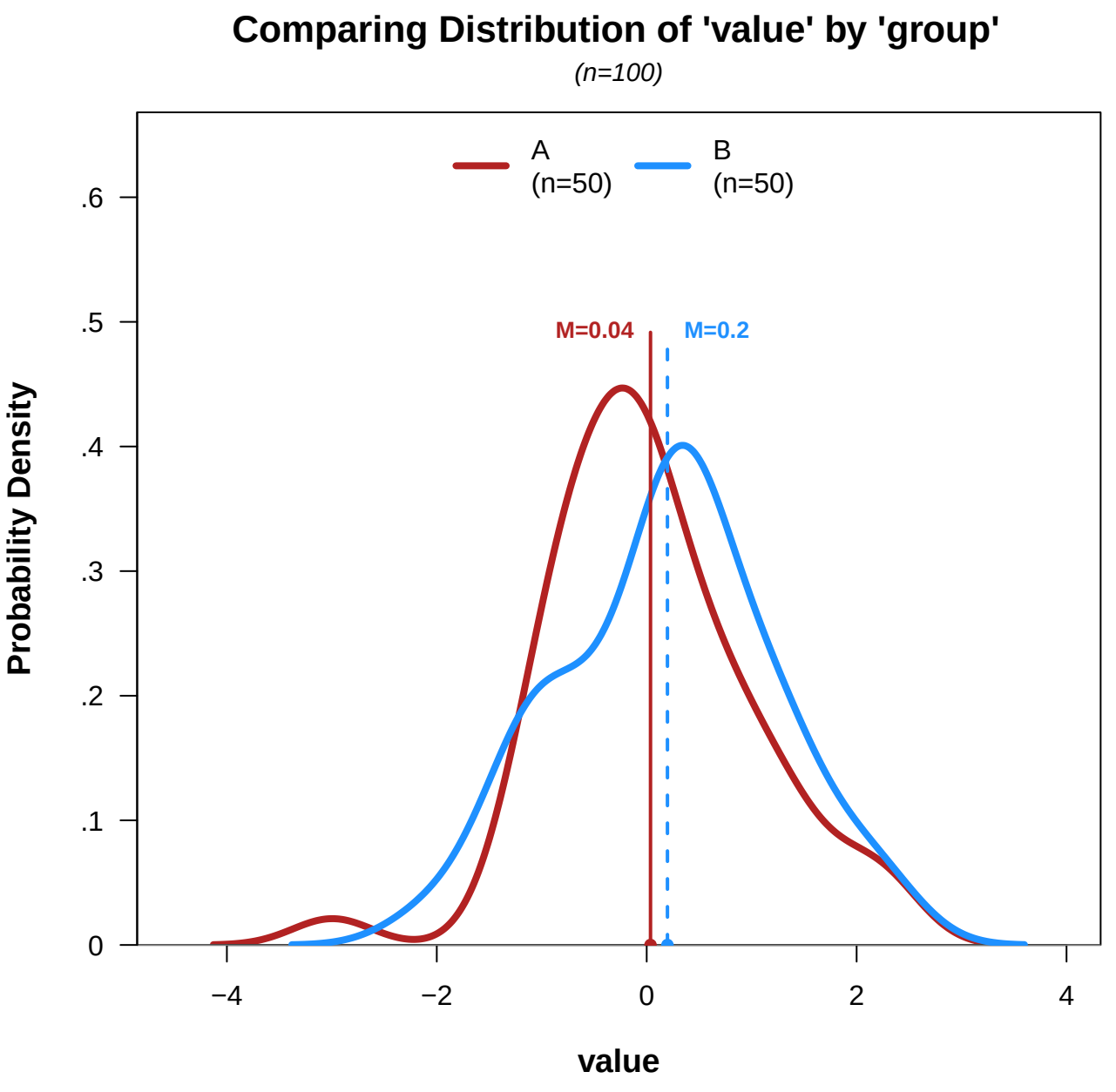

Comparing Distribution of 'value' by 'group'
y1
(n=5)
y2
(n=3)
100%
75%
50%
25%
0%
% of observations
p = .6171
p = .5651
p = .6171
2.5
2
1.5
2
3
4
Kolmogorov−Smirnov
D = 0.4
p = .7143
1
2
3
4
5
value

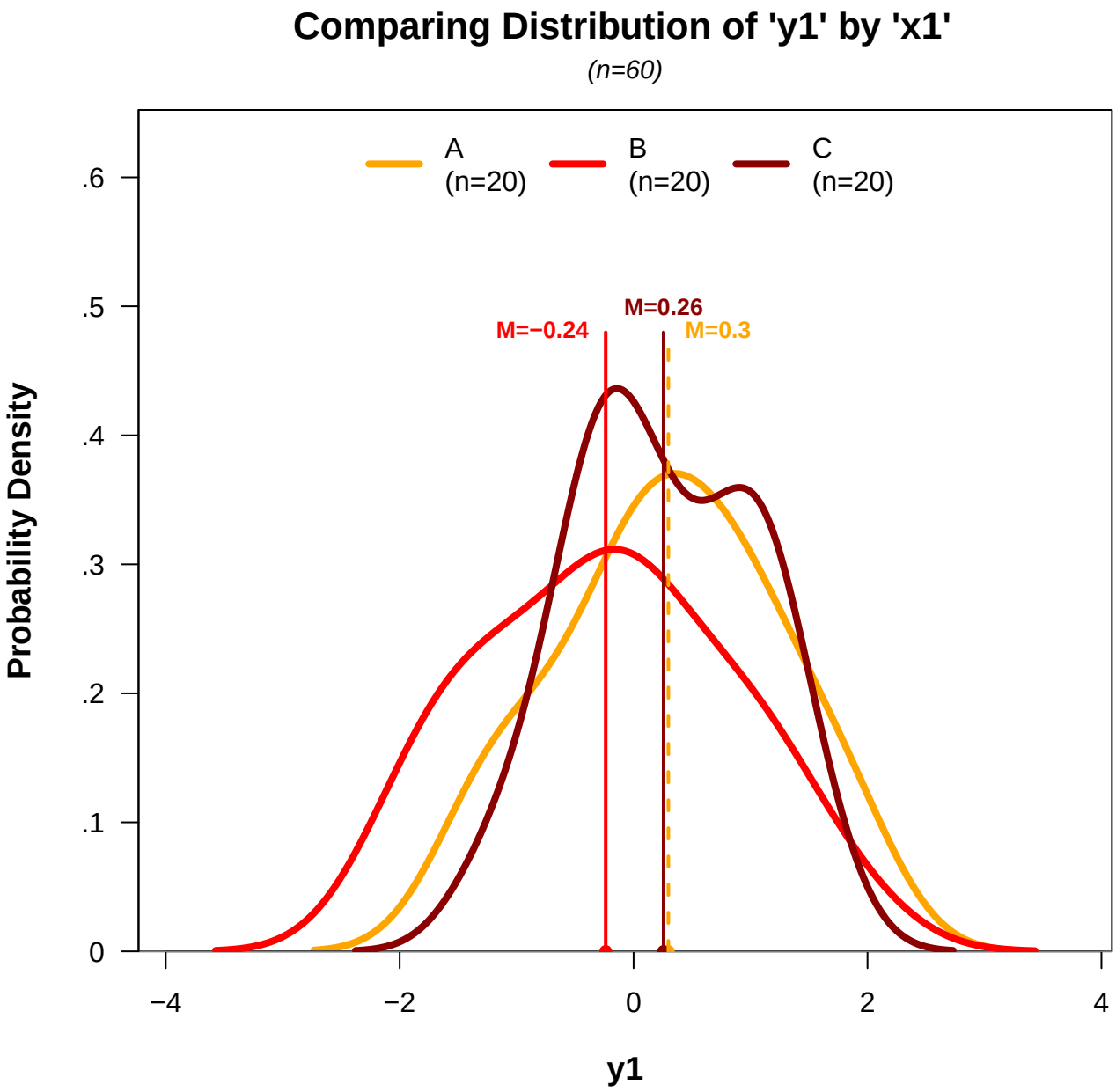

Distribution of 'x'
100%
75%
50%
25%
0%
% of observations
1
2
3
4
5
x

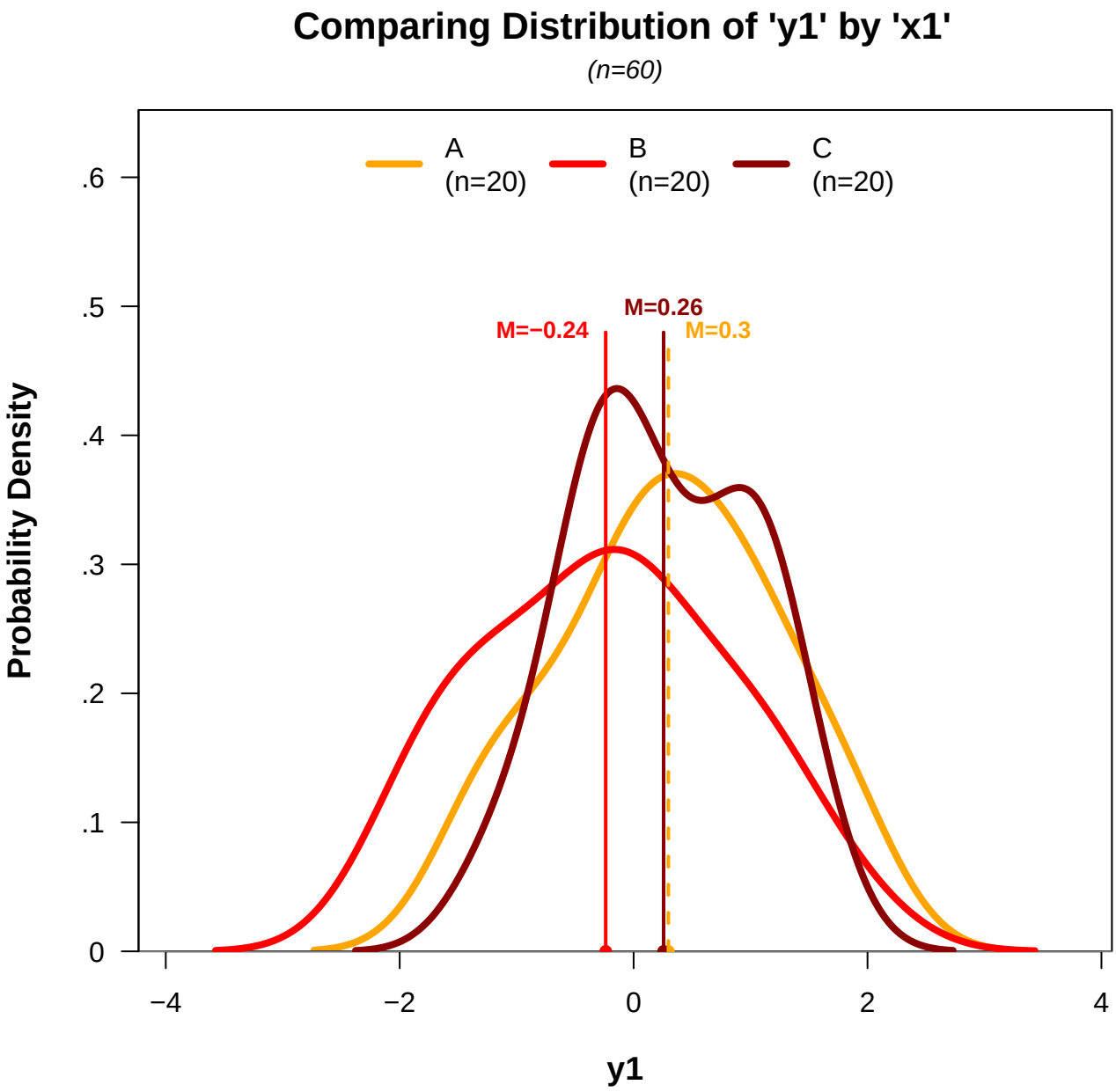

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.04
M=−0.17
M=0.09
Probability Density
.6
.4
.2
0
−4
−2
0
2
y

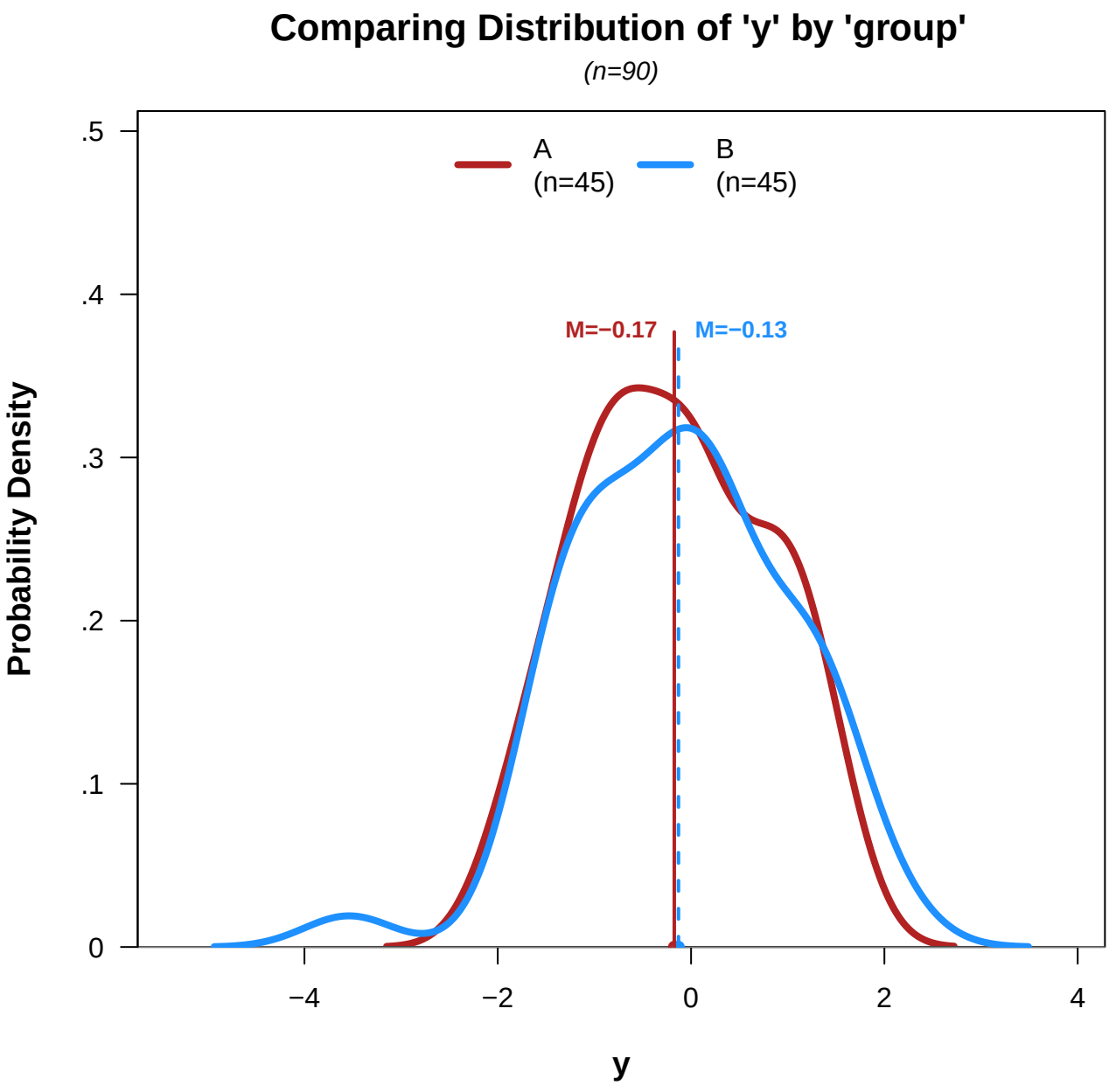

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.04
M=−0.17
M=0.09
Probability Density
.6
.4
.2
0
−4
−2
0
2
y

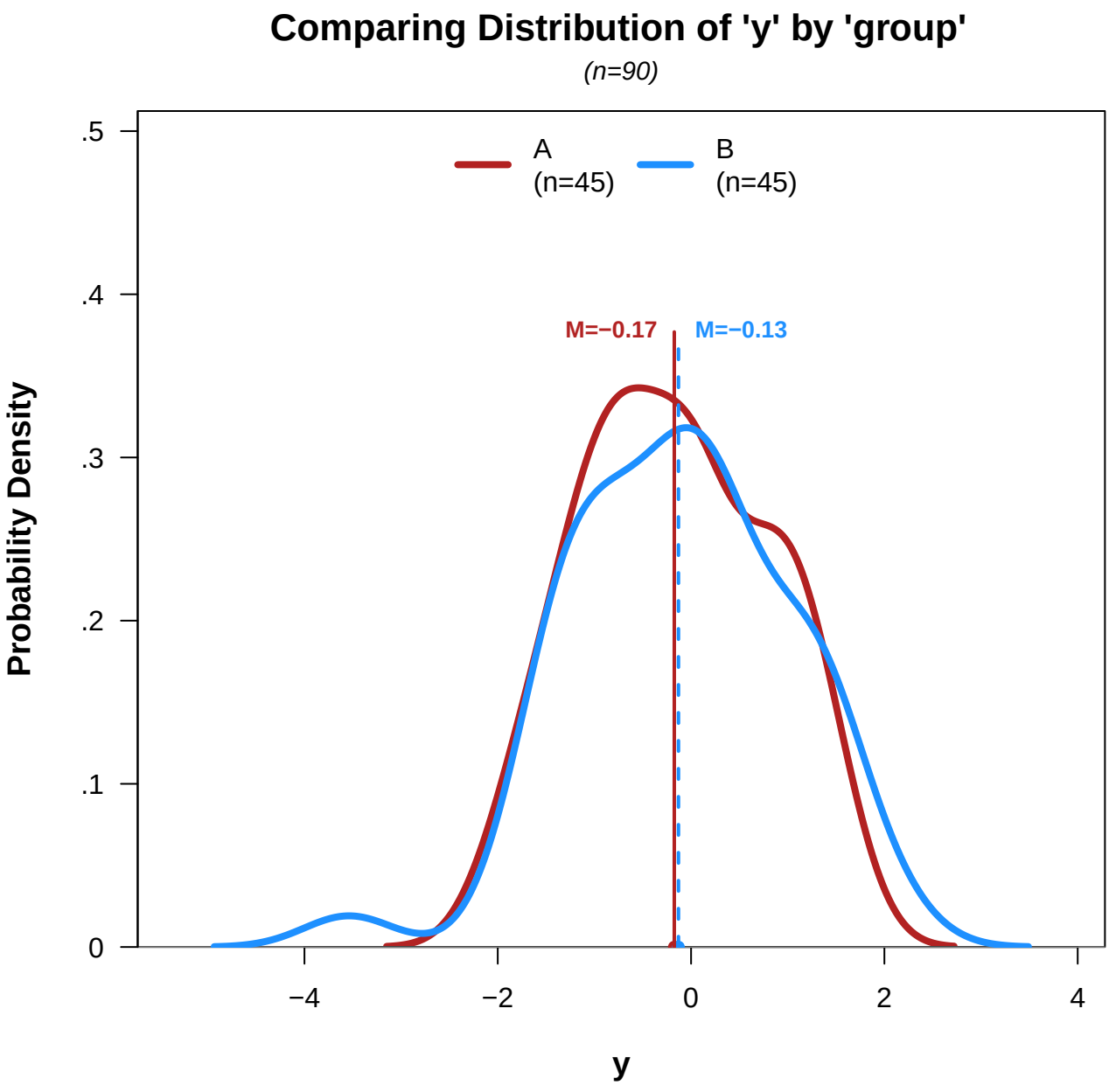

Comparing Distribution of 'value' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=0.04
M=0.2
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value

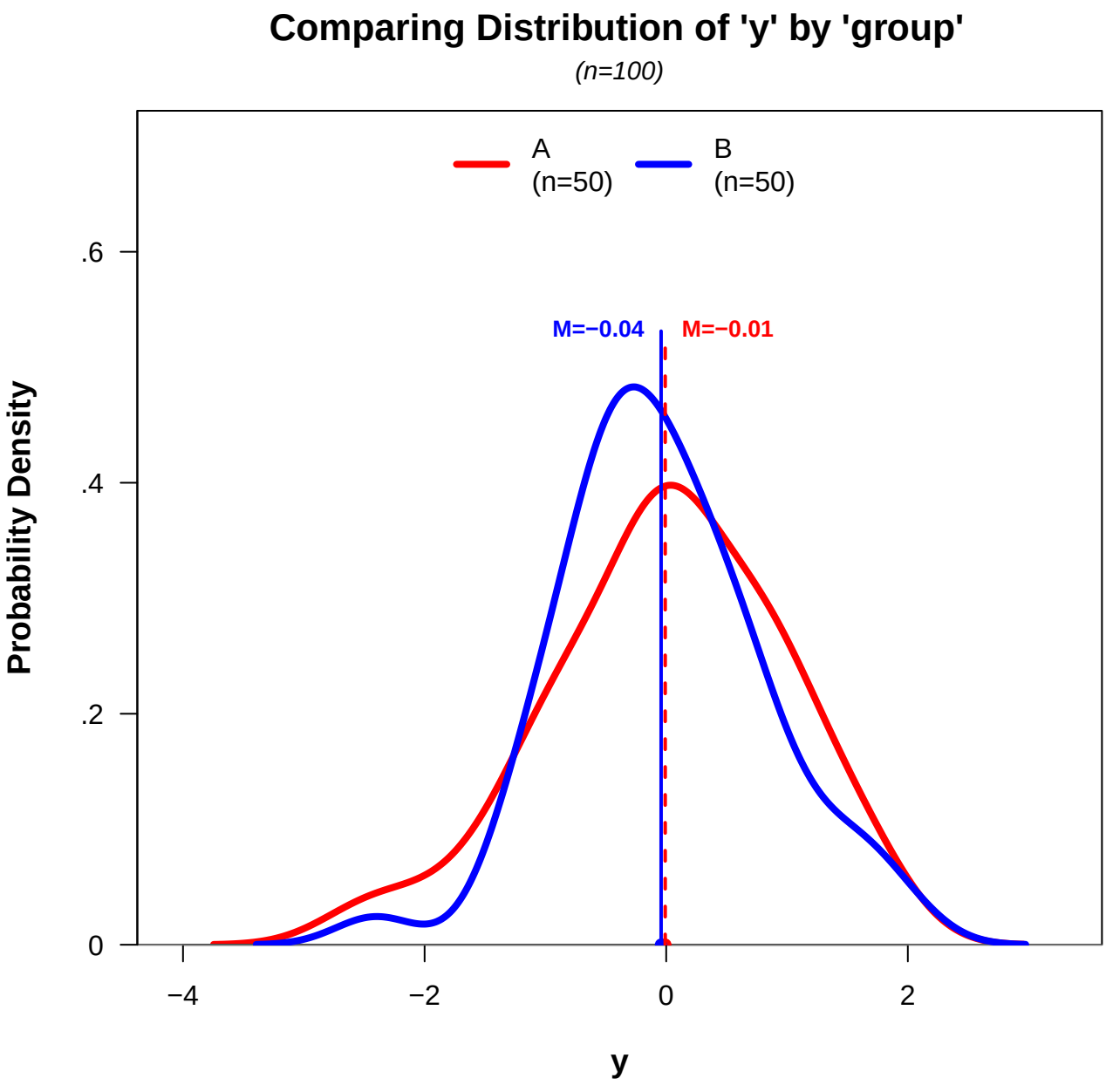

Comparing Distribution of 'value' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=0.04
M=0.2
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value

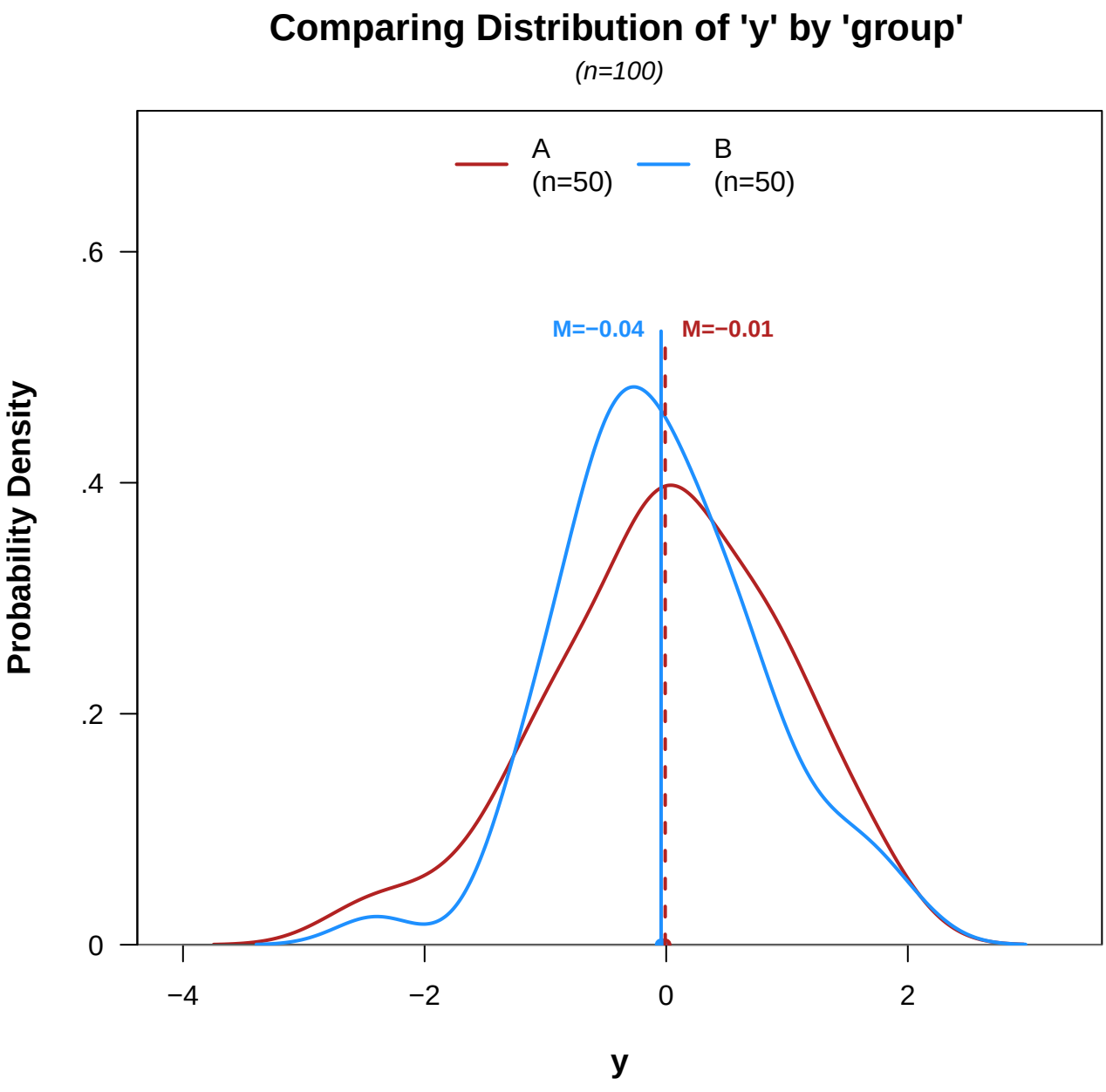

Comparing Distribution of 'y1' by 'x1'
(n=60)
A
(n=20)
B
(n=20)
C
(n=20)
M=0.26
M=−0.24
M=0.3
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y1

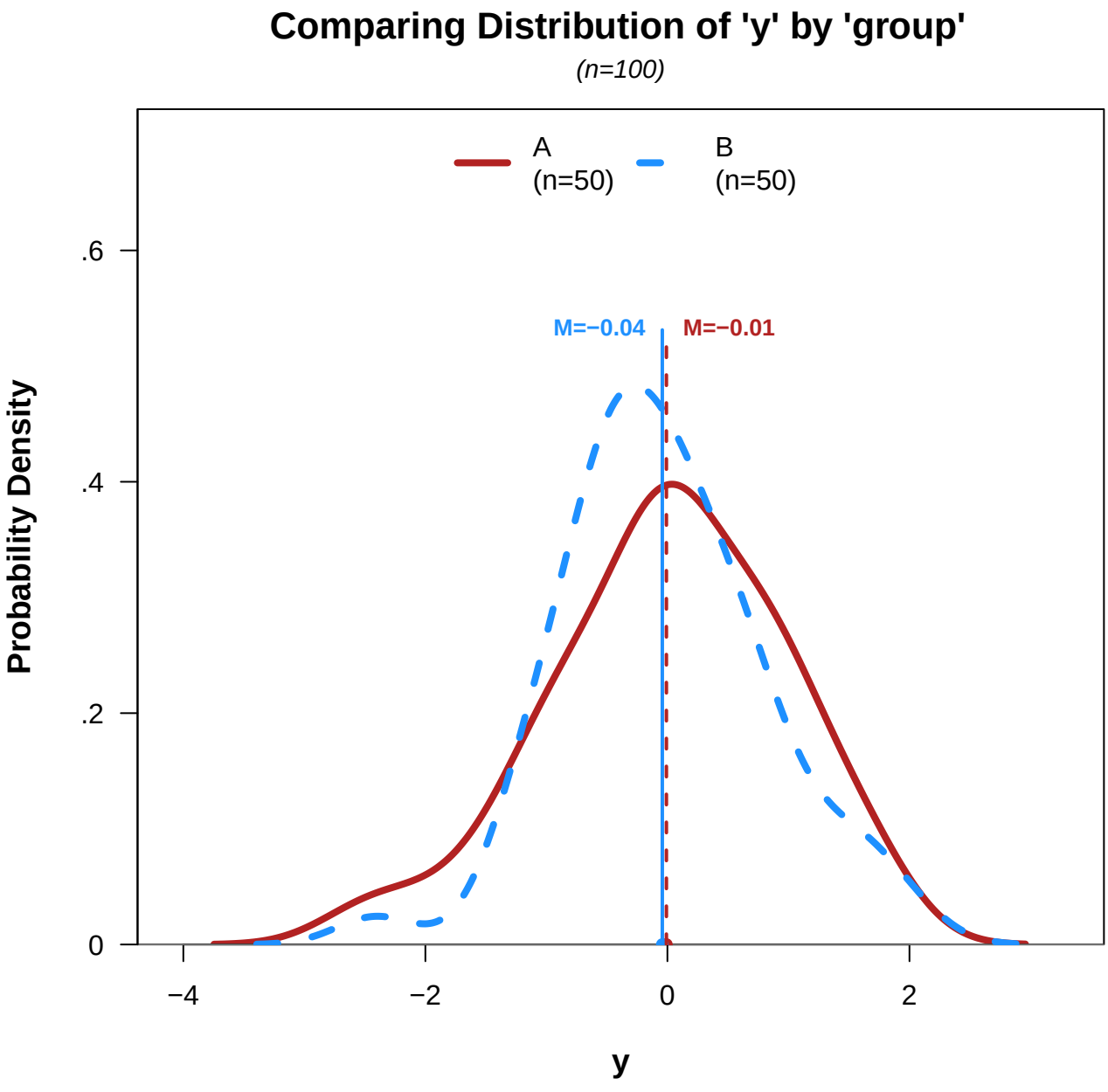

Comparing Distribution of 'y1' by 'x1'
(n=60)
A
(n=20)
B
(n=20)
C
(n=20)
M=0.26
M=−0.24
M=0.3
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y1

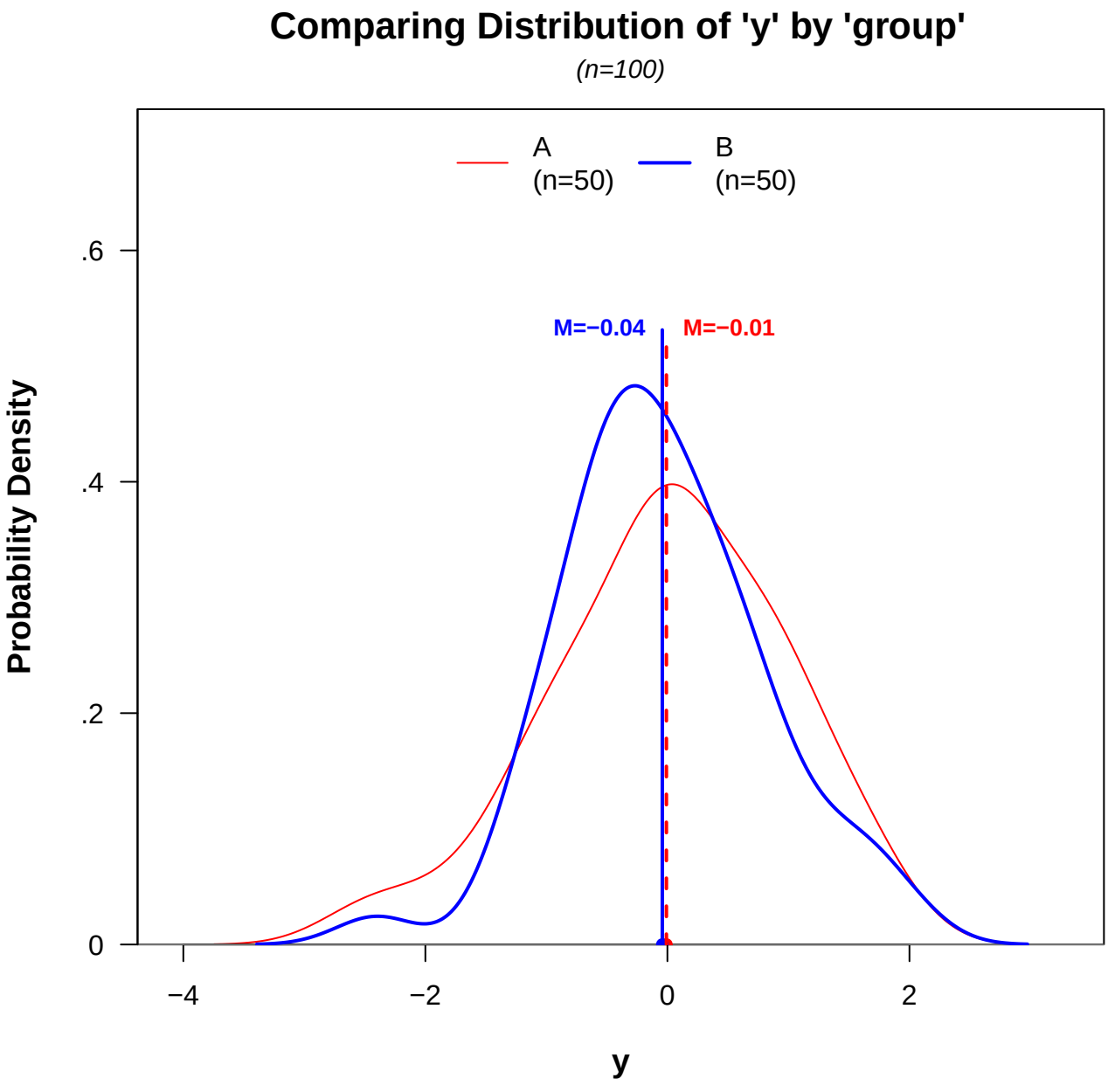

Comparing Distribution of 'y' by 'group'
(n=90)
A
(n=45)
B
(n=45)
M=−0.17
M=−0.13
Probability Density
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

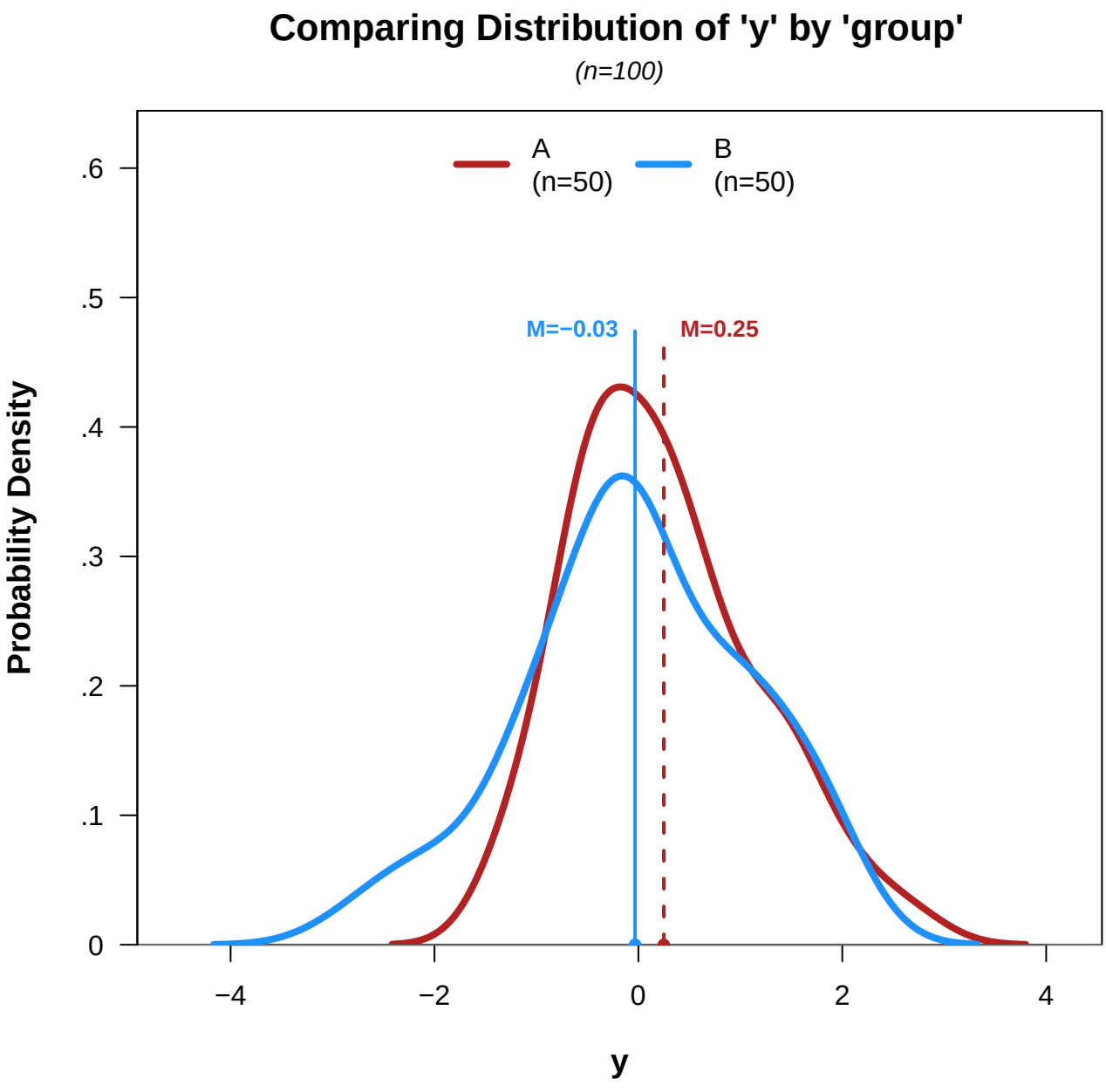

Comparing Distribution of 'y' by 'group'
(n=90)
A
(n=45)
B
(n=45)
M=−0.17
M=−0.13
Probability Density
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

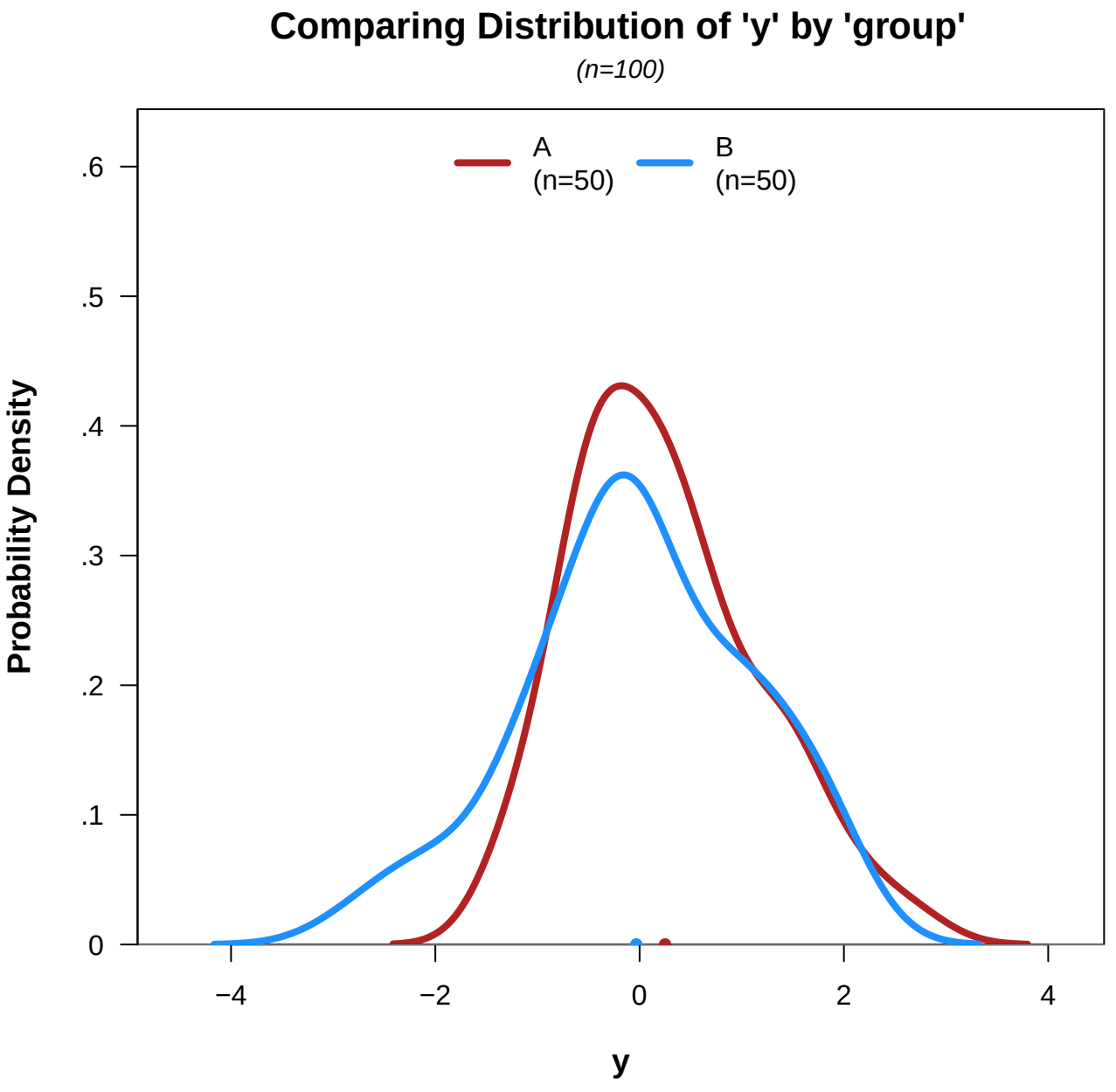

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.04
M=−0.01
Probability Density
.6
.4
.2
0
−4
−2
0
2
y

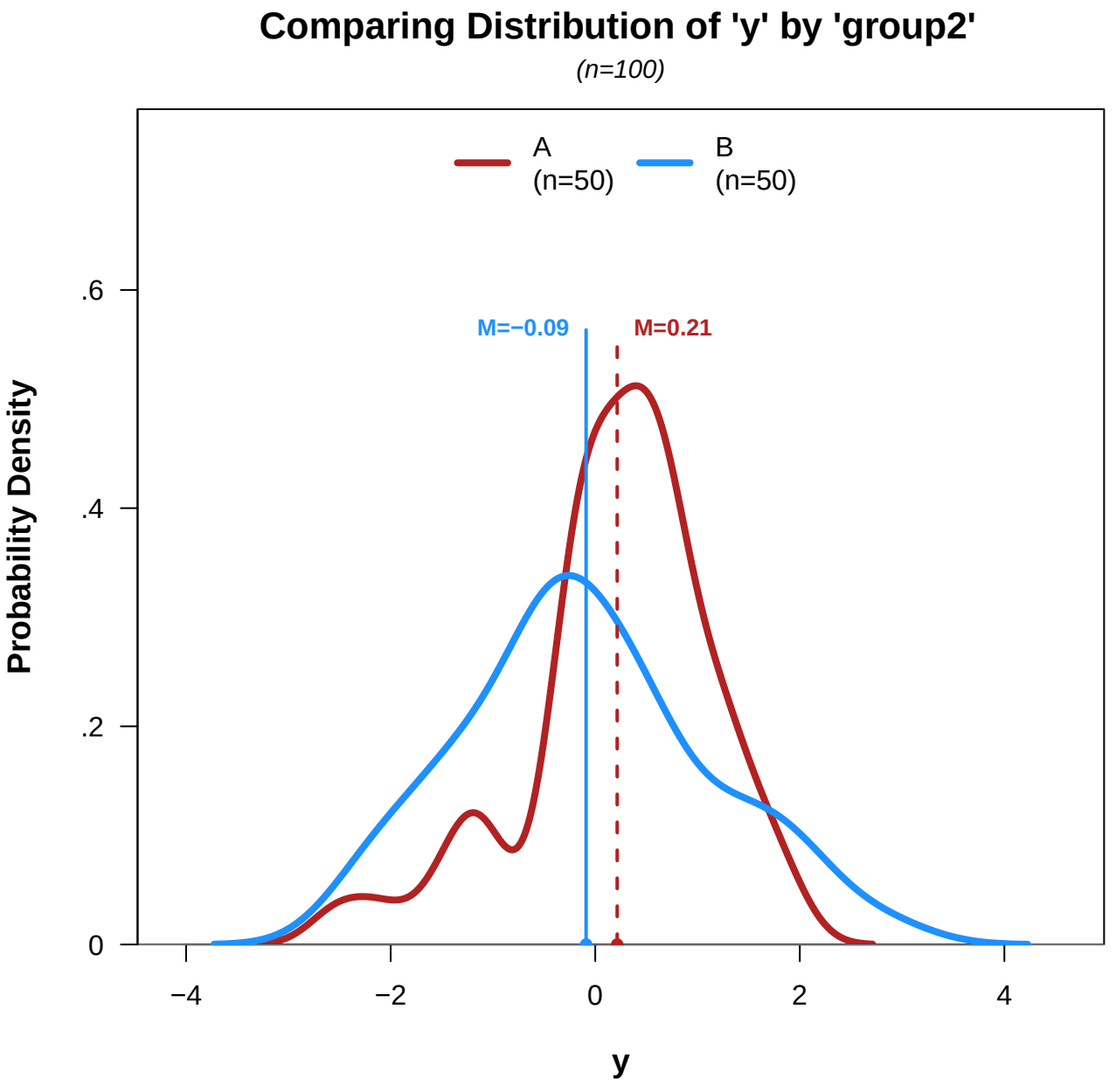

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.04
M=−0.01
Probability Density
.6
.4
.2
0
−4
−2
0
2
y

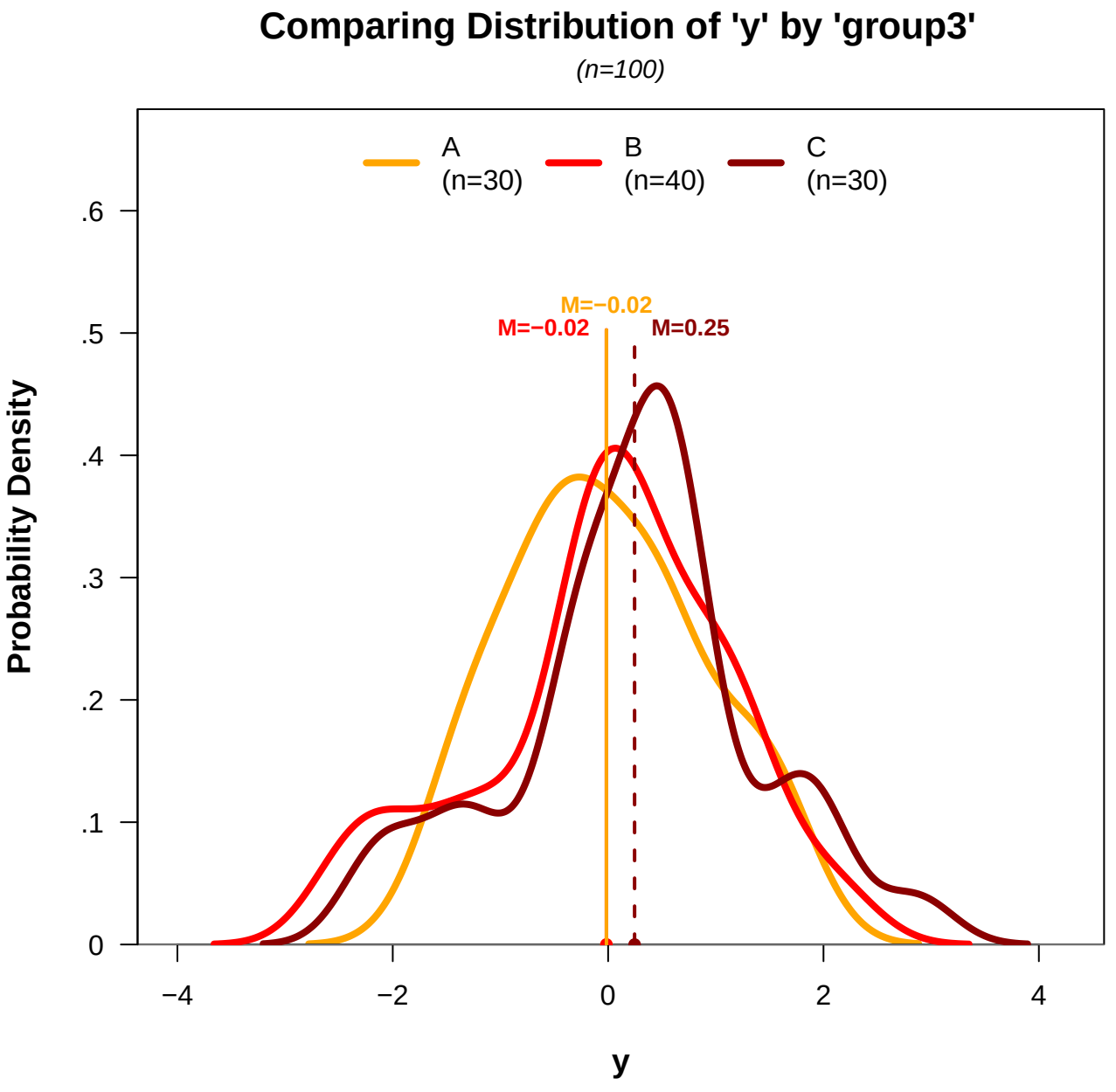

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.04
M=−0.01
Probability Density
.6
.4
.2
0
−4
−2
0
2
y

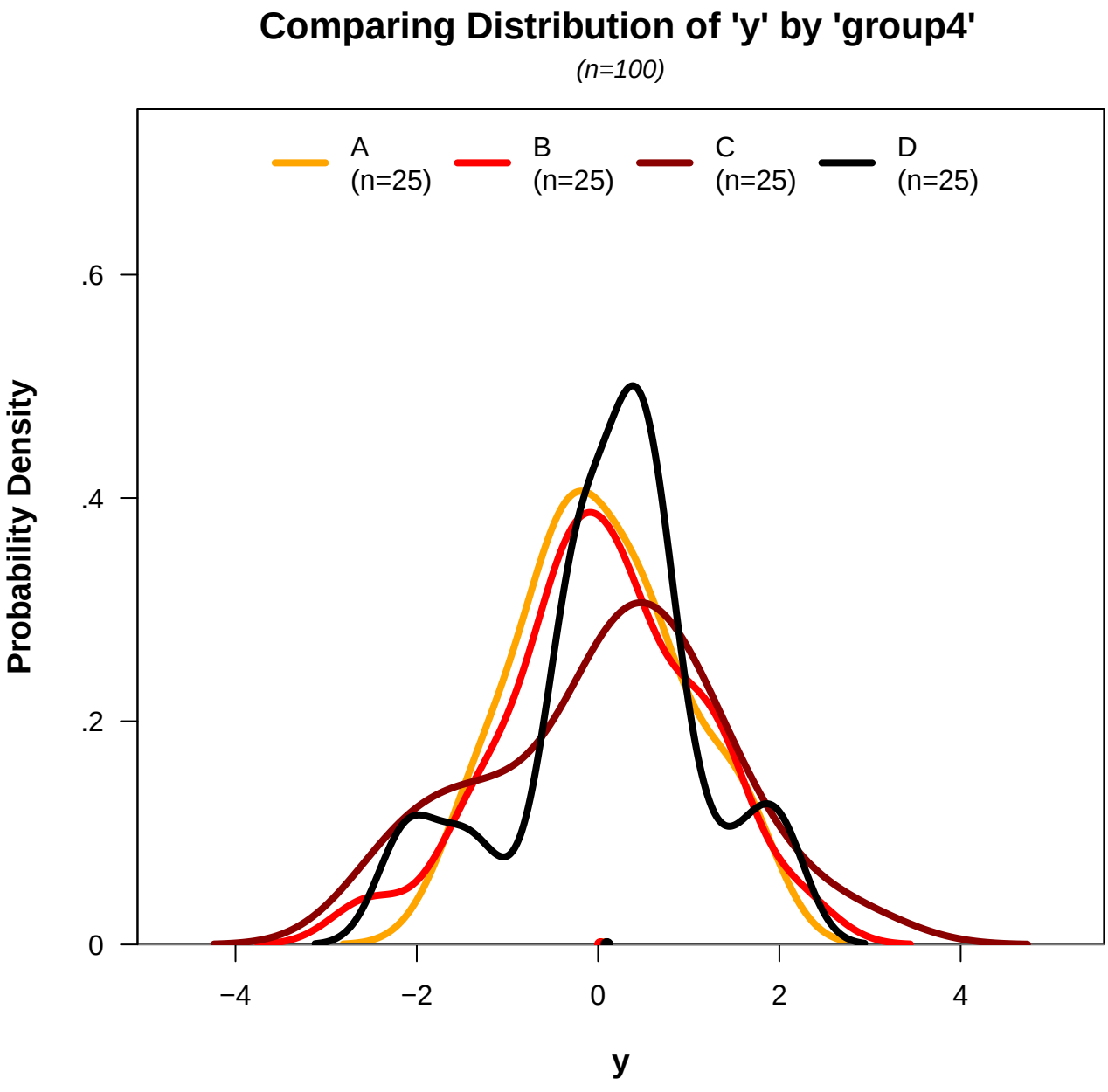

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.04
M=−0.01
Probability Density
.6
.4
.2
0
−4
−2
0
2
y

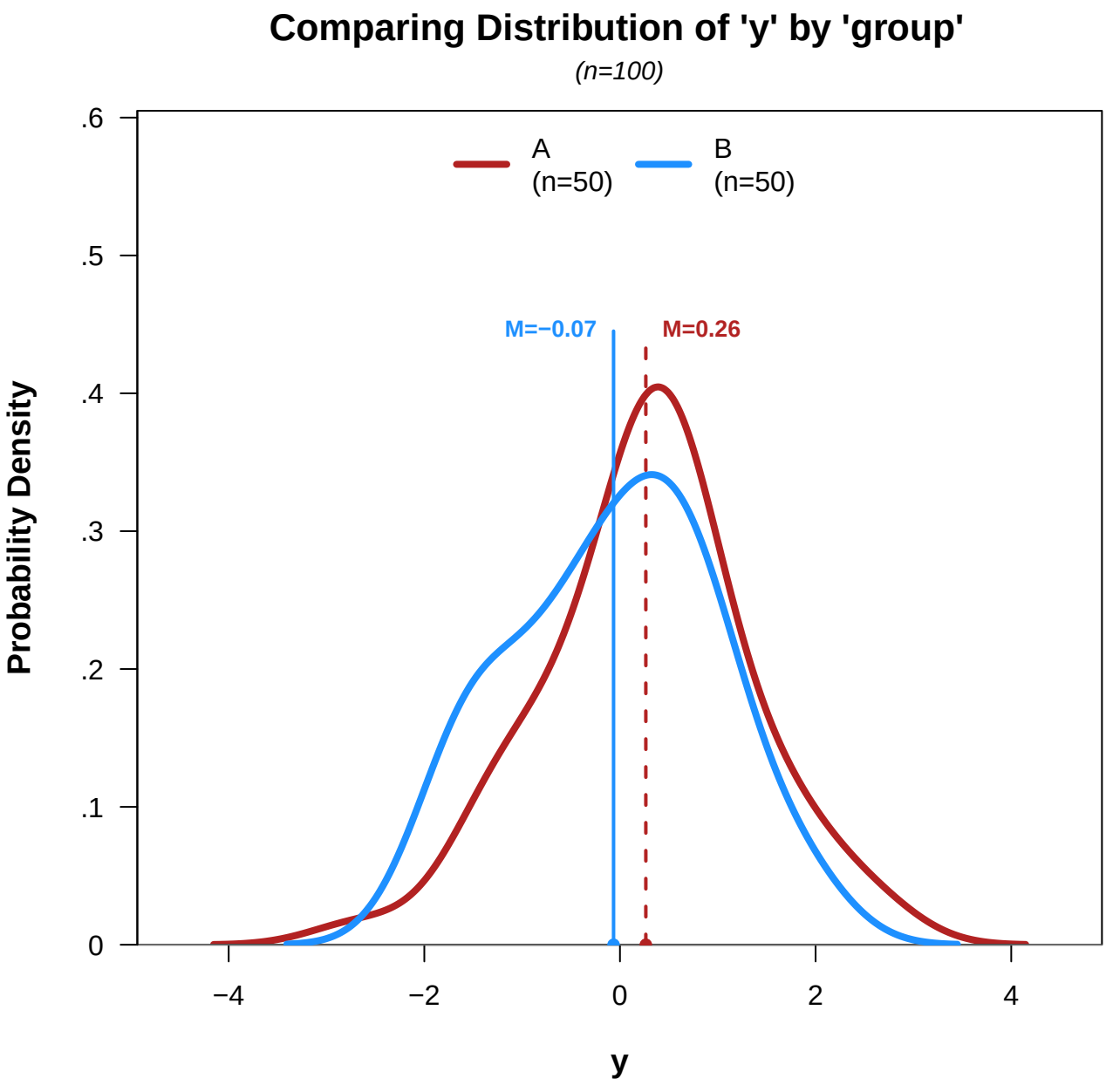

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.03
M=0.25
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

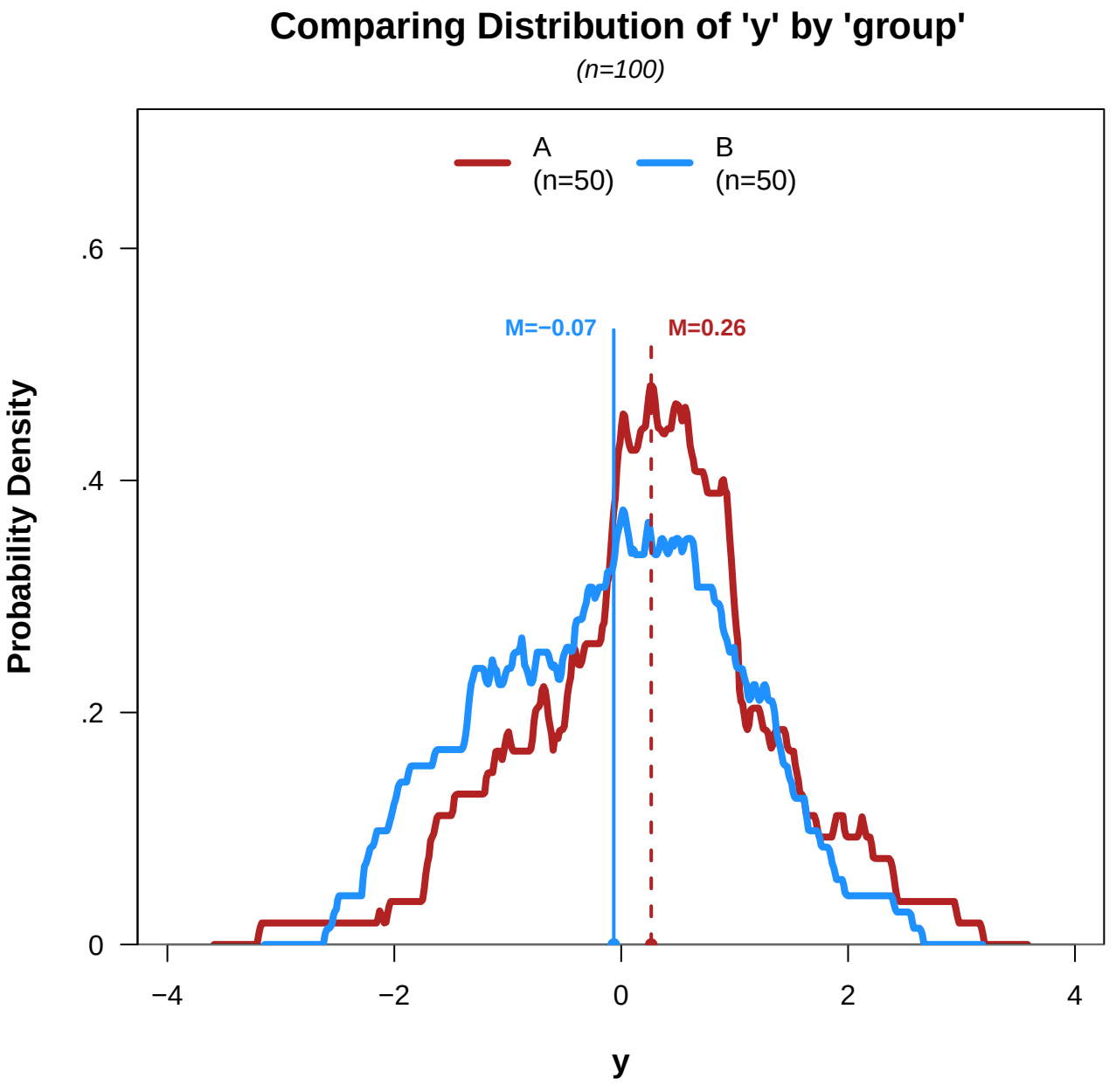

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

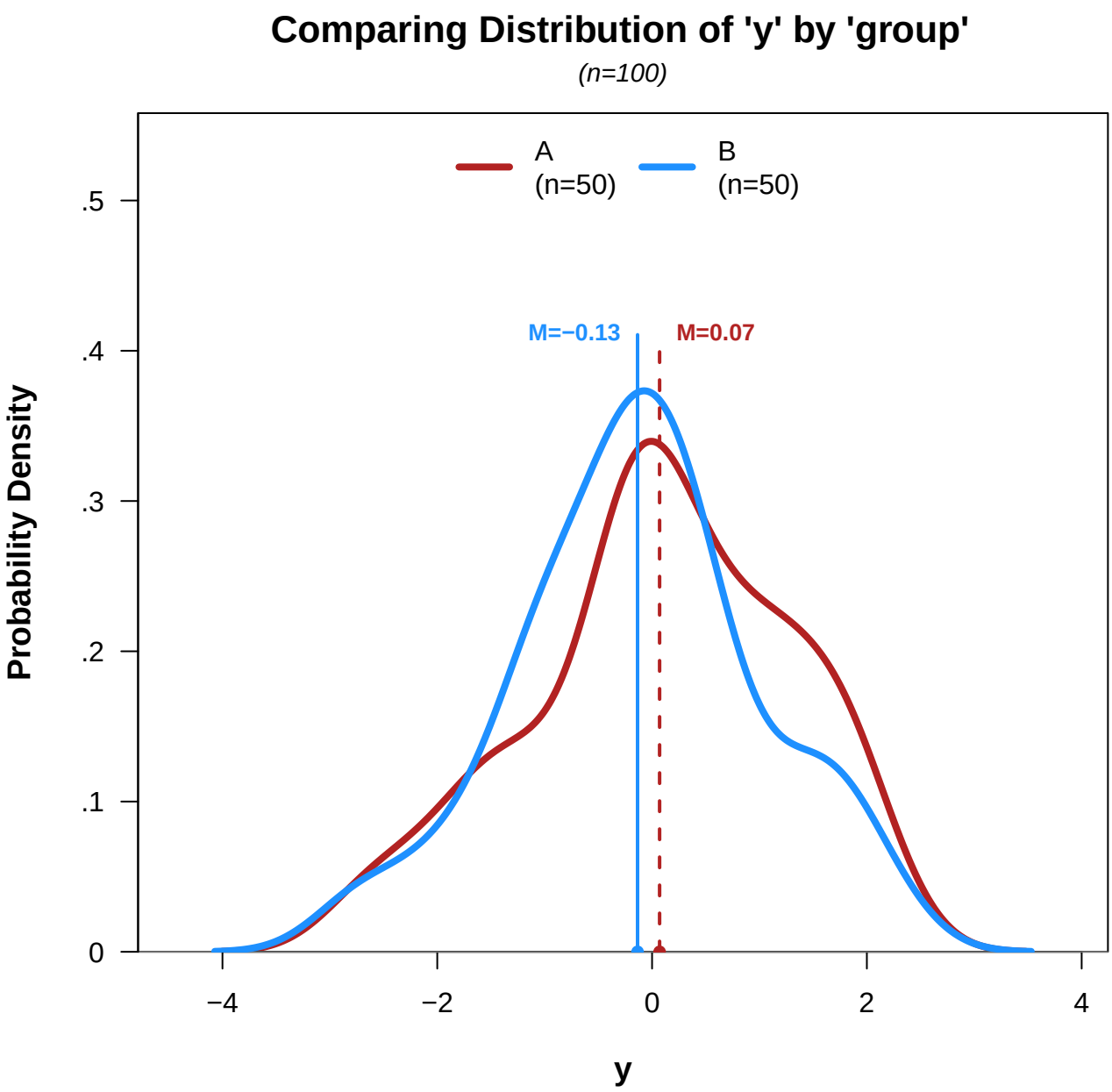

Comparing Distribution of 'y' by 'group2'
(n=100)
A
(n=50)
B
(n=50)
M=−0.09
M=0.21
Probability Density
.6
.4
.2
0
−4
−2
0
2
4
y

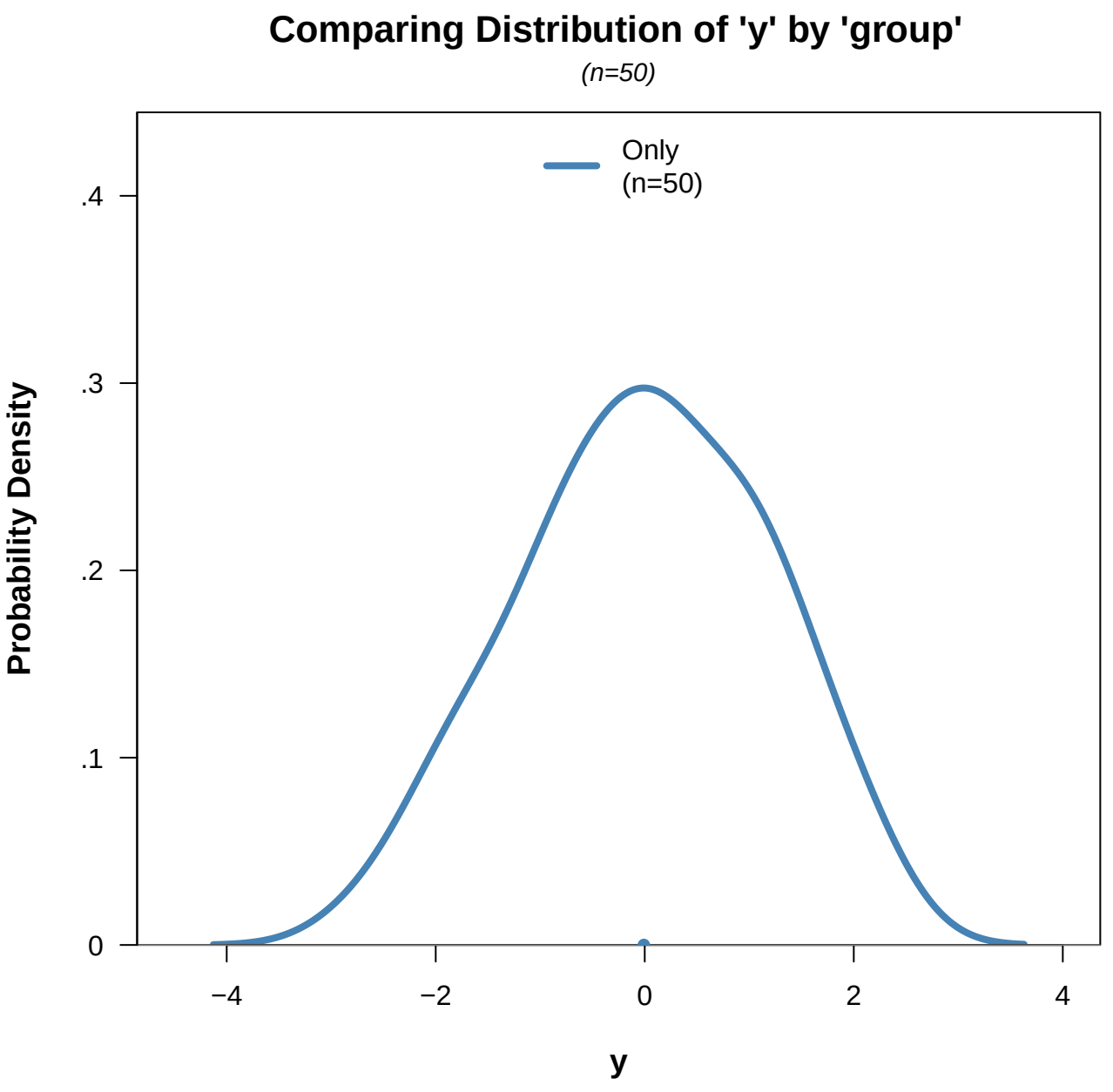

Comparing Distribution of 'y' by 'group3'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=−0.02
M=−0.02
M=0.25
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

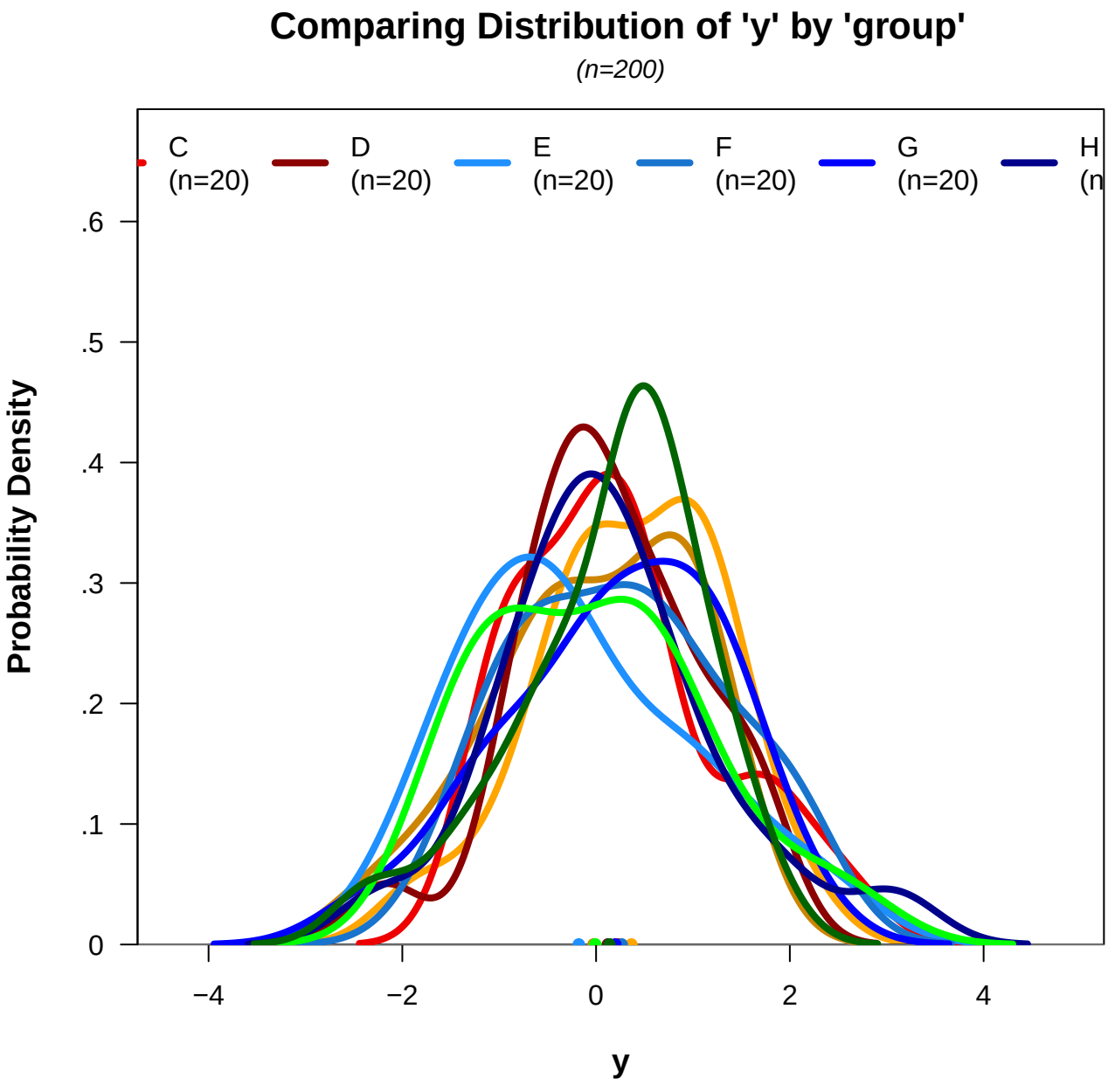

Comparing Distribution of 'y' by 'group4'
(n=100)
A
(n=25)
B
(n=25)
C
(n=25)
D
(n=25)
Probability Density
.6
.4
.2
0
−4
−2
0
2
4
y

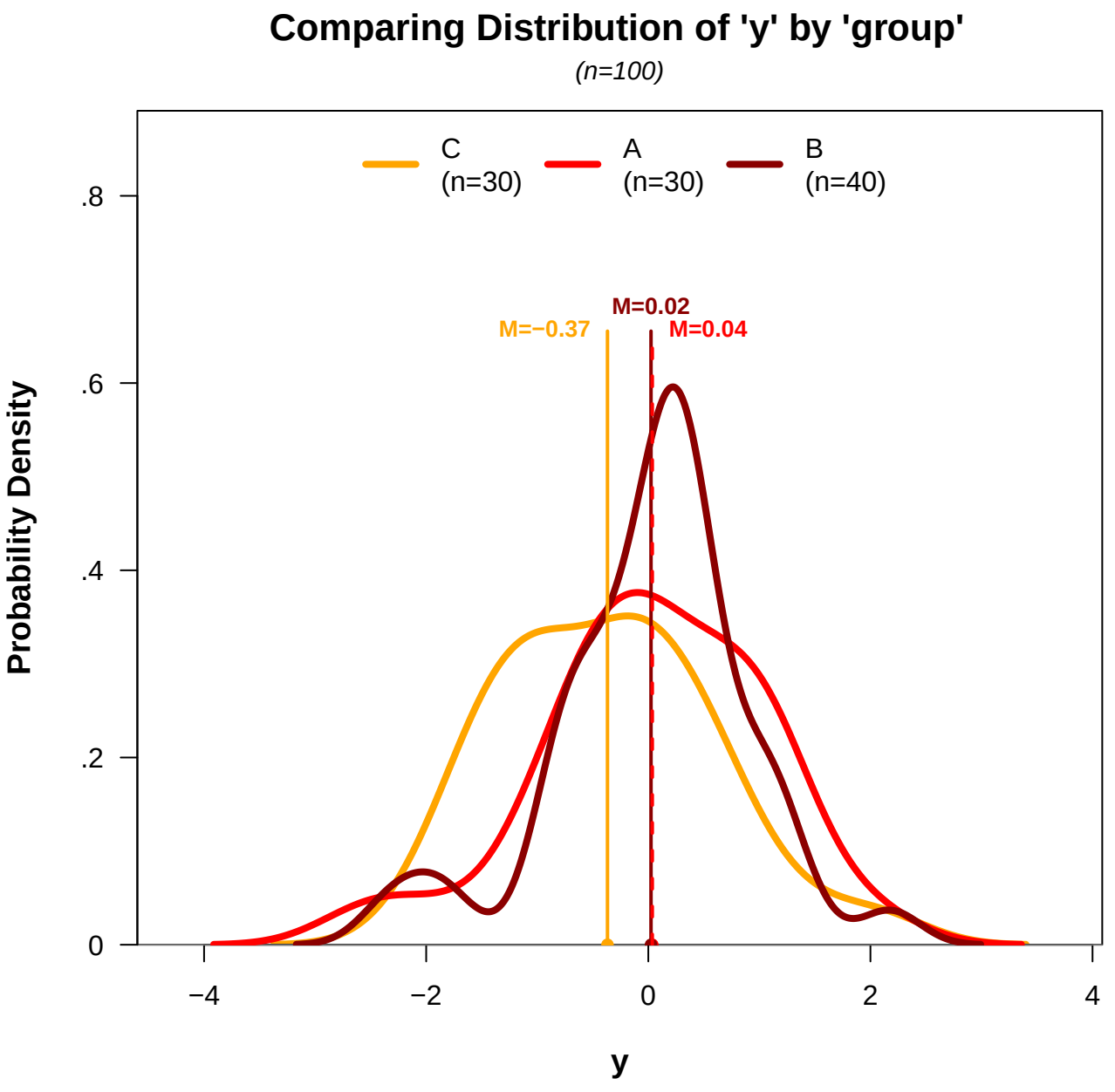

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.07
M=0.26
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

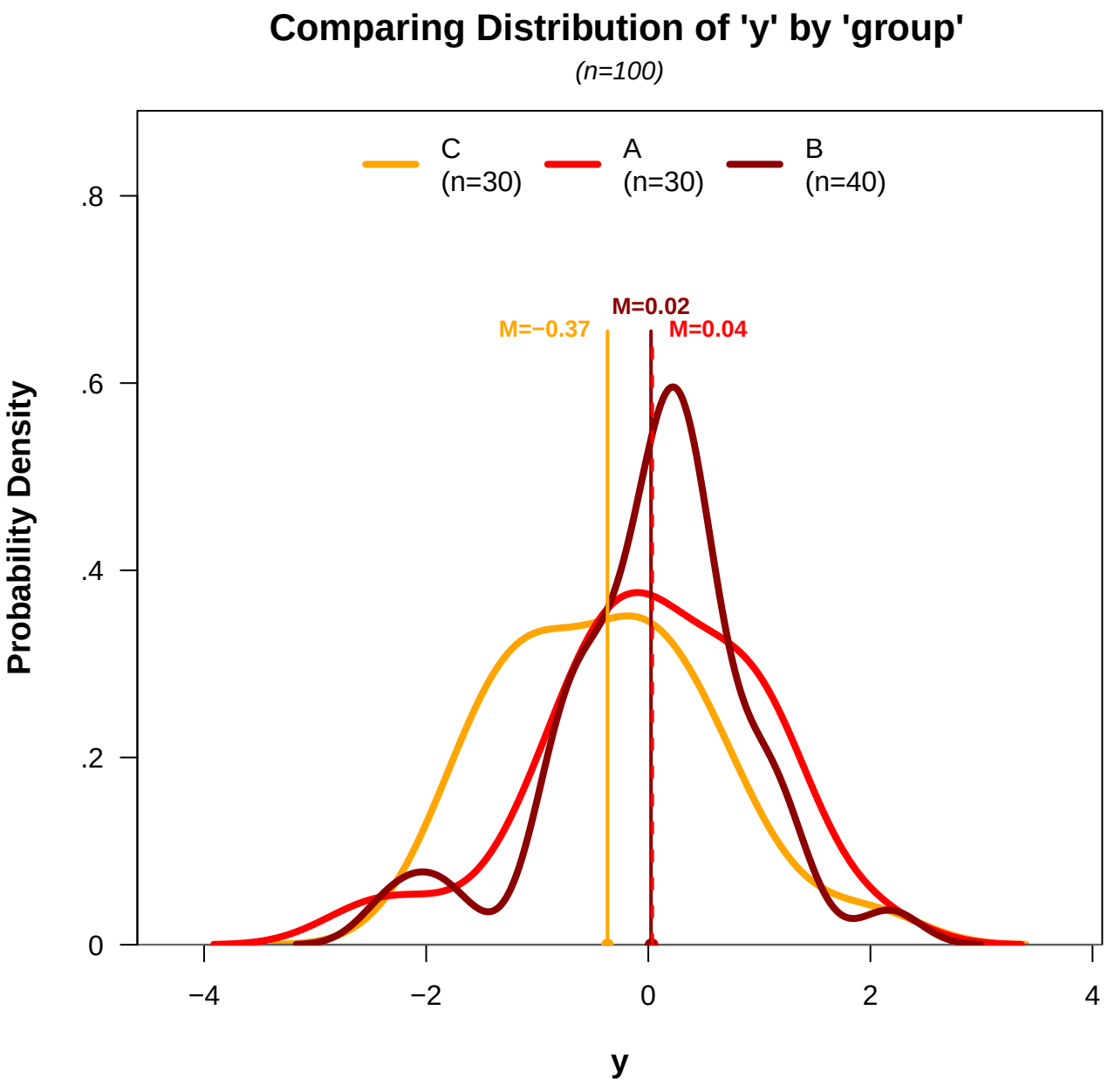

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.07
M=0.26
Probability Density
.6
.4
.2
0
−4
−2
0
2
4
y

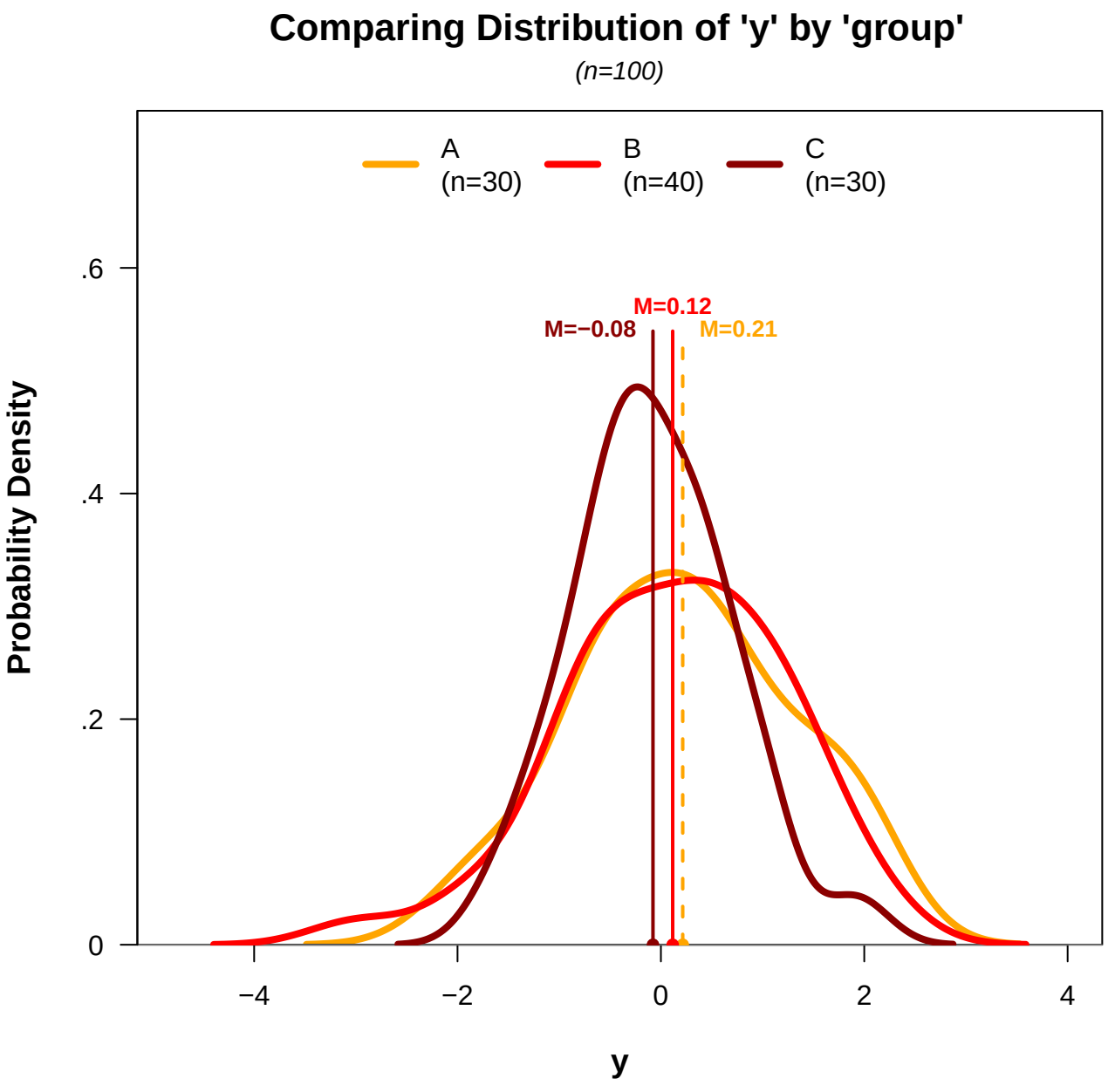

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.13
M=0.07
Probability Density
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

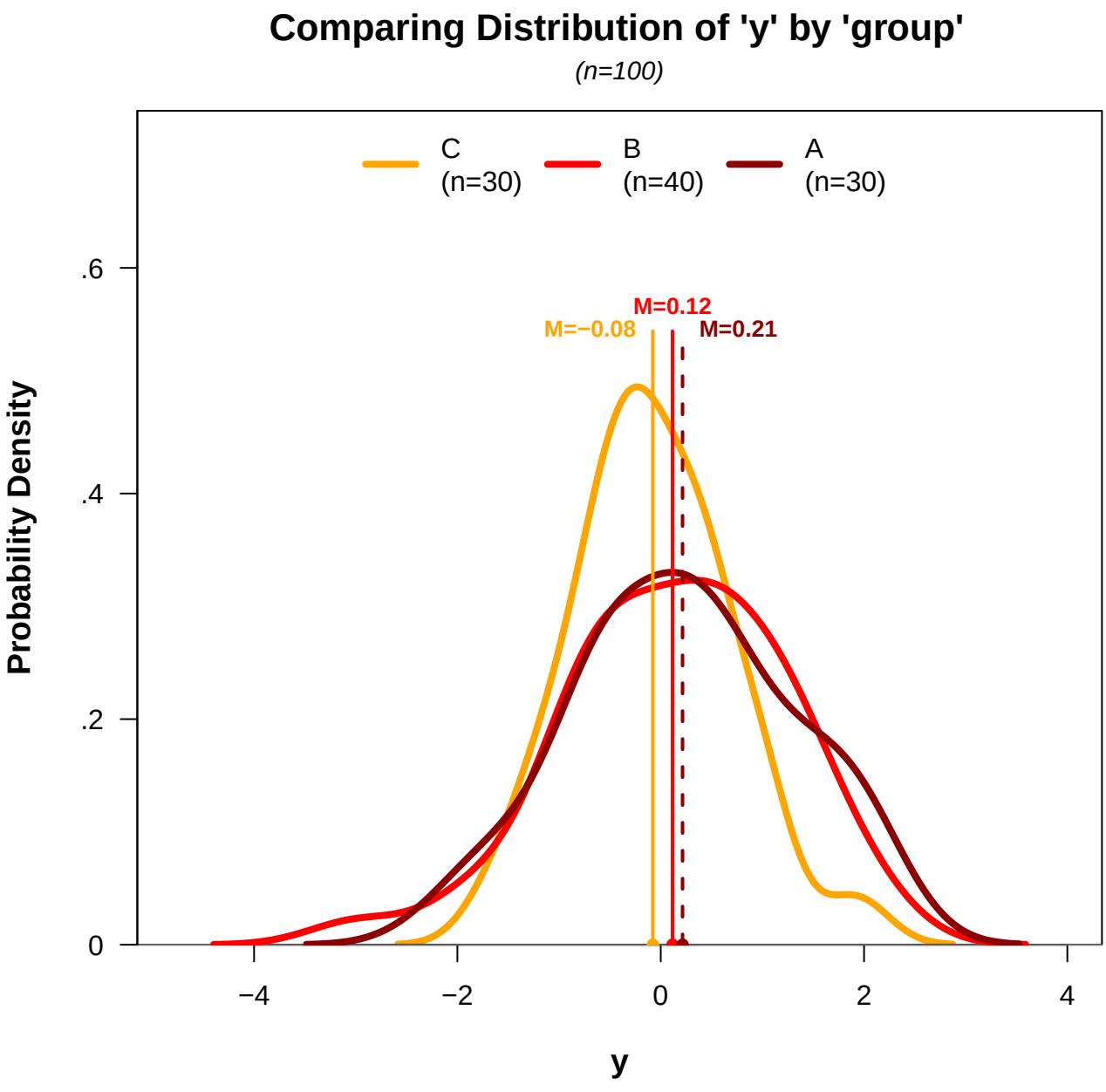

Comparing Distribution of 'y' by 'group'
(n=50)
Only
(n=50)
Probability Density
.4
.3
.2
.1
0
−4
−2
0
2
4
y

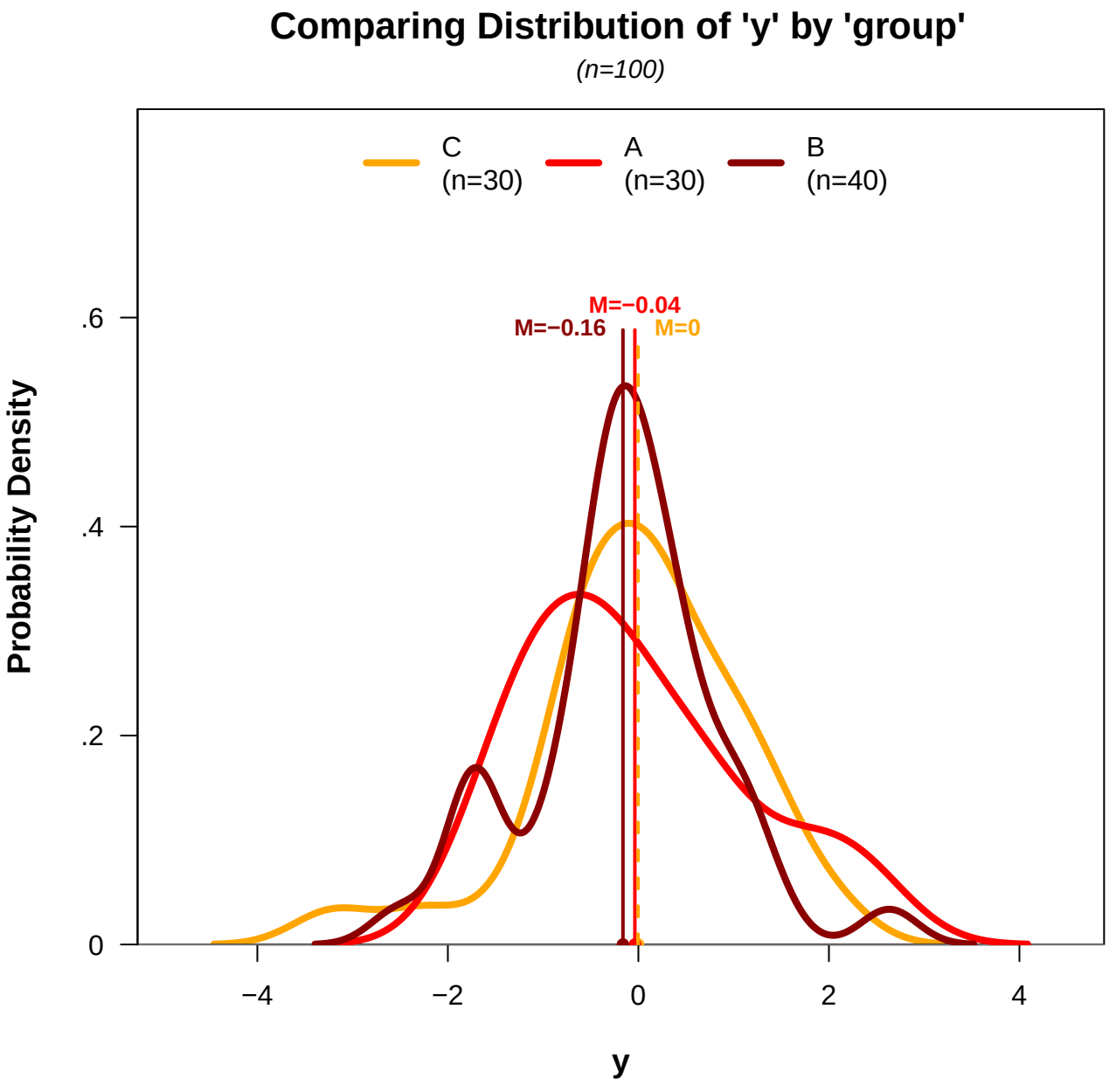

Comparing Distribution of 'y' by 'group'
(n=200)
C
(n=20)
D
(n=20)
E
(n=20)
F
(n=20)
G
(n=20)
H
(n
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

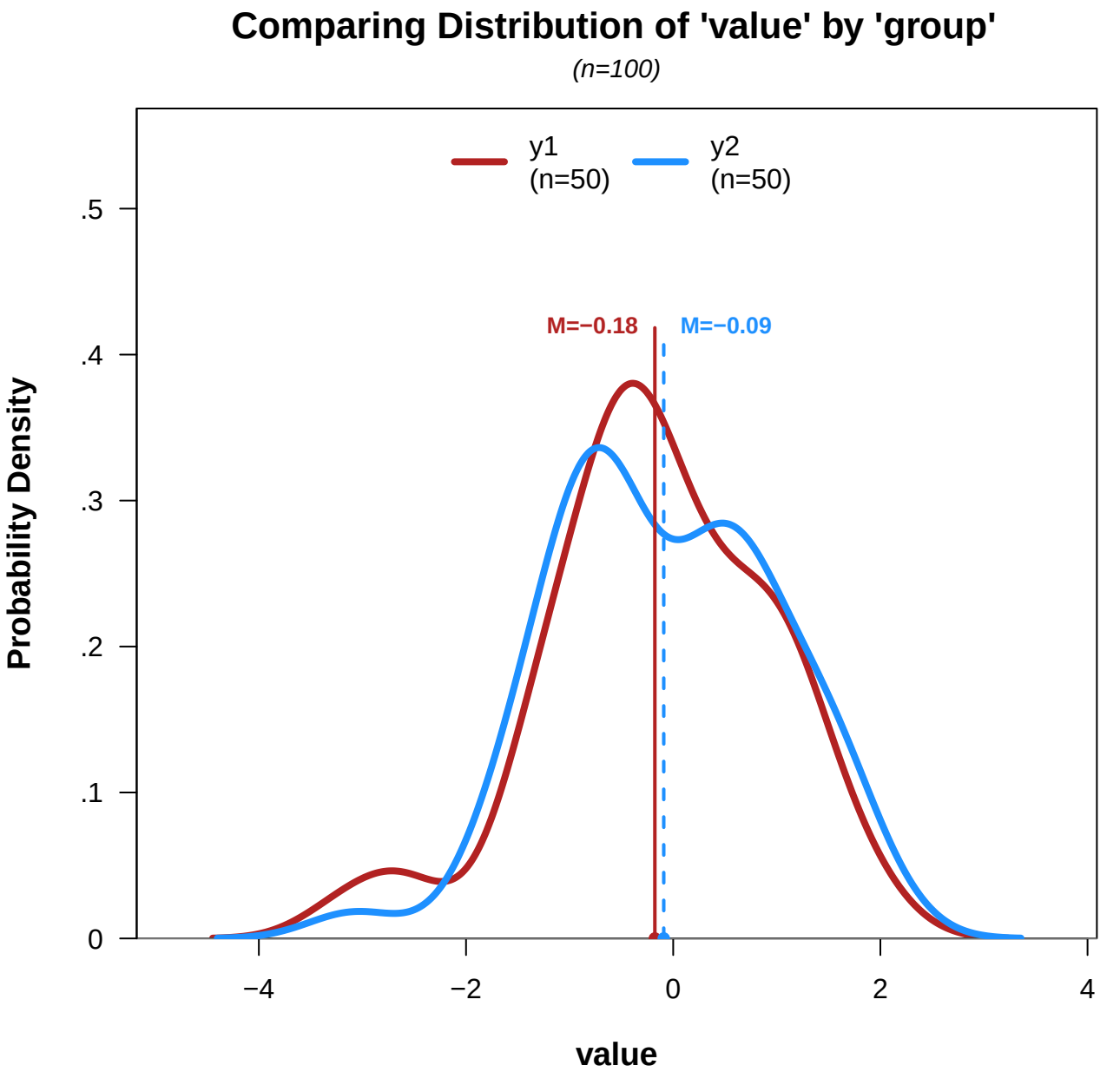

Comparing Distribution of 'y' by 'group'
(n=100)
C
(n=30)
A
(n=30)
B
(n=40)
M=0.02
M=−0.37
M=0.04
Probability Density
.8
.6
.4
.2
0
−4
−2
0
2
4
y

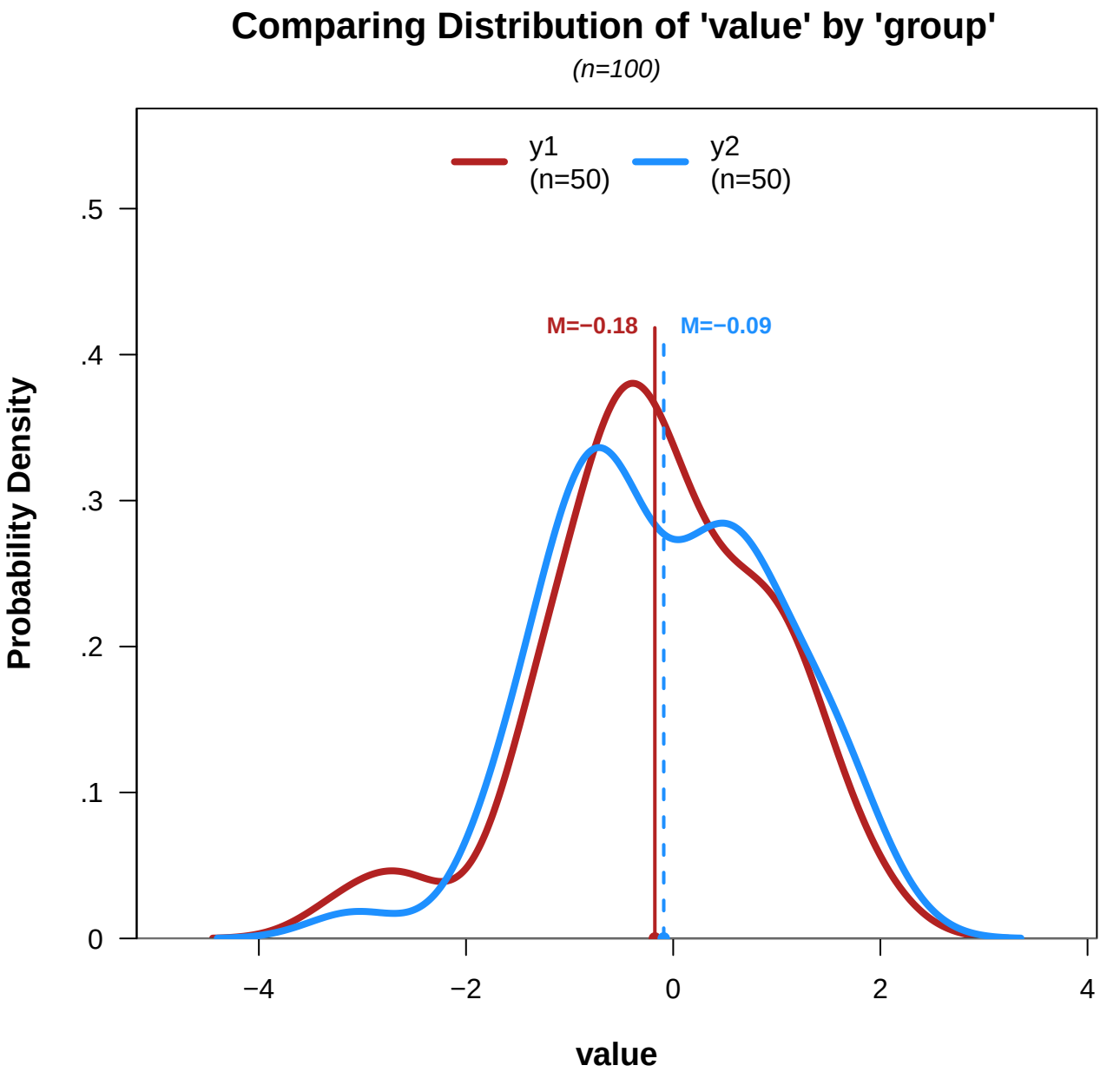

Comparing Distribution of 'y' by 'group'
(n=100)
C
(n=30)
A
(n=30)
B
(n=40)
M=0.02
M=−0.37
M=0.04
Probability Density
.8
.6
.4
.2
0
−4
−2
0
2
4
y

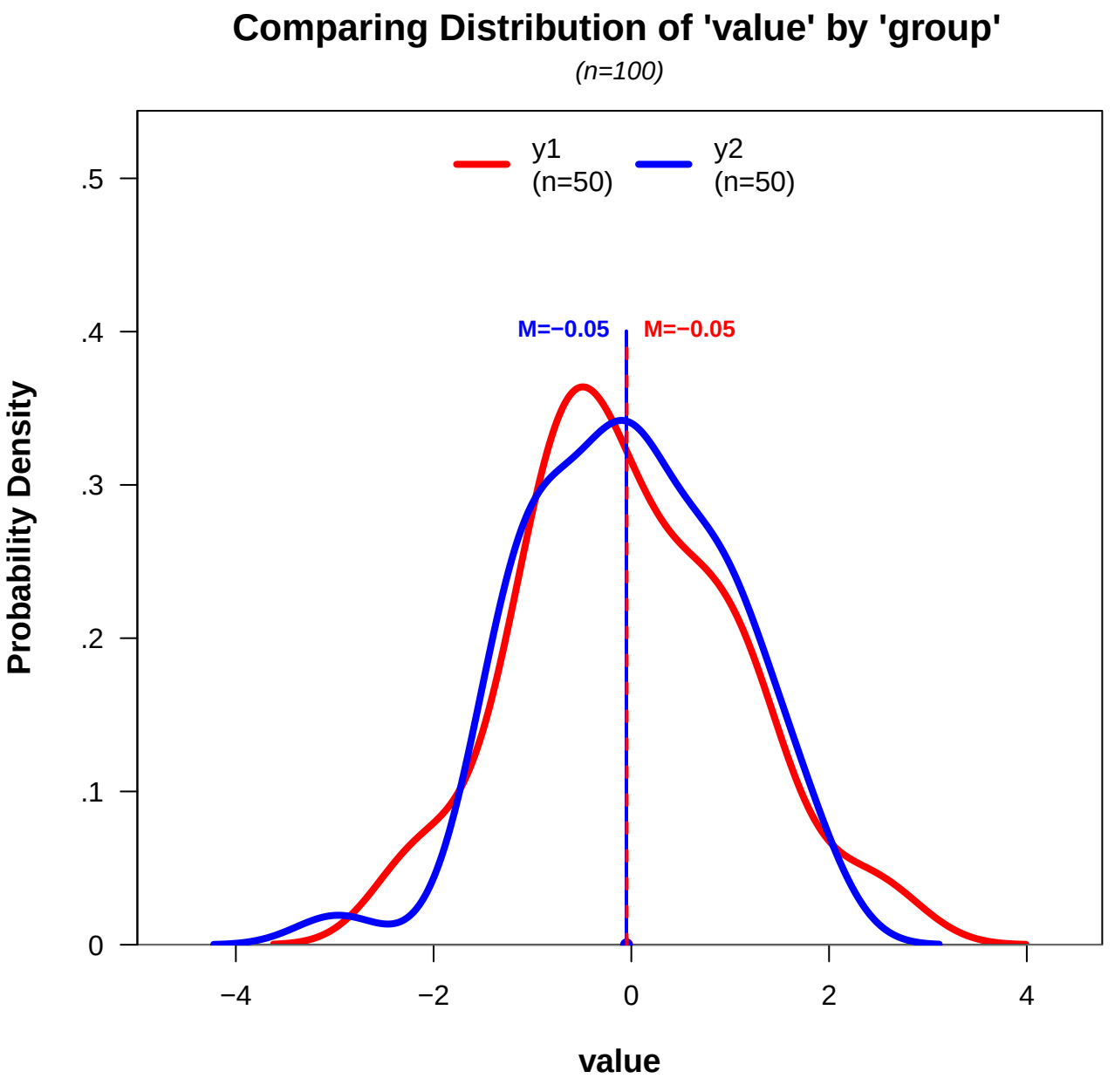

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.12
M=−0.08
M=0.21
Probability Density
.6
.4
.2
0
−4
−2
0
2
4
y

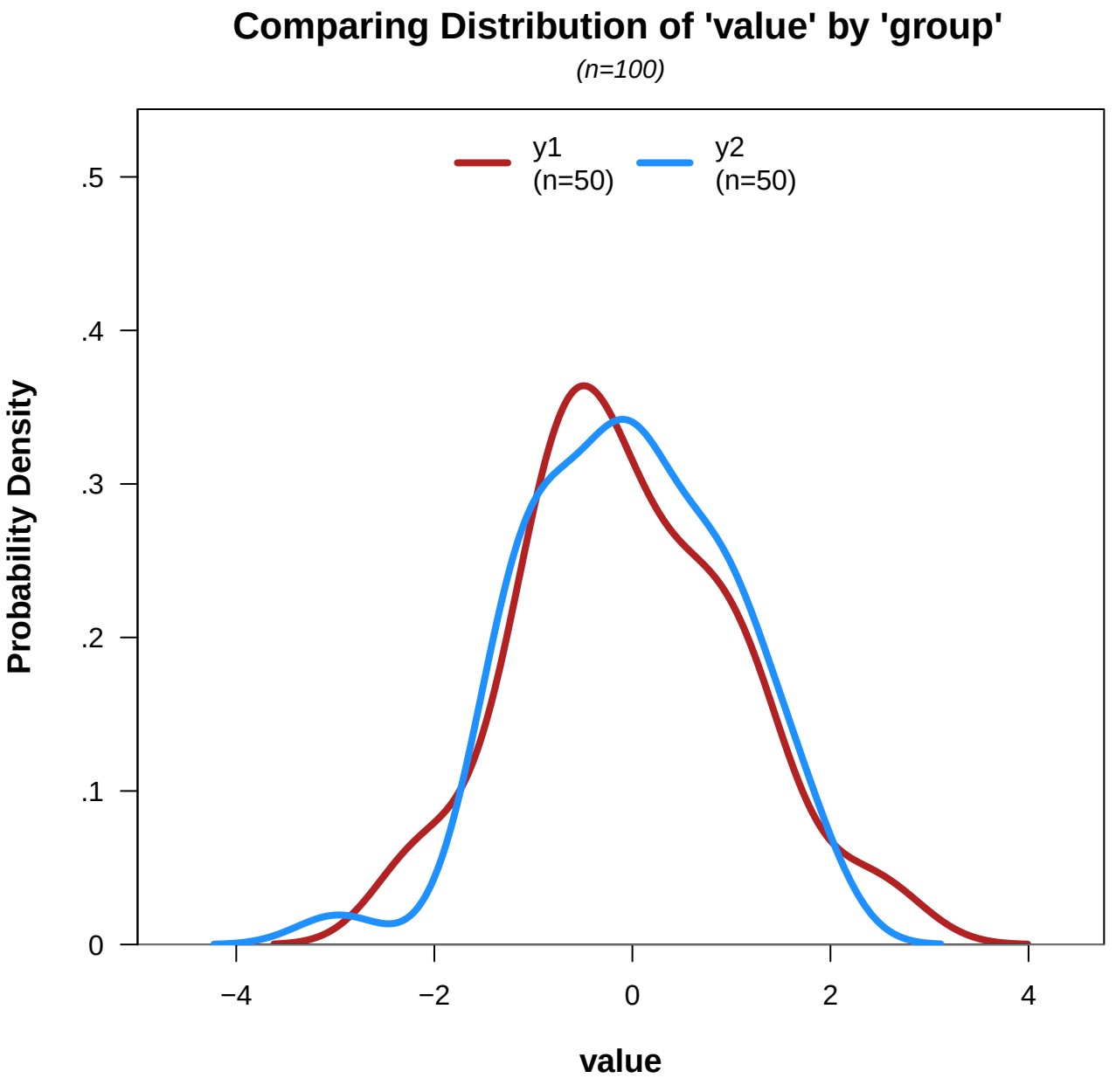

Comparing Distribution of 'y' by 'group'
(n=100)
C
(n=30)
B
(n=40)
A
(n=30)
M=0.12
M=−0.08
M=0.21
Probability Density
.6
.4
.2
0
−4
−2
0
2
4
y

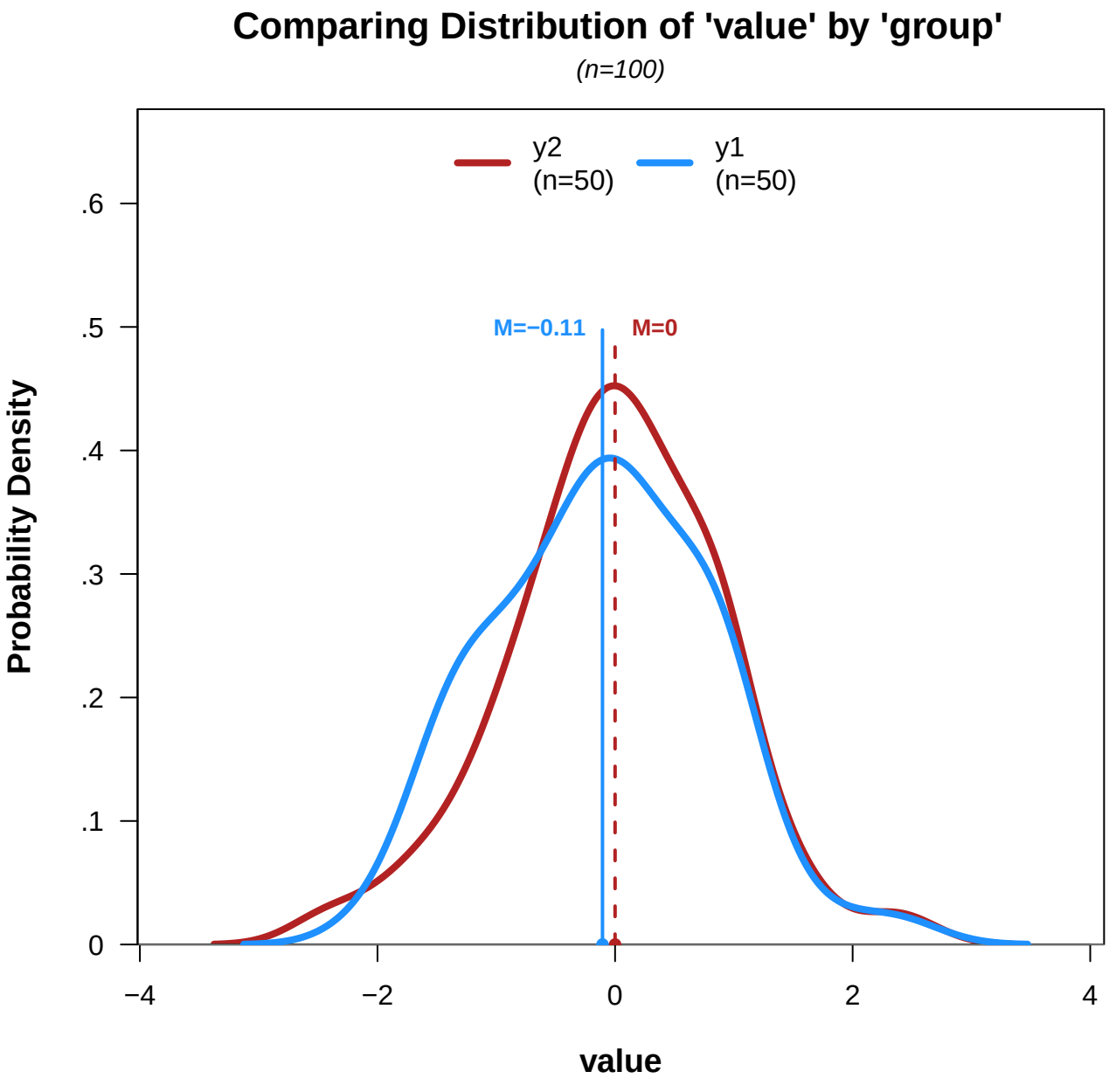

Comparing Distribution of 'y' by 'group'
(n=100)
C
(n=30)
A
(n=30)
B
(n=40)
M=−0.04
M=−0.16
M=0
Probability Density
.6
.4
.2
0
−4
−2
0
2
4
y

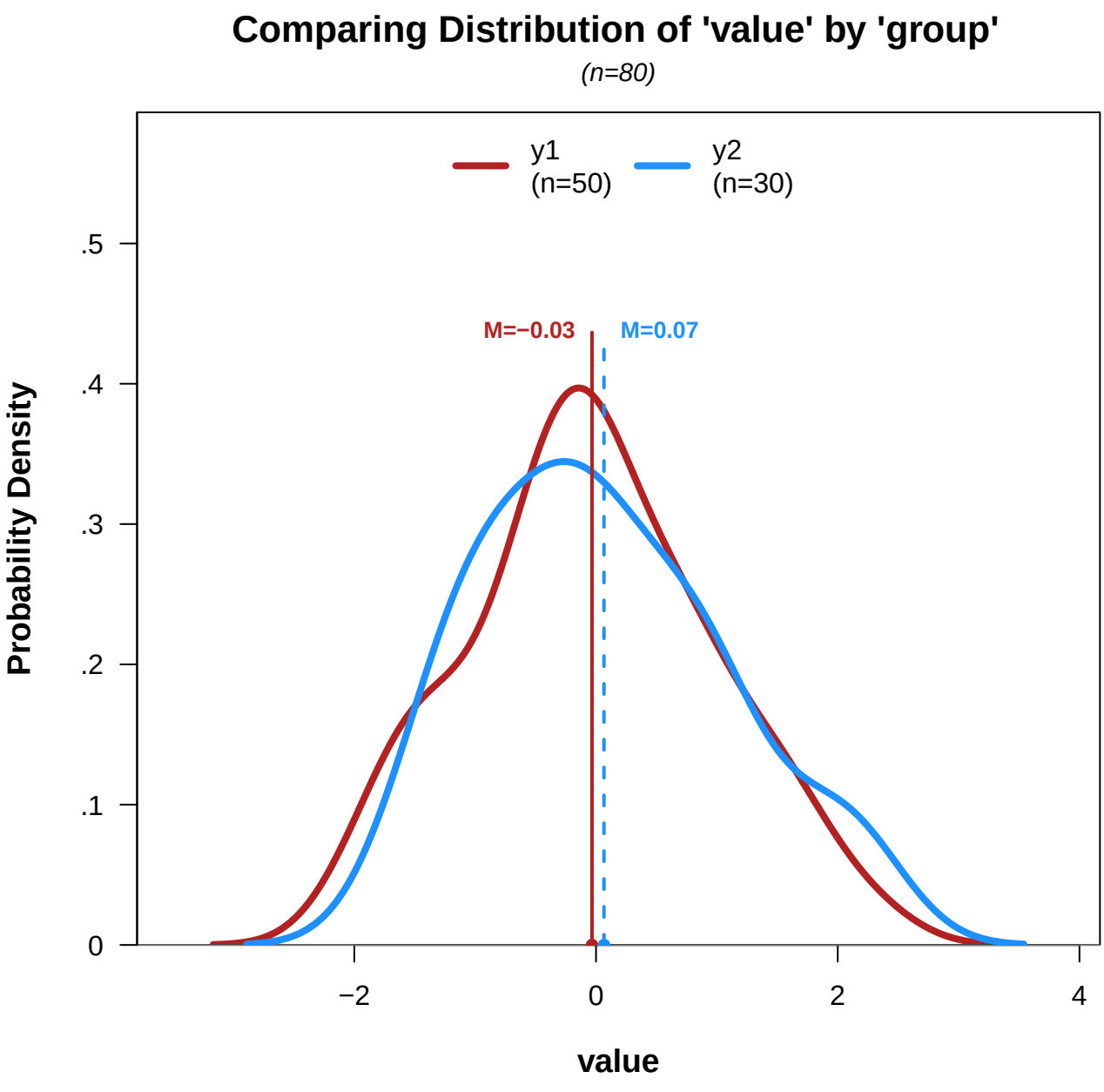

Comparing Distribution of 'value' by 'group'
(n=100)
y1
(n=50)
y2
(n=50)
M=−0.18
M=−0.09
Probability Density
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value

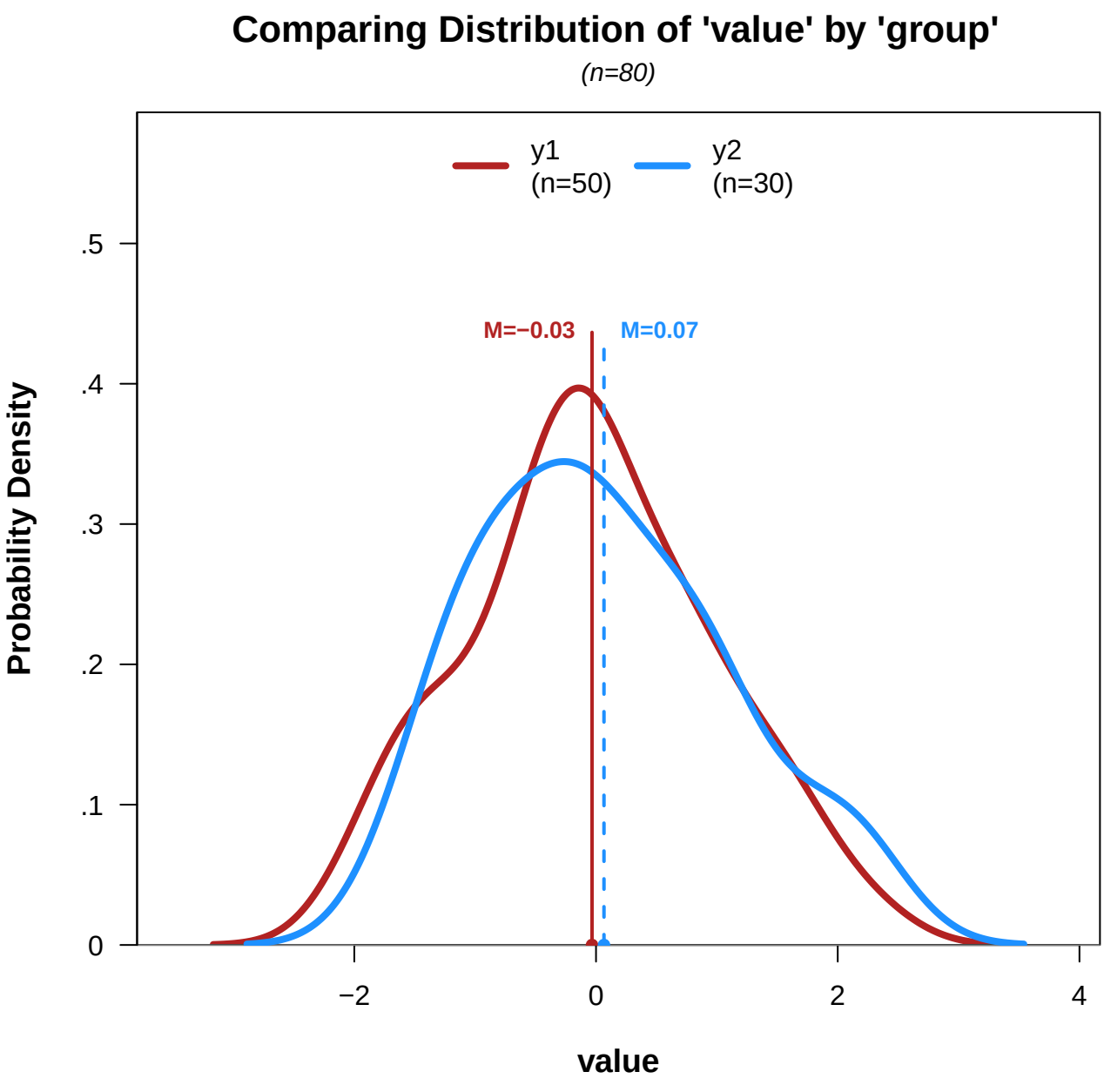

Comparing Distribution of 'value' by 'group'
(n=100)
y1
(n=50)
y2
(n=50)
M=−0.18
M=−0.09
Probability Density
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value

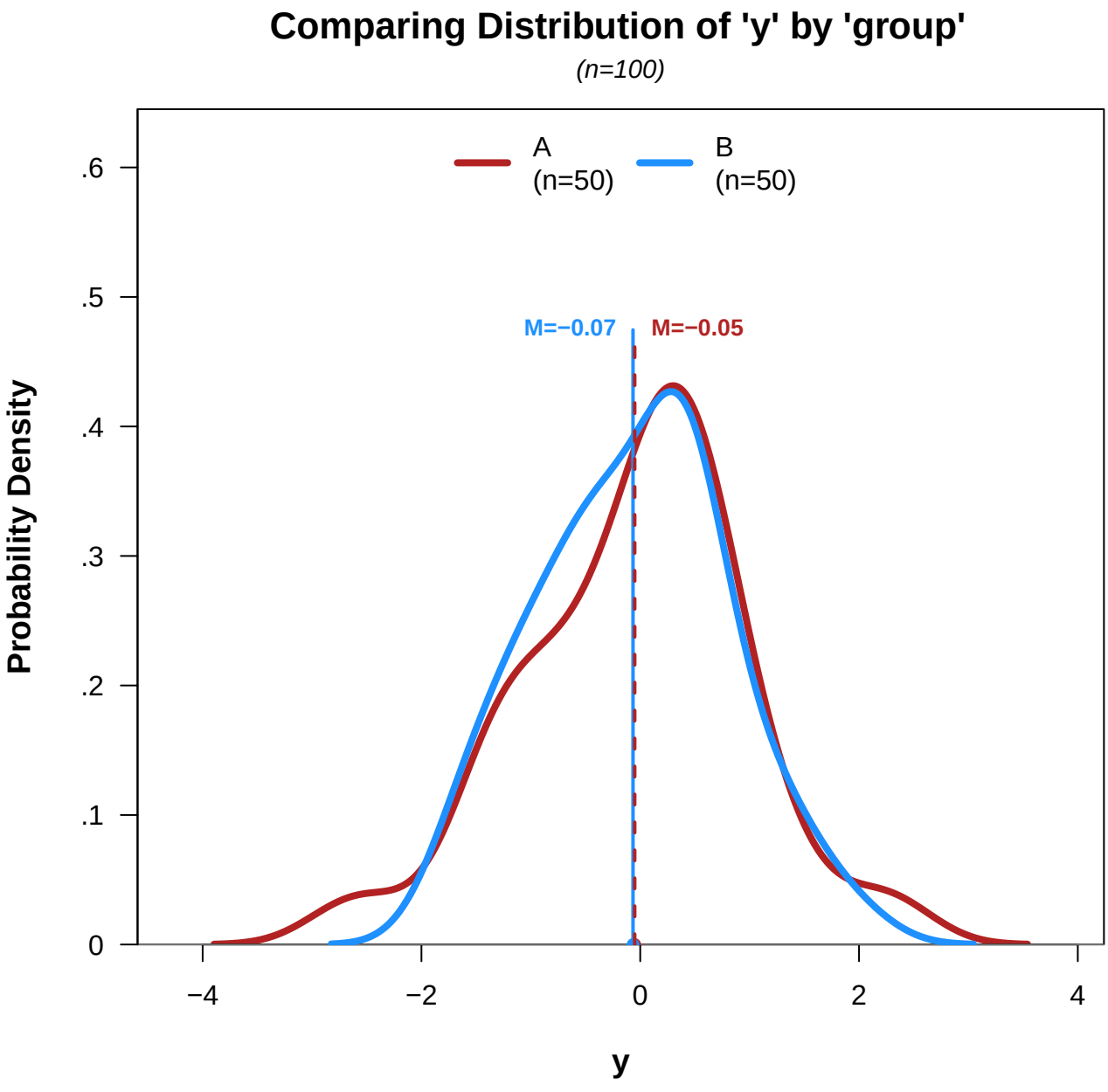

Comparing Distribution of 'value' by 'group'
(n=100)
y1
(n=50)
y2
(n=50)
M=−0.05
M=−0.05
Probability Density
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value

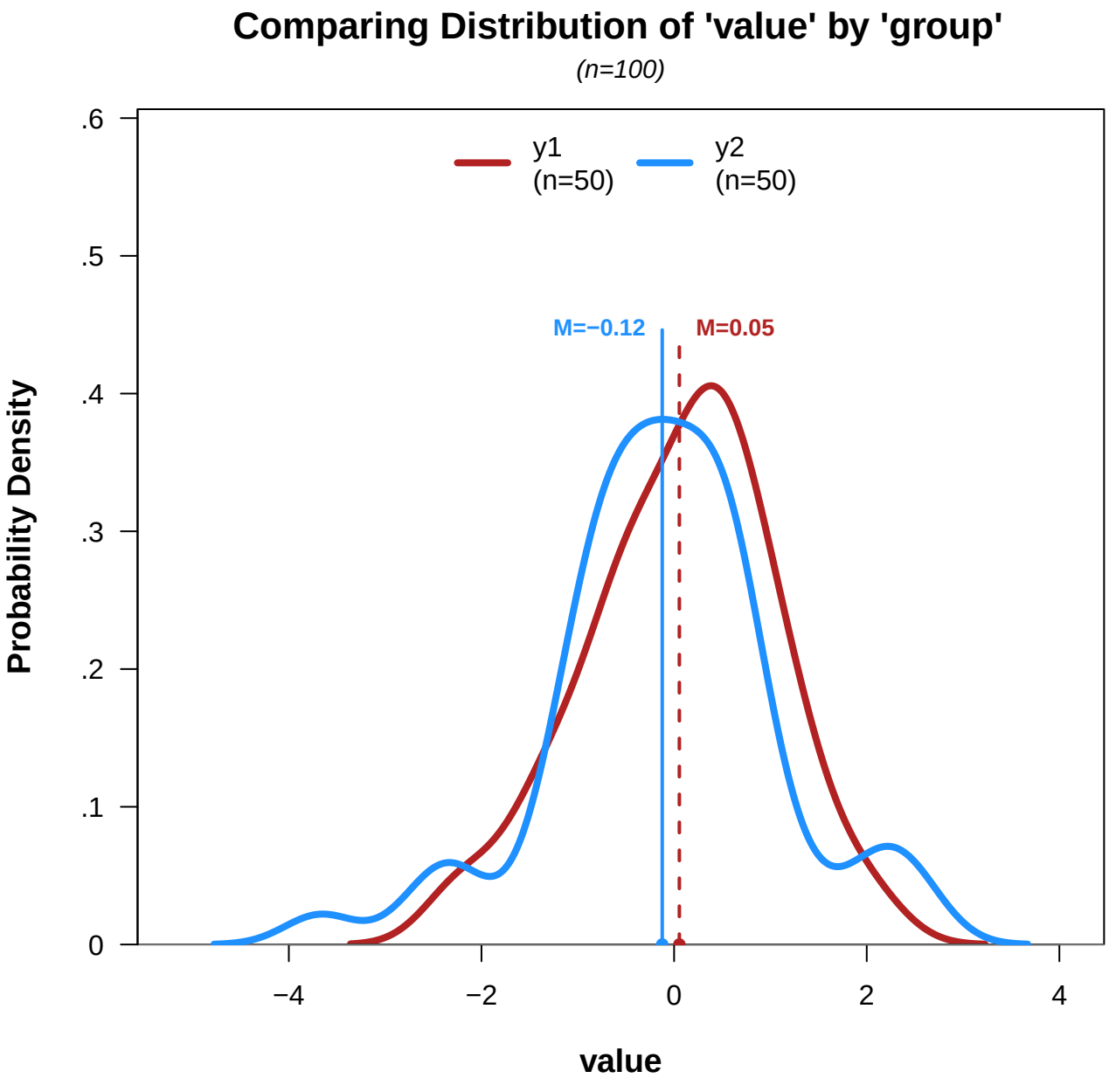

Comparing Distribution of 'value' by 'group'
(n=100)
y1
(n=50)
y2
(n=50)
Probability Density
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value

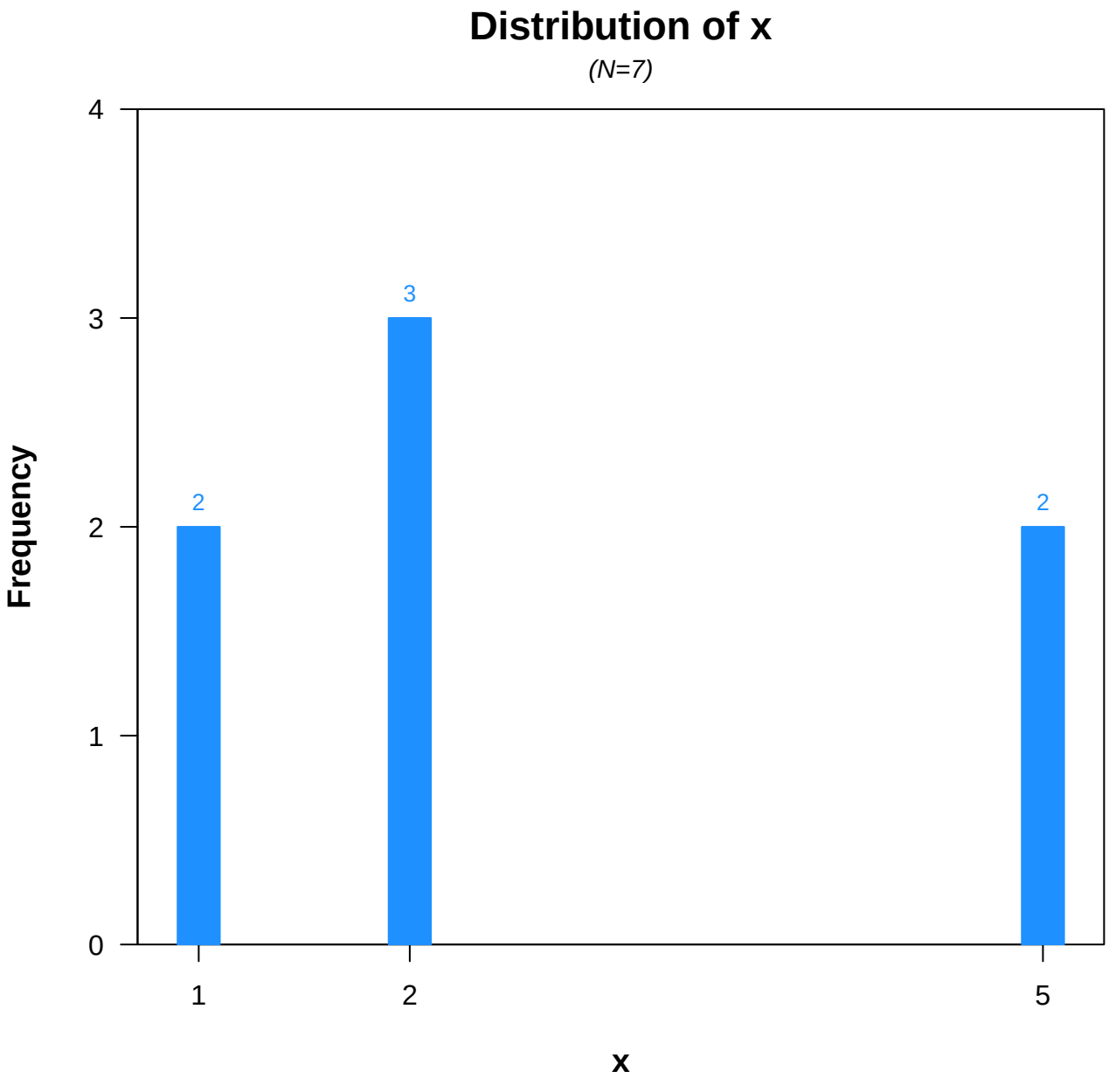

Comparing Distribution of 'value' by 'group'
(n=100)
y2
(n=50)
y1
(n=50)
M=−0.11
M=0
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value

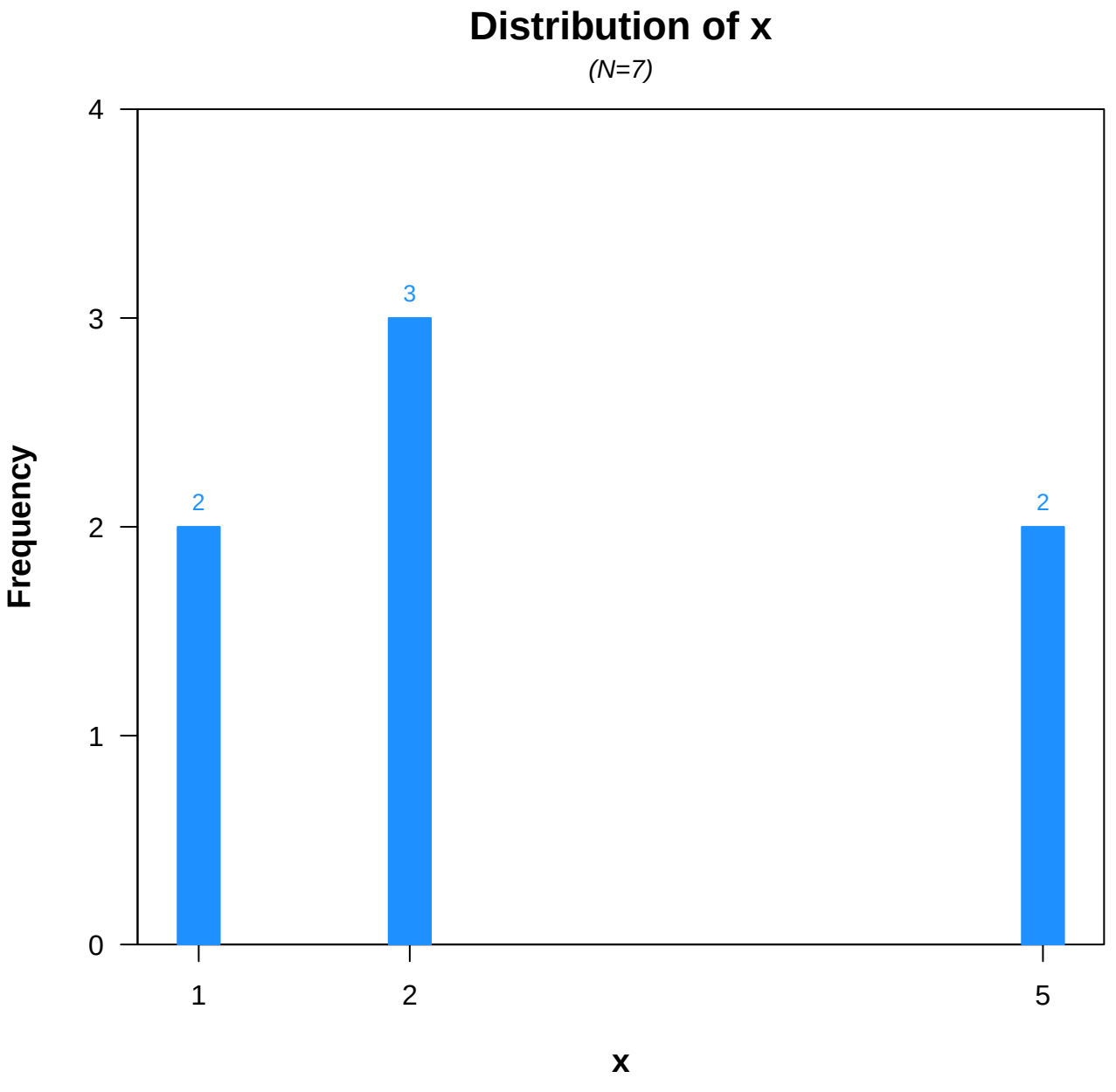

Comparing Distribution of 'value' by 'group'
(n=80)
y1
(n=50)
y2
(n=30)
M=−0.03
M=0.07
Probability Density
.5
.4
.3
.2
.1
0
−2
0
2
4
value

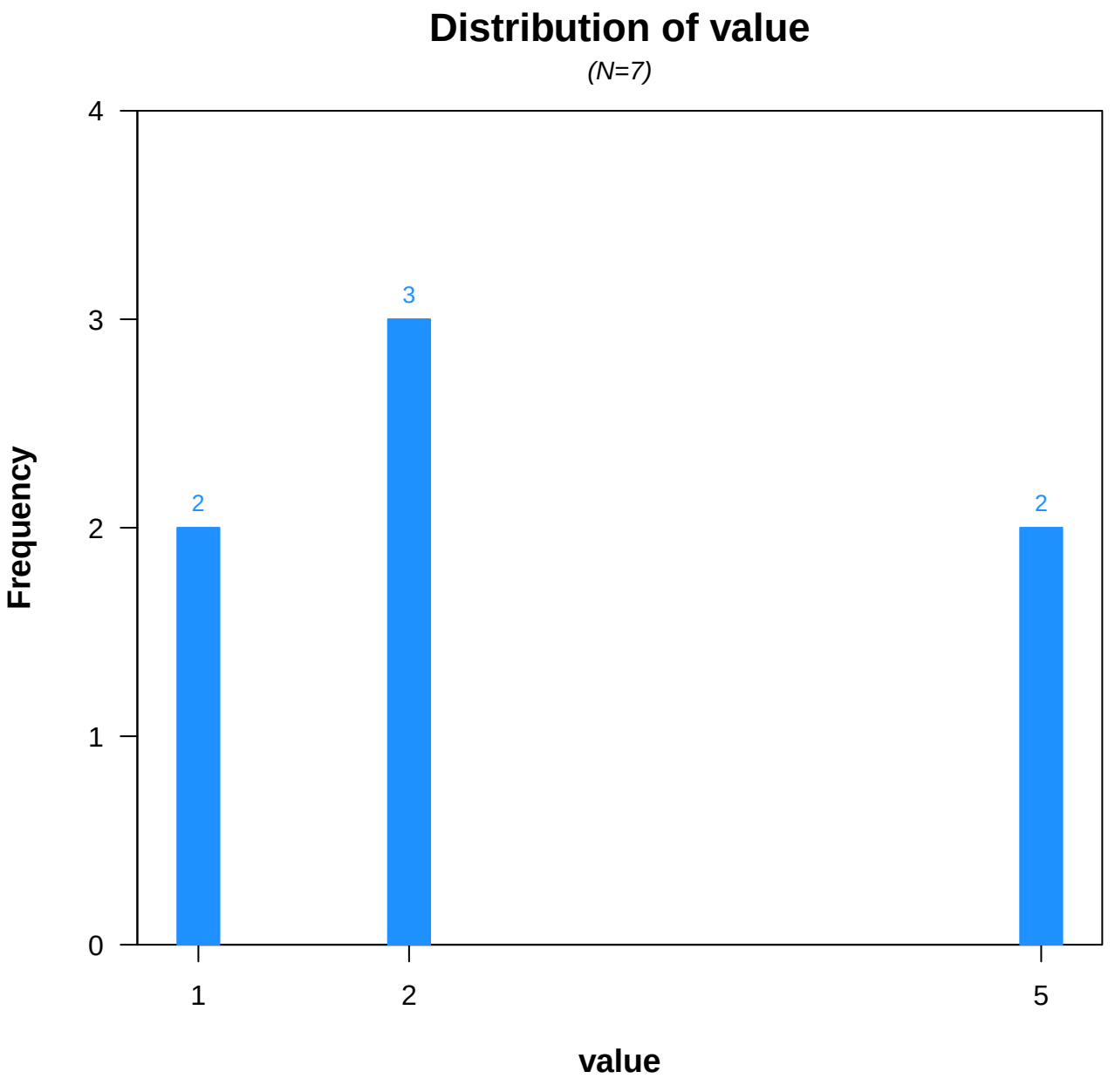

Comparing Distribution of 'value' by 'group'
(n=80)
y1
(n=50)
y2
(n=30)
M=−0.03
M=0.07
Probability Density
.5
.4
.3
.2
.1
0
−2
0
2
4
value

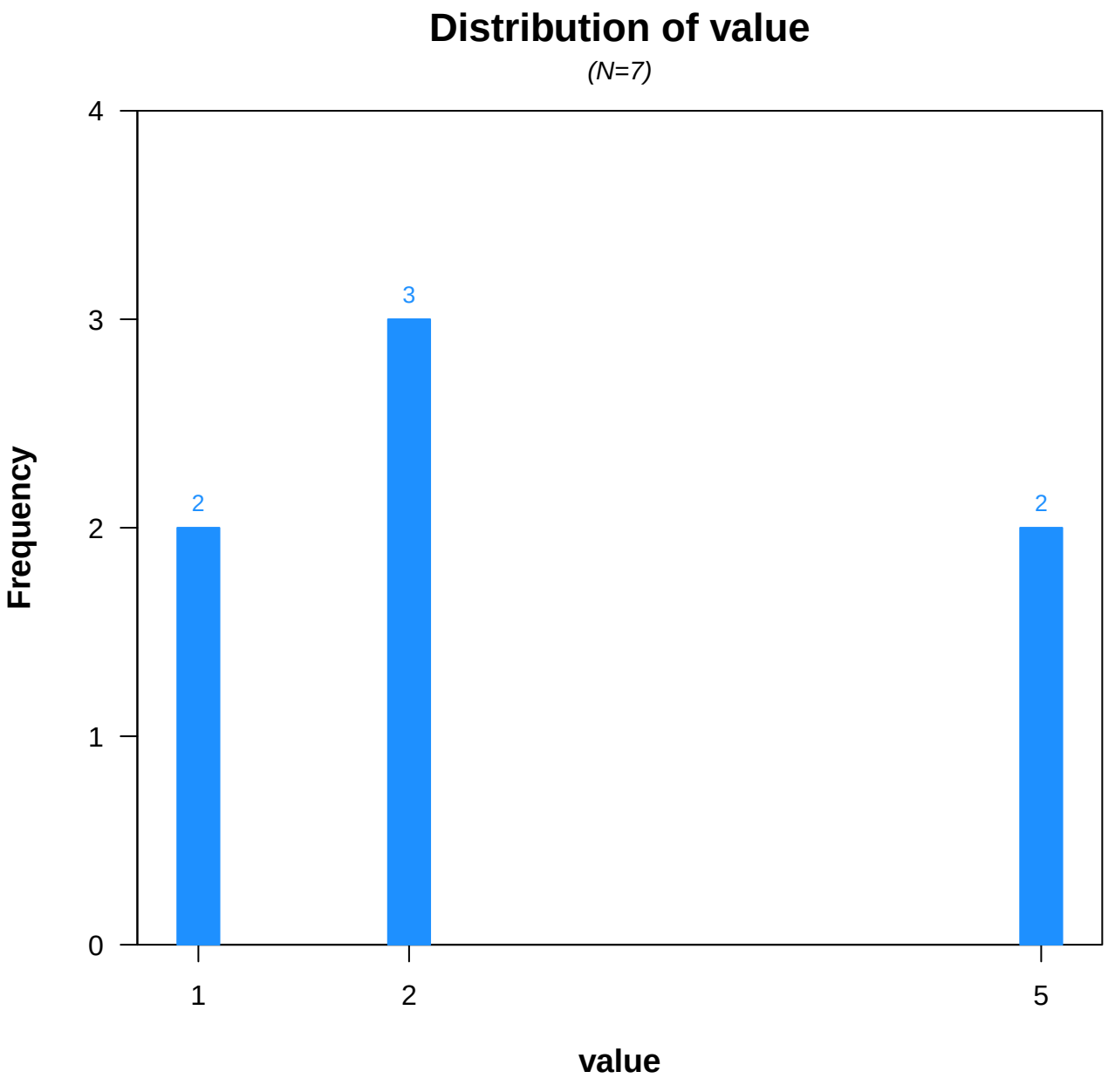

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.07
M=−0.05
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y

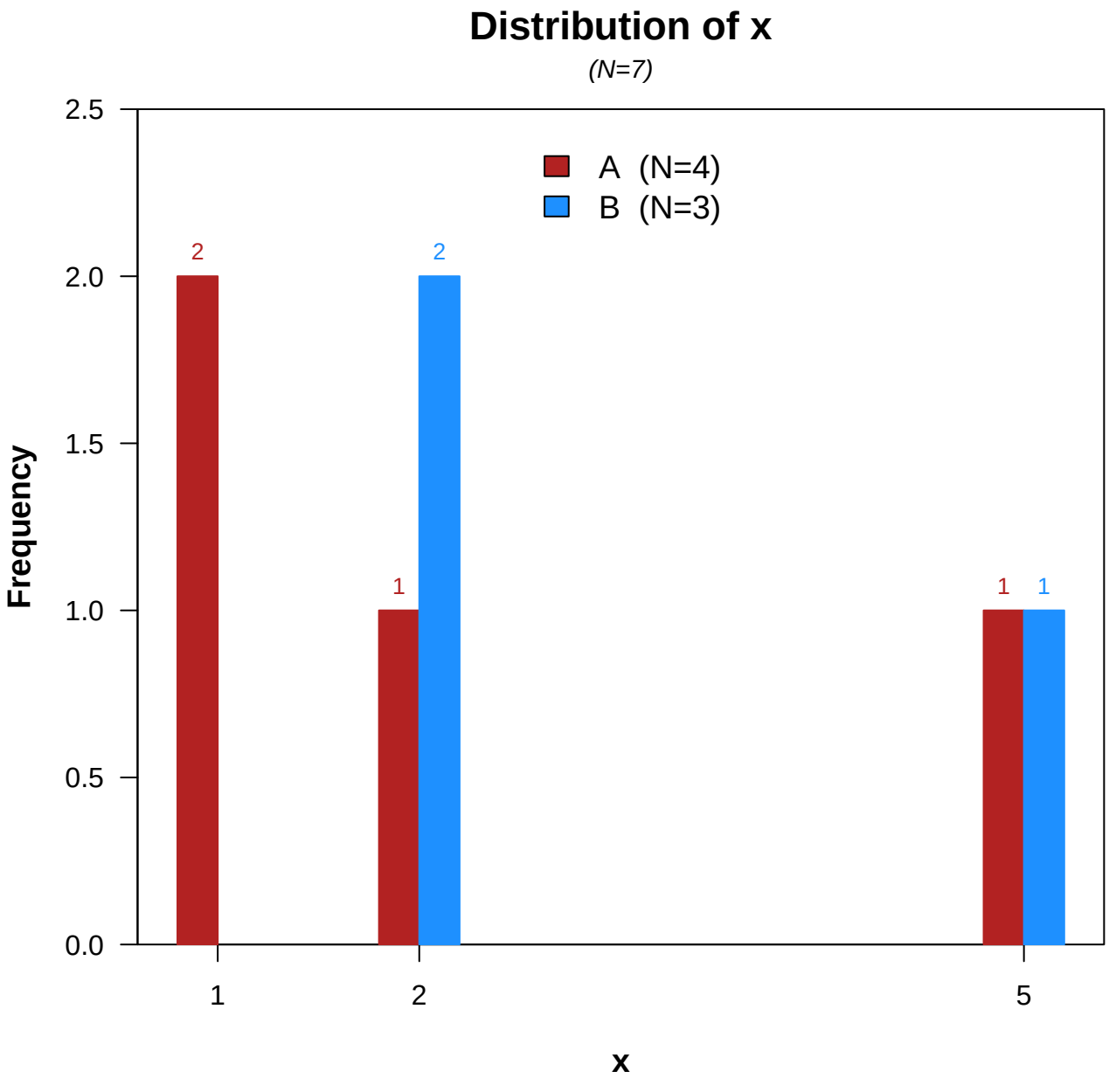

Comparing Distribution of 'value' by 'group'
(n=100)
y1
(n=50)
y2
(n=50)
M=−0.12
M=0.05
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value

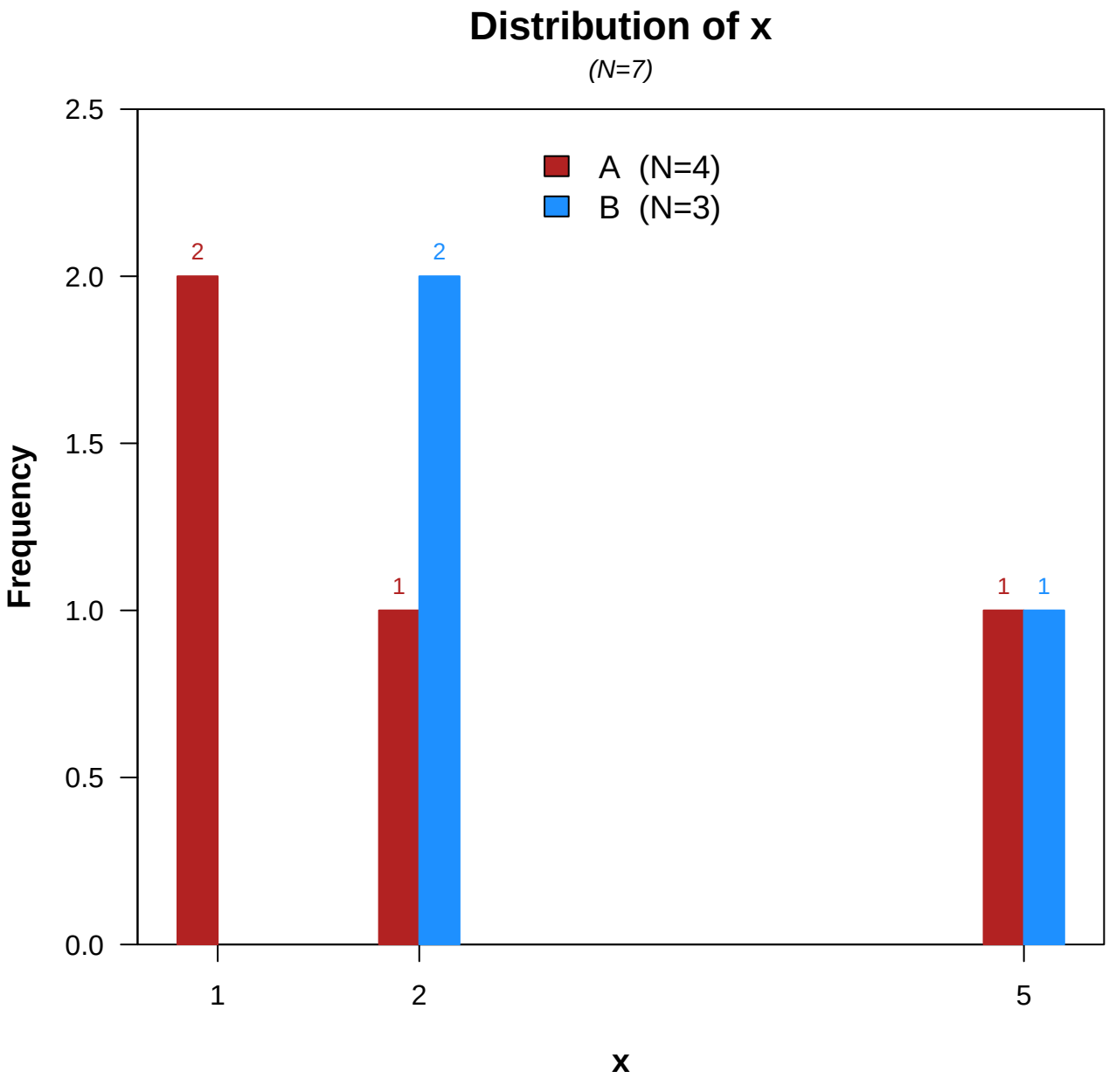

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
5
x

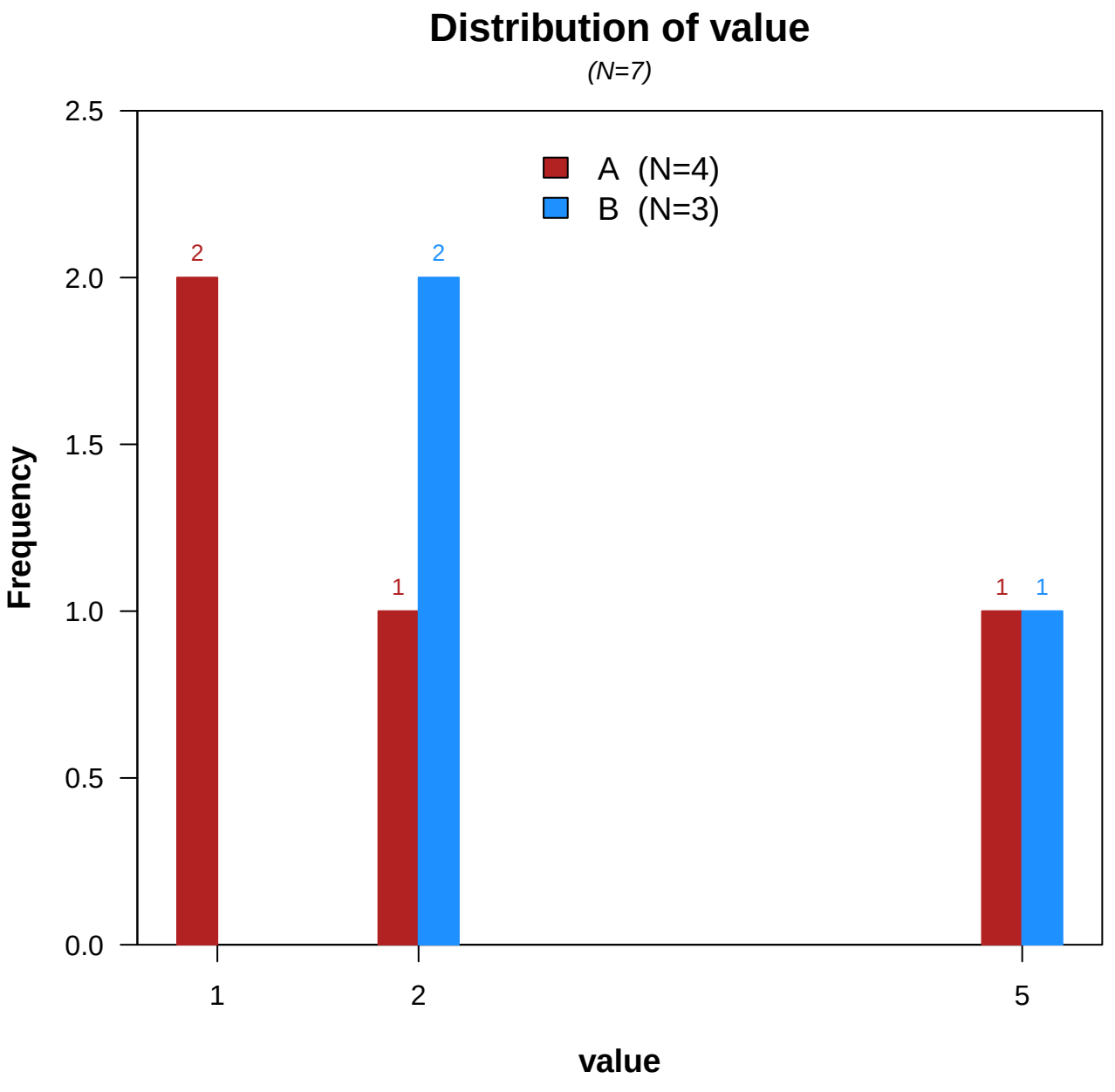

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
5
x

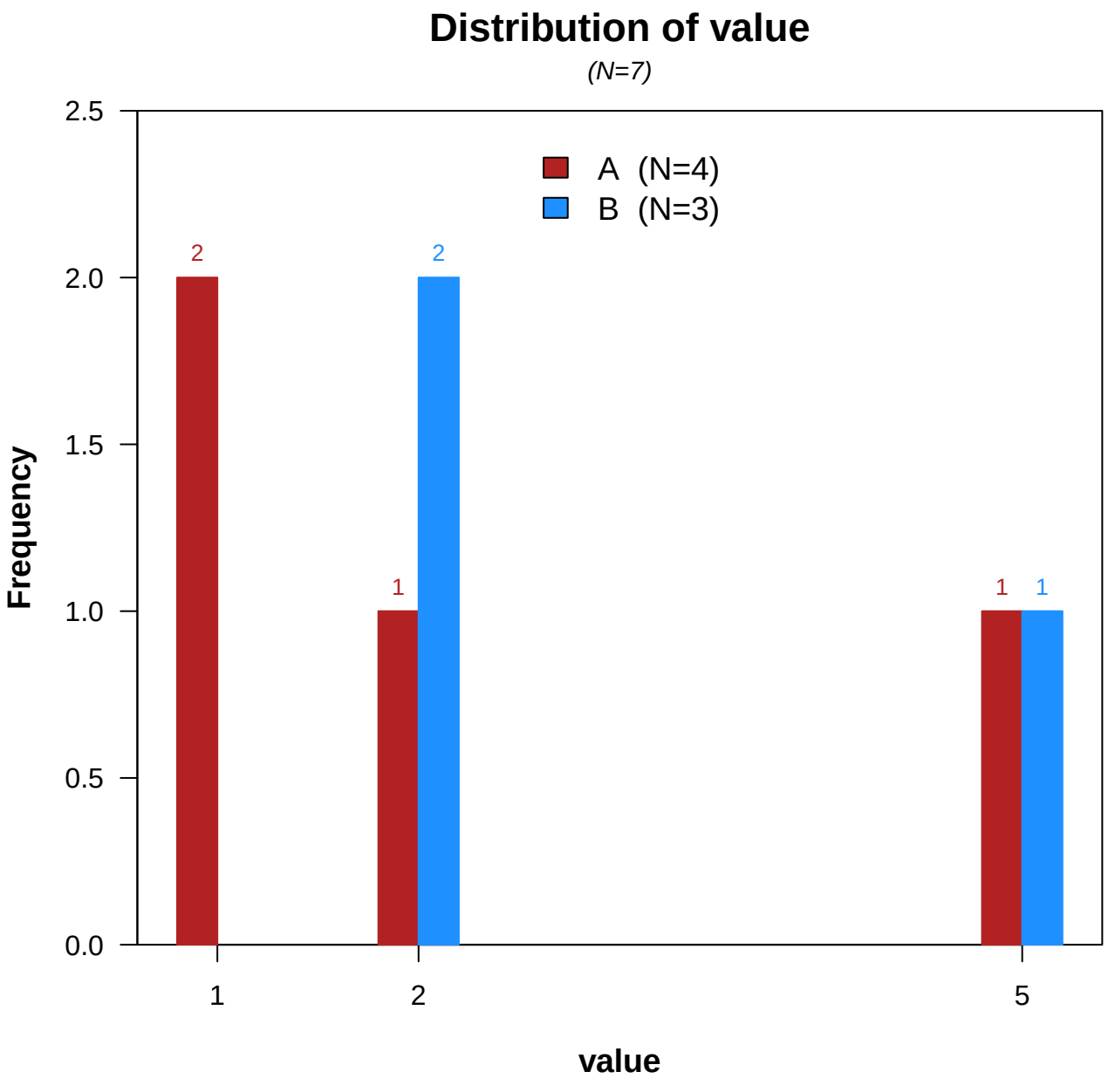

Distribution of value
(N=7)
Frequency
value
0
1
2
3
4
1
2
5
2
3
2

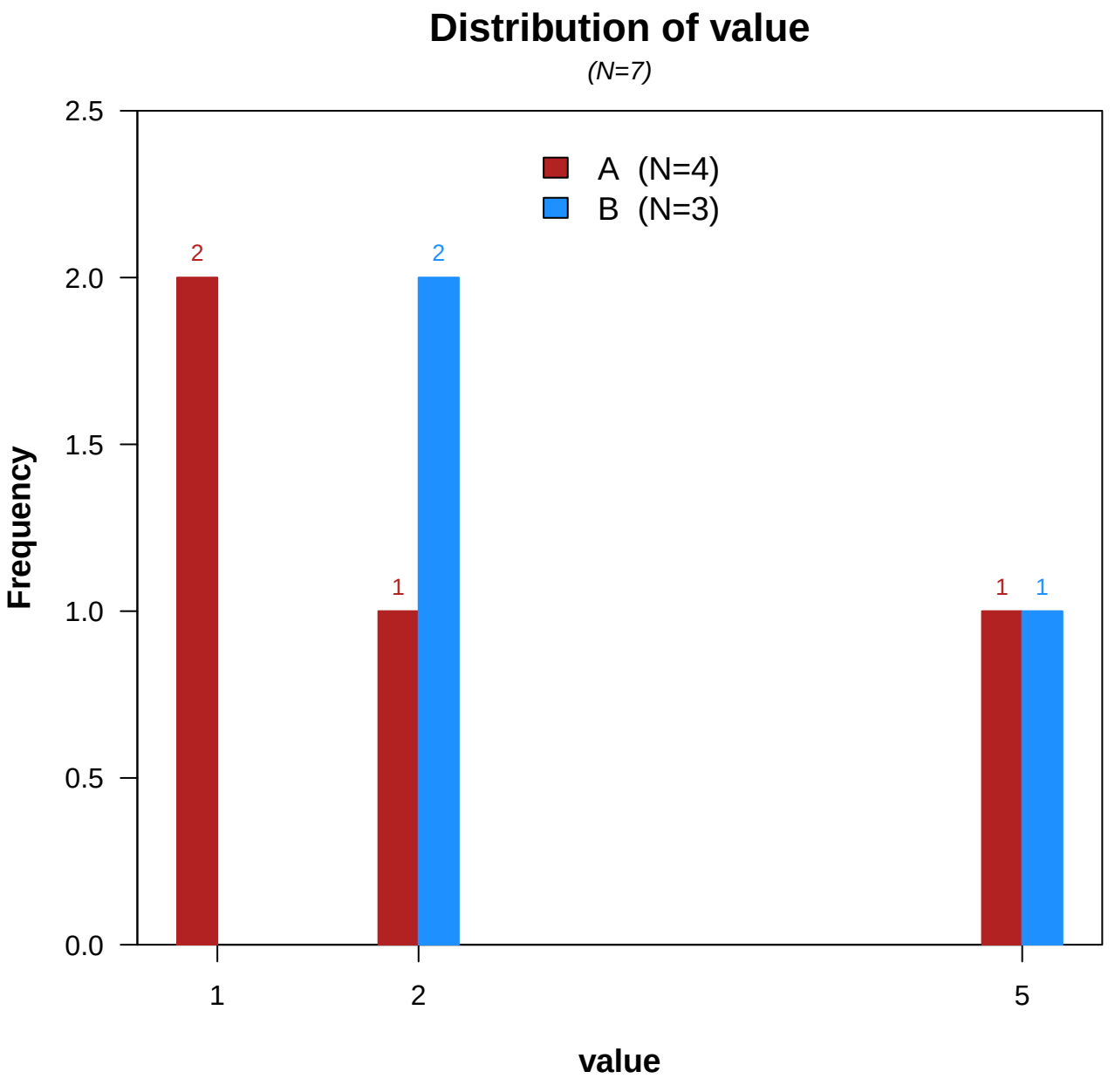

Distribution of value
(N=7)
Frequency
value
0
1
2
3
4
1
2
5
2
3
2

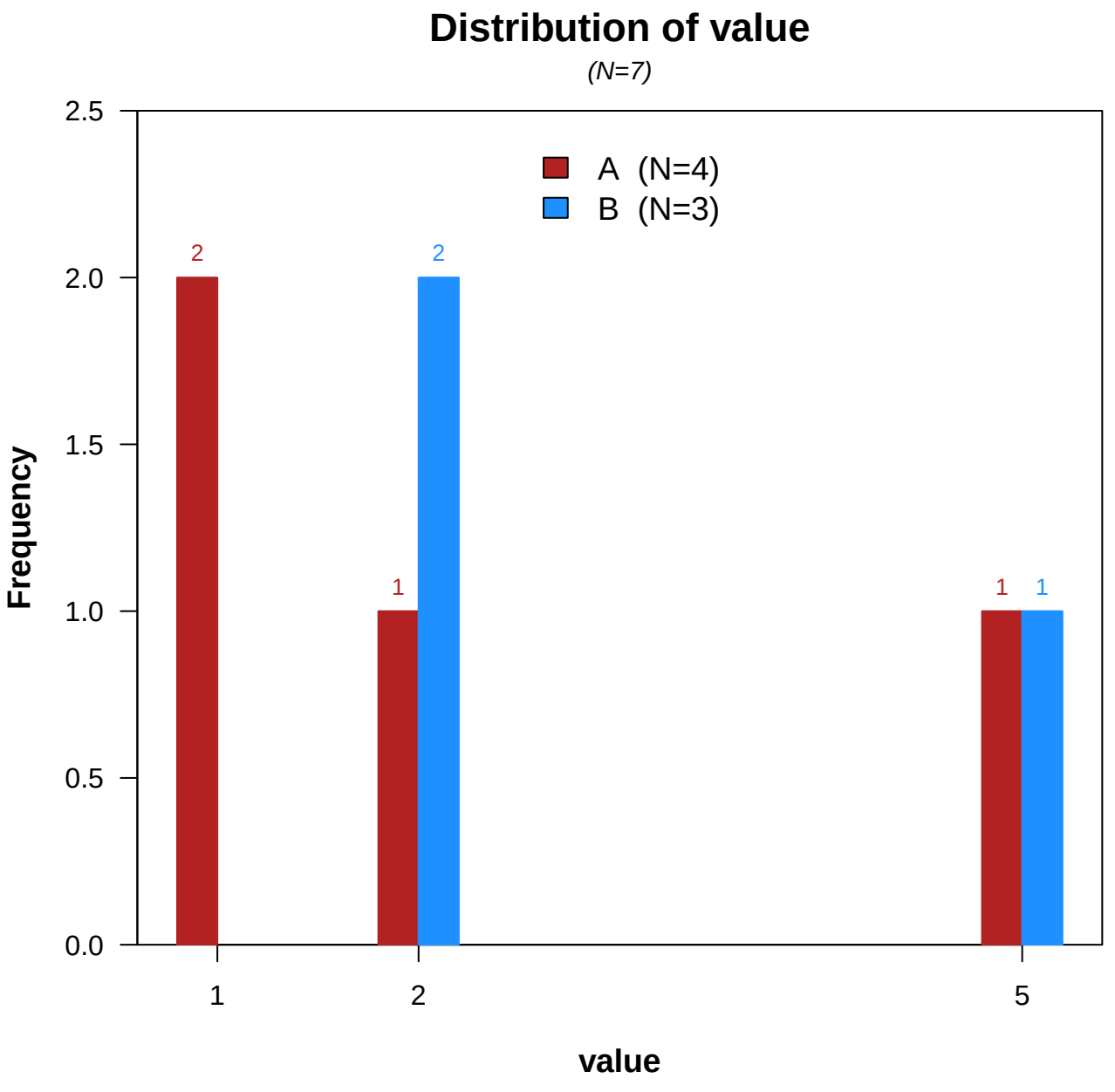

Distribution of x
(N=7)
A (N=4)
B (N=3)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
2
1
1
1
1
2
5
x

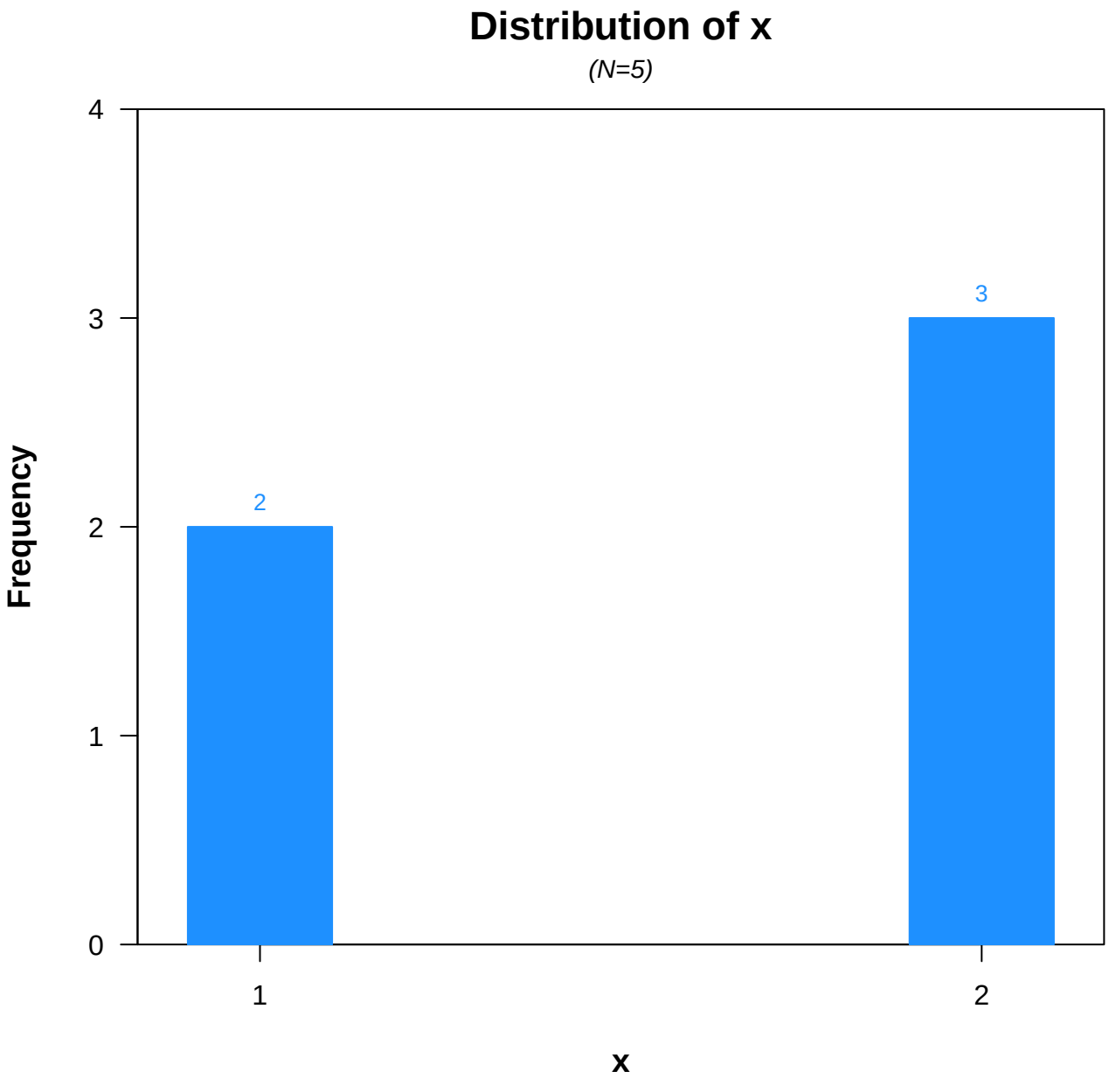

Distribution of x
(N=7)
A (N=4)
B (N=3)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
2
1
1
1
1
2
5
x

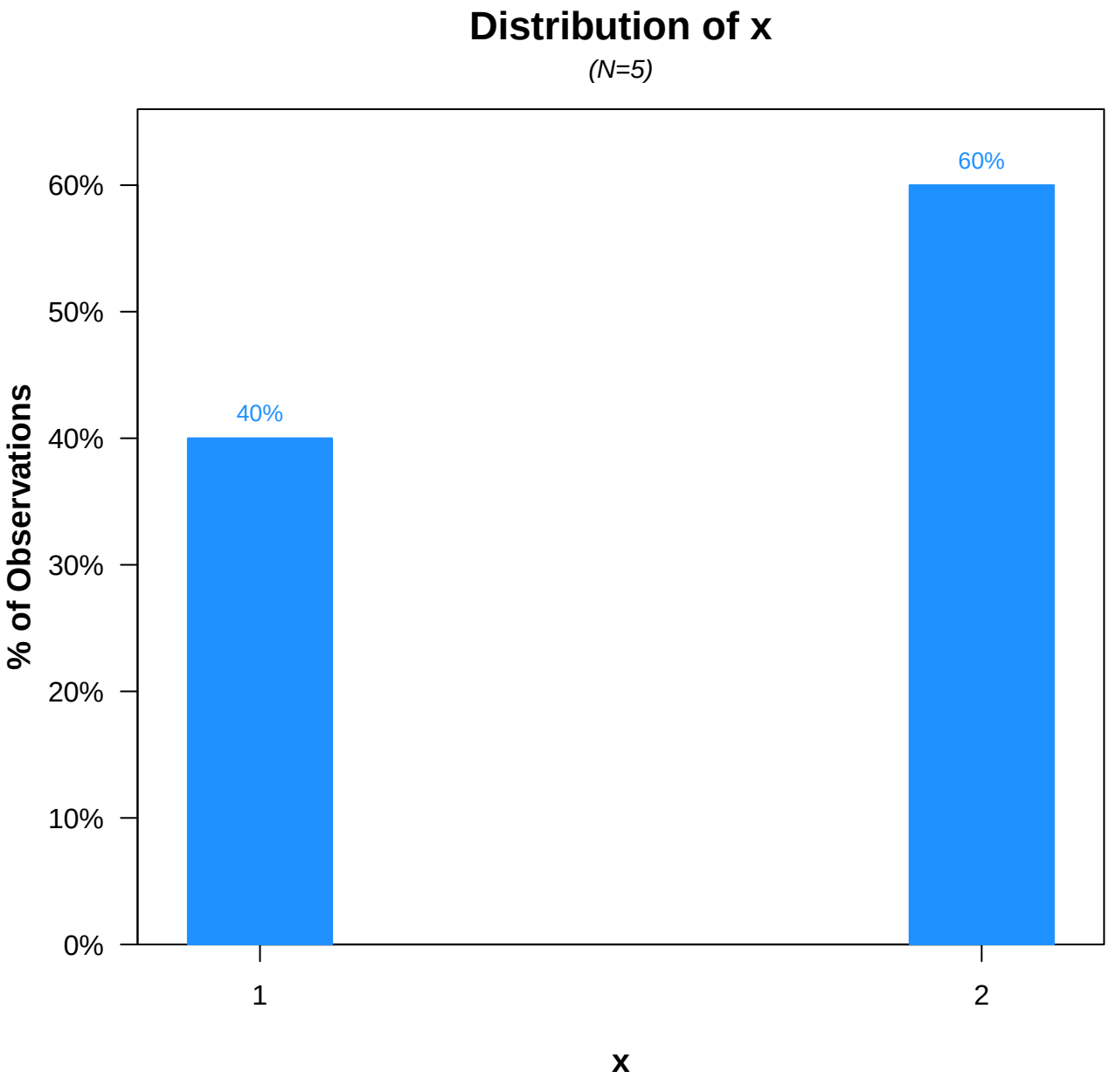

Distribution of value
(N=7)
A (N=4)
B (N=3)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
2
1
1
1
1
2
5
value

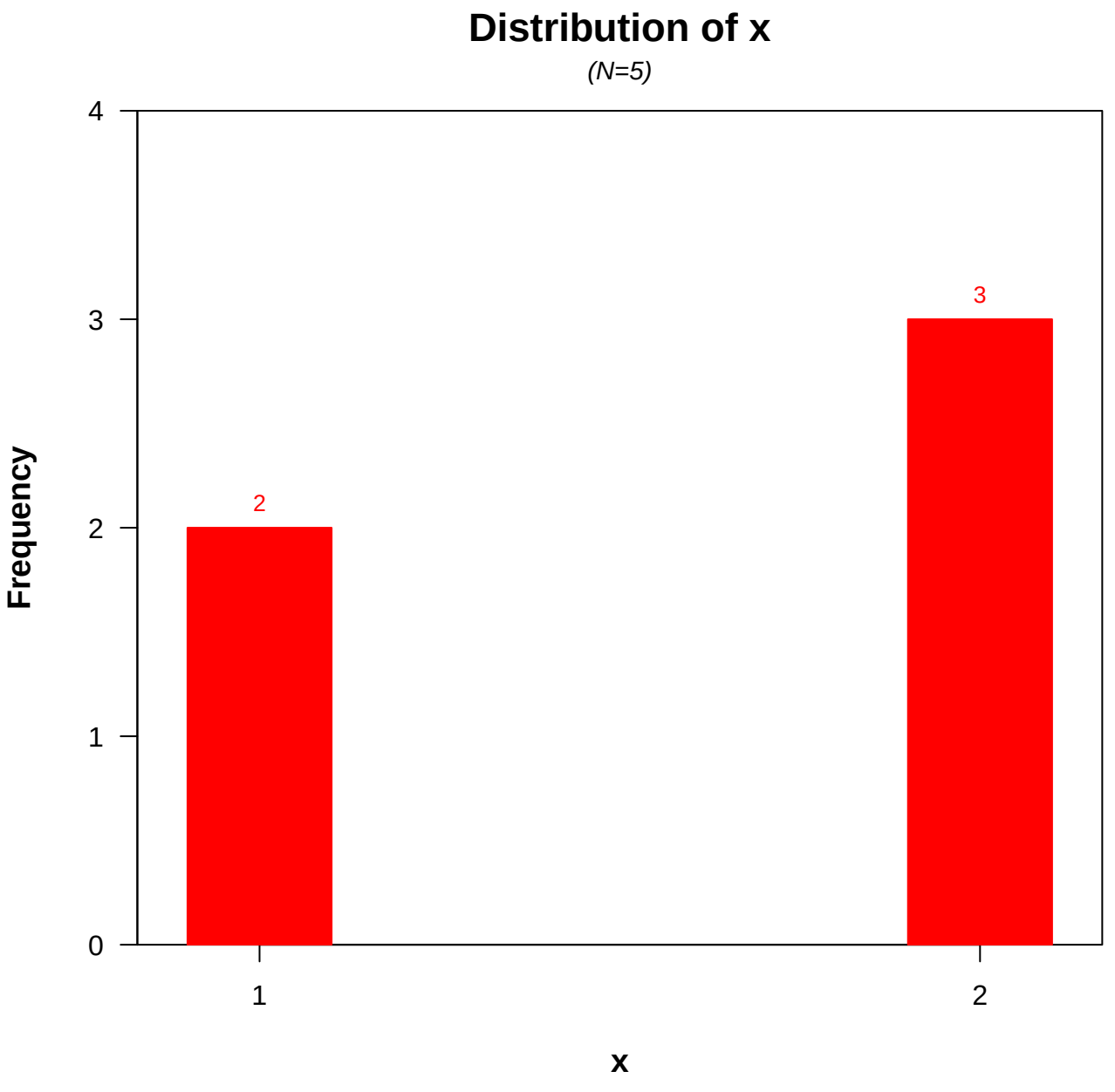

Distribution of value
(N=7)
A (N=4)
B (N=3)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
2
1
1
1
1
2
5
value

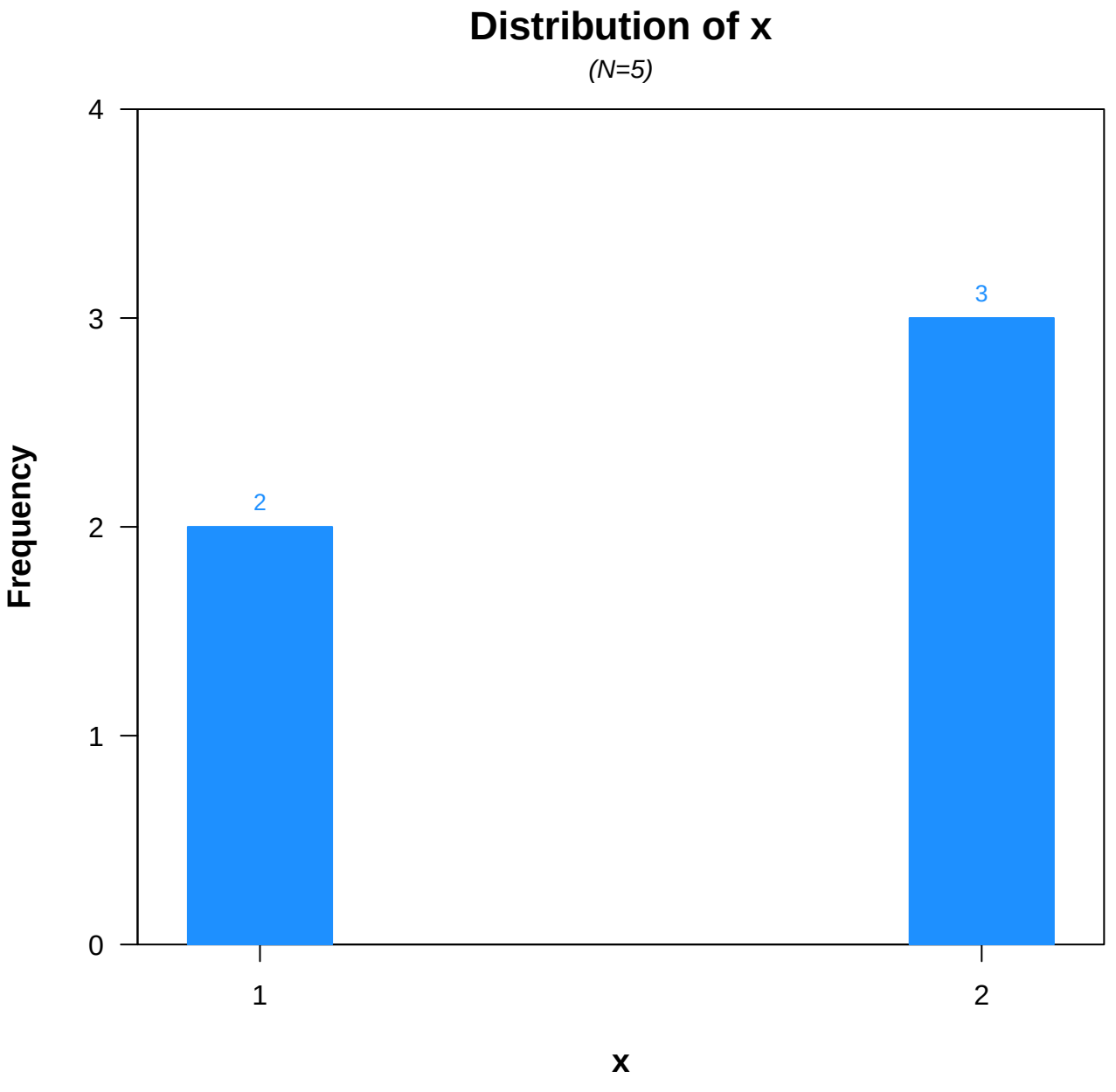

Distribution of value
(N=7)
A (N=4)
B (N=3)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
2
1
1
1
1
2
5
value

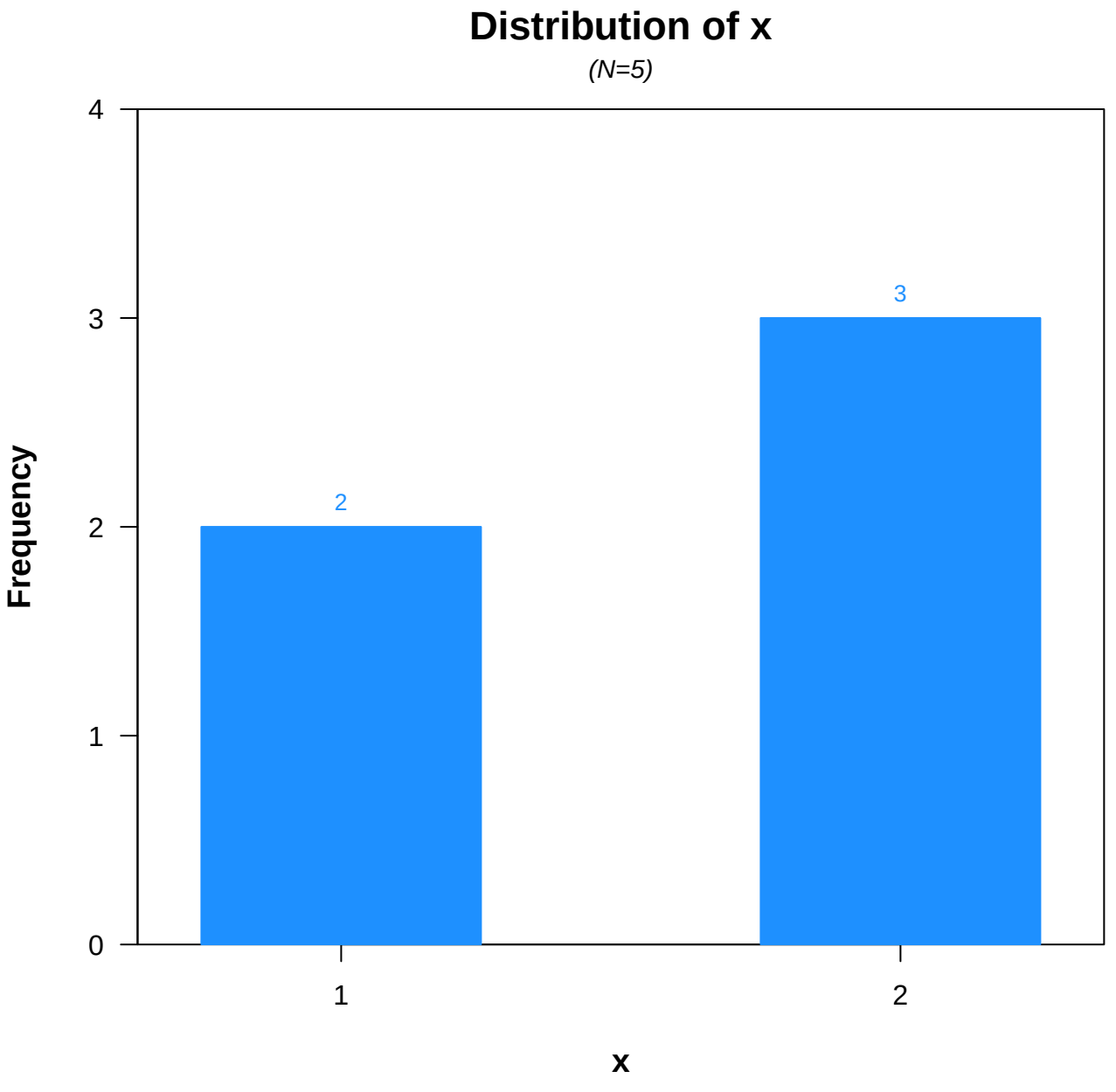

Distribution of value
(N=7)
A (N=4)
B (N=3)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
2
1
1
1
1
2
5
value

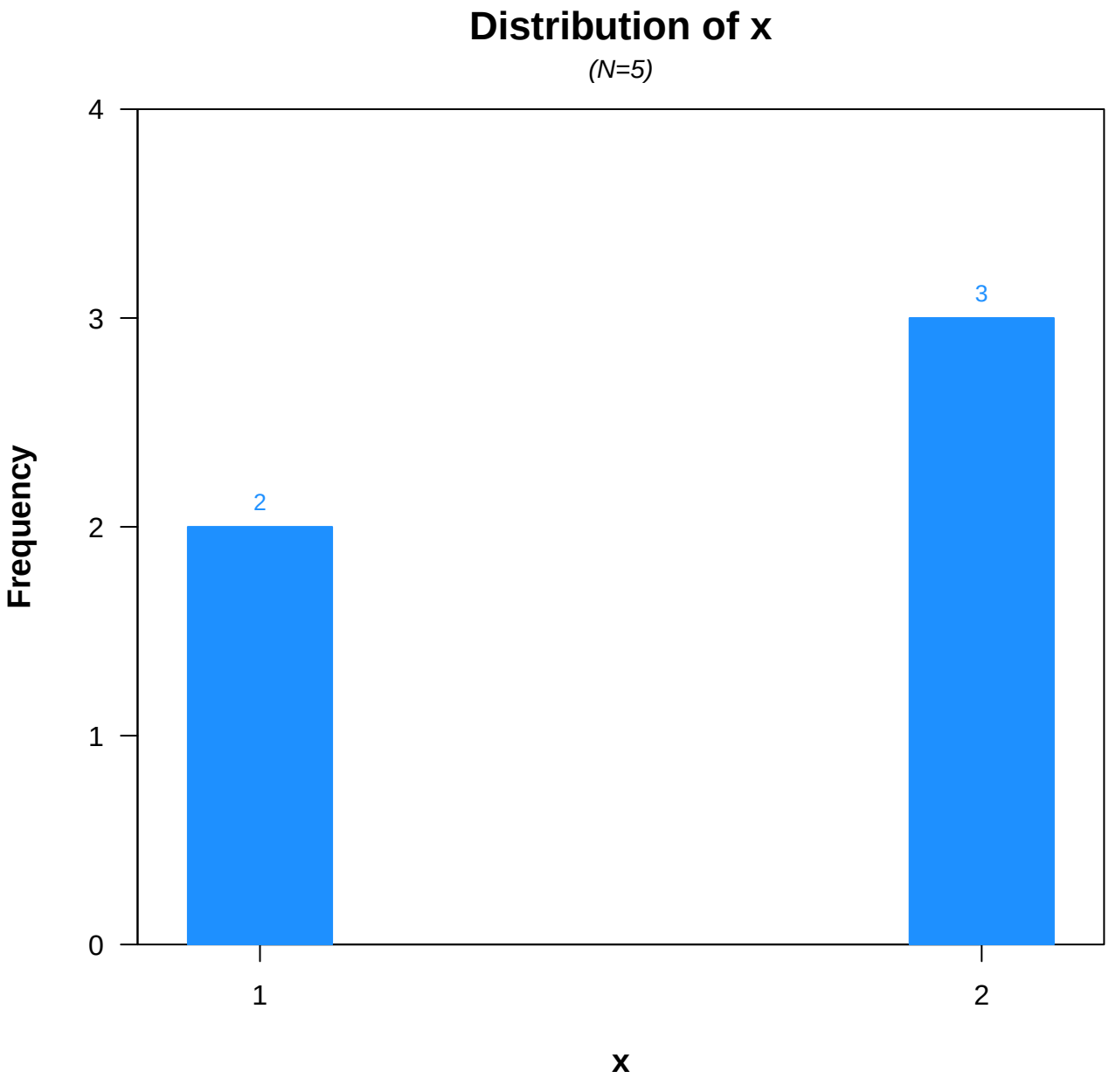

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

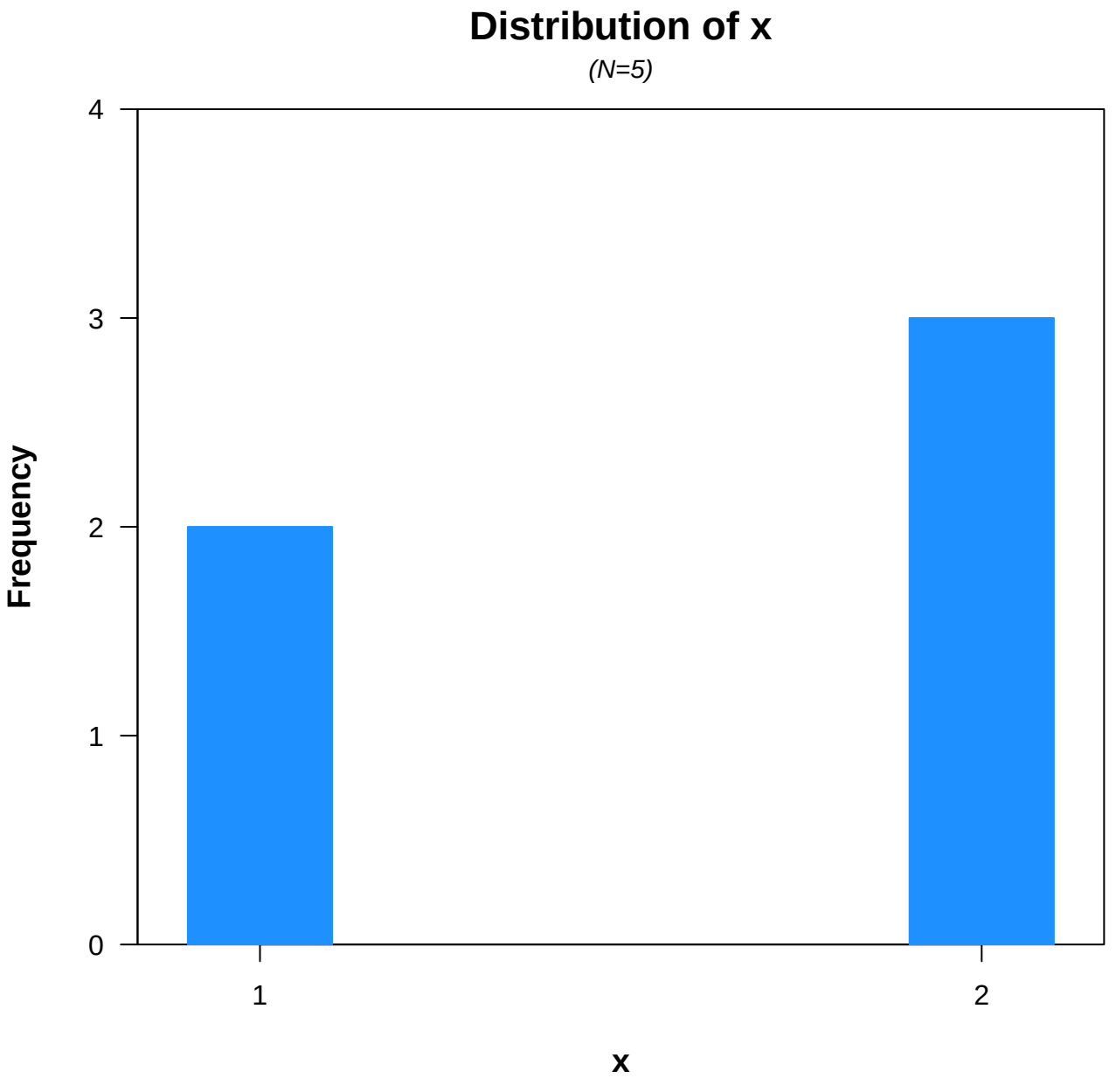

Distribution of x
(N=5)
% of Observations
60%
50%
40%
30%
20%
10%
0%
40%
60%
1
2
x

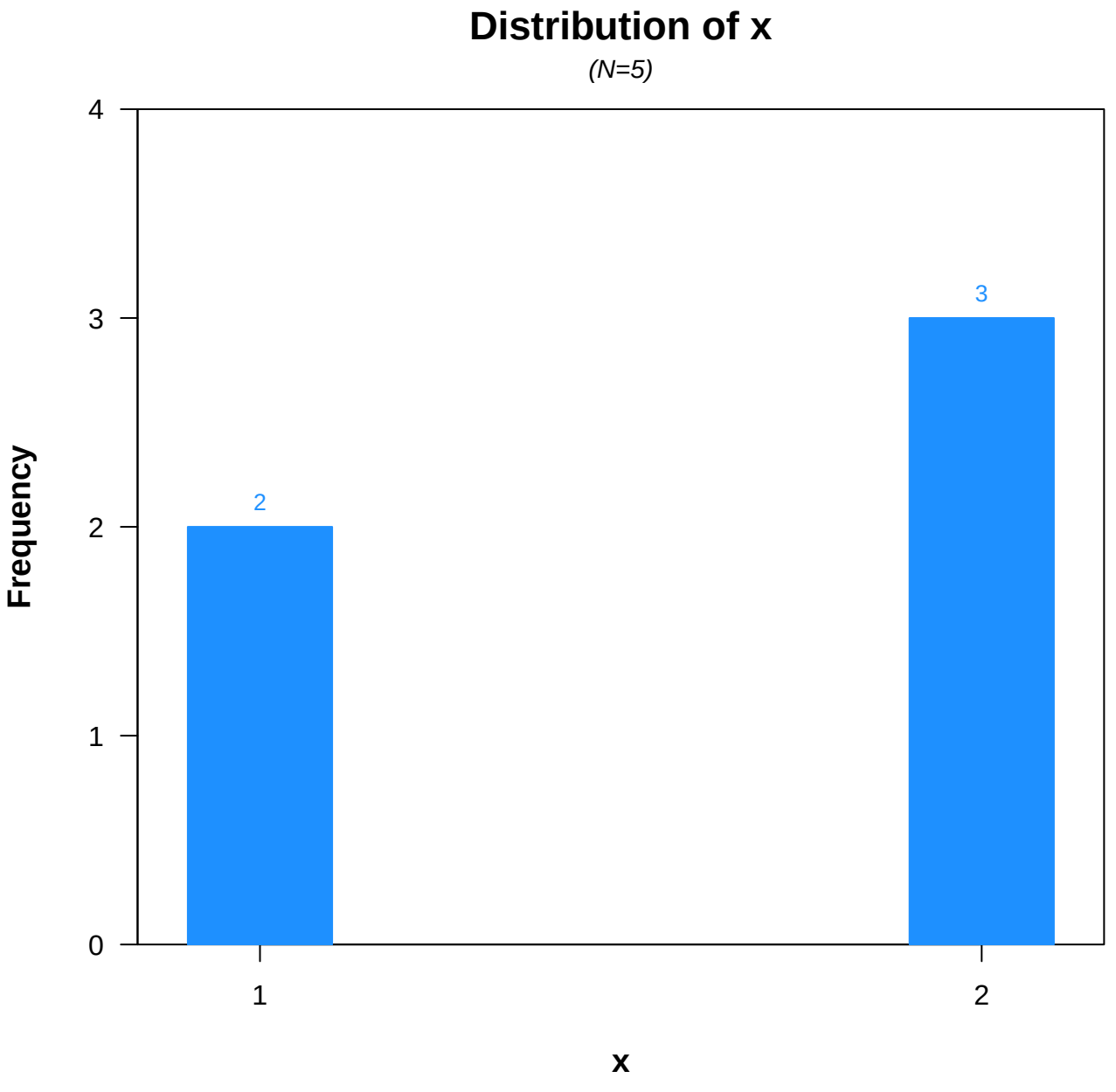

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

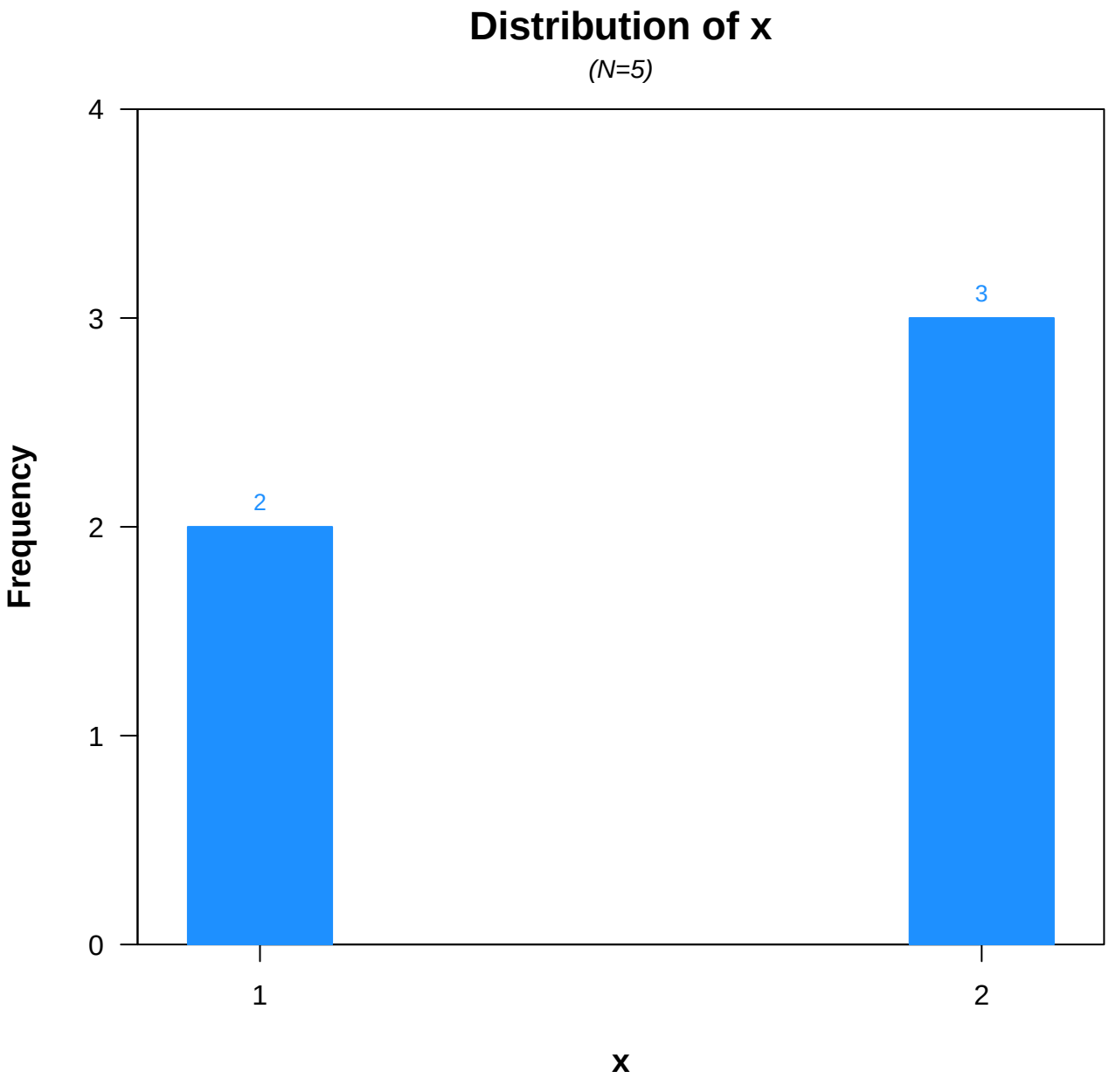

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

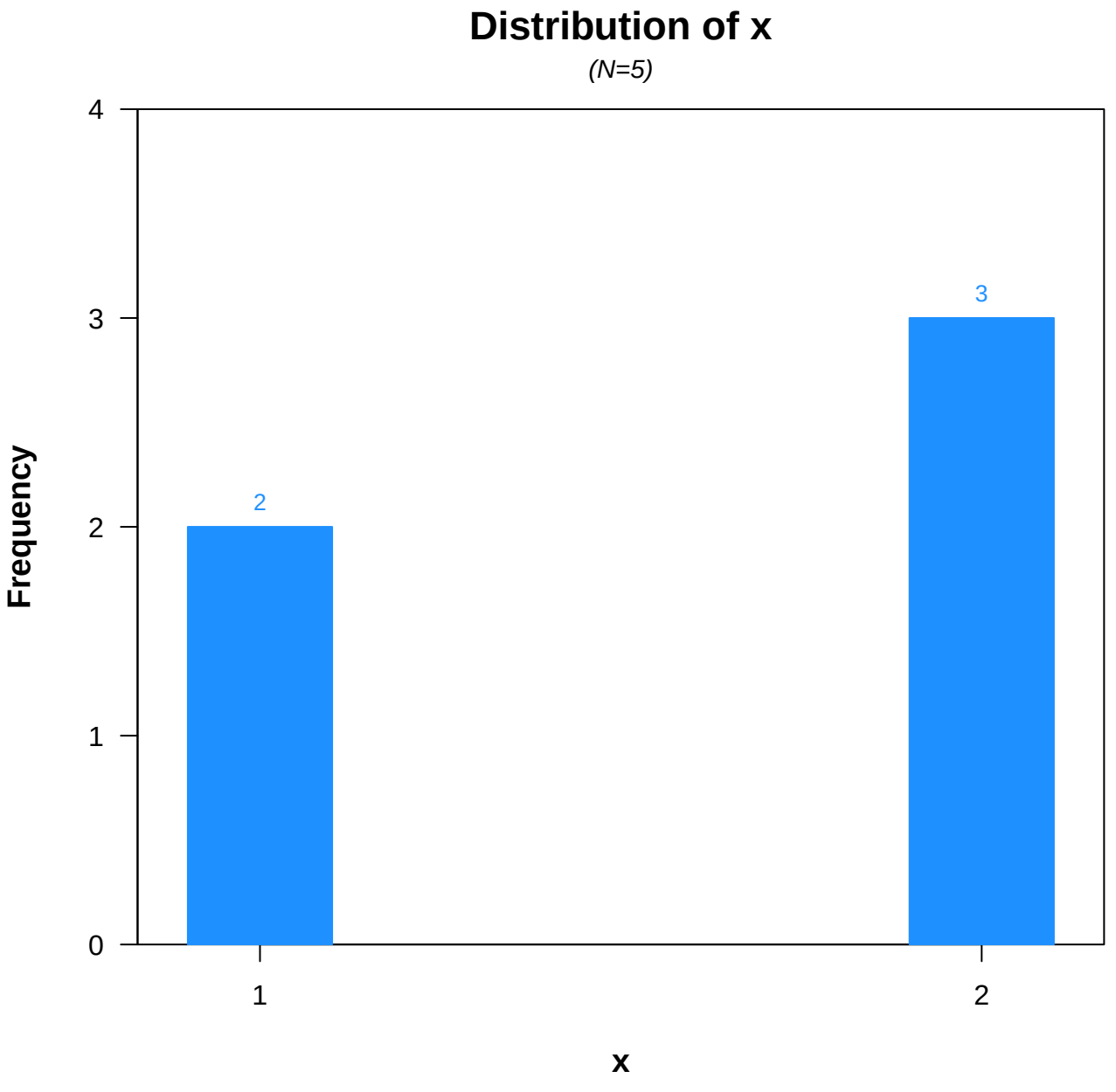

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

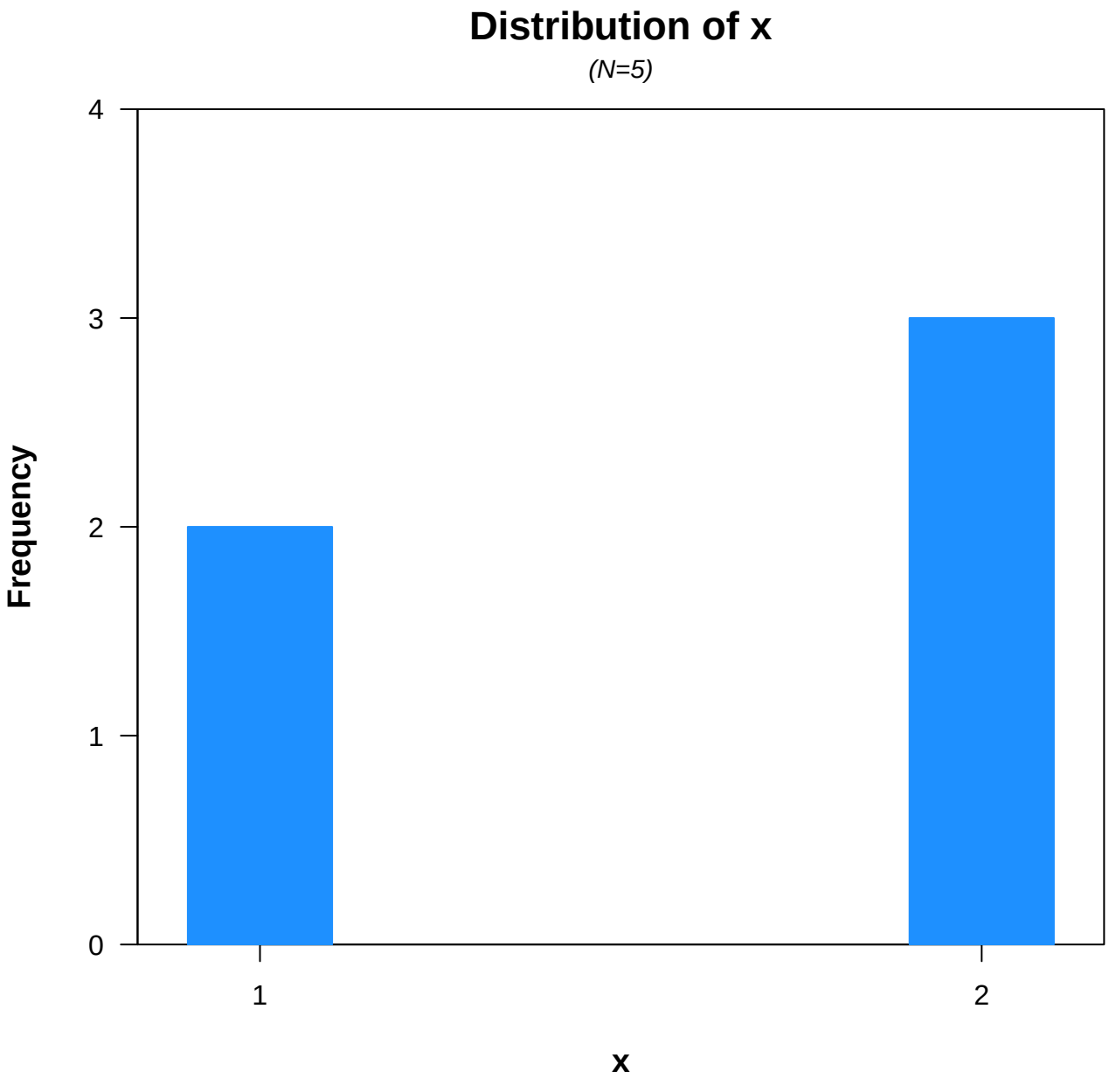

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

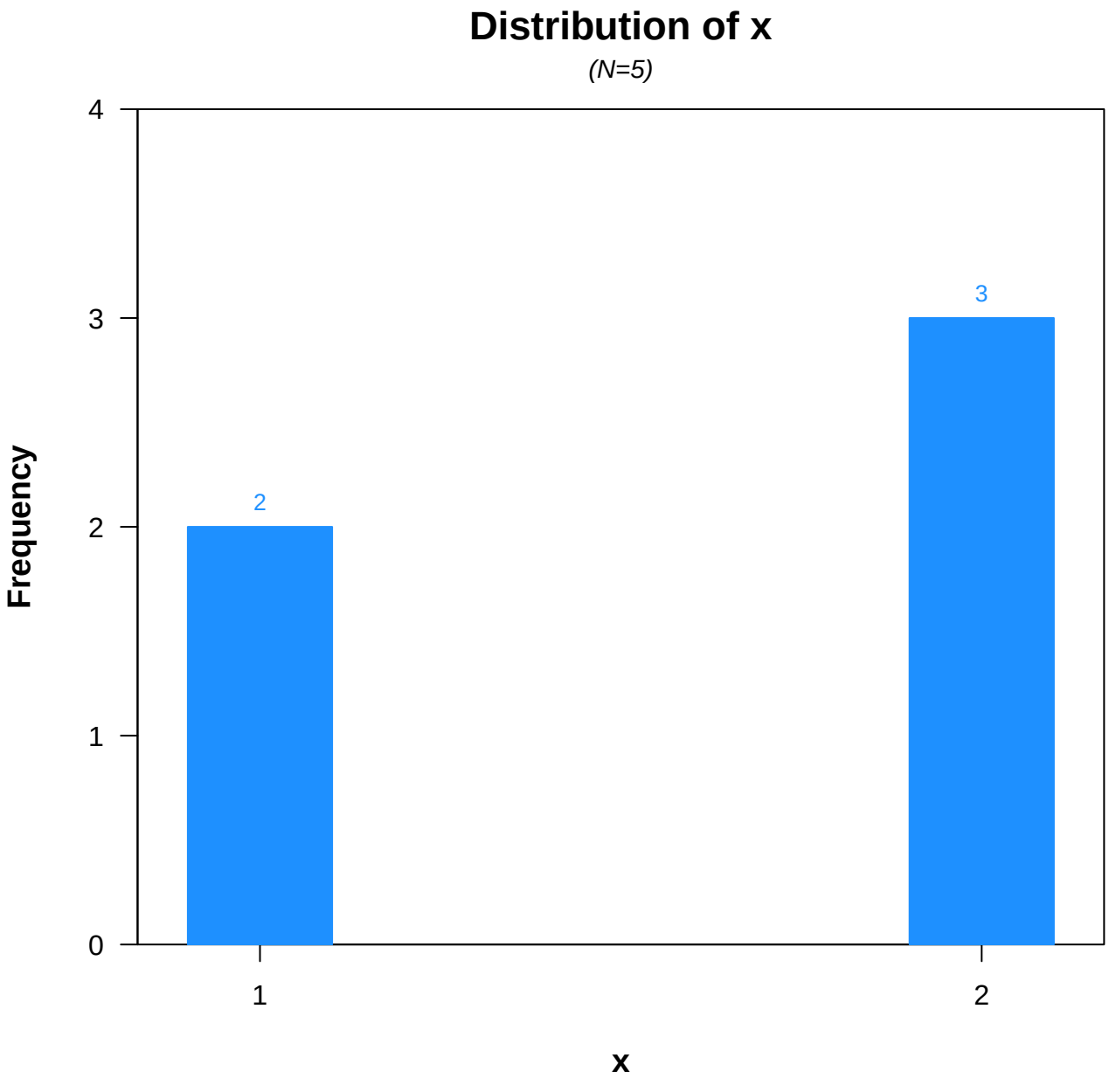

Distribution of x
(N=5)
Frequency
4
3
2
1
0
1
2
x

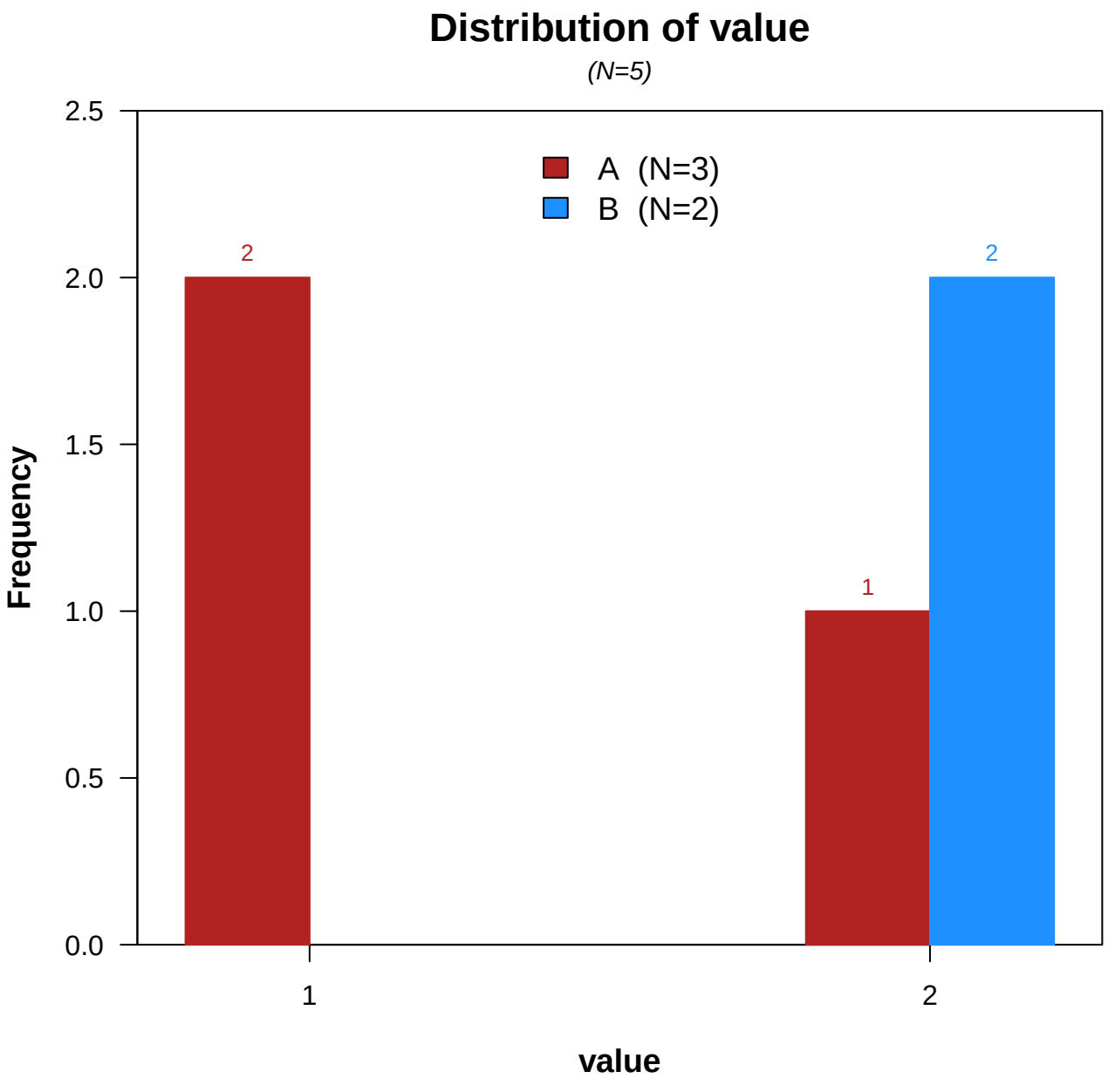

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

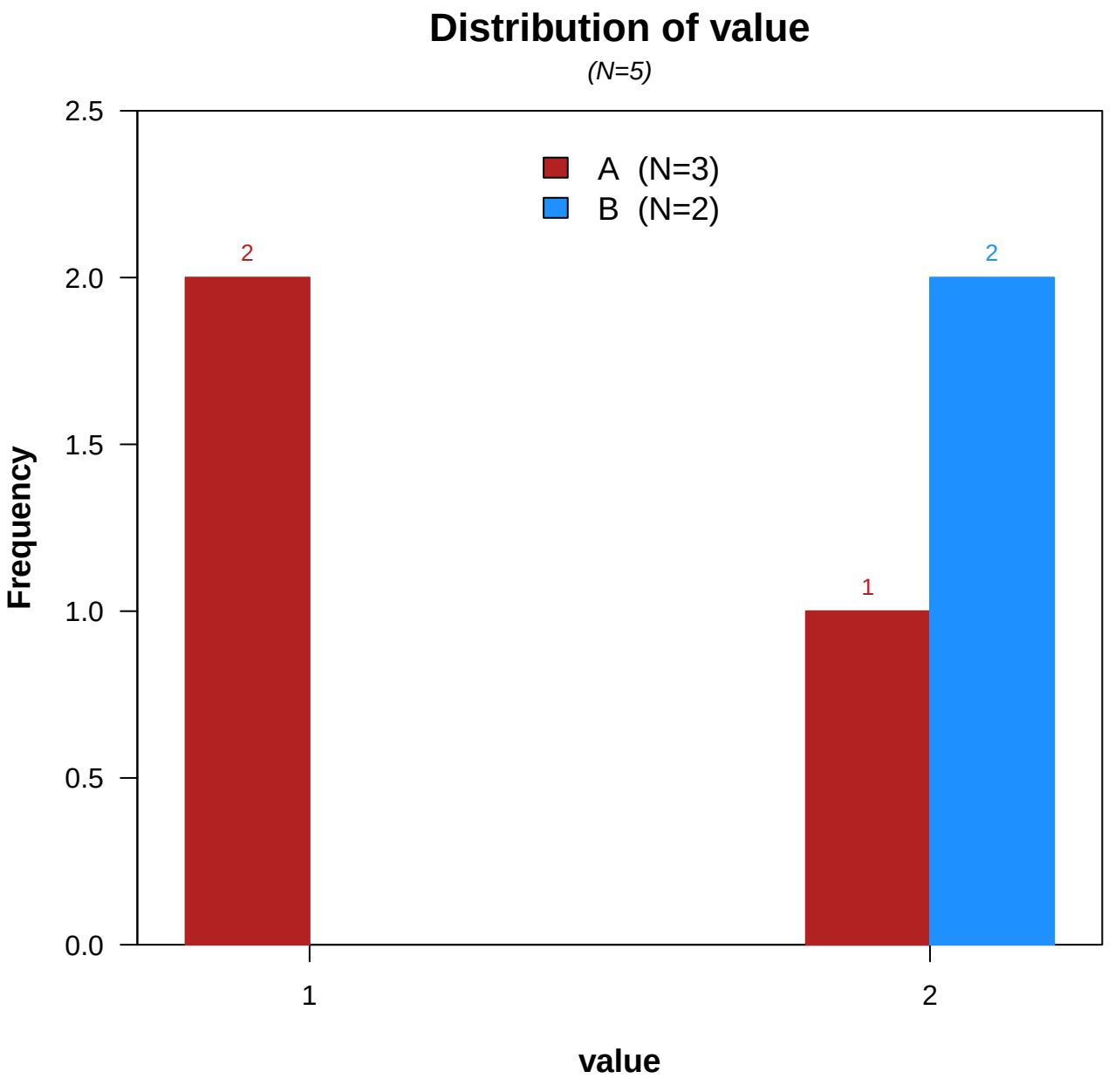

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

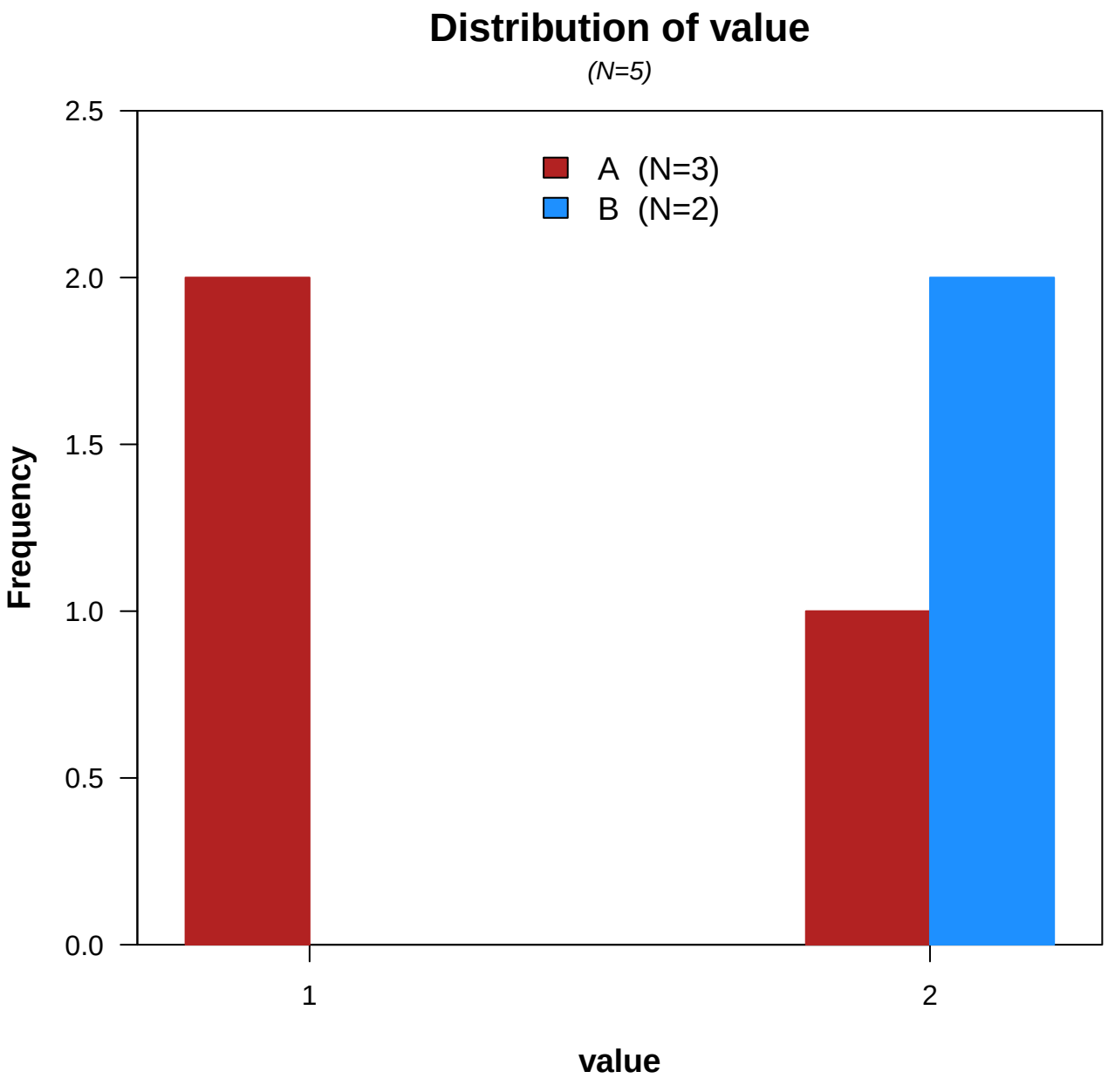

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

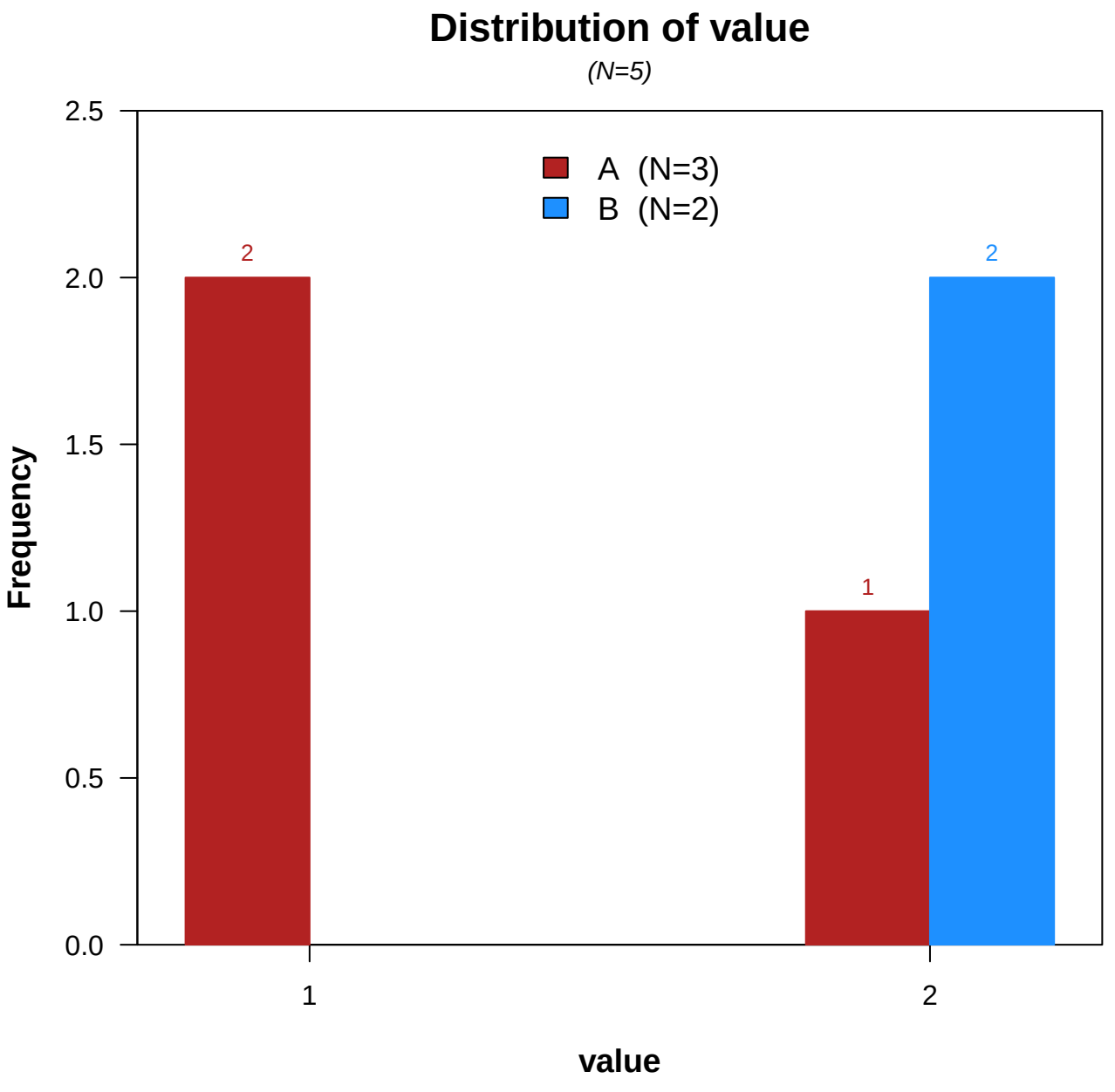

Distribution of x
(N=5)
Frequency
4
3
2
1
0
1
2
x

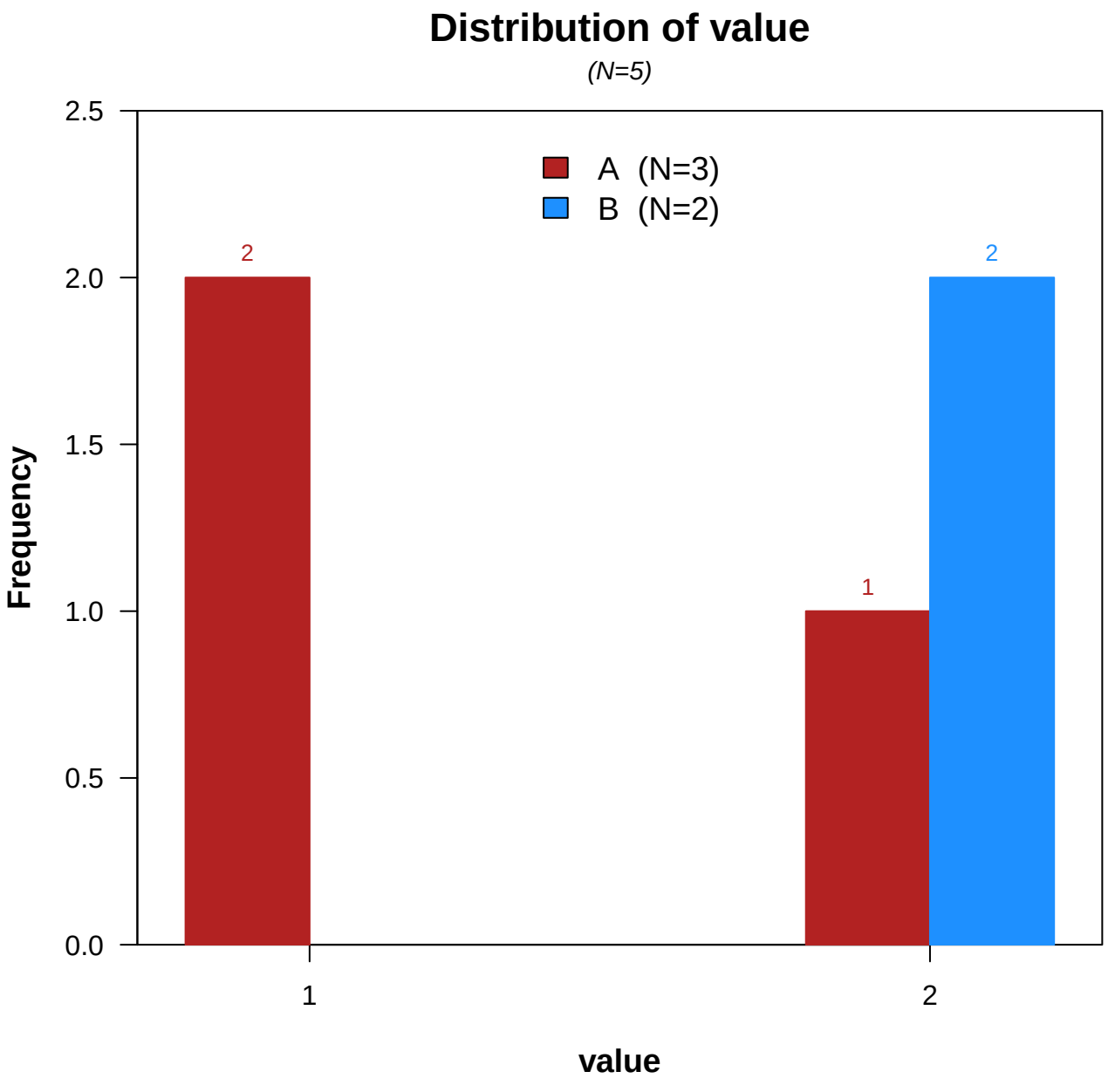

Distribution of x
(N=5)
Frequency
4
3
2
1
0
2
3
1
2
x

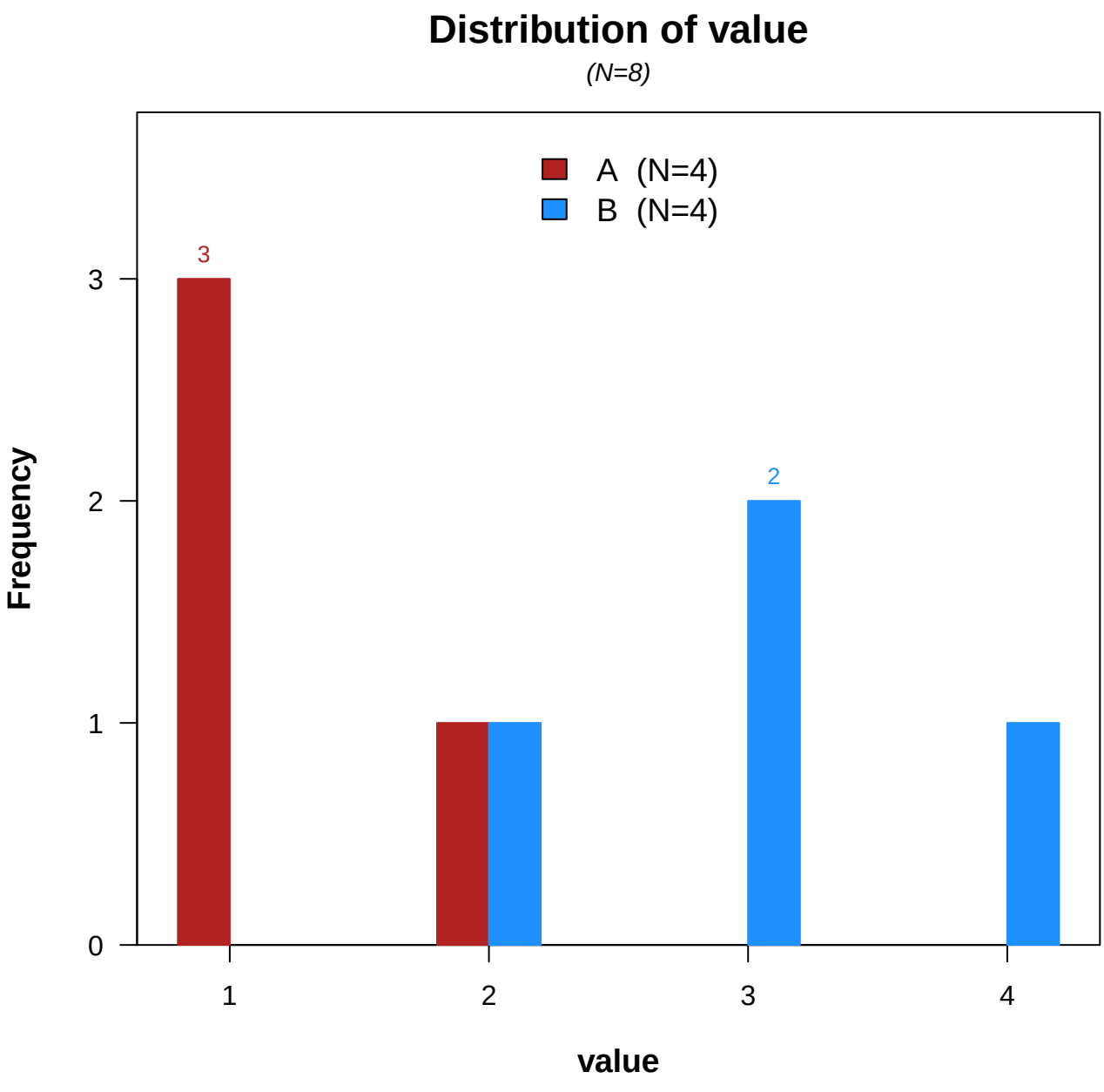

Distribution of value
(N=5)
A (N=3)
B (N=2)
2
2
1
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
value

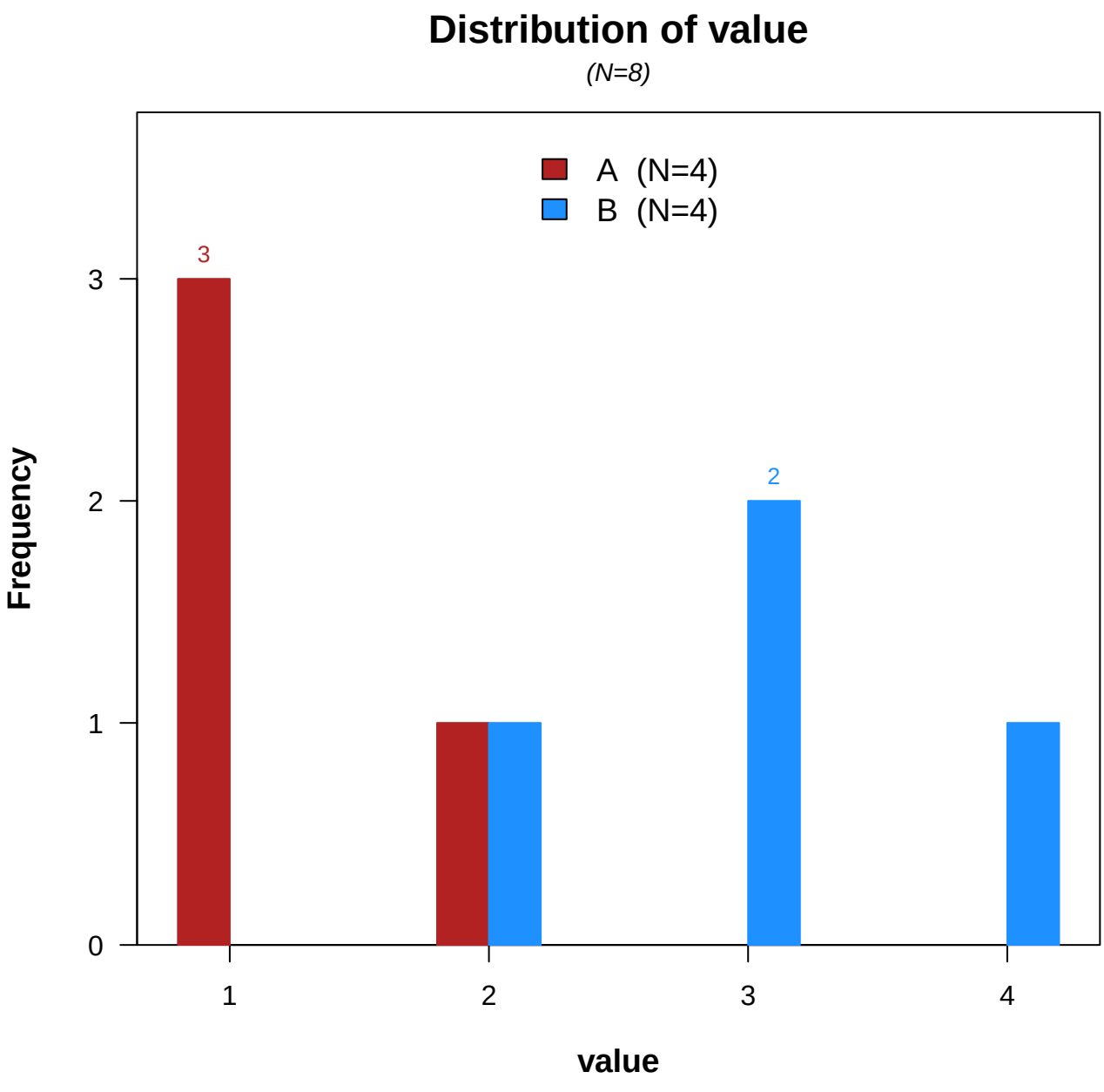

Distribution of value
(N=5)
A (N=3)
B (N=2)
2
2
1
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
value

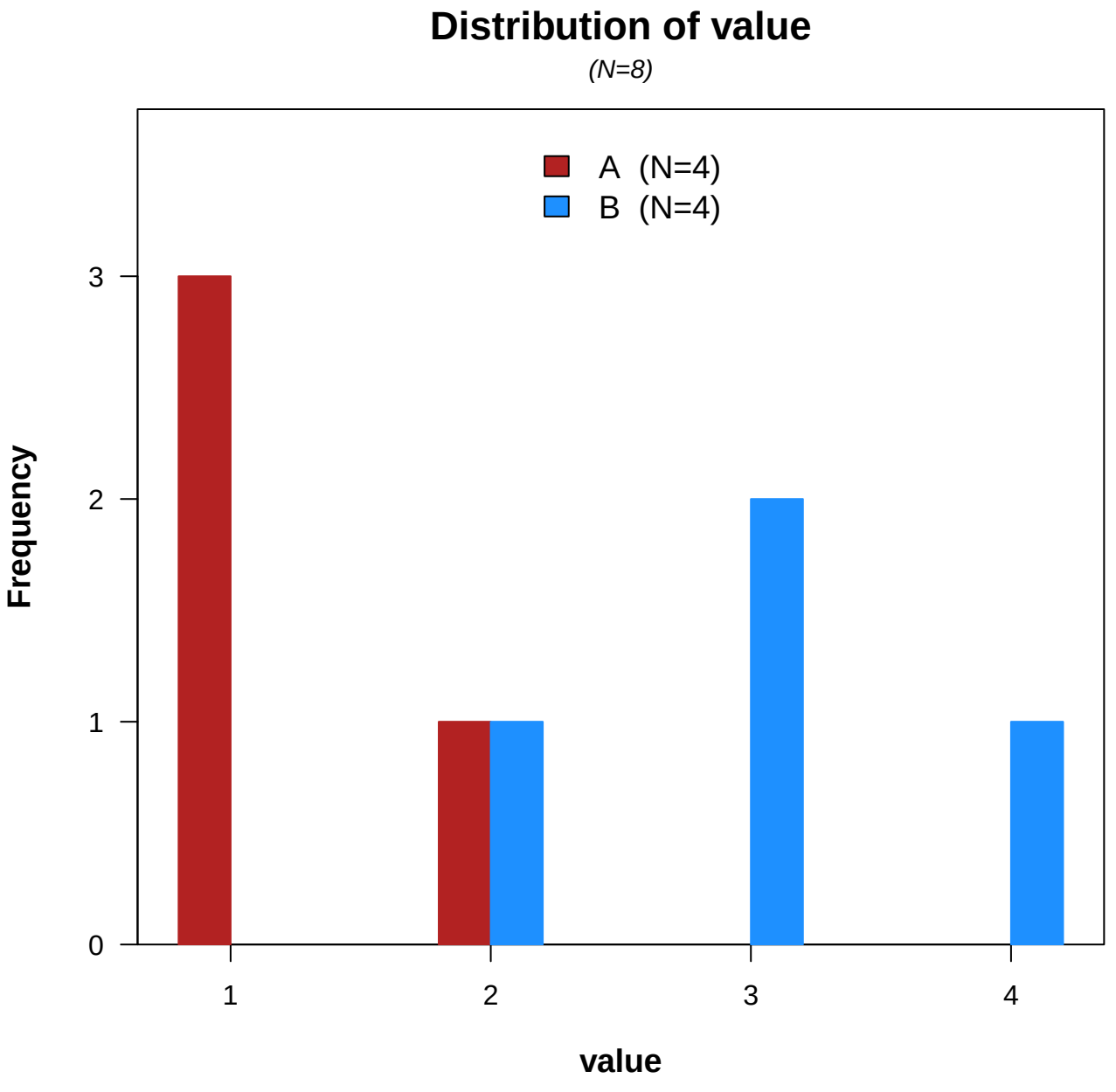

Distribution of value
(N=5)
A (N=3)
B (N=2)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
1
2
value

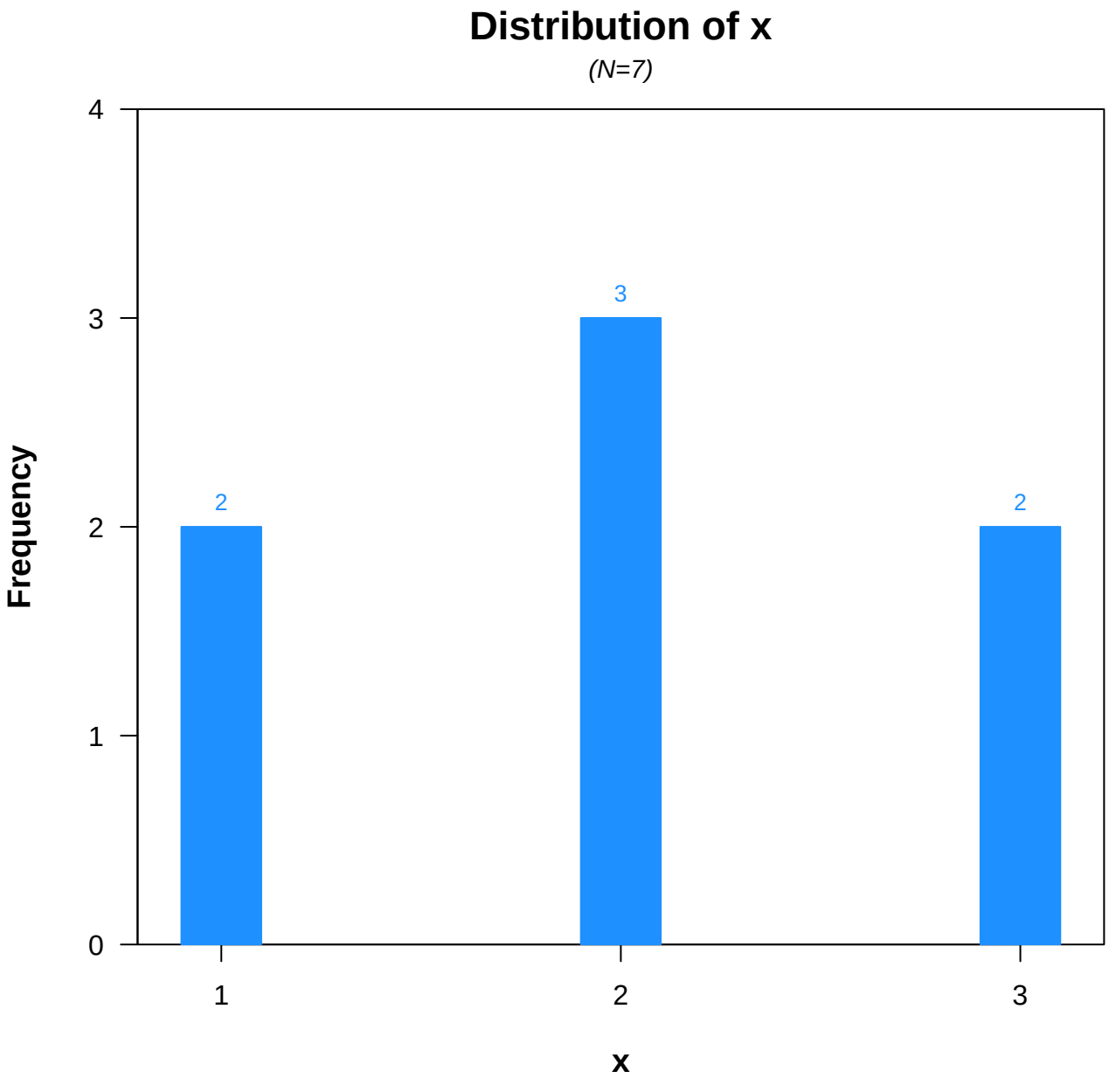

Distribution of value
(N=5)
A (N=3)
B (N=2)
2
2
1
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
value

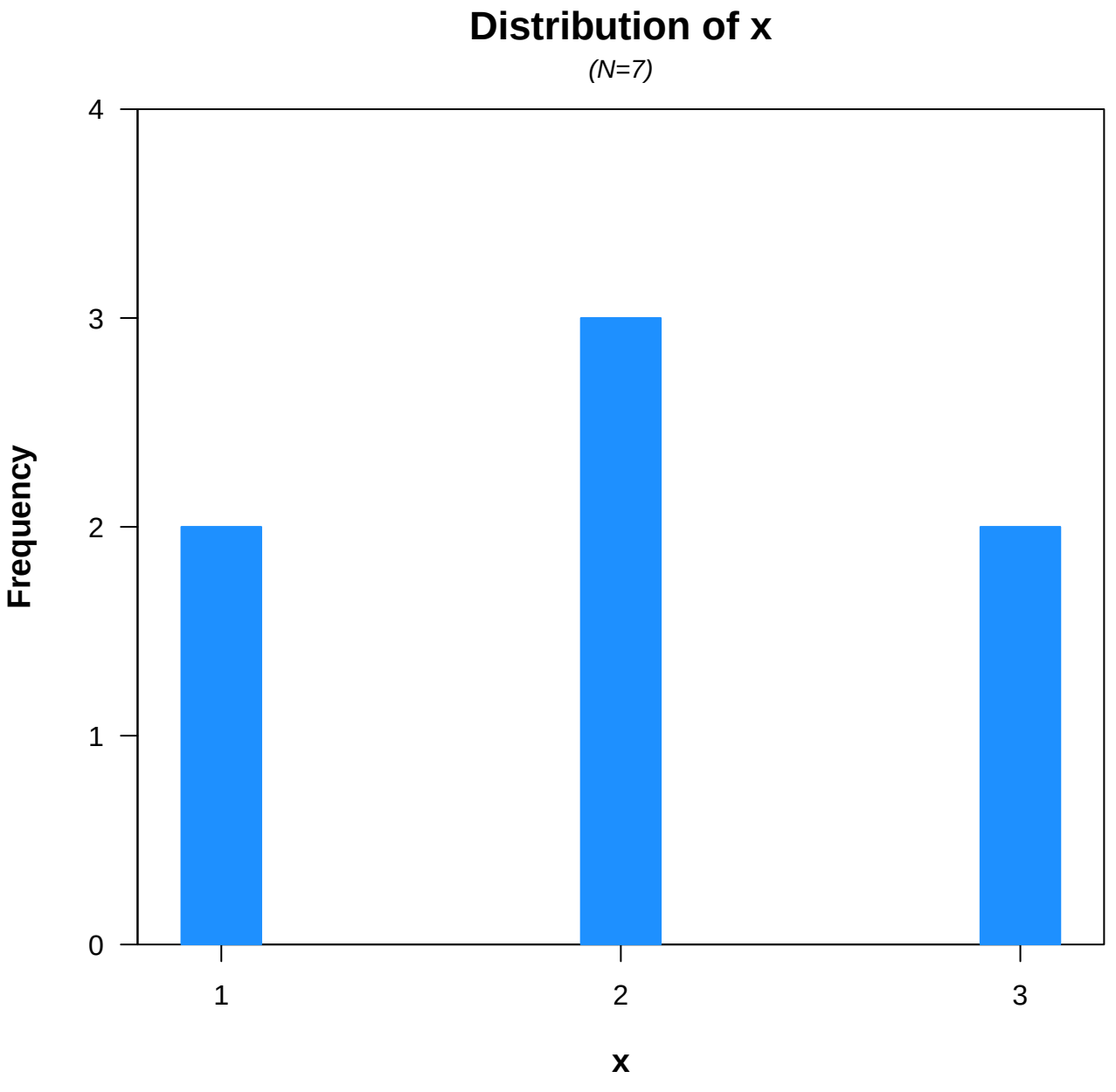

Distribution of value
(N=5)
A (N=3)
B (N=2)
2
2
1
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
value

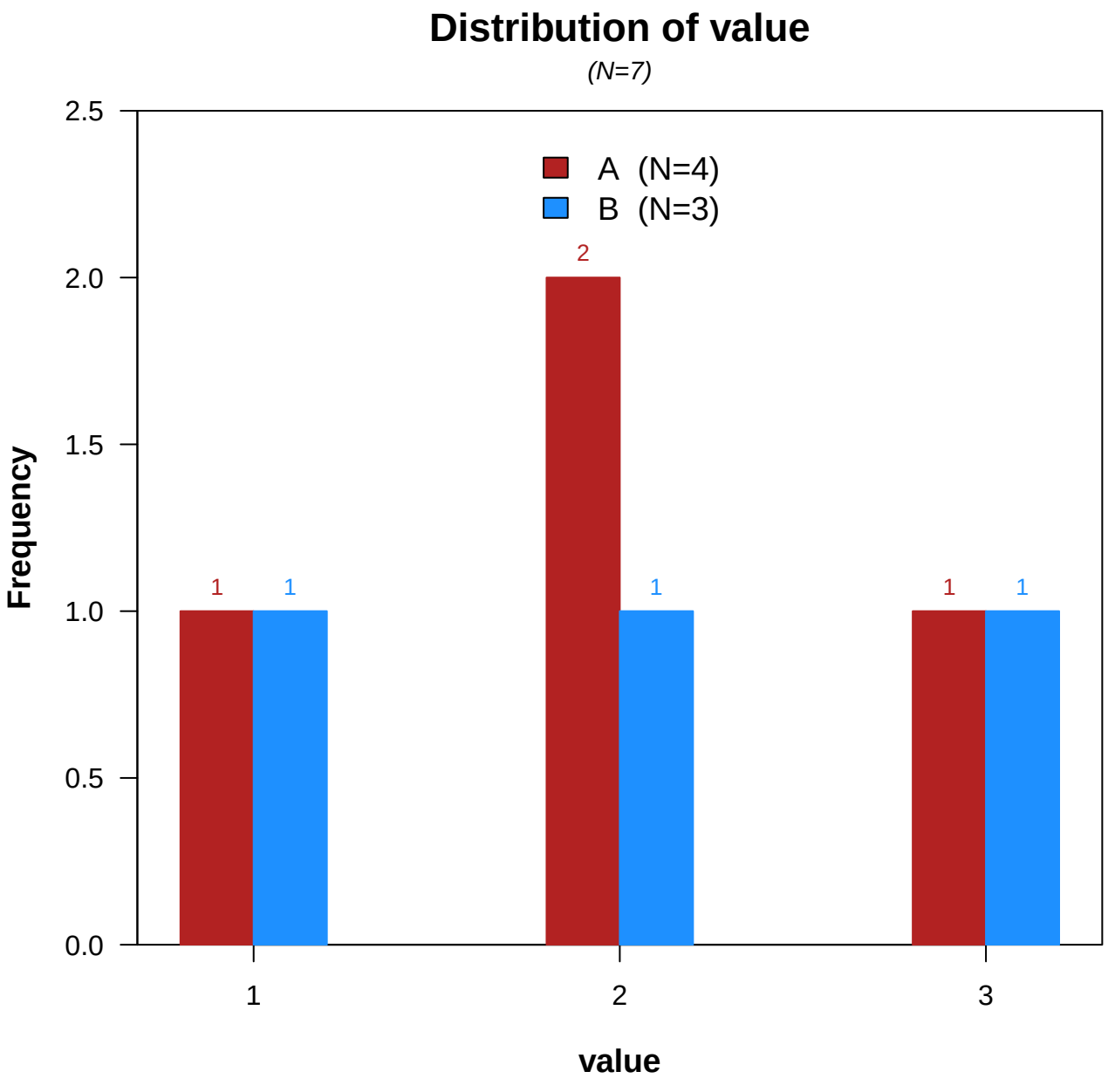

Distribution of value
(N=8)
A (N=4)
B (N=4)
3
2
Frequency
0
1
2
3
1
2
3
4
value

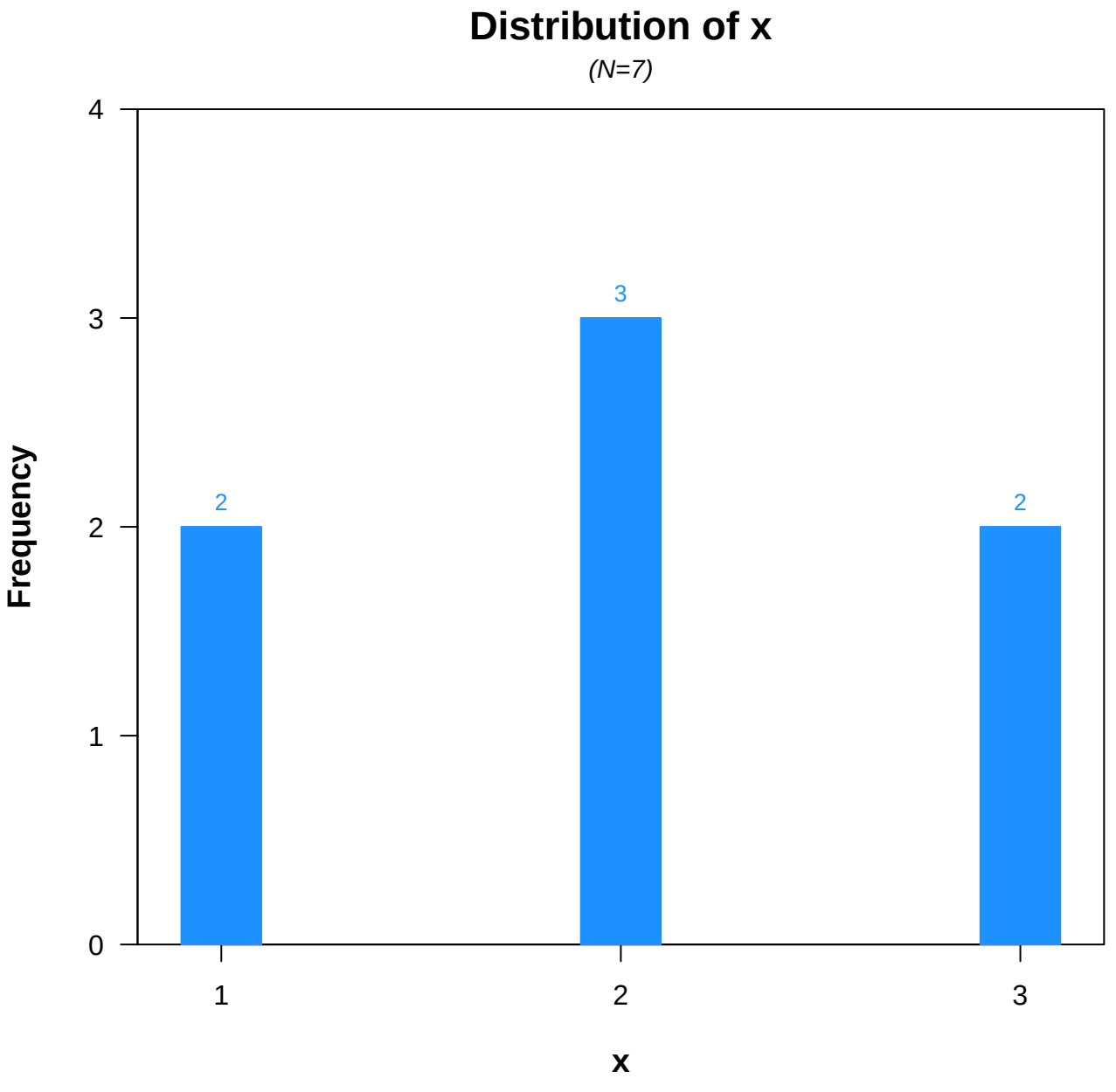

Distribution of value
(N=8)
A (N=4)
B (N=4)
3
2
Frequency
0
1
2
3
1
2
3
4
value

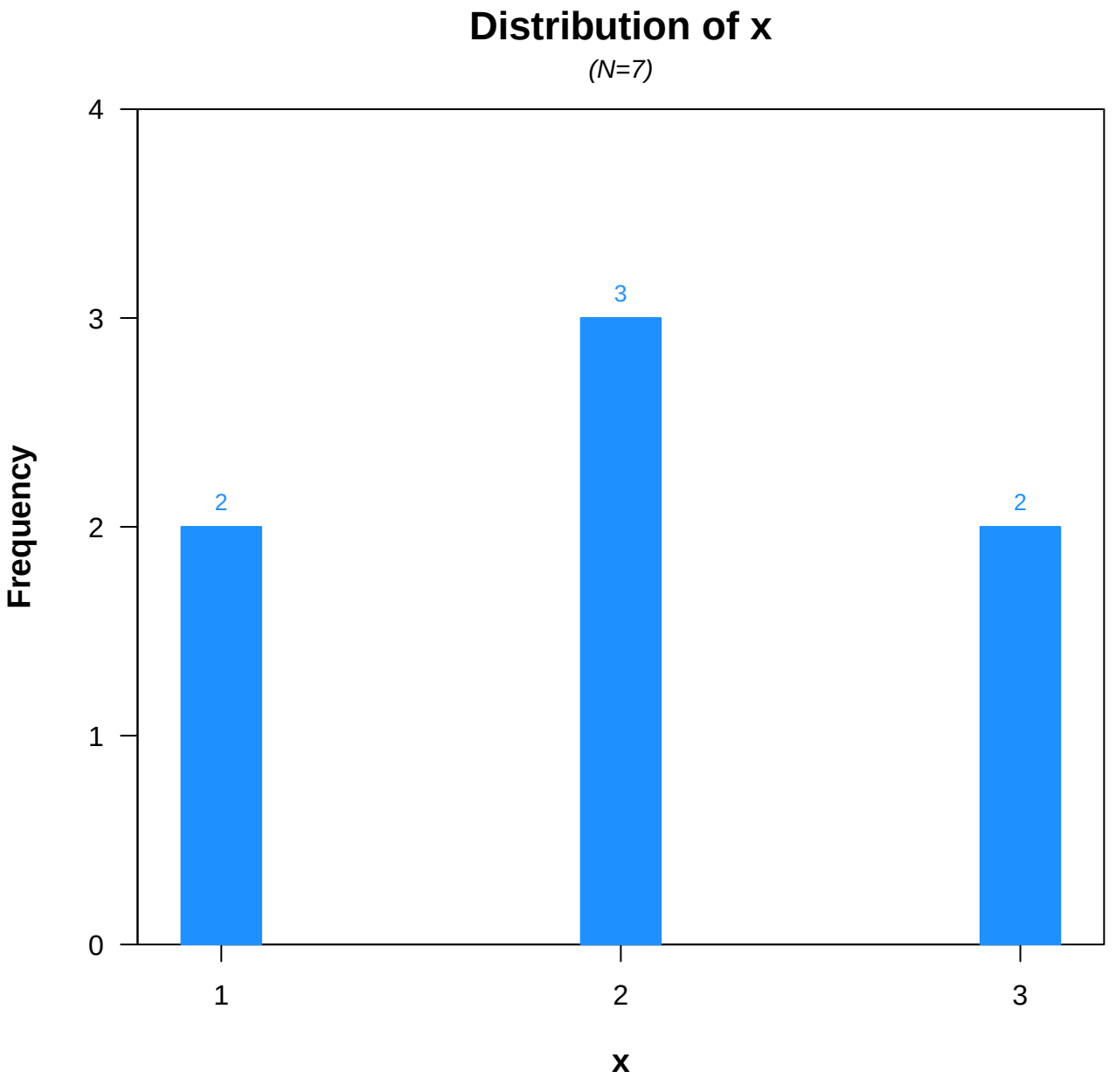

Distribution of value
(N=8)
A (N=4)
B (N=4)
Frequency
0
1
2
3
1
2
3
4
value

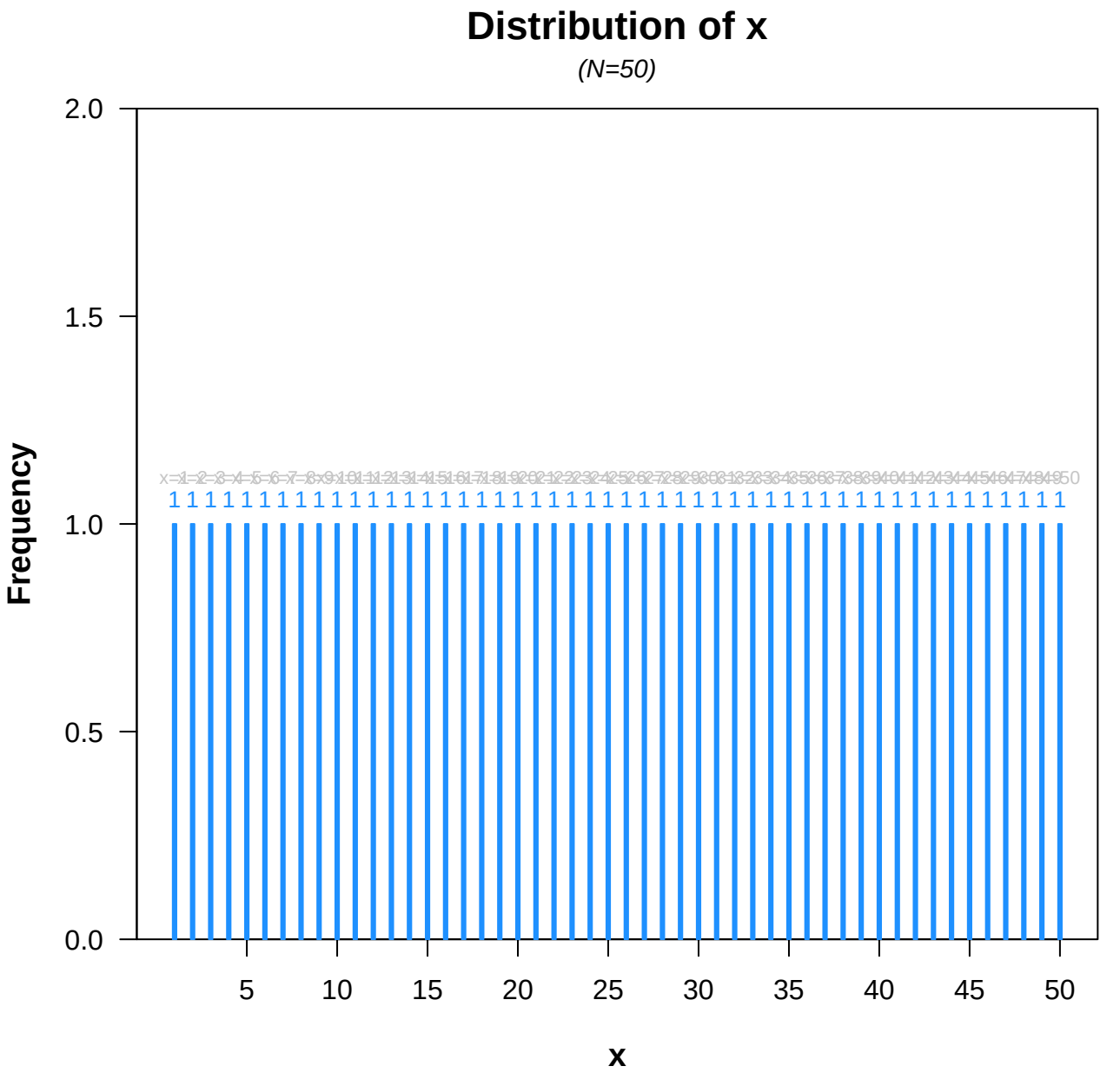

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
3
x

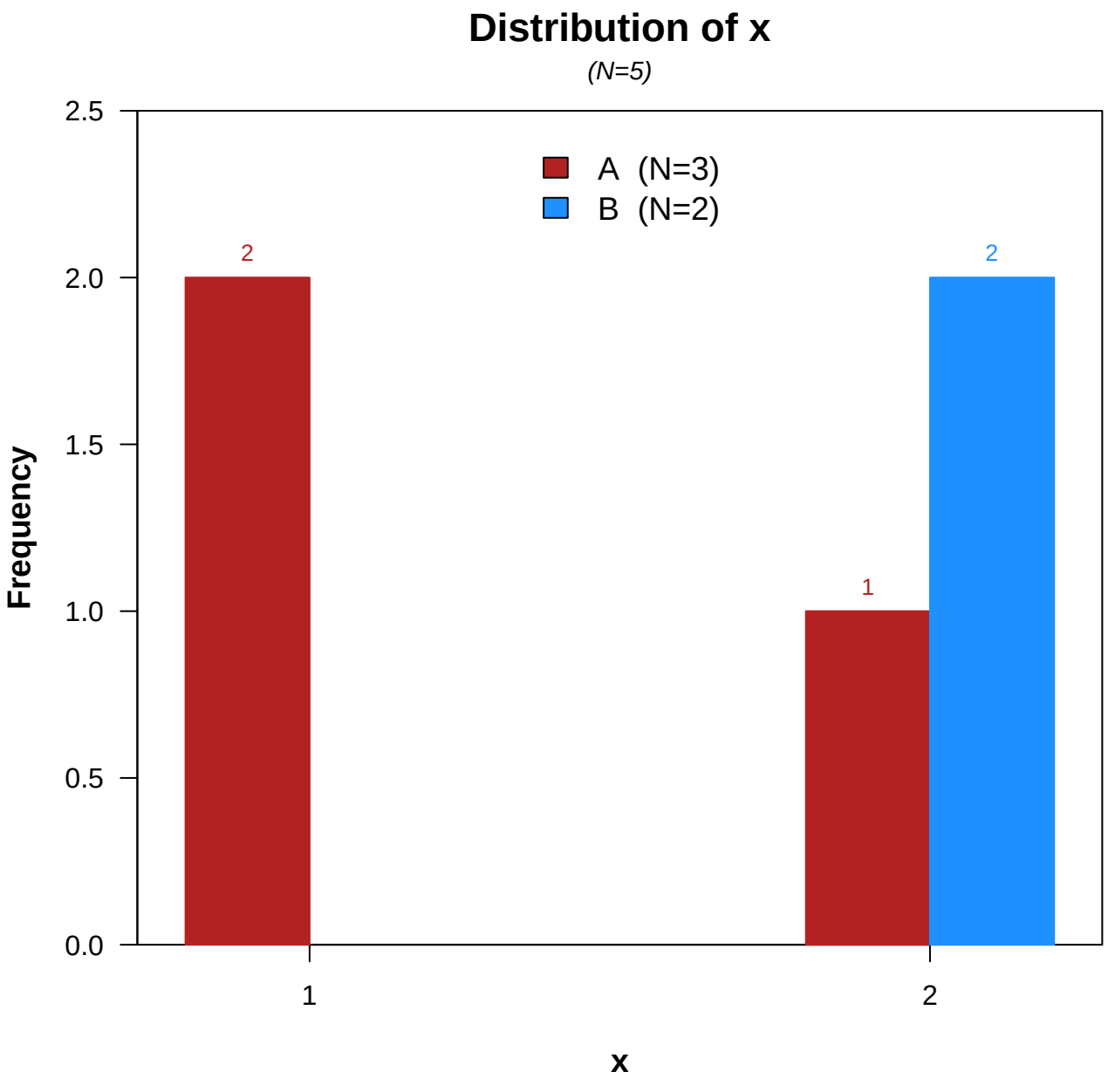

Distribution of x
(N=7)
Frequency
4
3
2
1
0
1
2
3
x

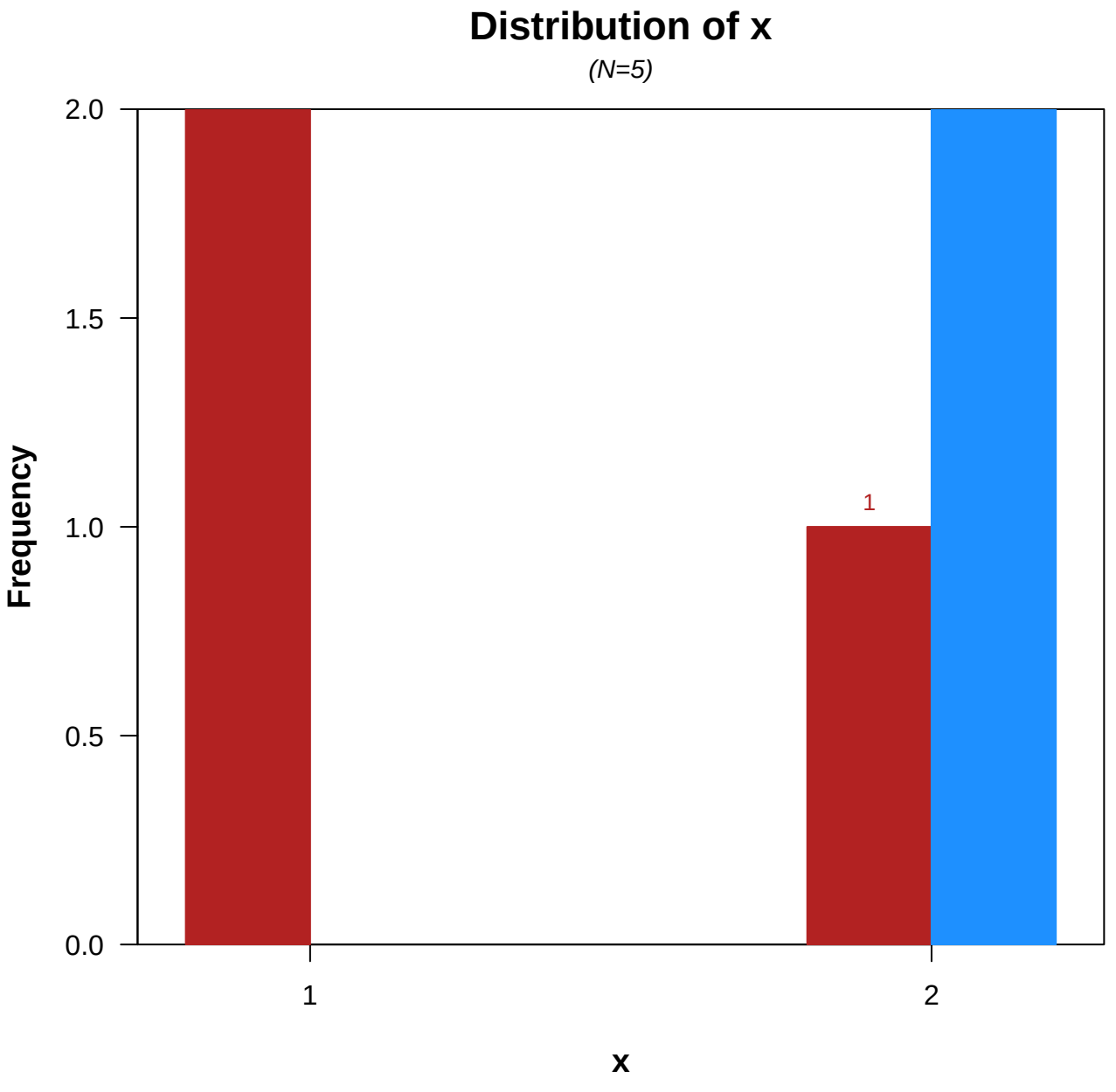

Distribution of value
(N=7)
A (N=4)
B (N=3)
Frequency
value
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
1
1
2
1
1
1

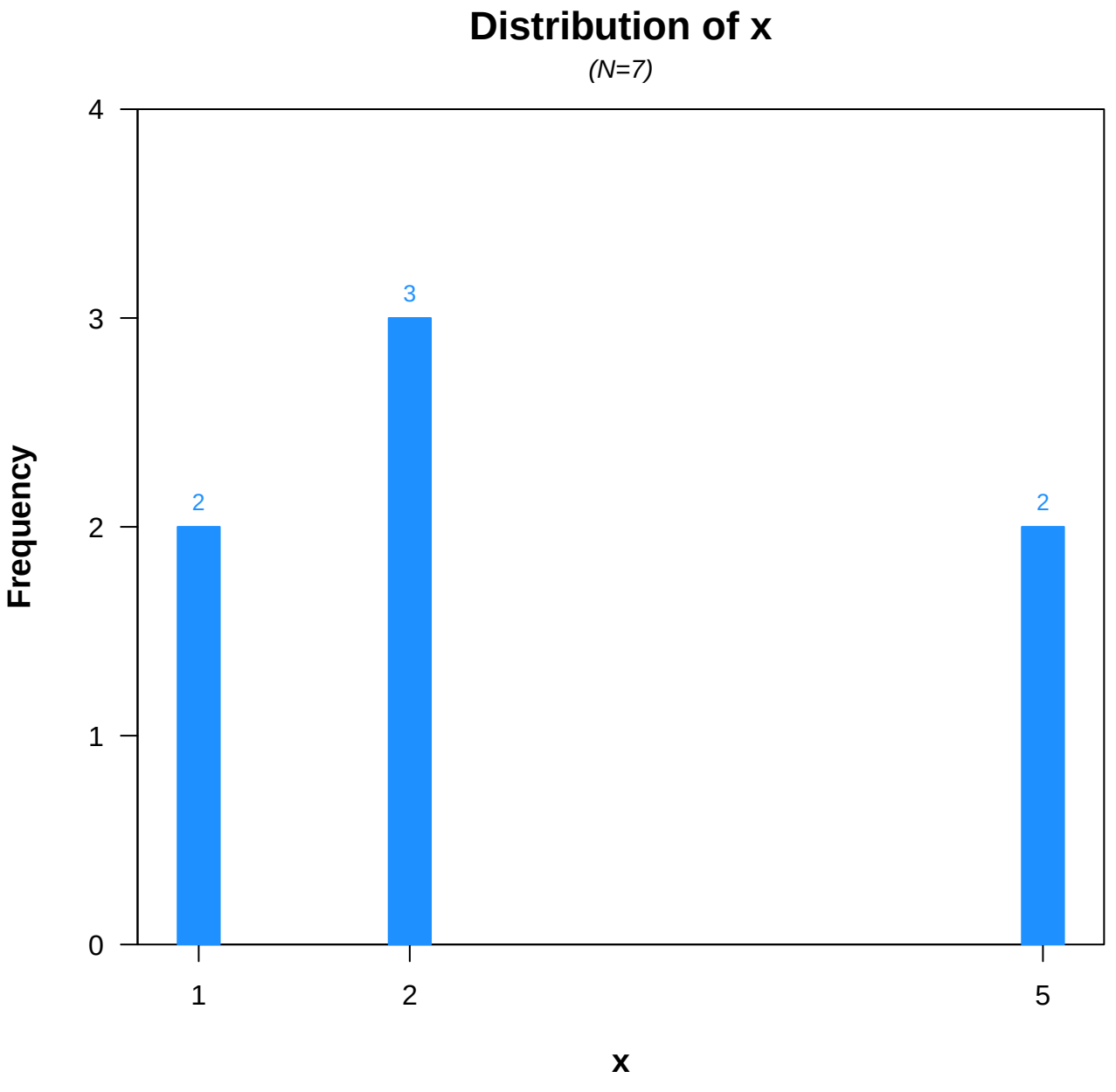

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
3
x

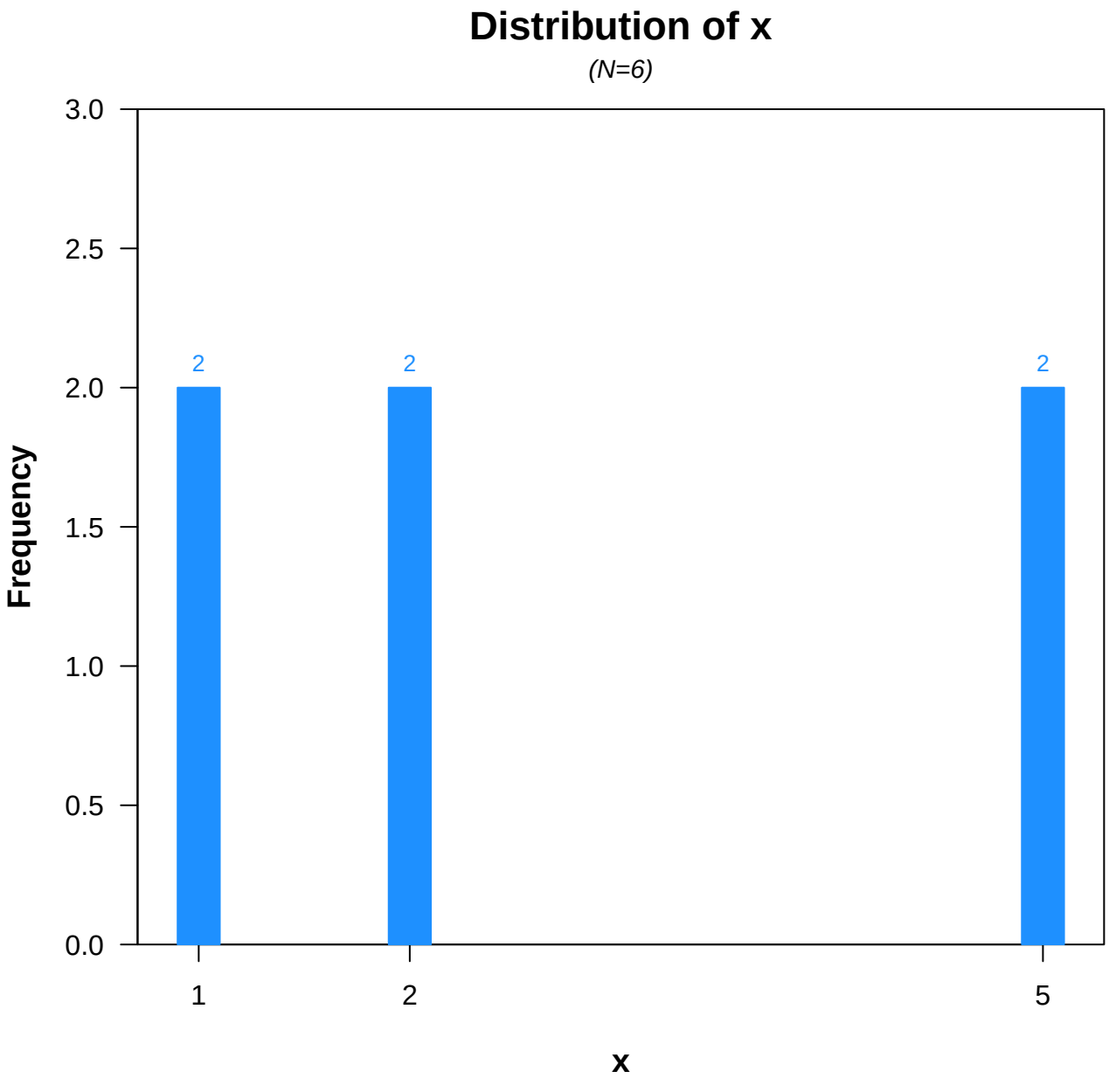

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
3
x

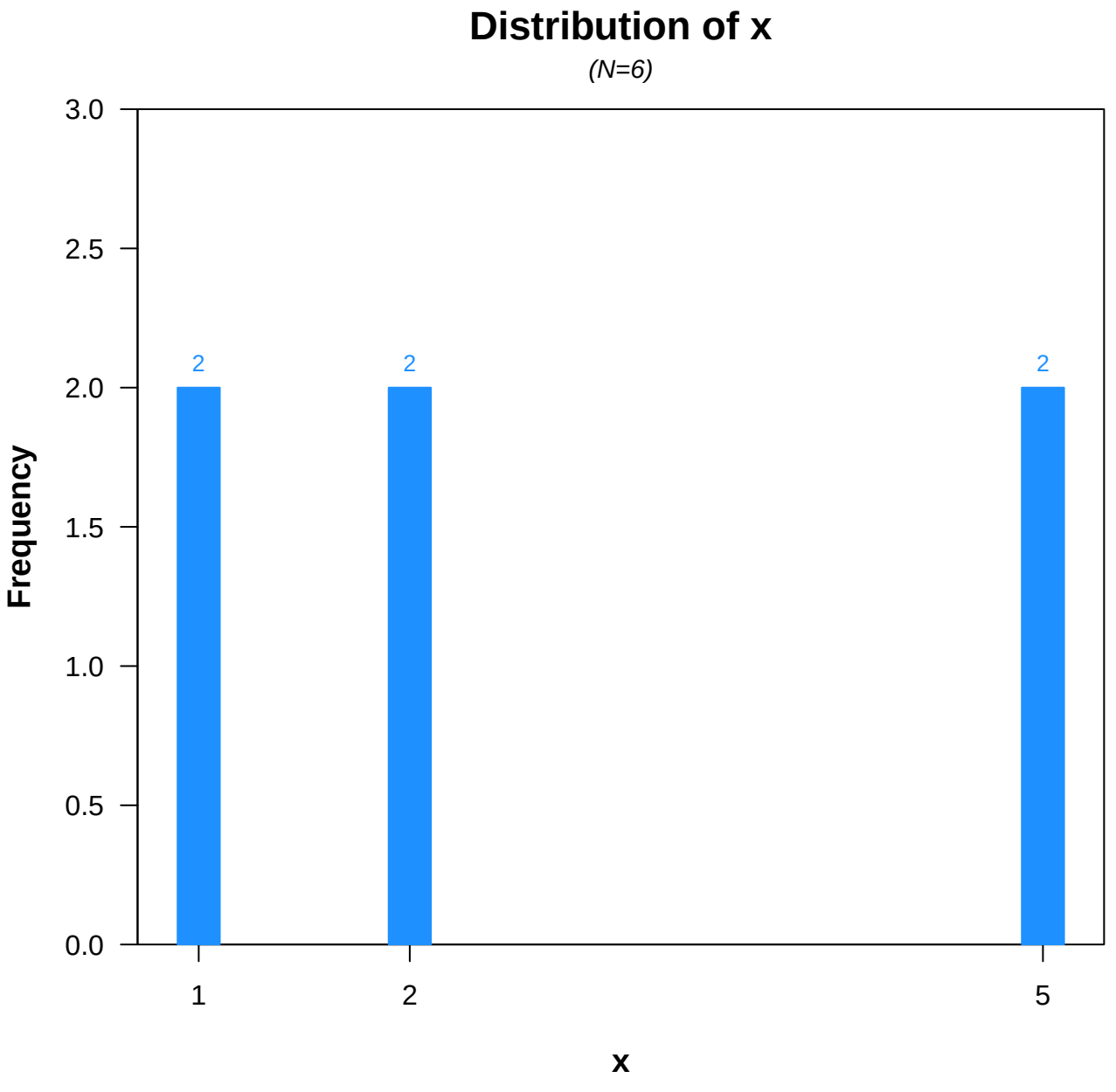

Distribution of x
(N=50)
Frequency
2.0
1.5
1.0
0.5
0.0
5
10
15
20
25
30
35
40
45
50
x

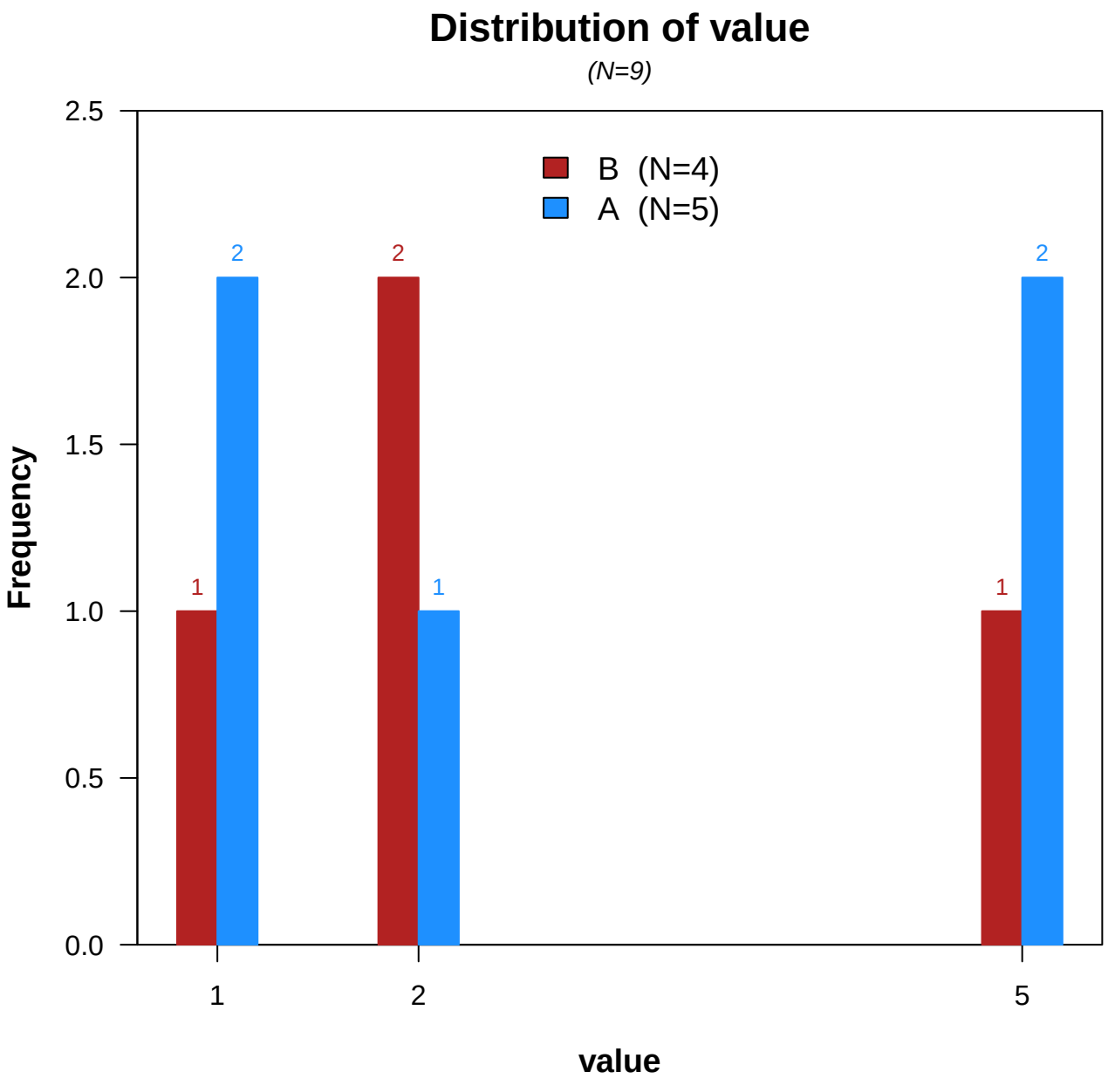

Distribution of x
(N=5)
A (N=3)
B (N=2)
2
2
1
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
x

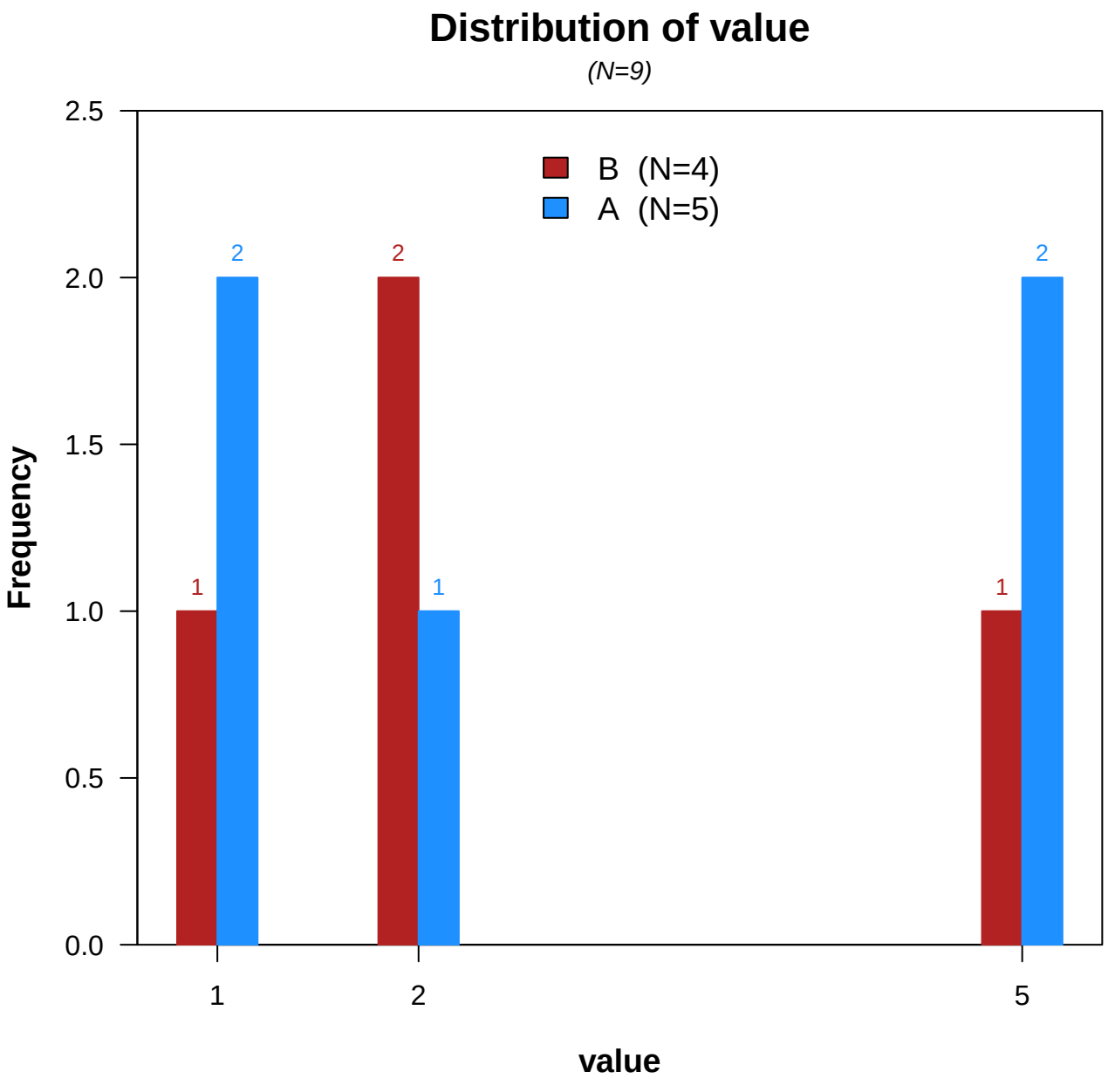

Distribution of x
(N=5)
Frequency
2.0
1.5
1.0
0.5
0.0
1
1
2
x

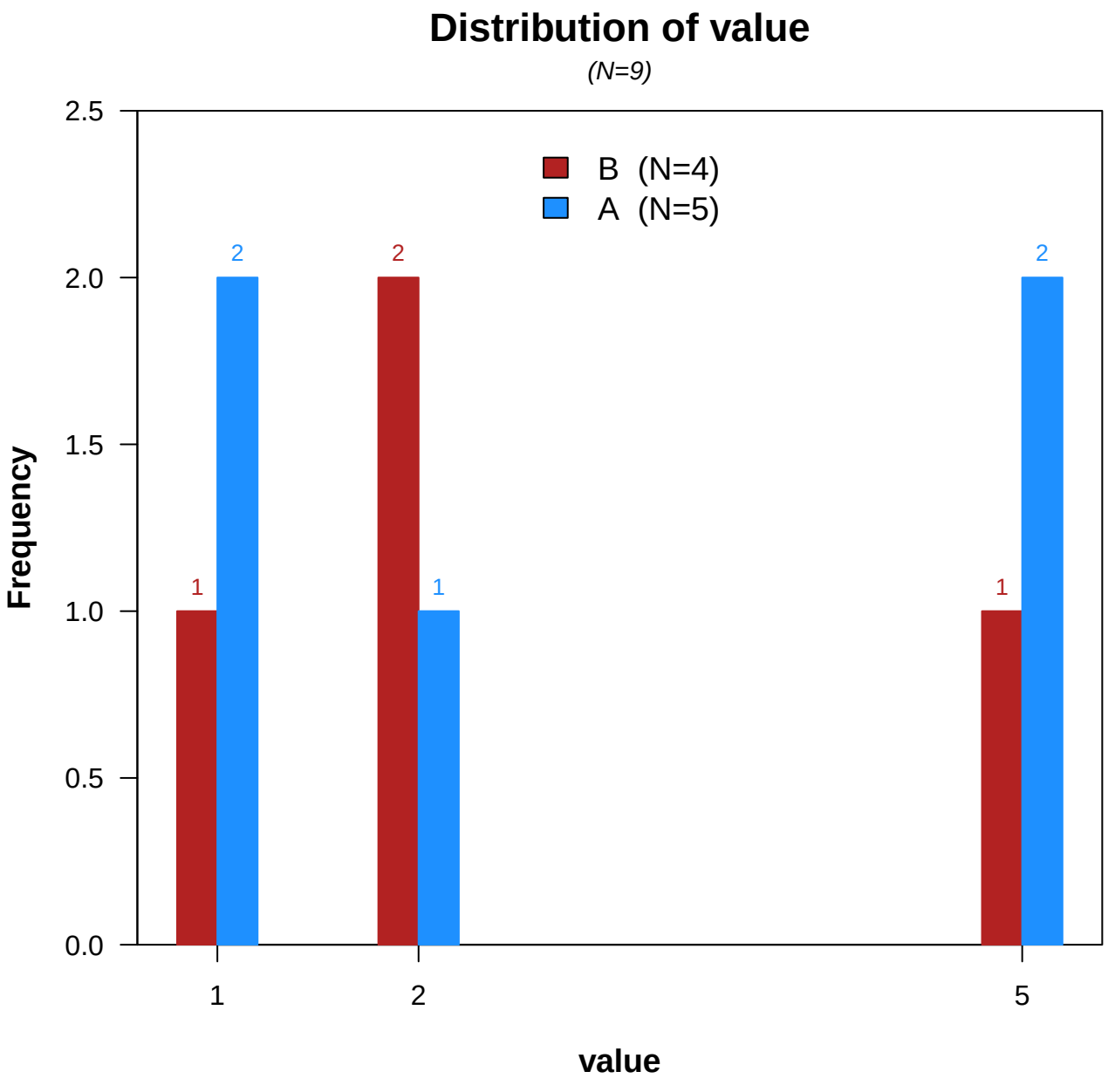

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
5
x

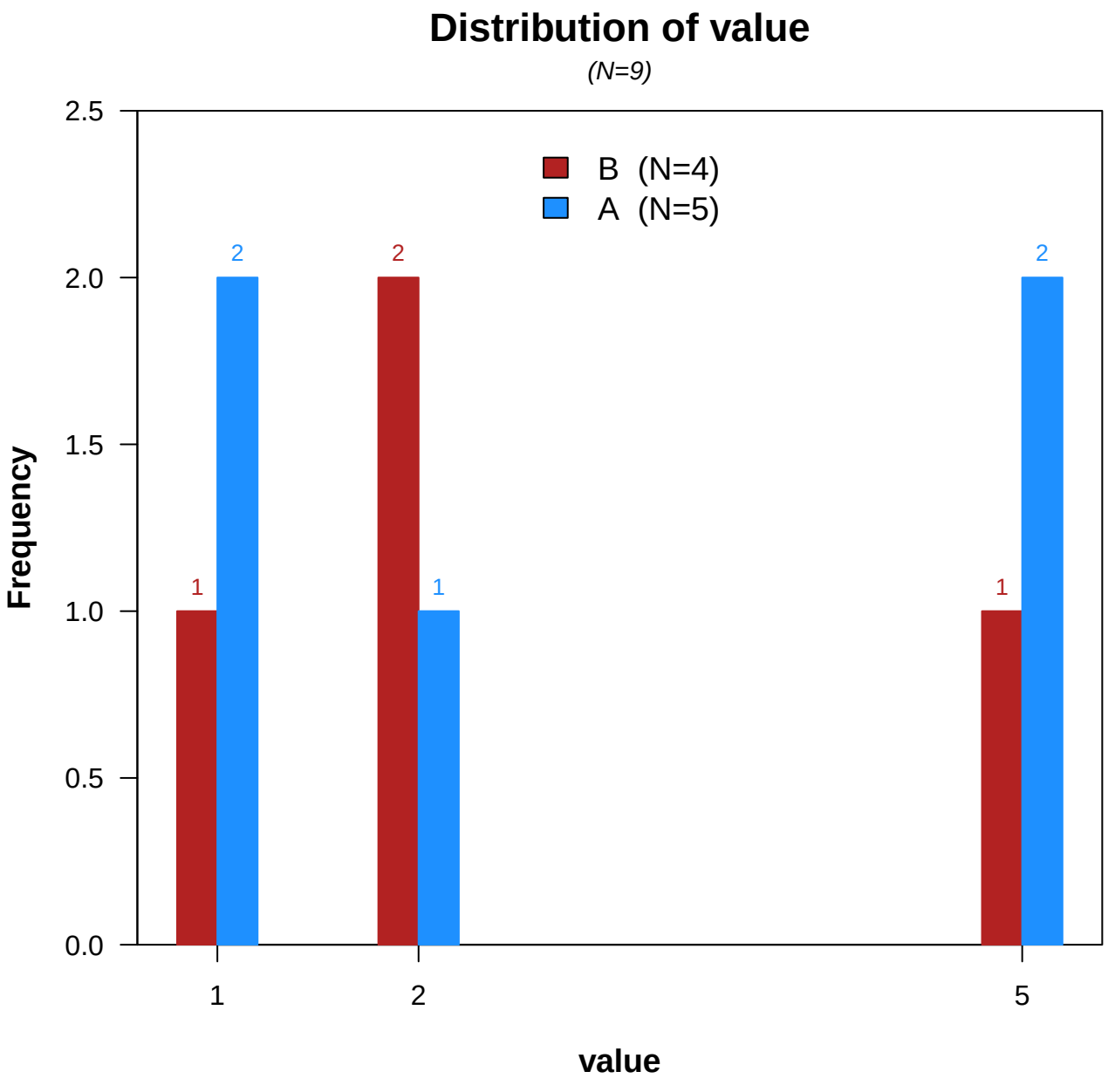

Distribution of x
(N=6)
Frequency
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2
2
2
1
2
5
x

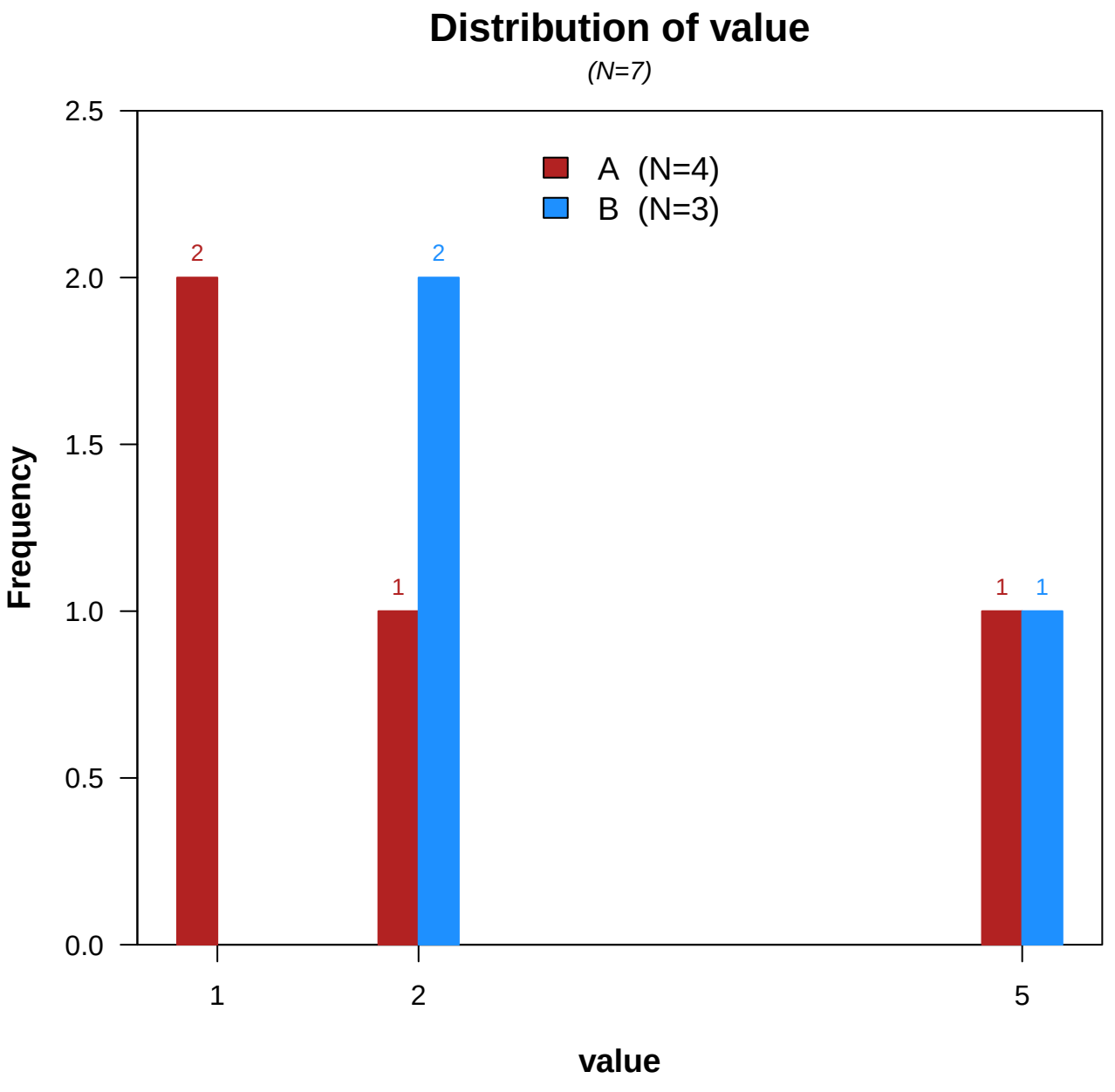

Distribution of x
(N=6)
Frequency
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2
2
2
1
2
5
x

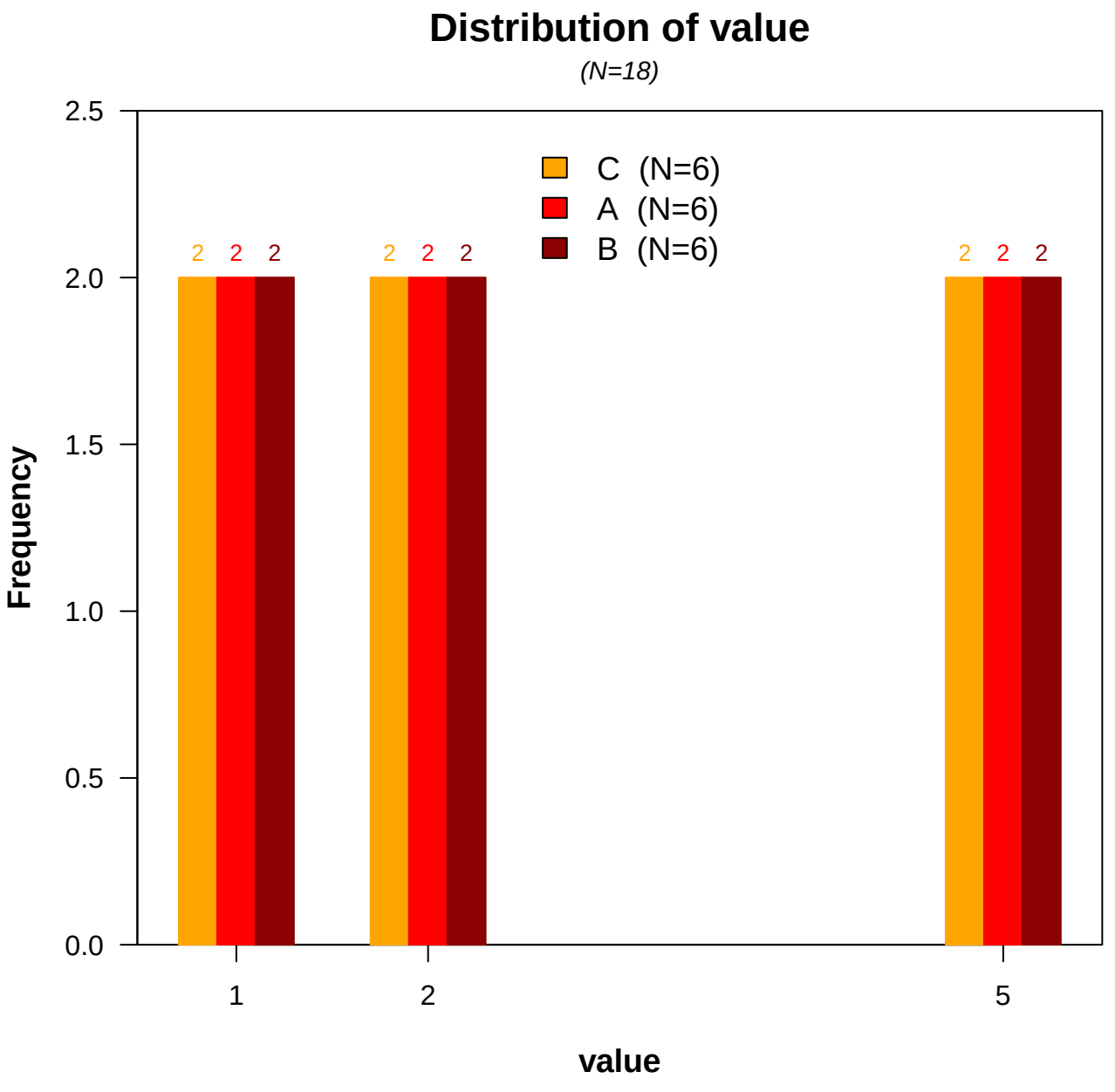

Distribution of value
(N=9)
B (N=4)
A (N=5)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
value
1
2
1
2
1
2

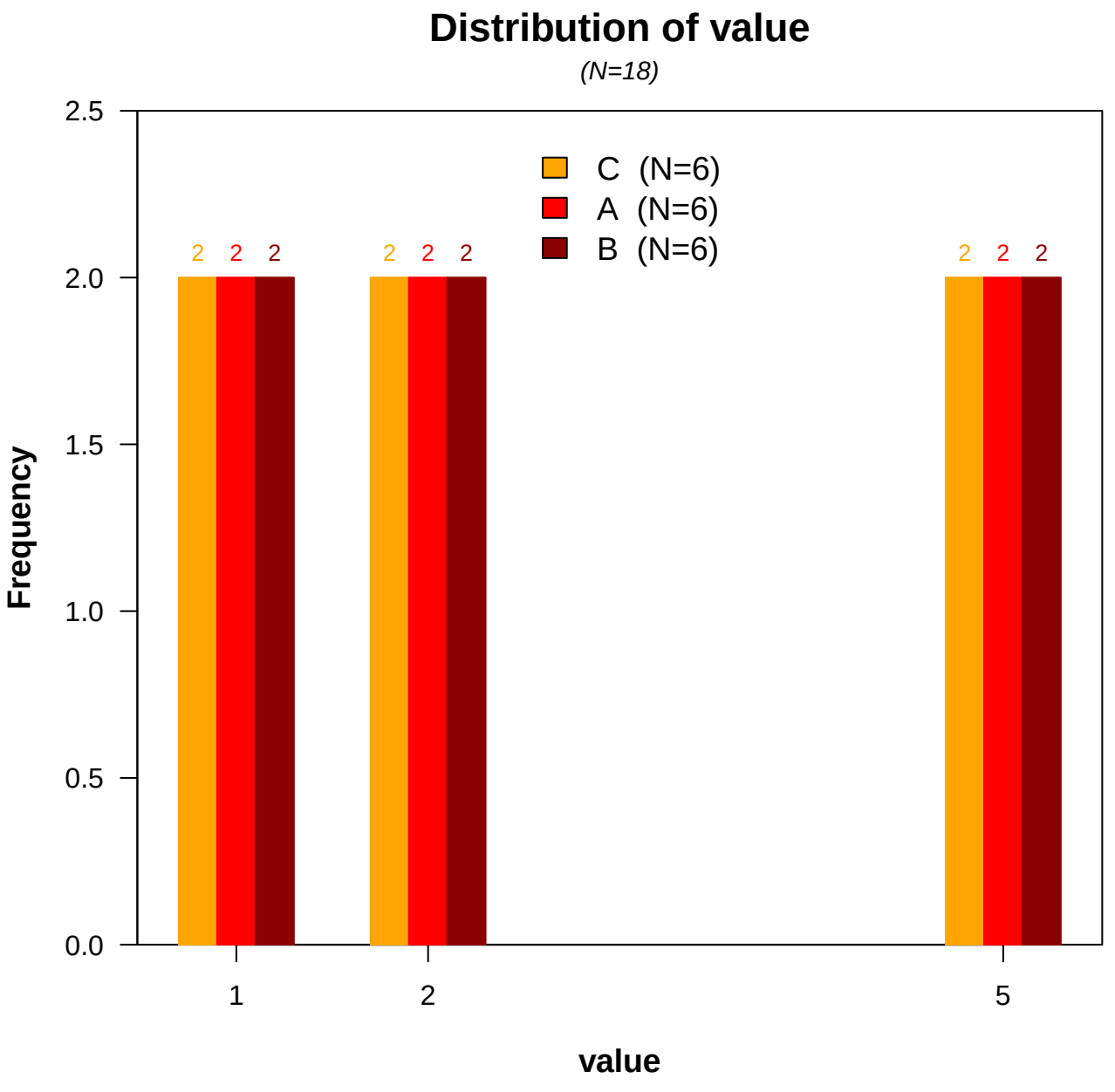

Distribution of value
(N=9)
B (N=4)
A (N=5)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
value
1
2
1
2
1
2

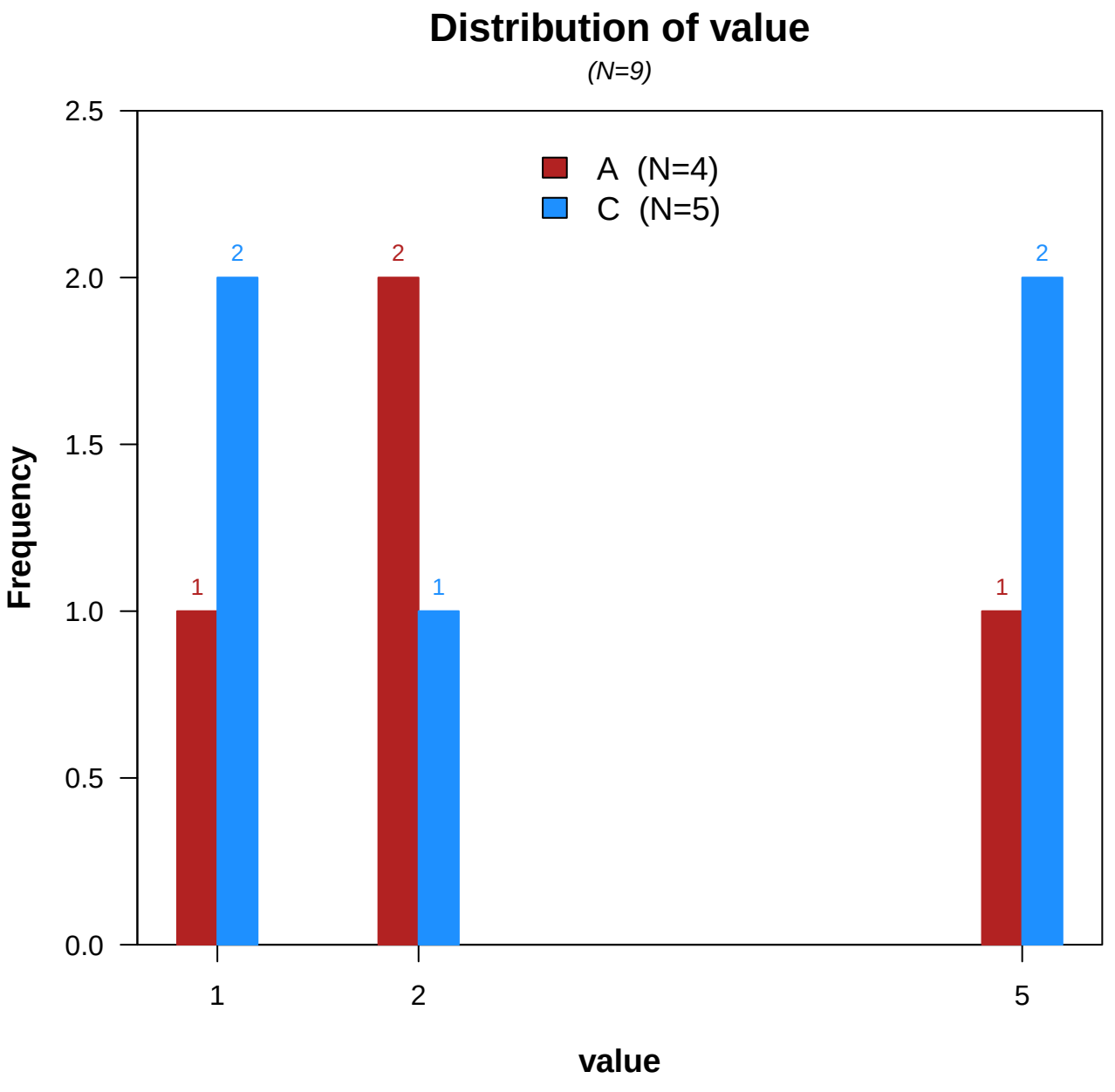

Distribution of value
(N=9)
B (N=4)
A (N=5)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
value
1
2
1
2
1
2

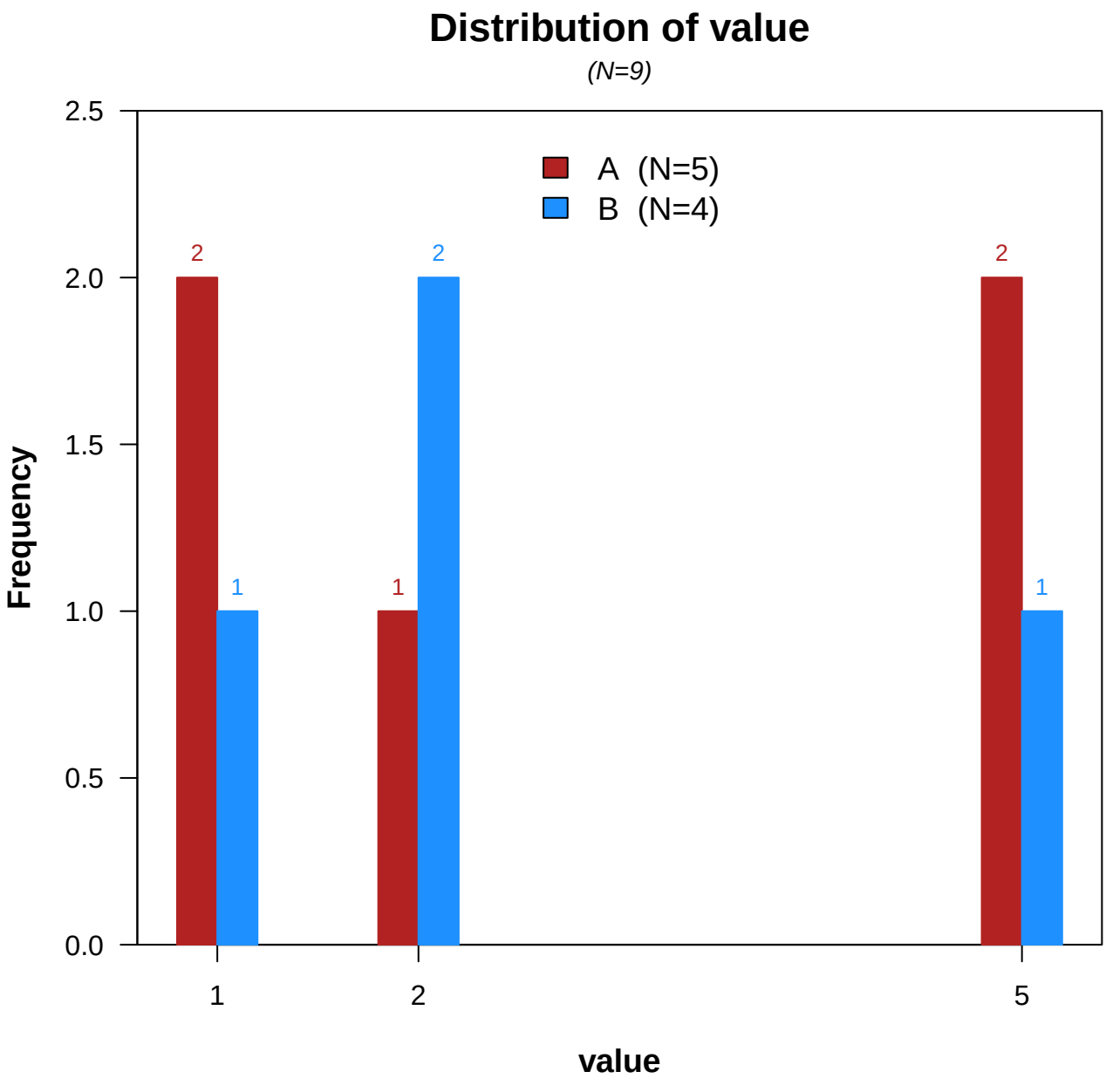

Distribution of value
(N=9)
B (N=4)
A (N=5)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
value
1
2
1
2
1
2

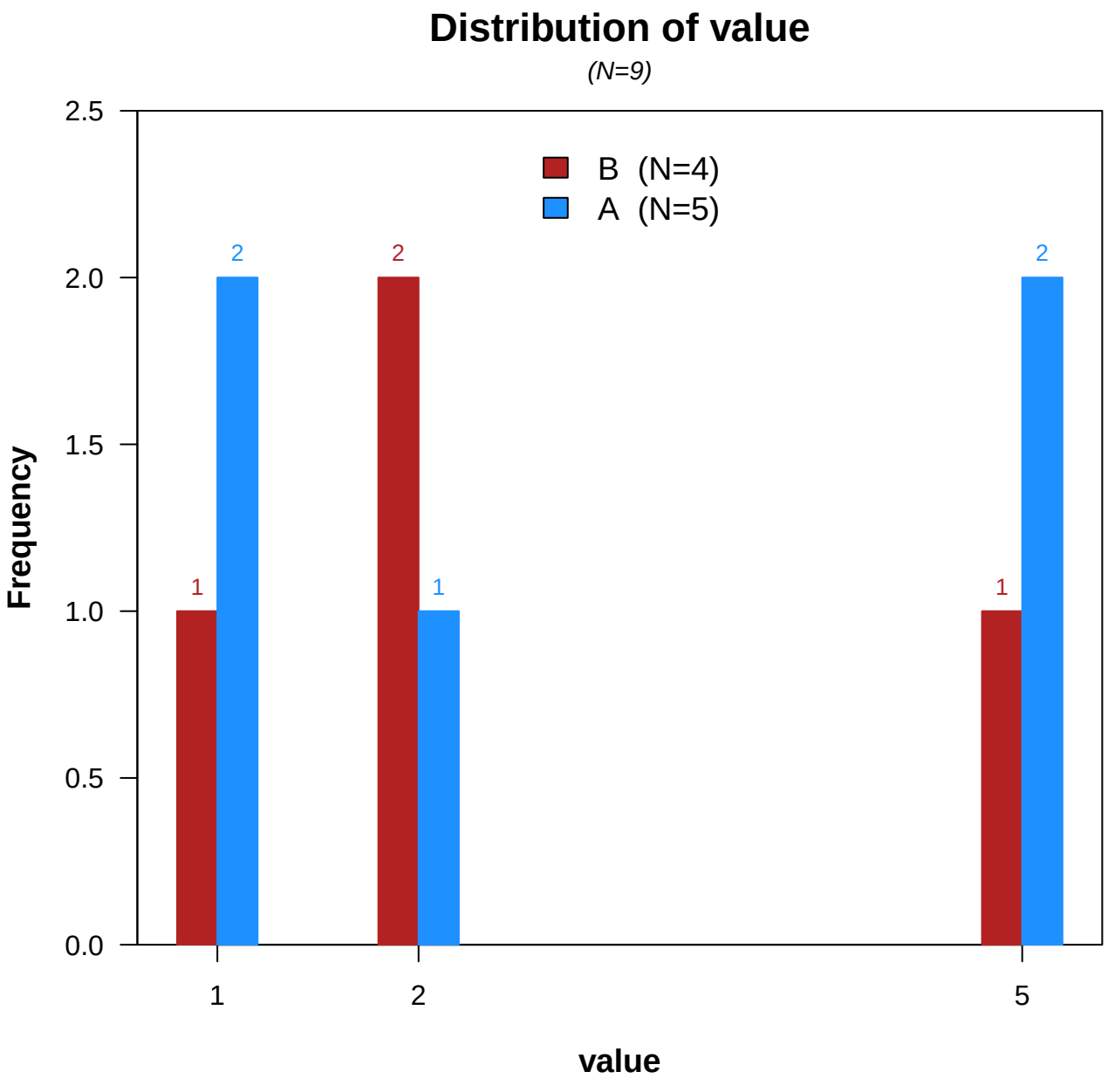

Distribution of value
(N=7)
A (N=4)
B (N=3)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
2
1
1
1
1
2
5
value

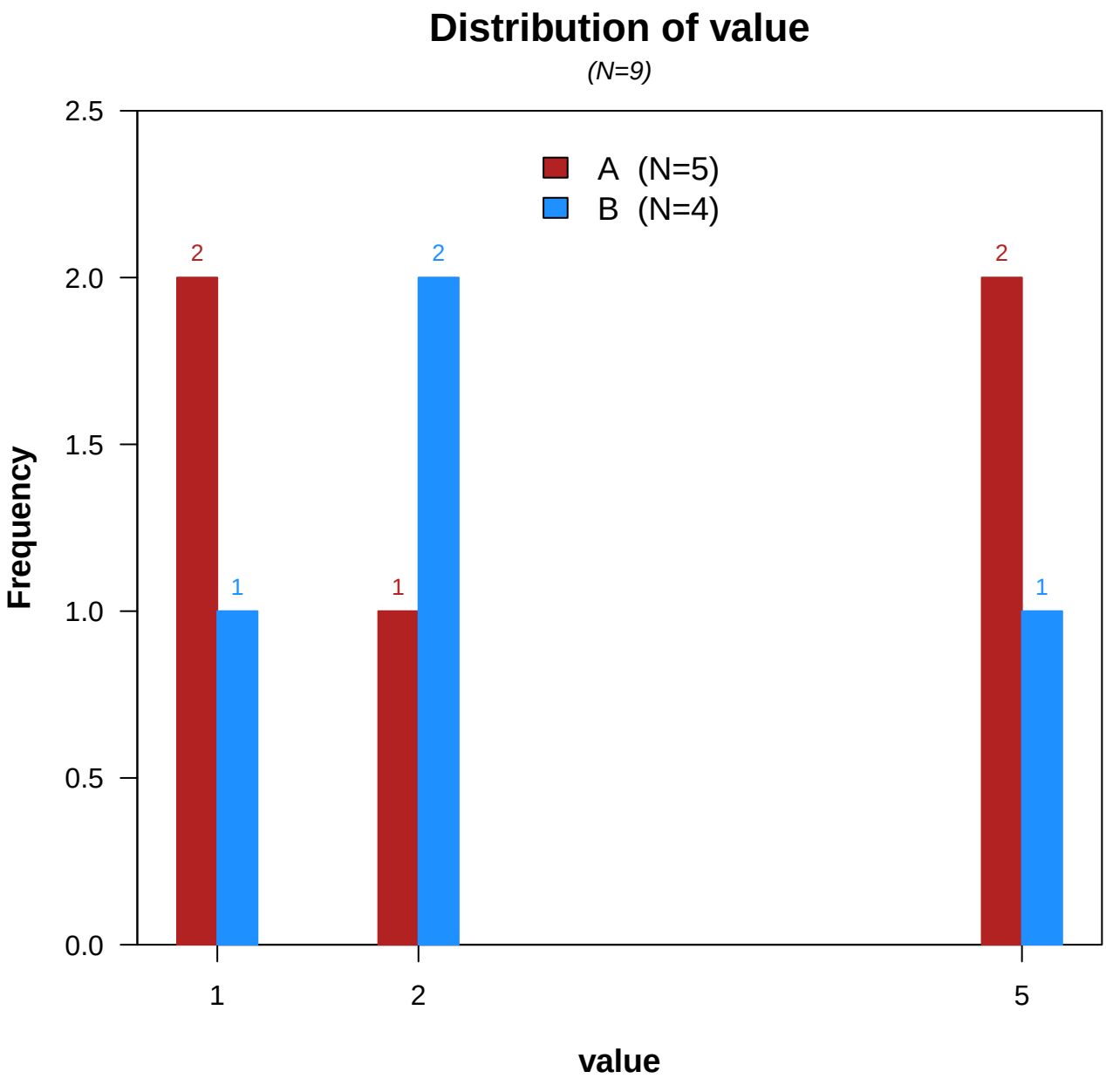

Distribution of value
(N=18)
C (N=6)
A (N=6)
B (N=6)
2 2 2
2 2 2
2 2 2
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
5
value

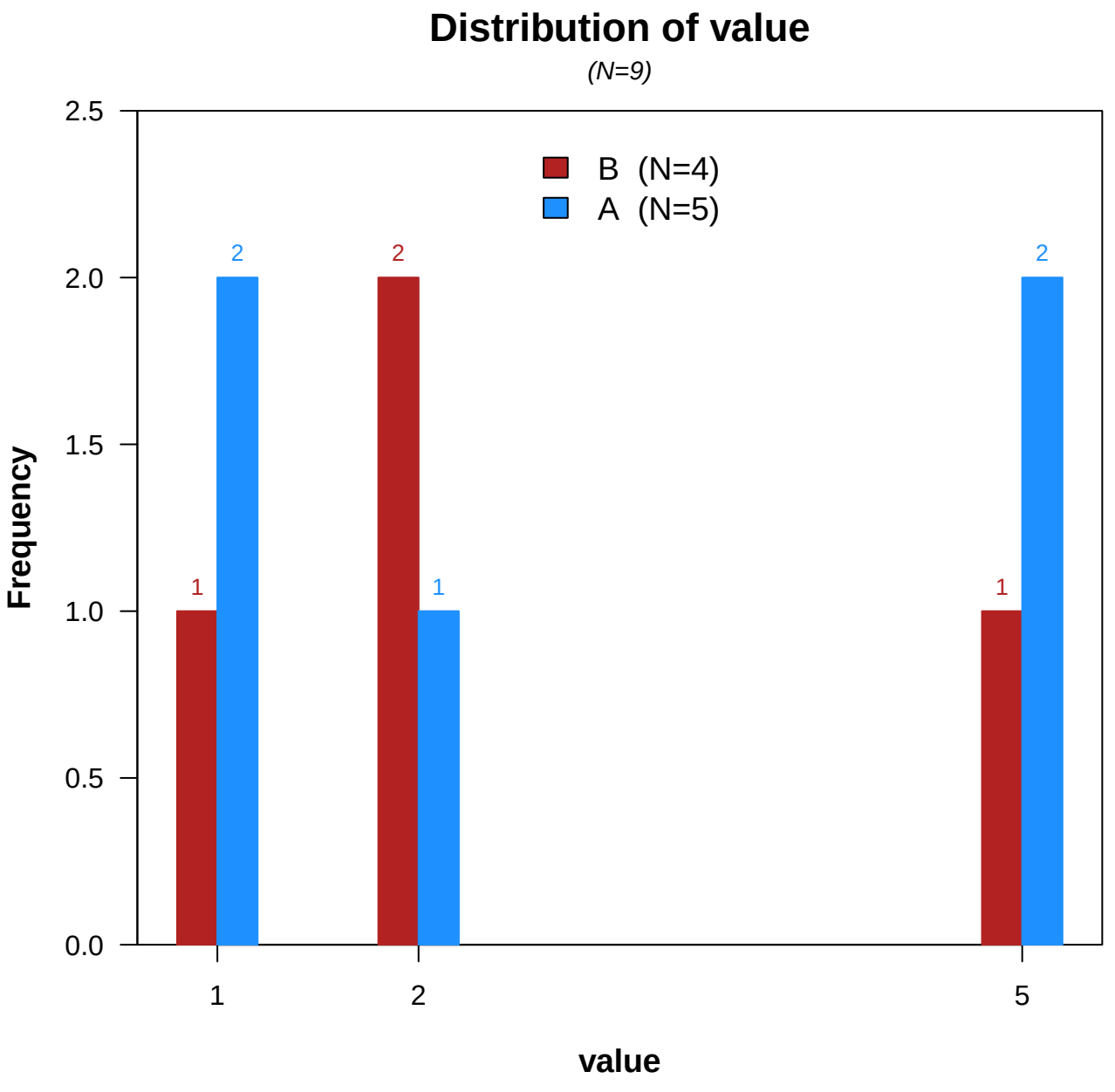

Distribution of value
(N=18)
C (N=6)
A (N=6)
B (N=6)
2 2 2
2 2 2
2 2 2
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
5
value

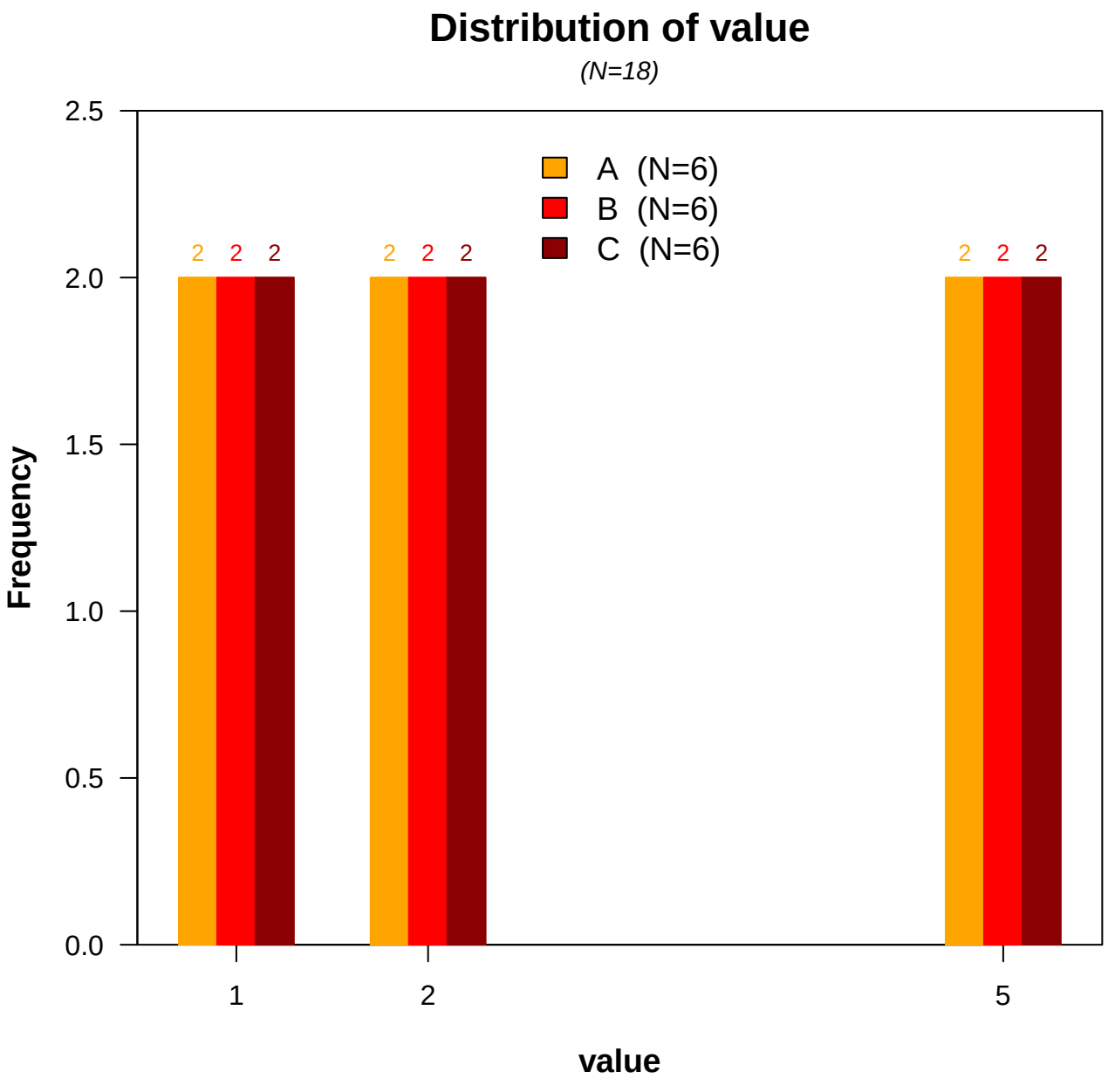

Distribution of value
(N=9)
A (N=4)
C (N=5)
Frequency
value
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
1
2
1
2
1
2

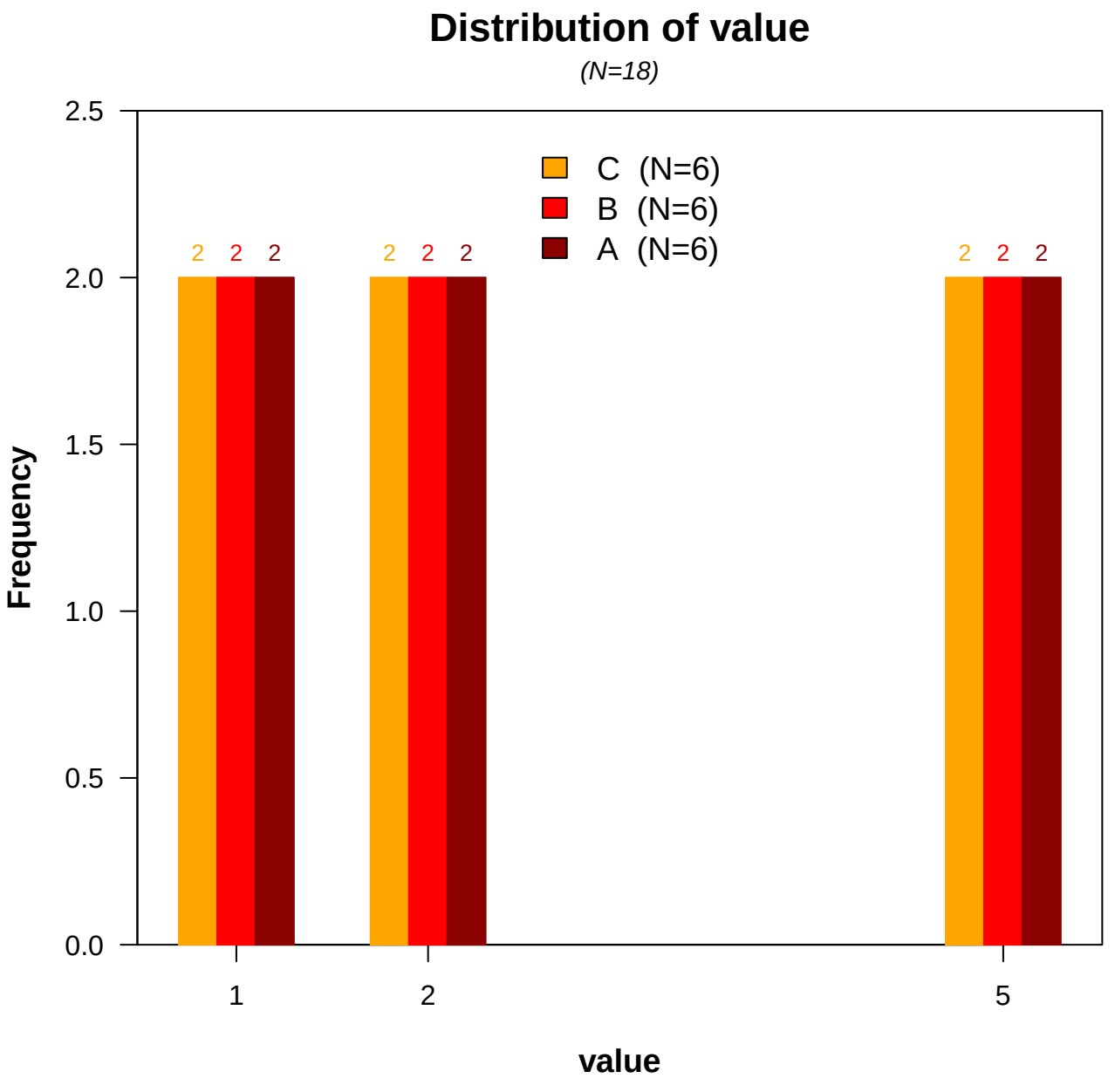

Distribution of value
(N=9)
A (N=5)
B (N=4)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
1
1
2
2
1
1
2
5
value

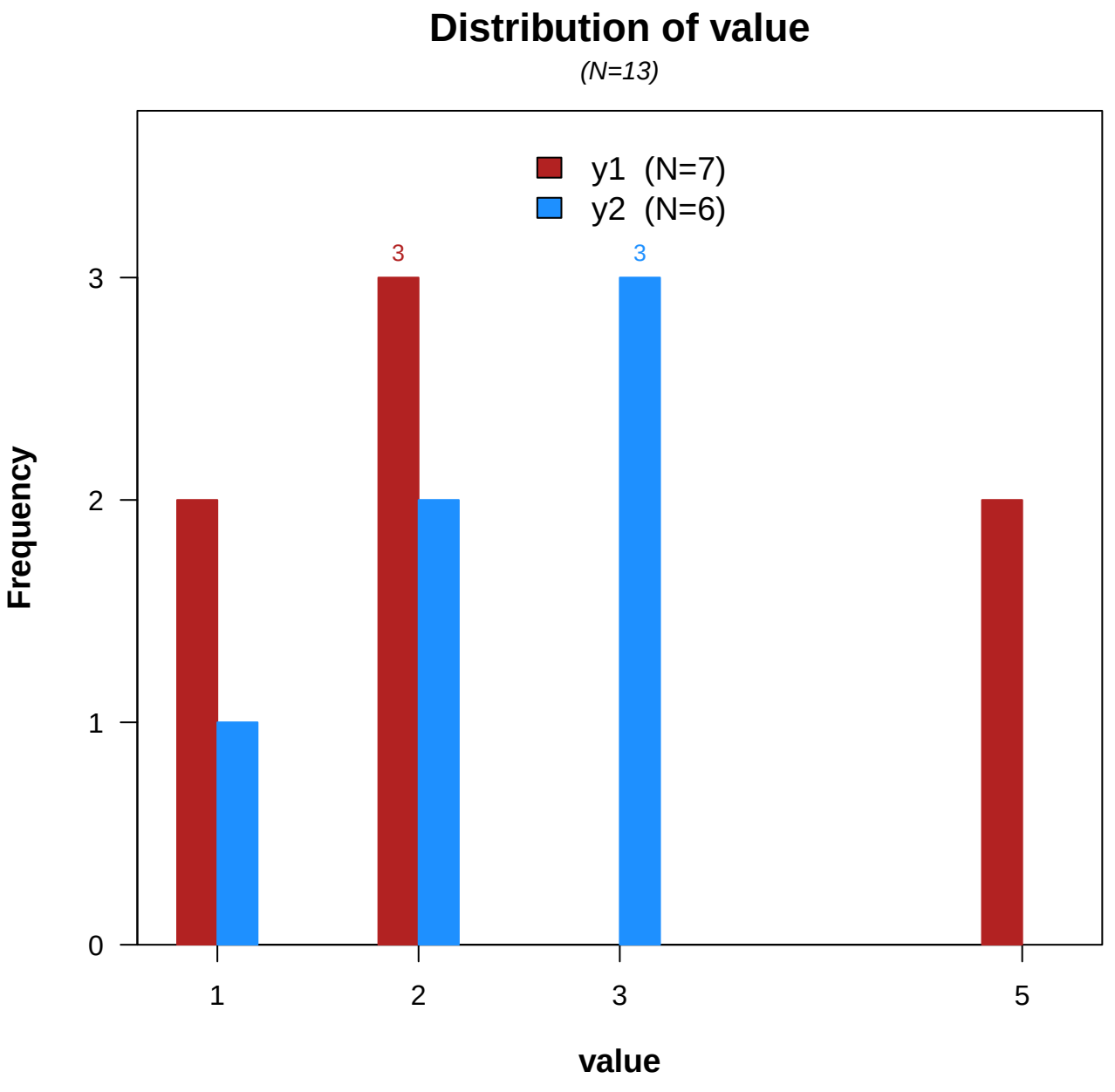

Distribution of value
(N=9)
B (N=4)
A (N=5)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
value
1
2
1
2
1
2

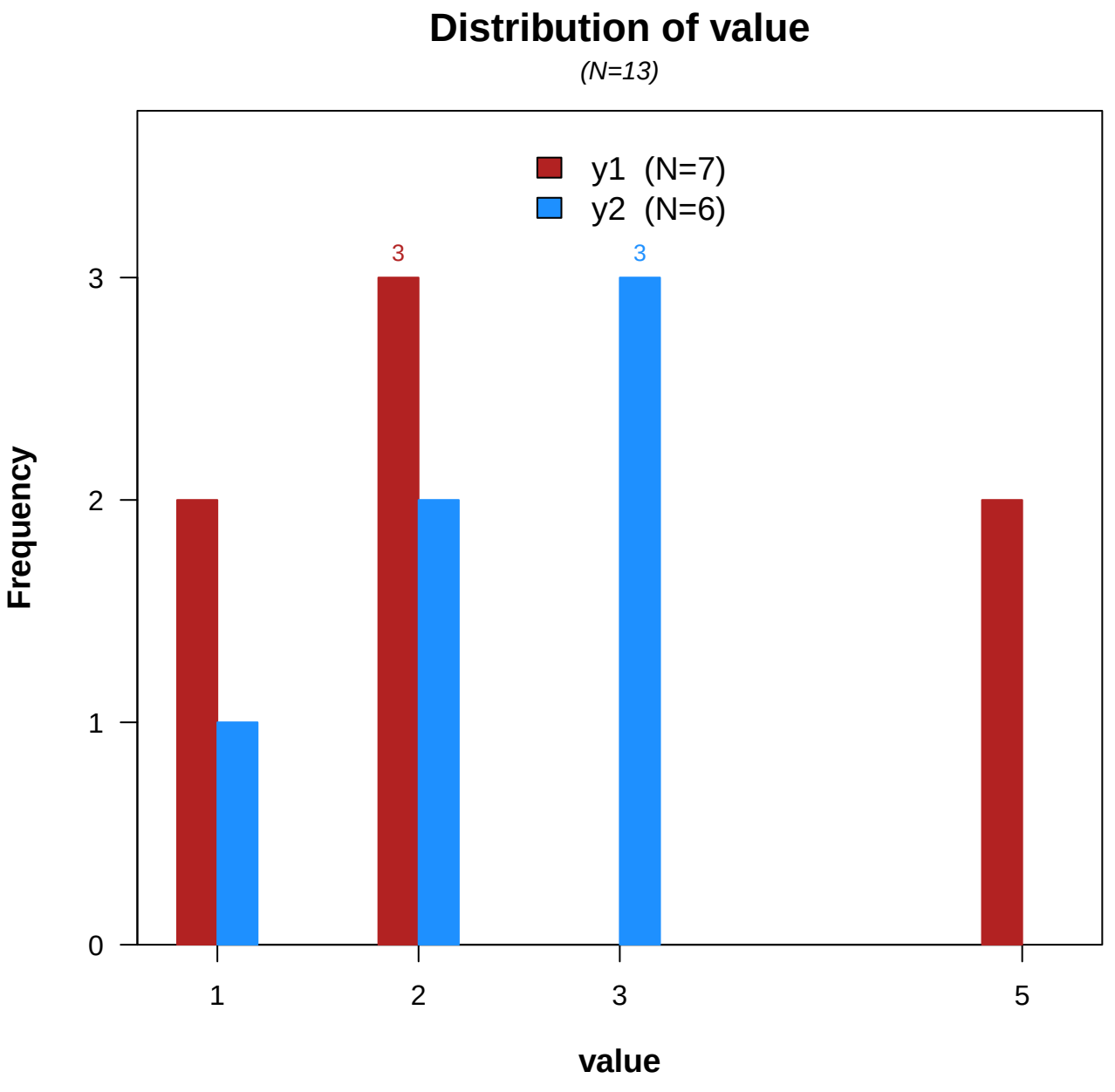

Distribution of value
(N=9)
A (N=5)
B (N=4)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
2
1
1
2
2
1
1
2
5
value

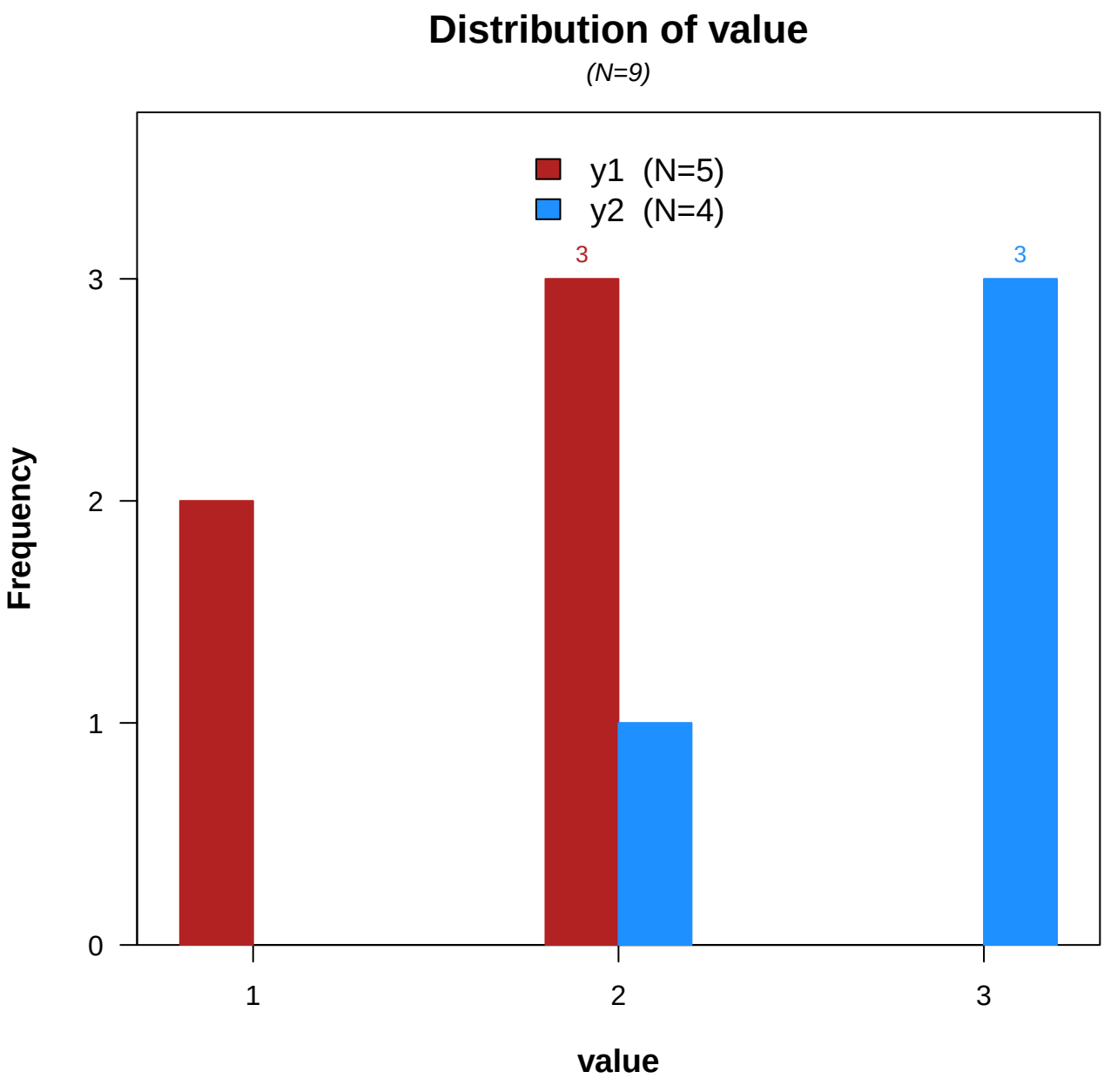

Distribution of value
(N=9)
B (N=4)
A (N=5)
Frequency
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
value
1
2
1
2
1
2

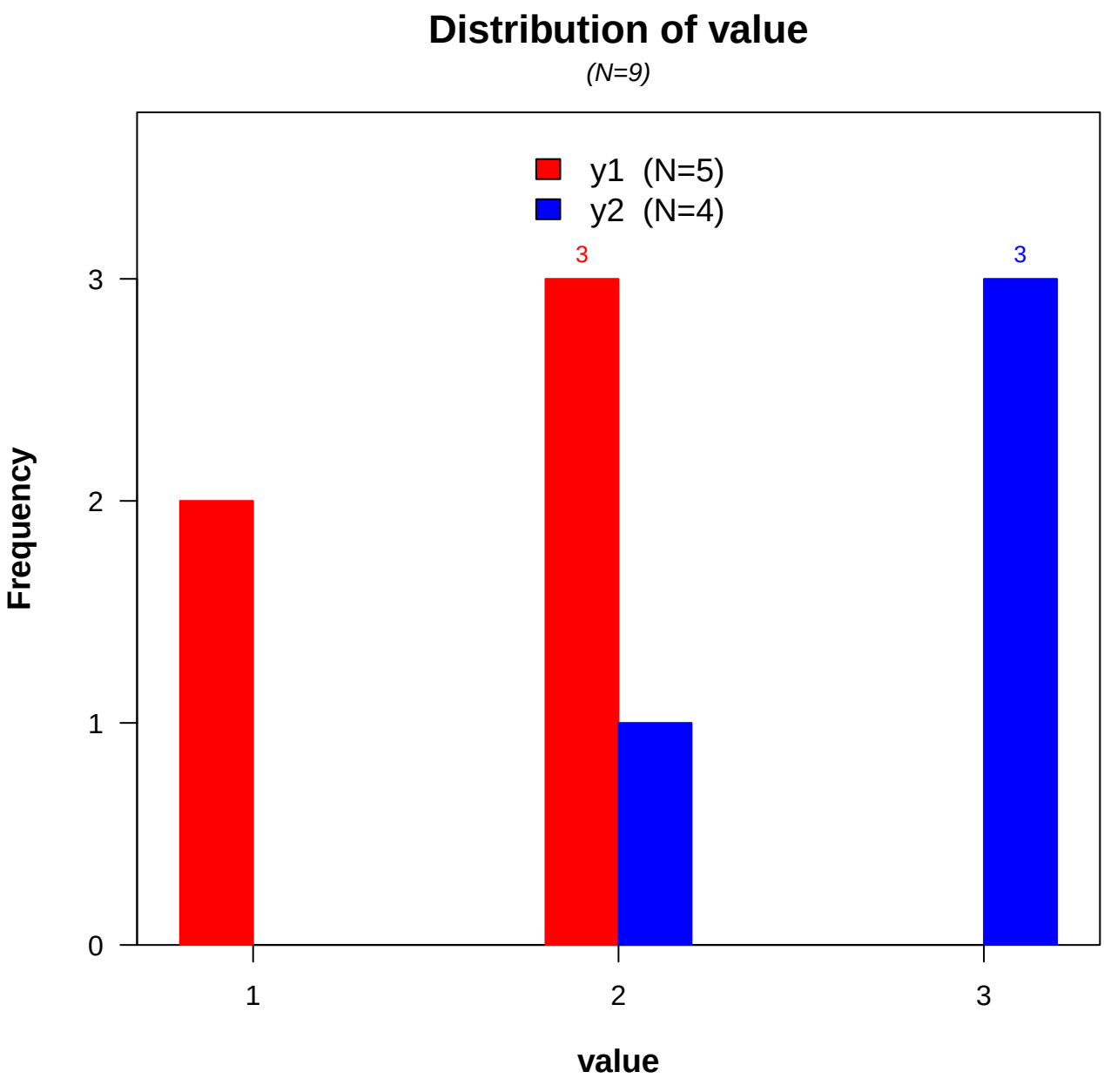

Distribution of value
(N=18)
A (N=6)
B (N=6)
C (N=6)
2 2 2
2 2 2
2 2 2
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
5
value

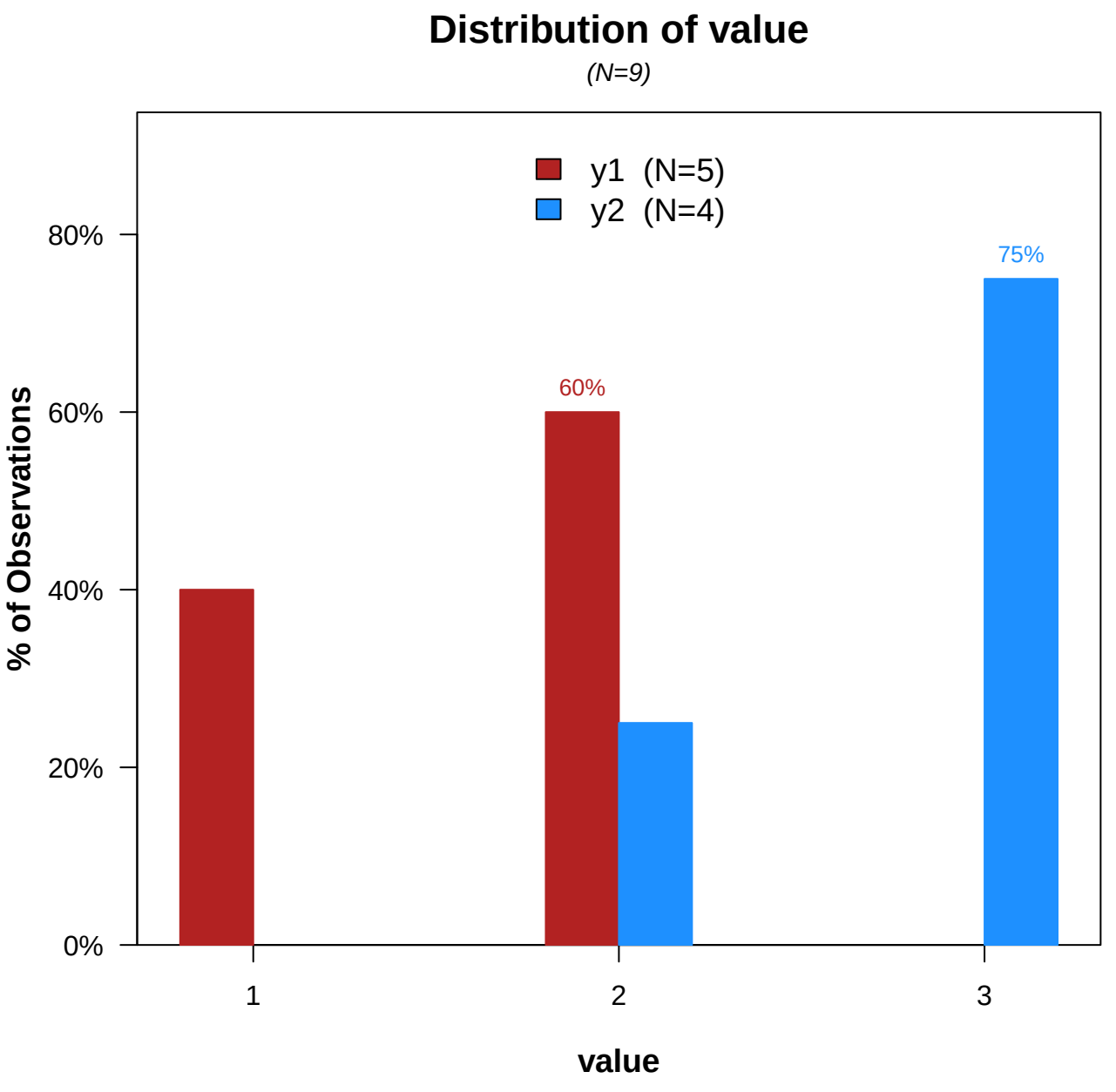

Distribution of value
(N=18)
C (N=6)
B (N=6)
A (N=6)
2 2 2
2 2 2
2 2 2
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
5
value

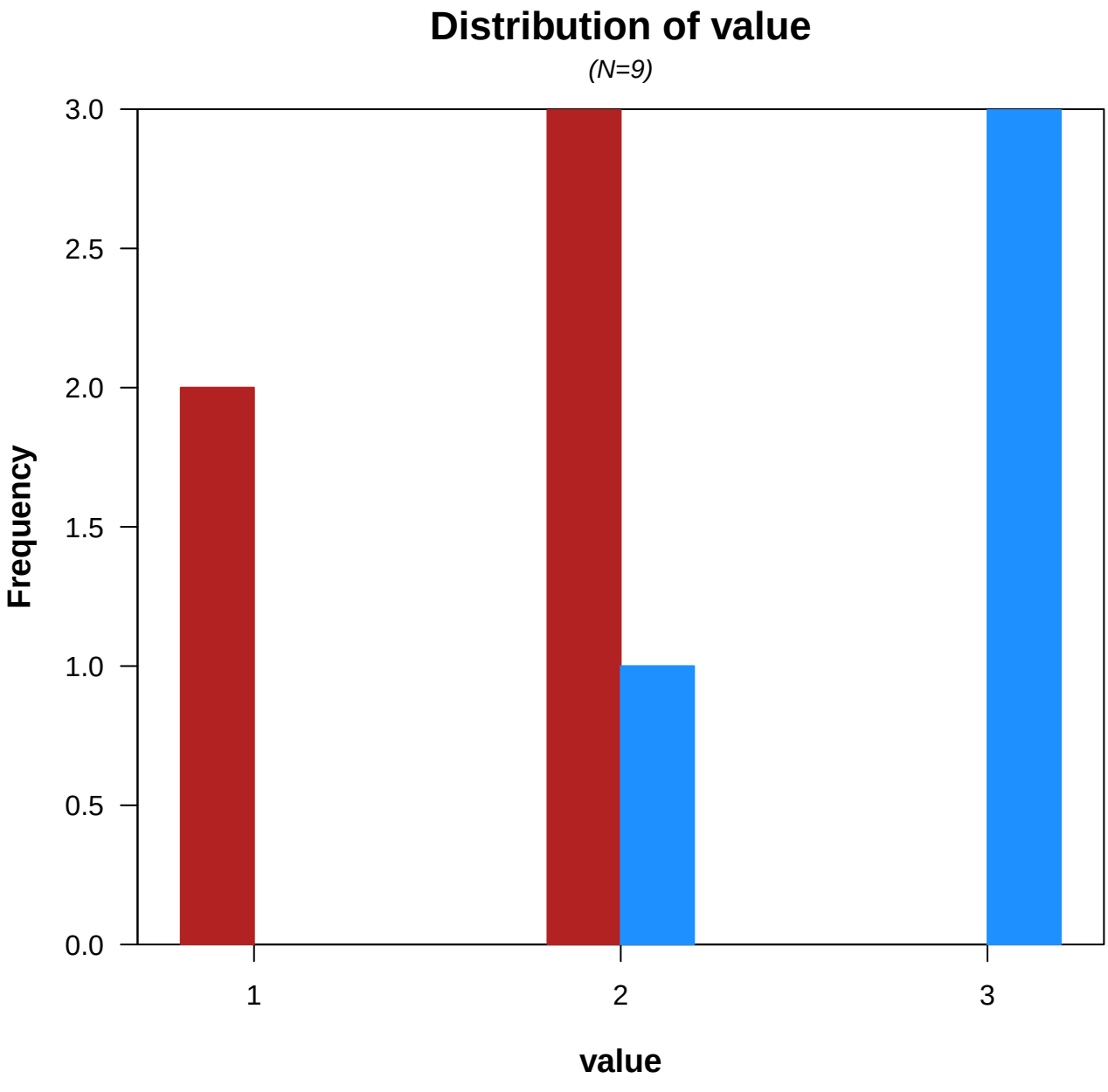

Distribution of value
(N=13)
y1 (N=7)
y2 (N=6)
3
3
Frequency
0
1
2
3
1
2
3
5
value

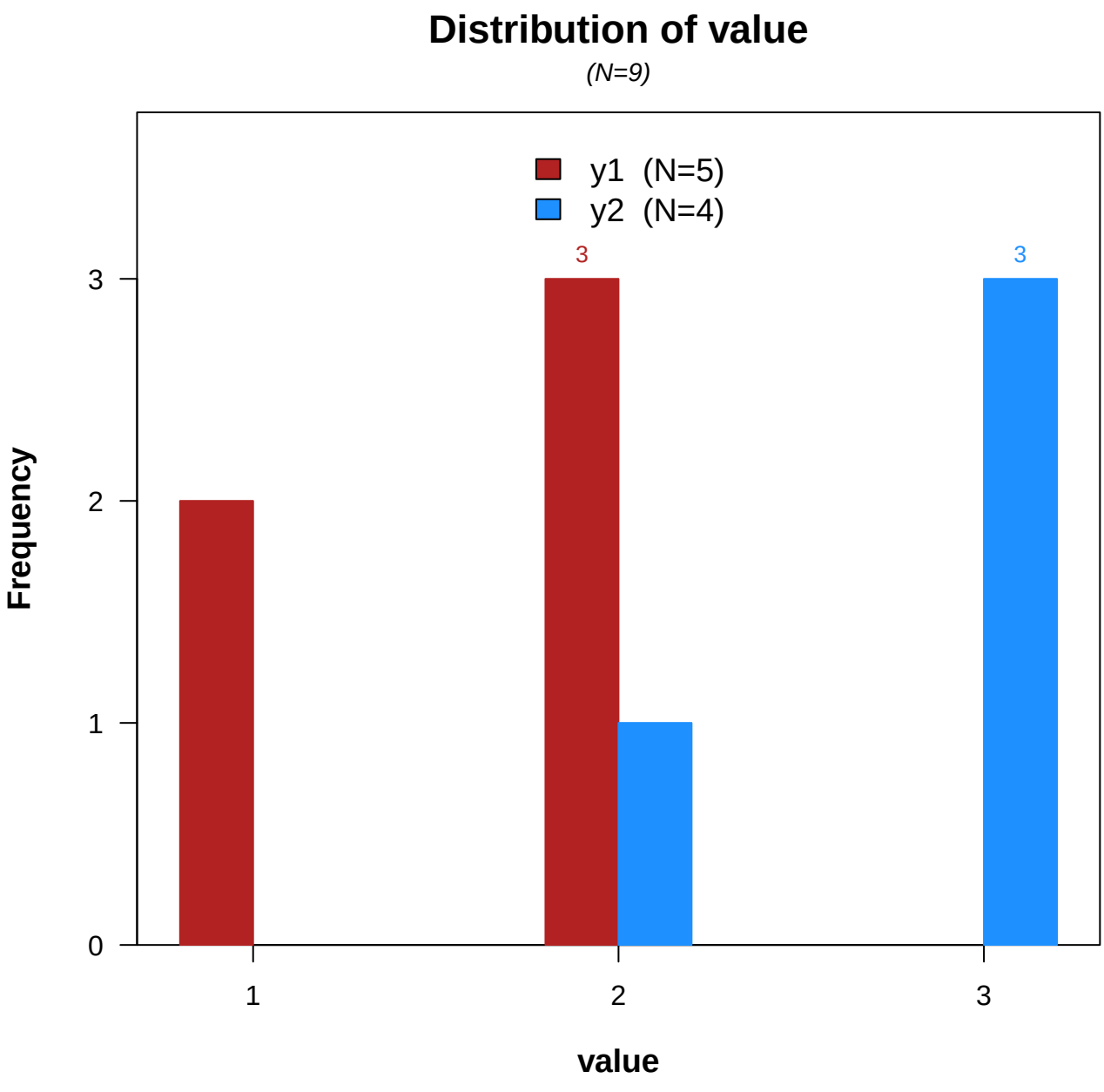

Distribution of value
(N=13)
y1 (N=7)
y2 (N=6)
3
3
Frequency
0
1
2
3
1
2
3
5
value

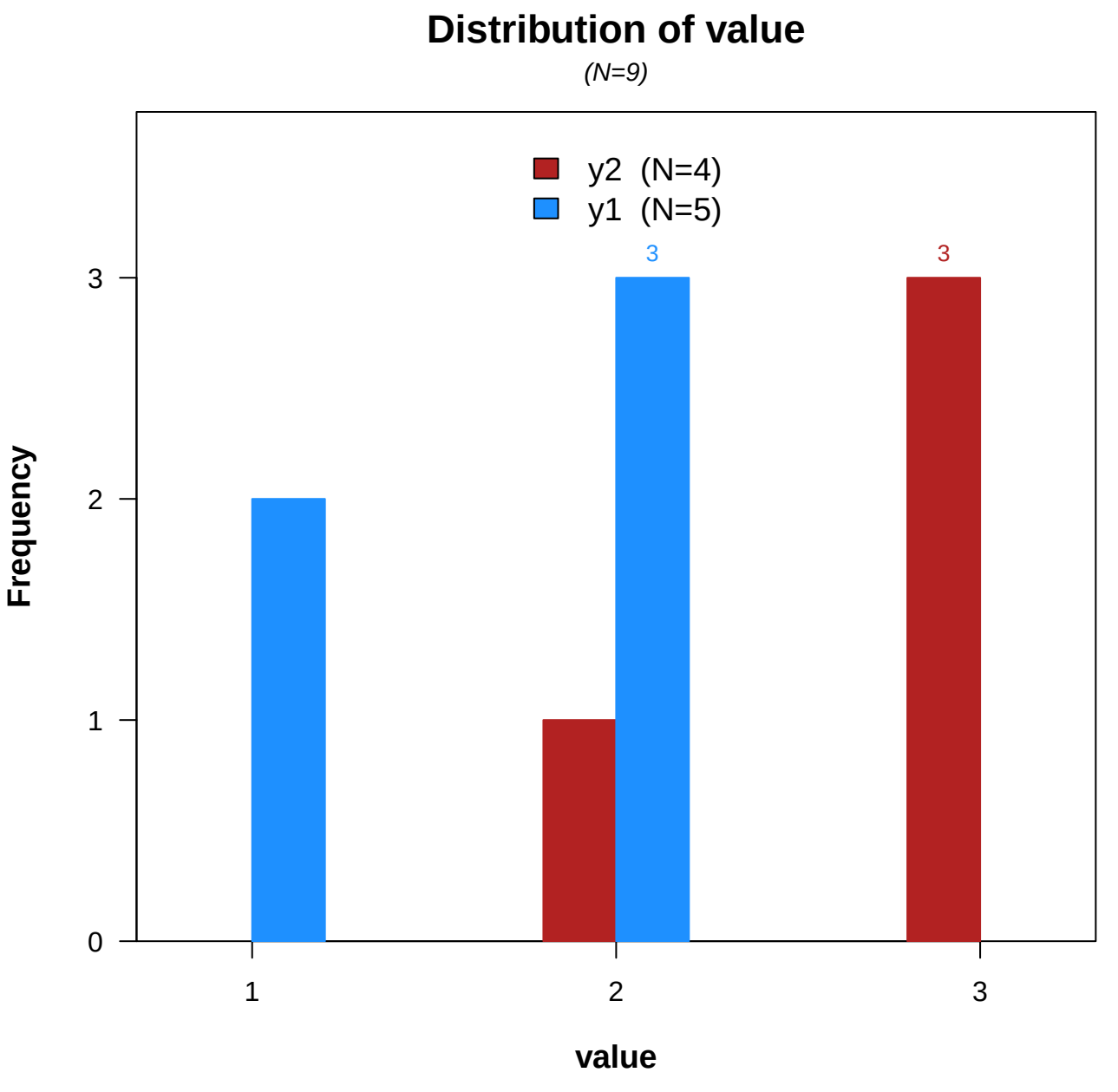

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
3
3
Frequency
0
1
2
3
1
2
3
value

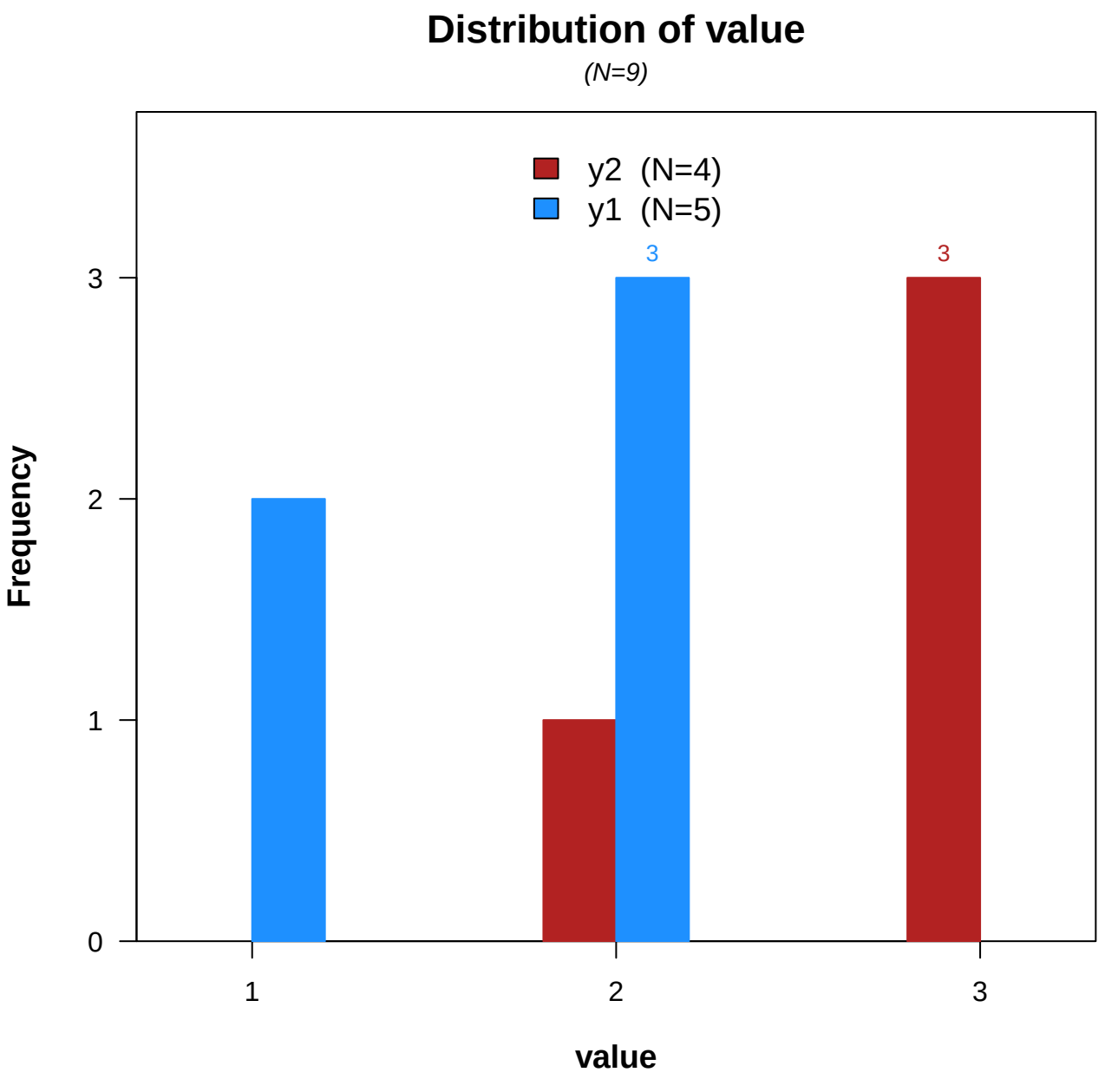

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
3
3
Frequency
0
1
2
3
1
2
3
value

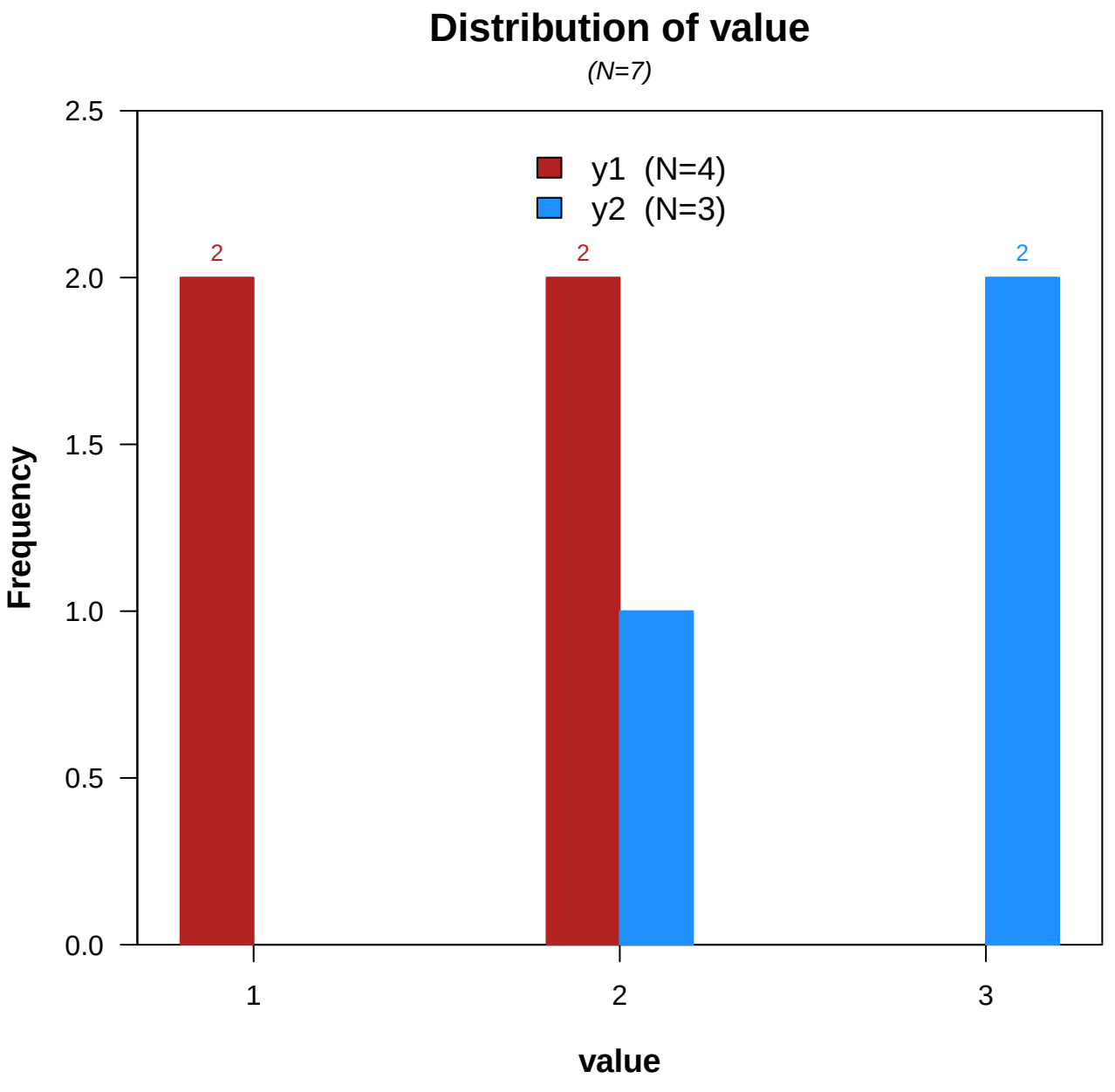

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
80%
60%
40%
20%
0%
% of Observations
60%
75%
1
2
3
value

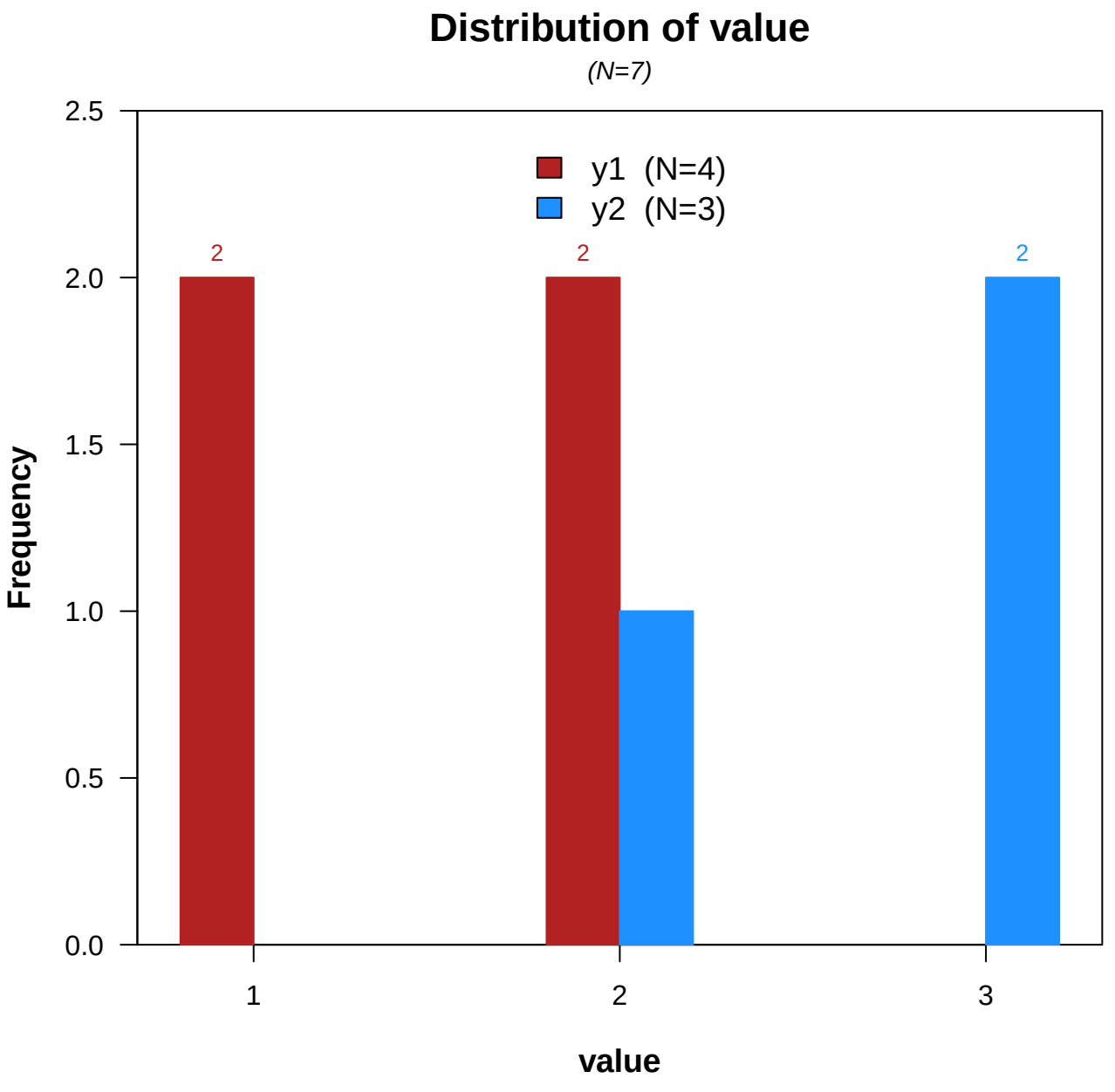

Distribution of value
(N=9)
Frequency
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1
2
3
value

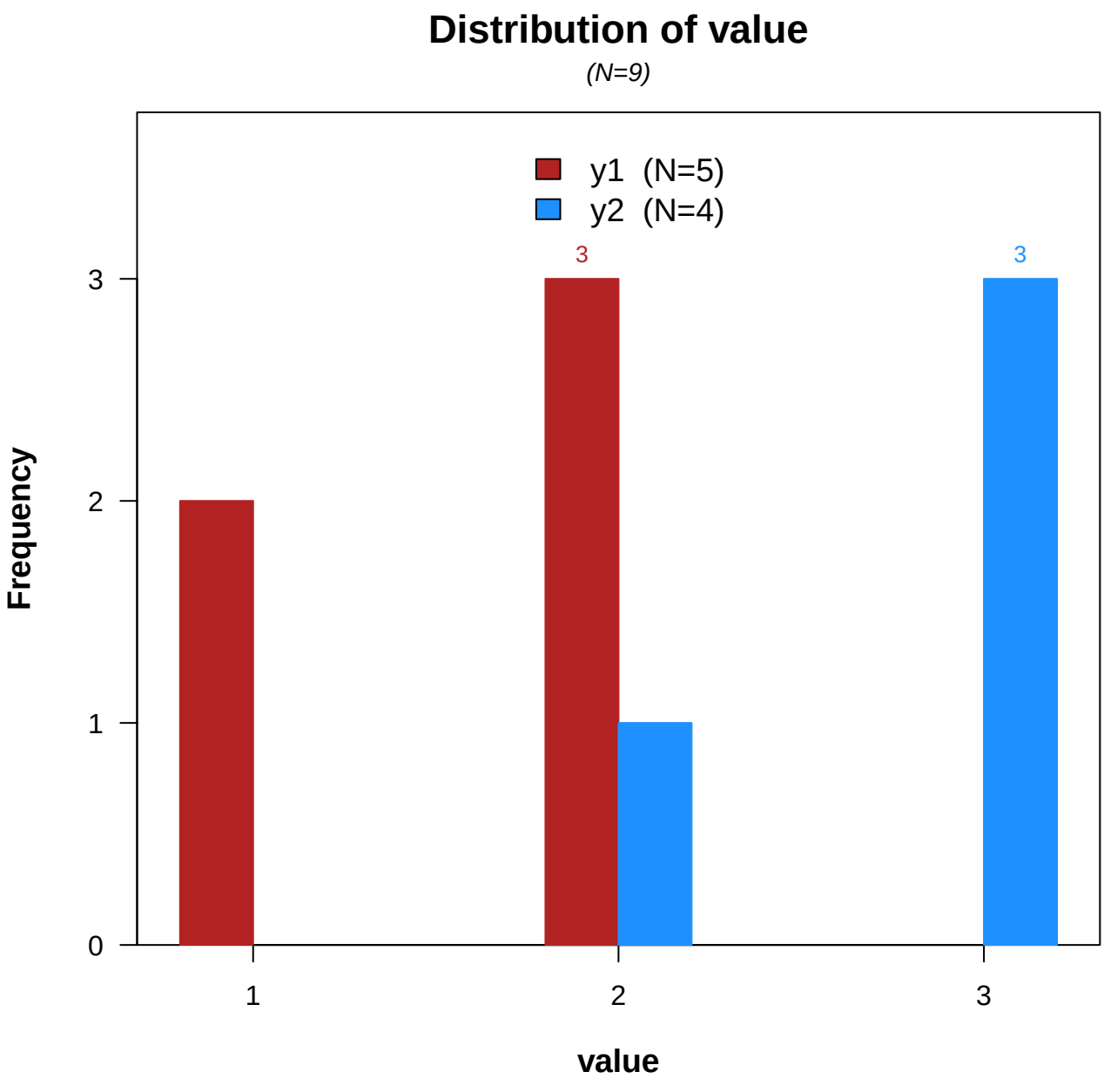

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
3
3
Frequency
0
1
2
3
1
2
3
value

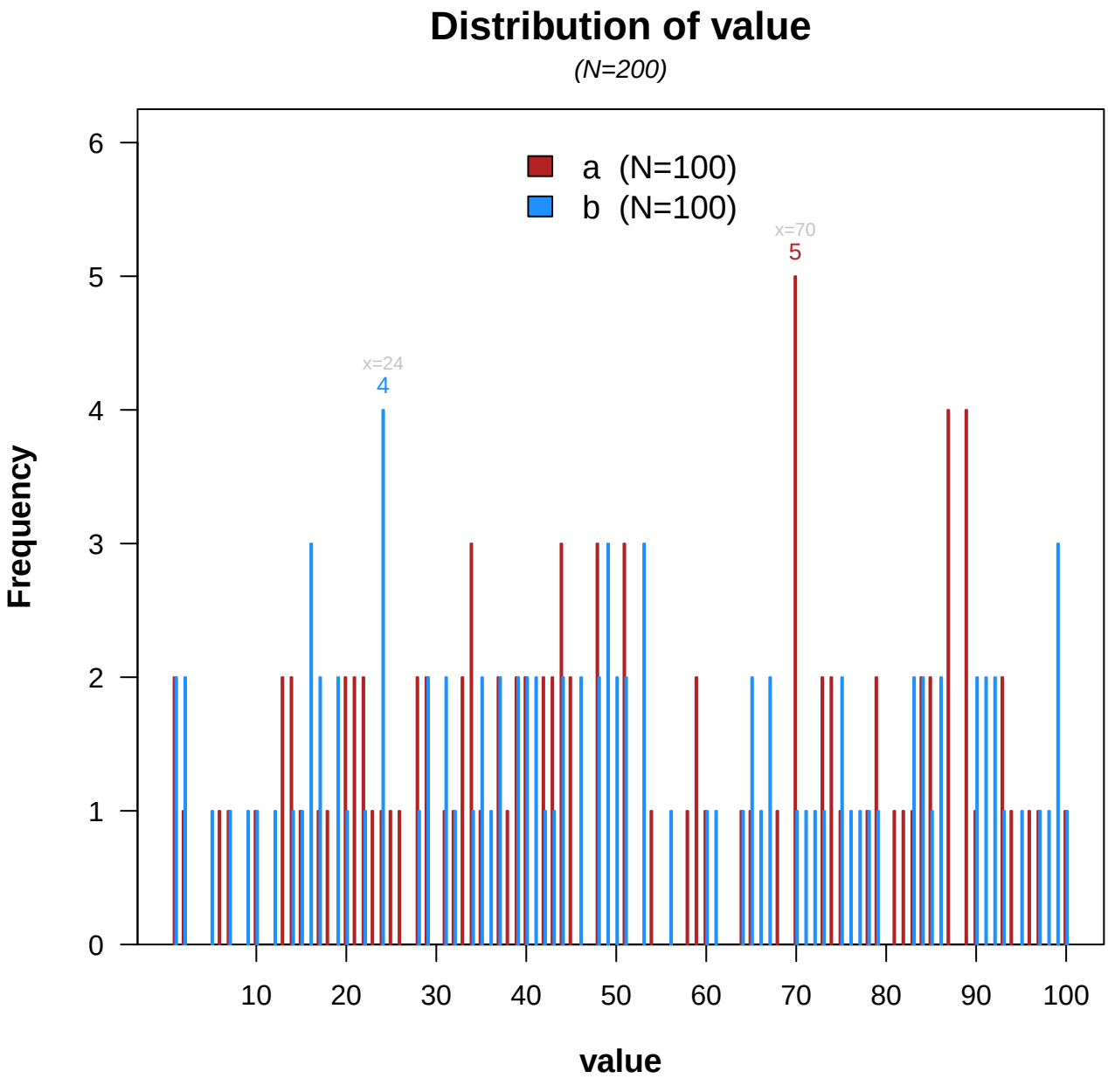

Distribution of value
(N=9)
y2 (N=4)
y1 (N=5)
3
3
Frequency
0
1
2
3
1
2
3
value

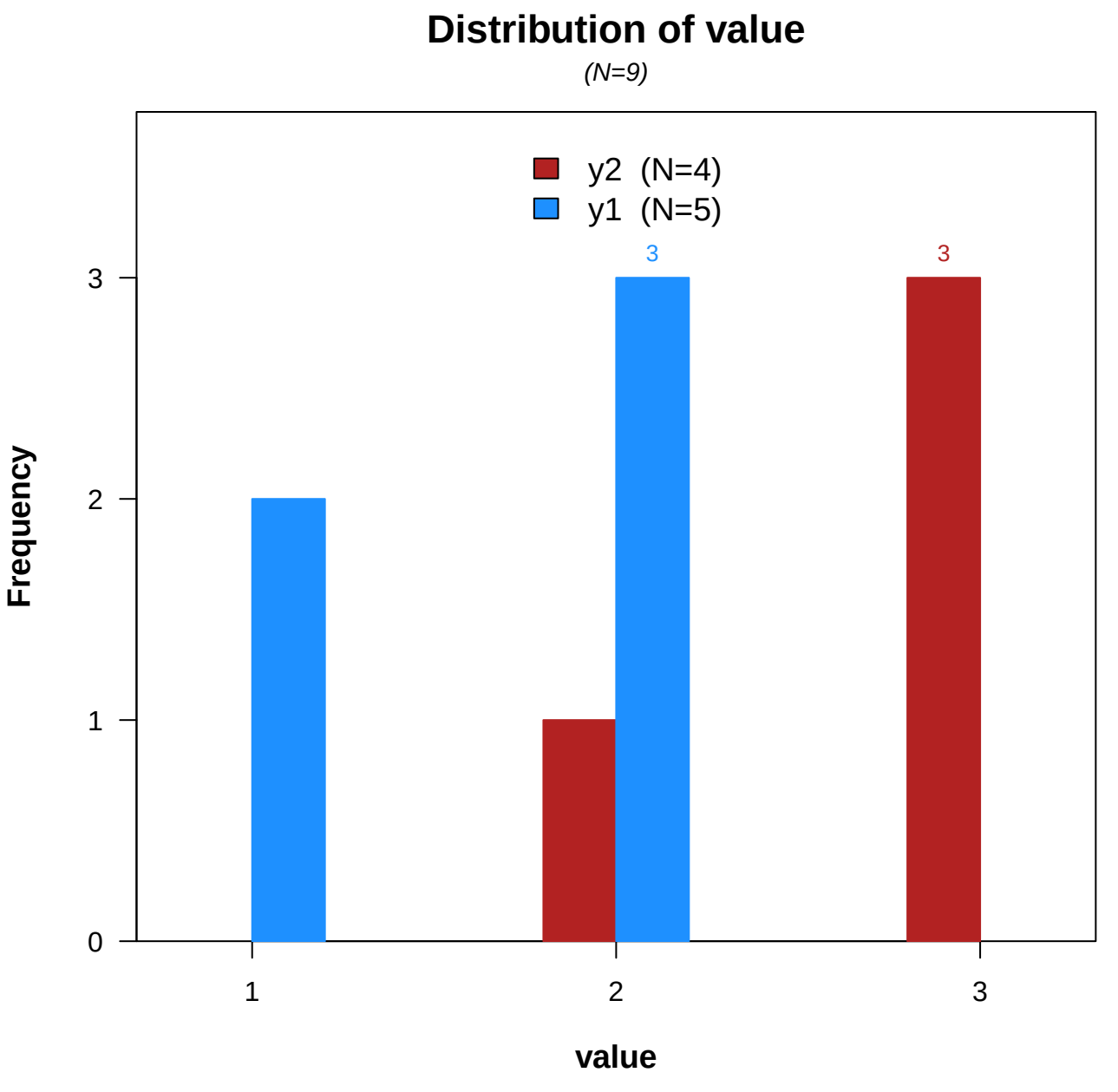

Distribution of value
(N=9)
y2 (N=4)
y1 (N=5)
3
3
Frequency
0
1
2
3
1
2
3
value

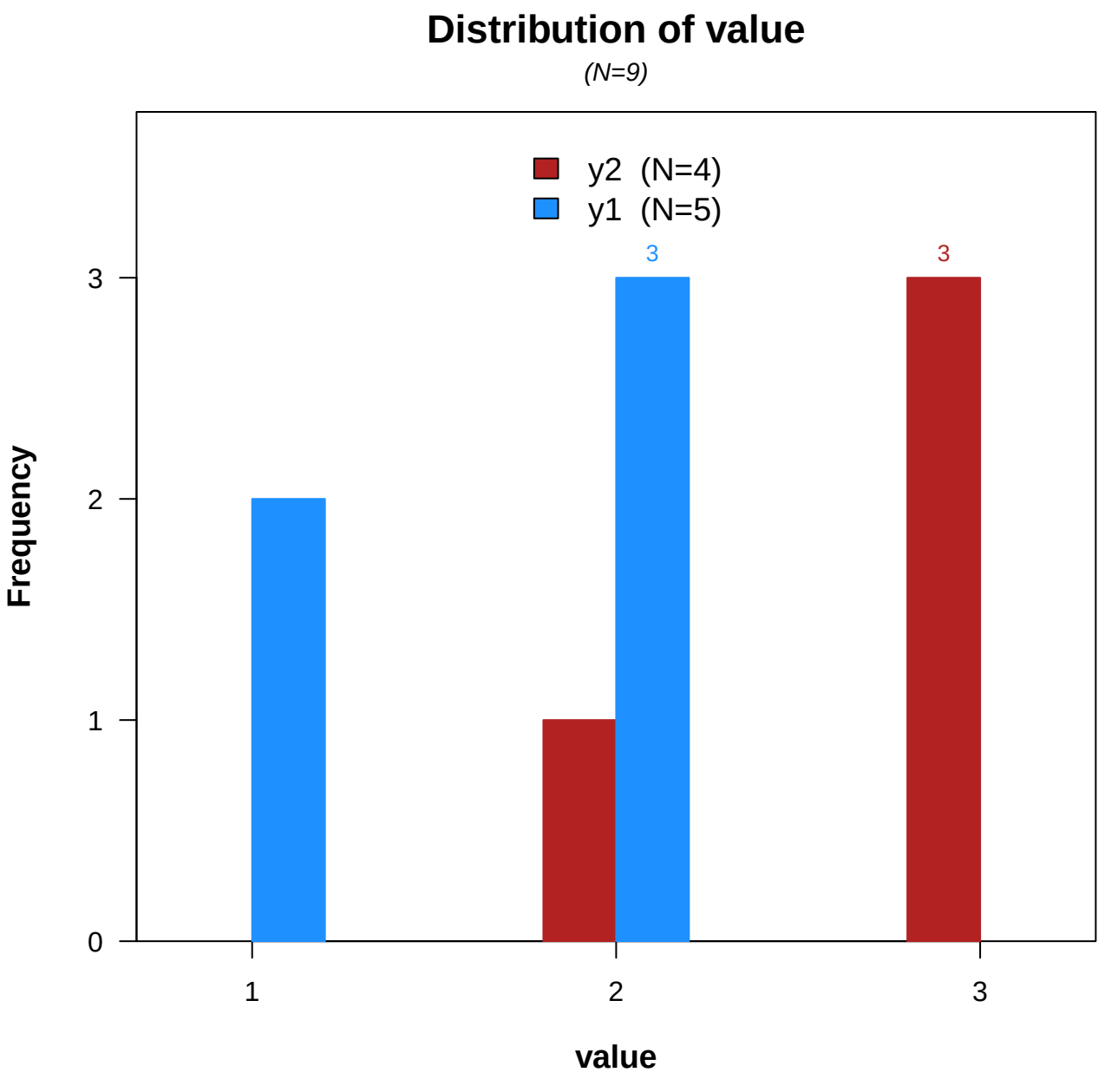

Distribution of value
(N=7)
y1 (N=4)
y2 (N=3)
2
2
2
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
3
value

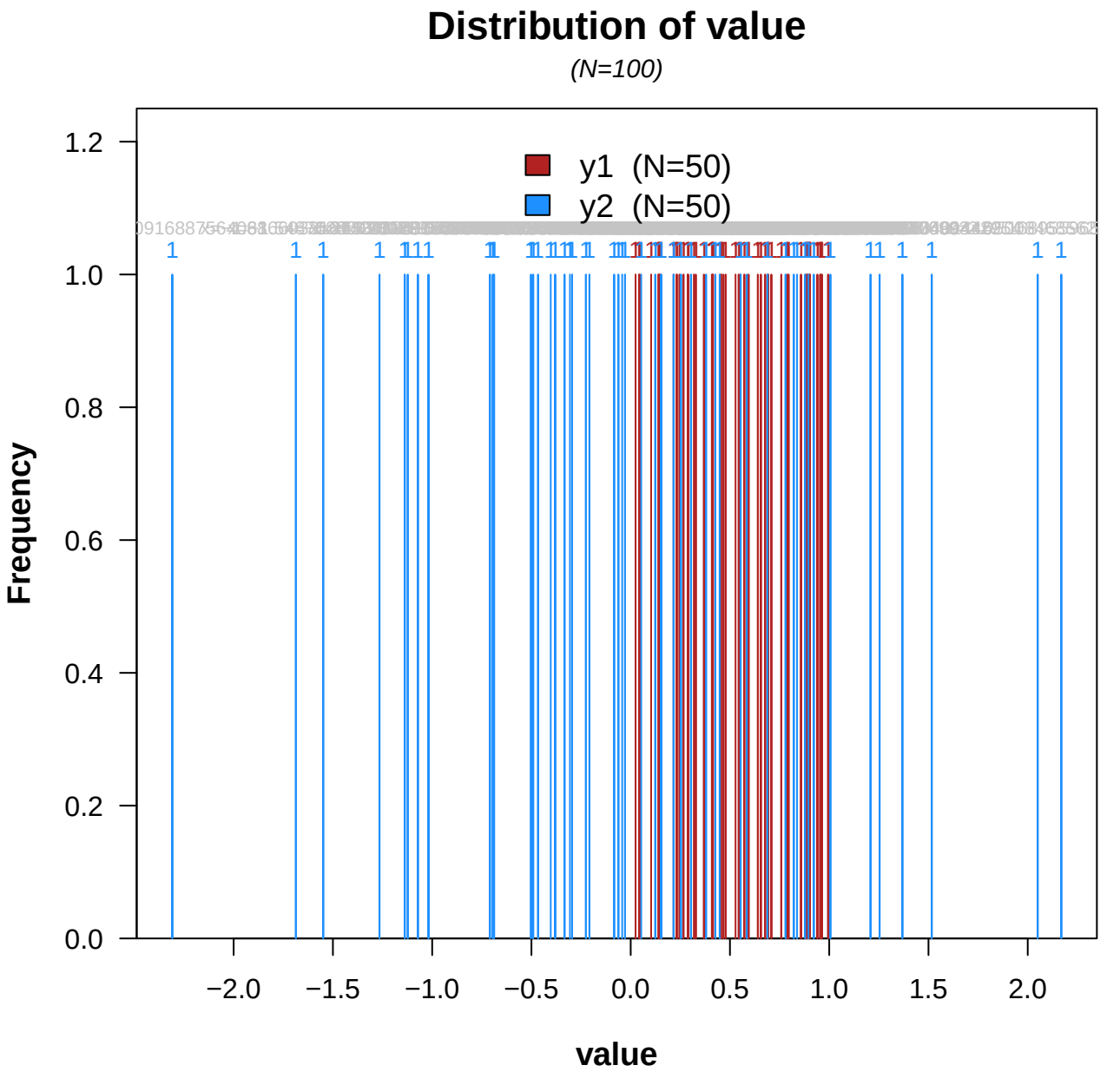

Distribution of value
(N=7)
y1 (N=4)
y2 (N=3)
2
2
2
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
1
2
3
value

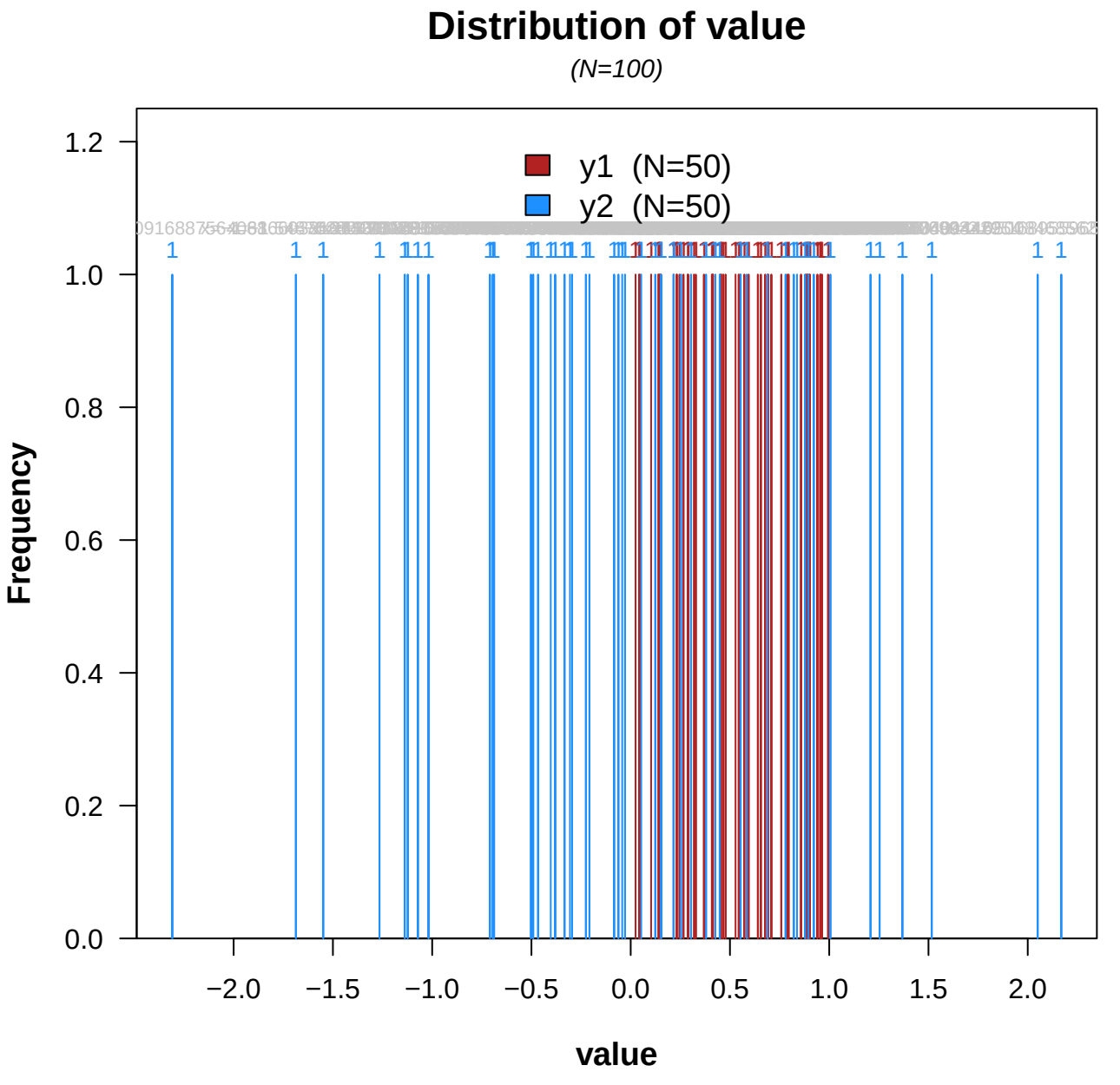

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
3
3
Frequency
0
1
2
3
1
2
3
value

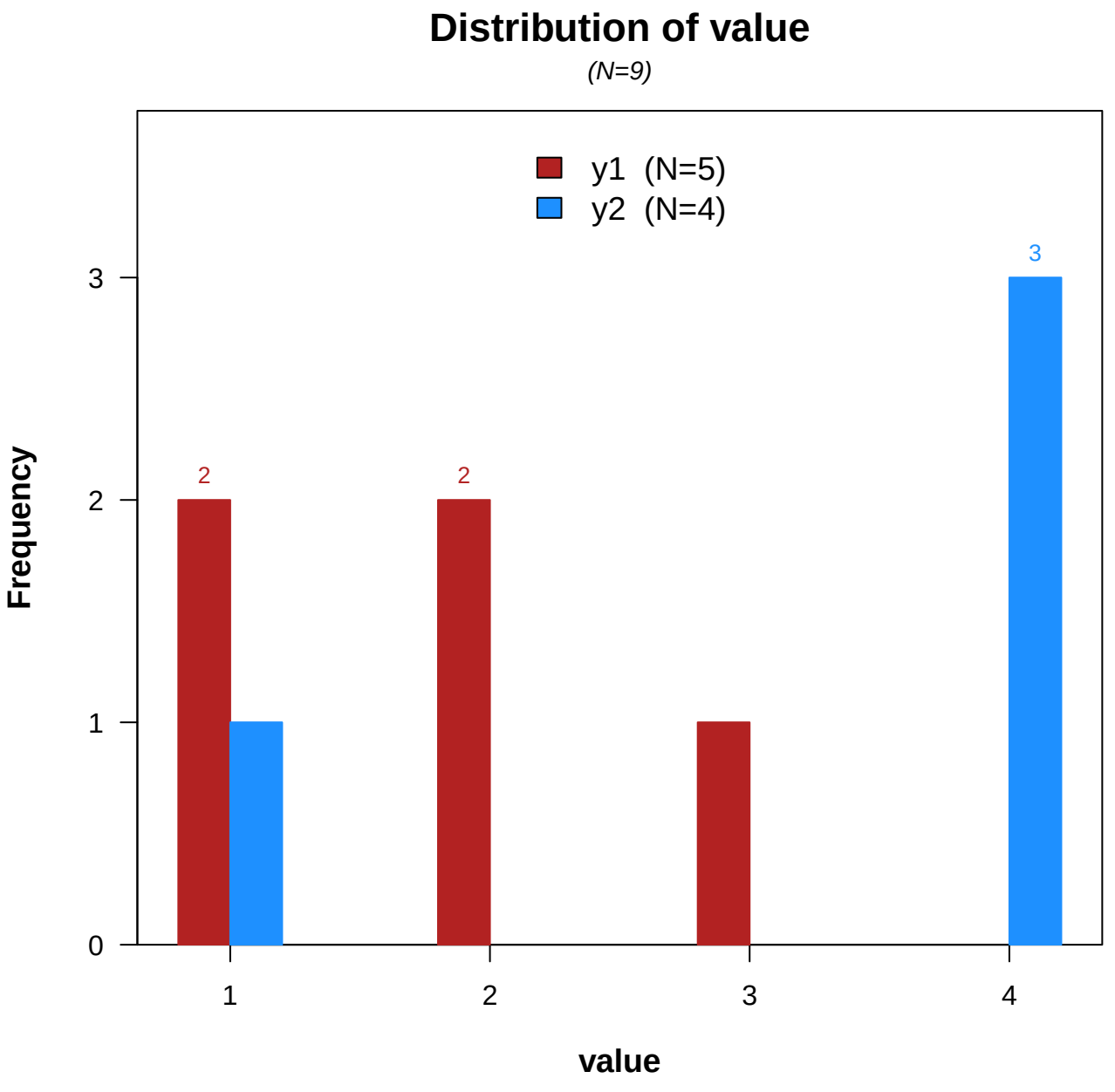

Distribution of value
(N=200)
a (N=100)
b (N=100)
x=70
5
x=24
4
Frequency
value
0
1
2
3
4
5
6
10
20
30
40
50
60
70
80
90
100

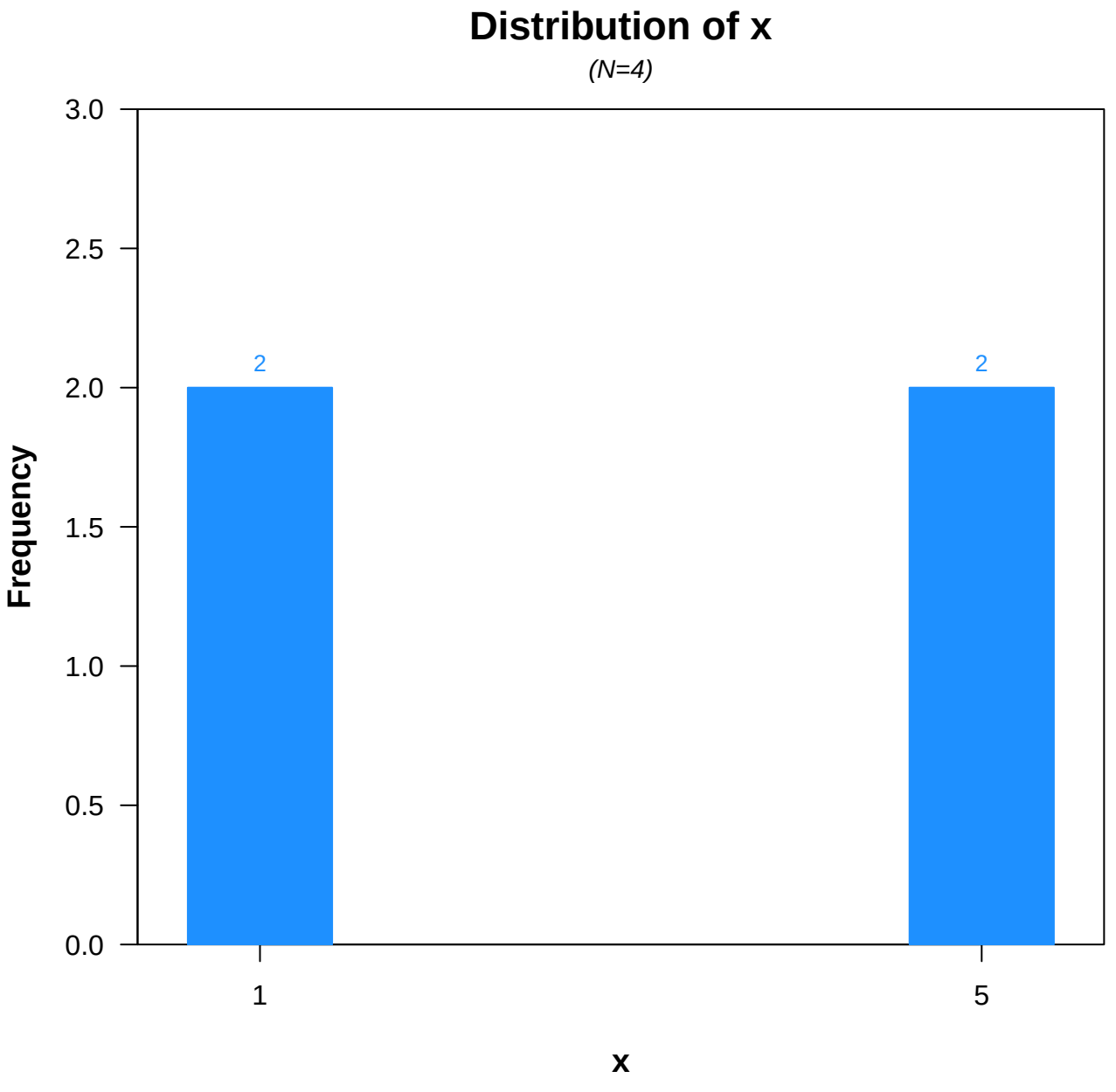

Distribution of value
(N=9)
y2 (N=4)
y1 (N=5)
3
3
Frequency
0
1
2
3
1
2
3
value

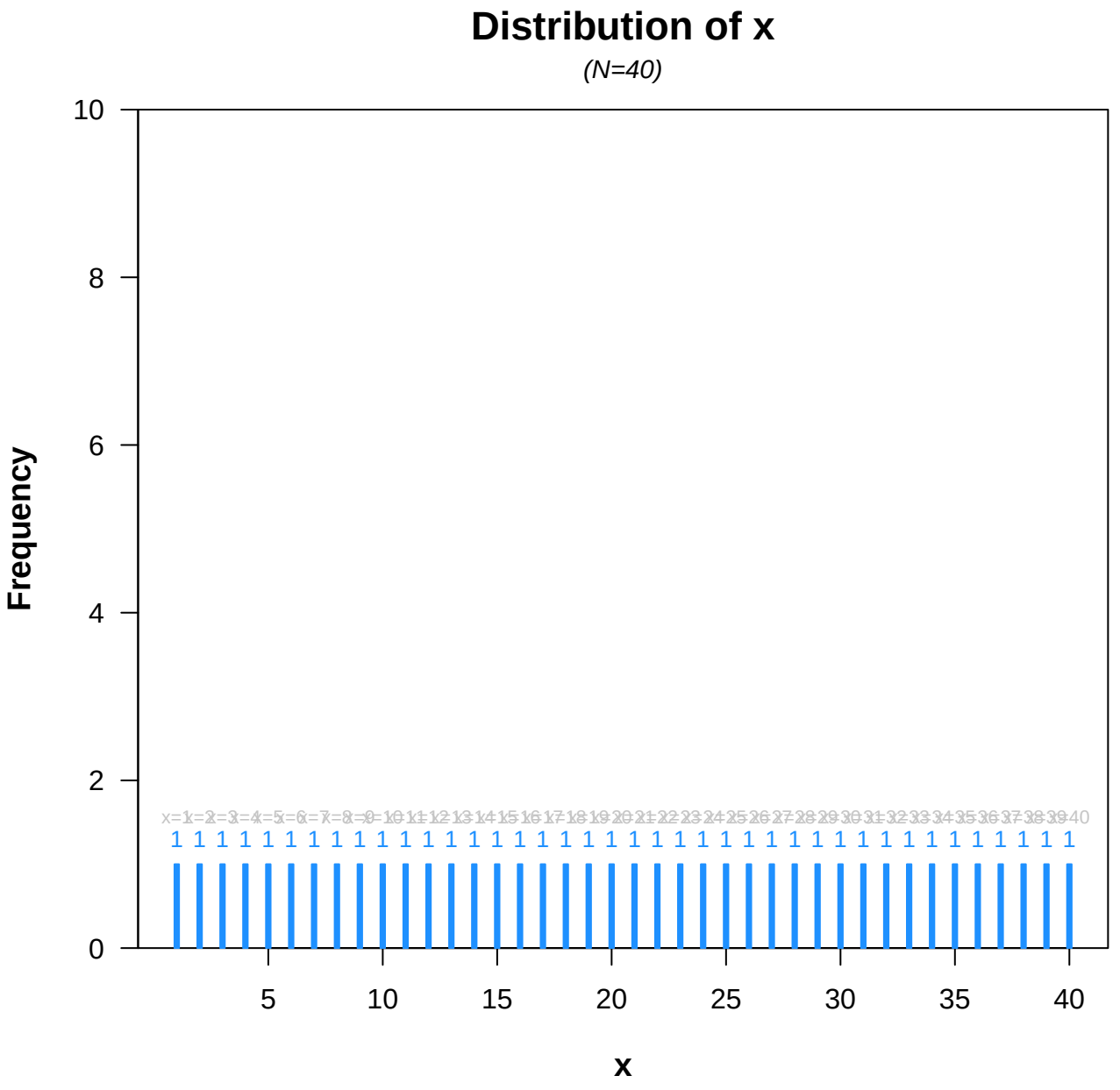

Distribution of value
(N=9)
y2 (N=4)
y1 (N=5)
3
3
Frequency
0
1
2
3
1
2
3
value

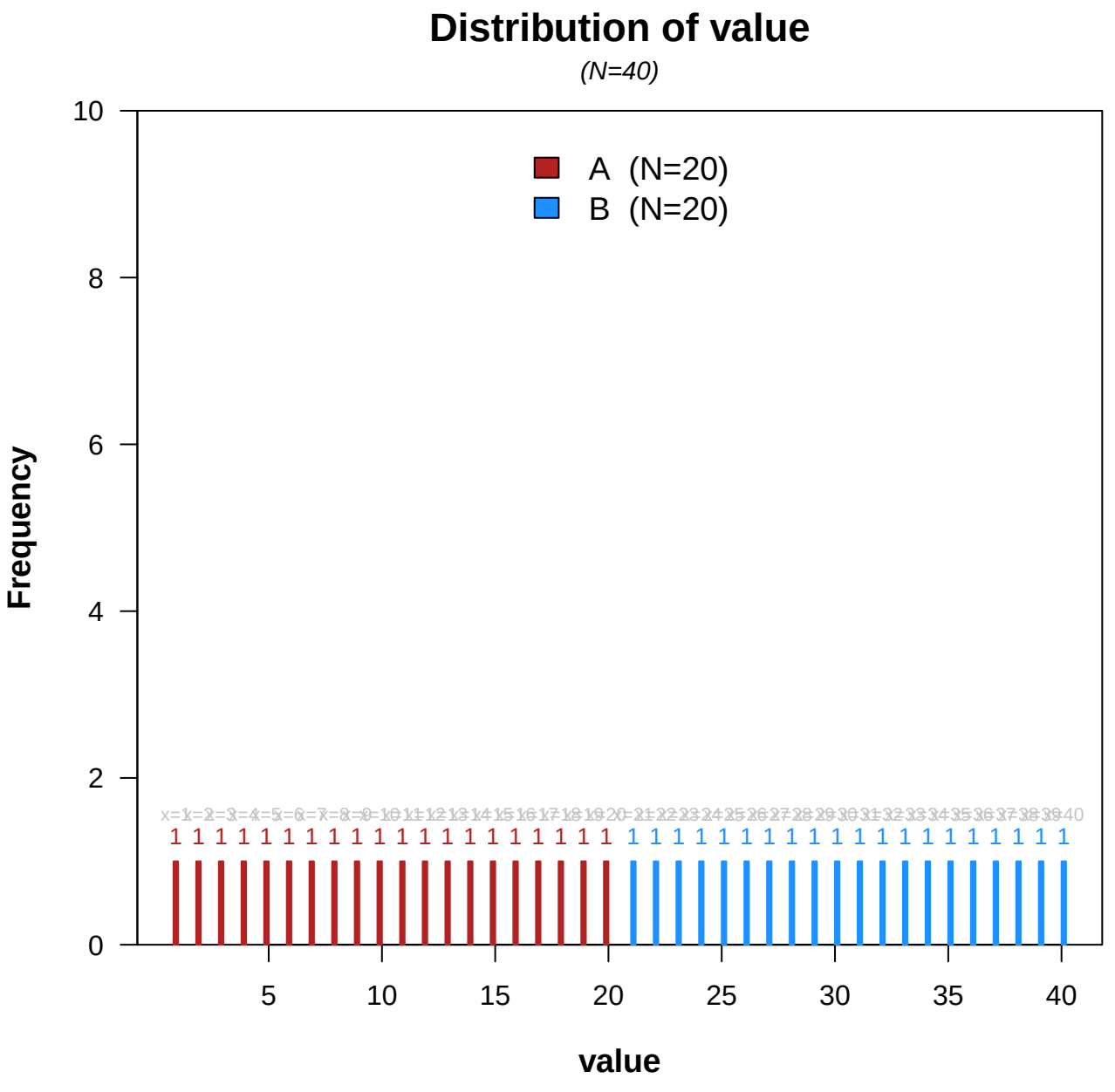

Distribution of value
(N=100)
y1 (N=50)
y2 (N=50)
Frequency
value
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

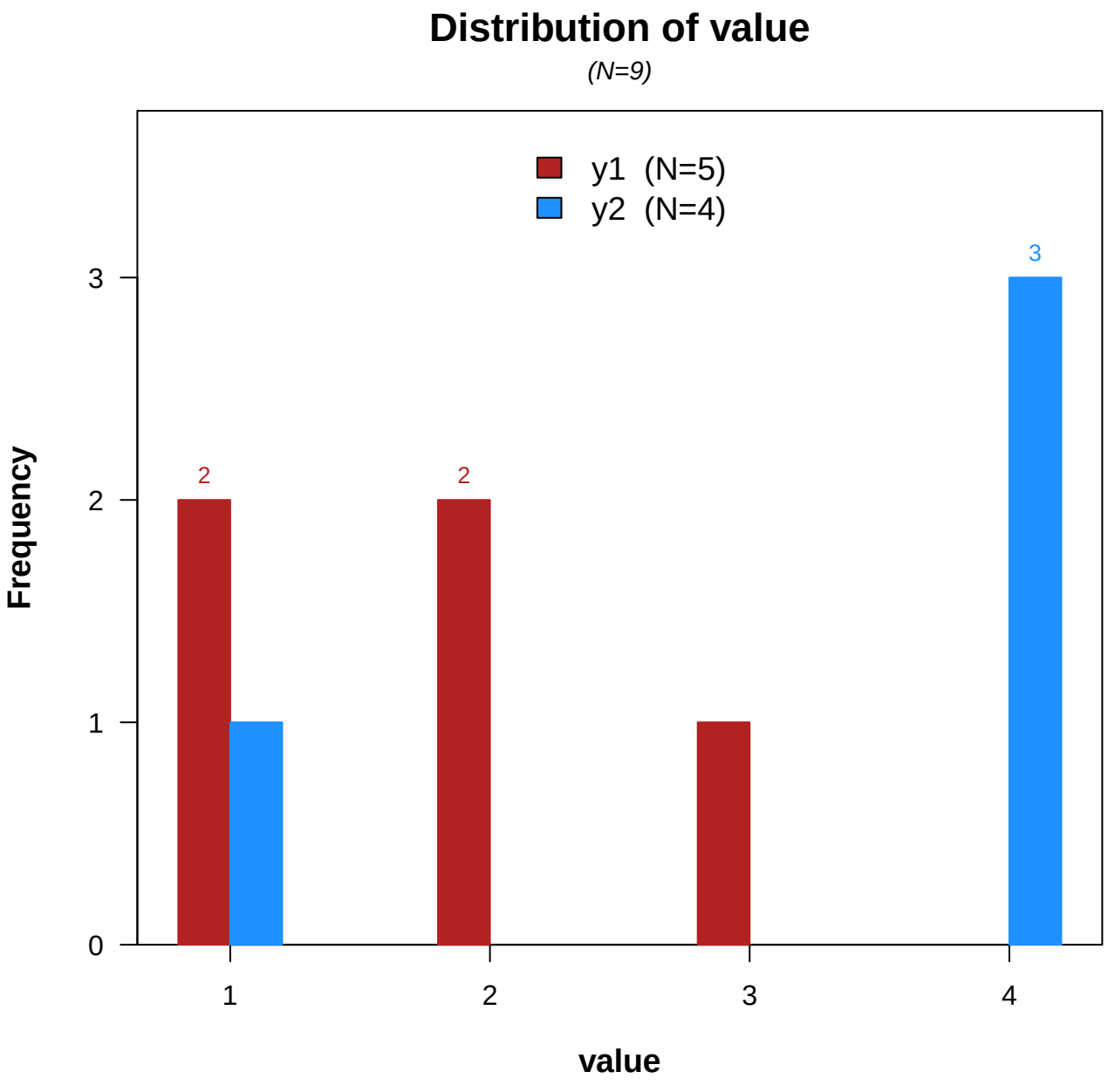

Distribution of value
(N=100)
y1 (N=50)
y2 (N=50)
Frequency
value
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0

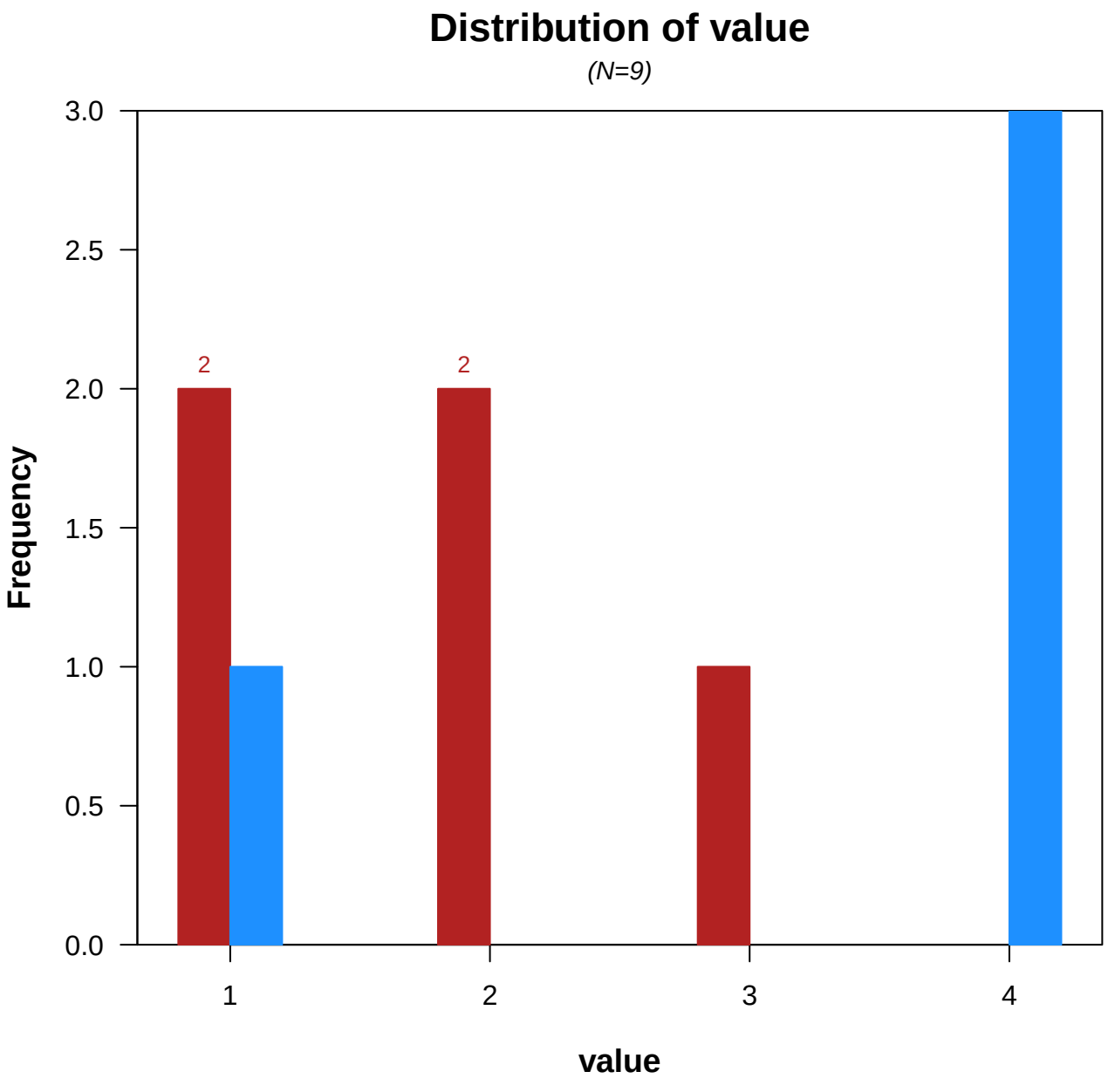

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
Frequency
value
0
1
2
3
1
2
3
4
2
2
3

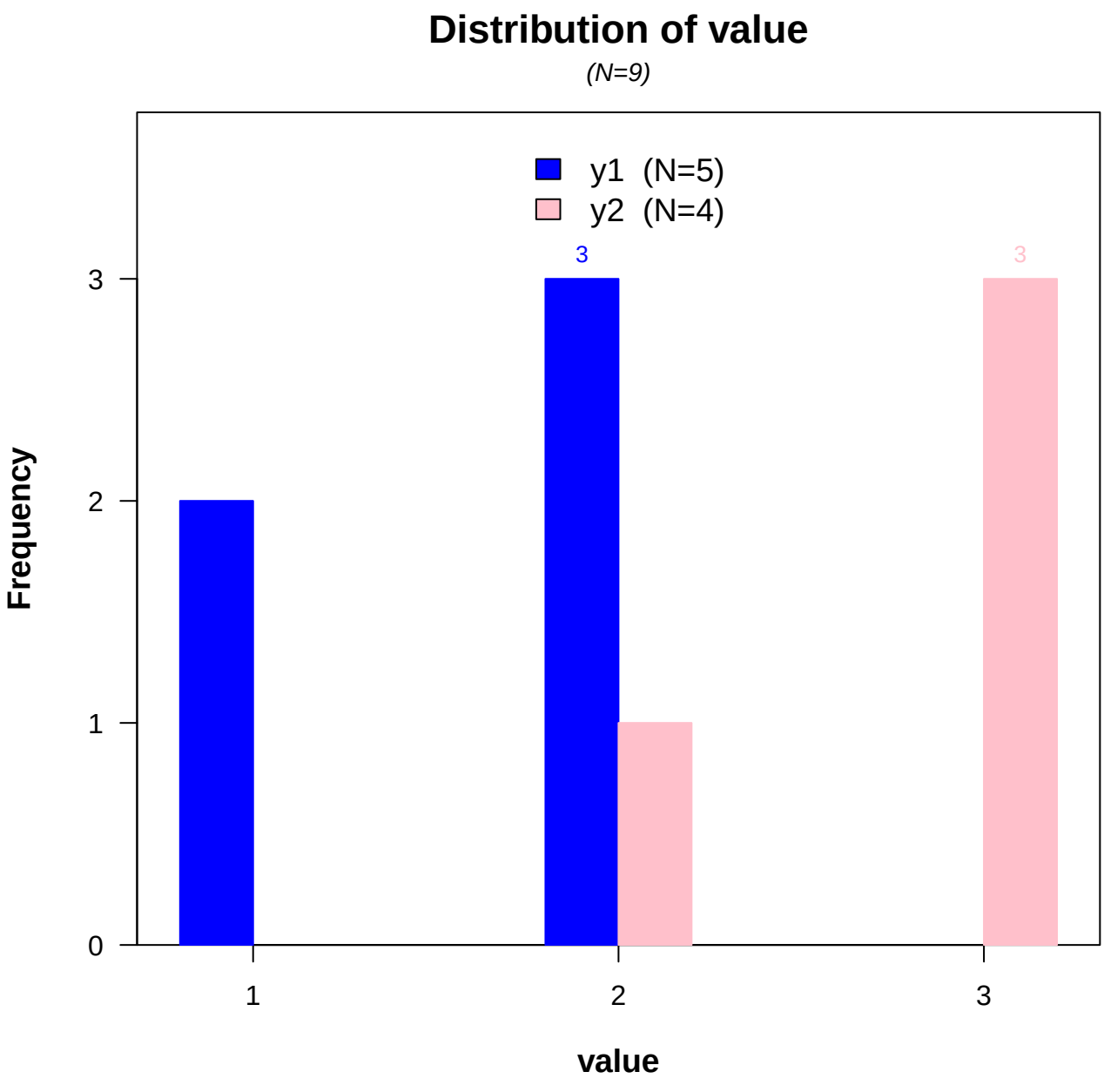

Distribution of x
(N=4)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Frequency
2
2
1
5
x

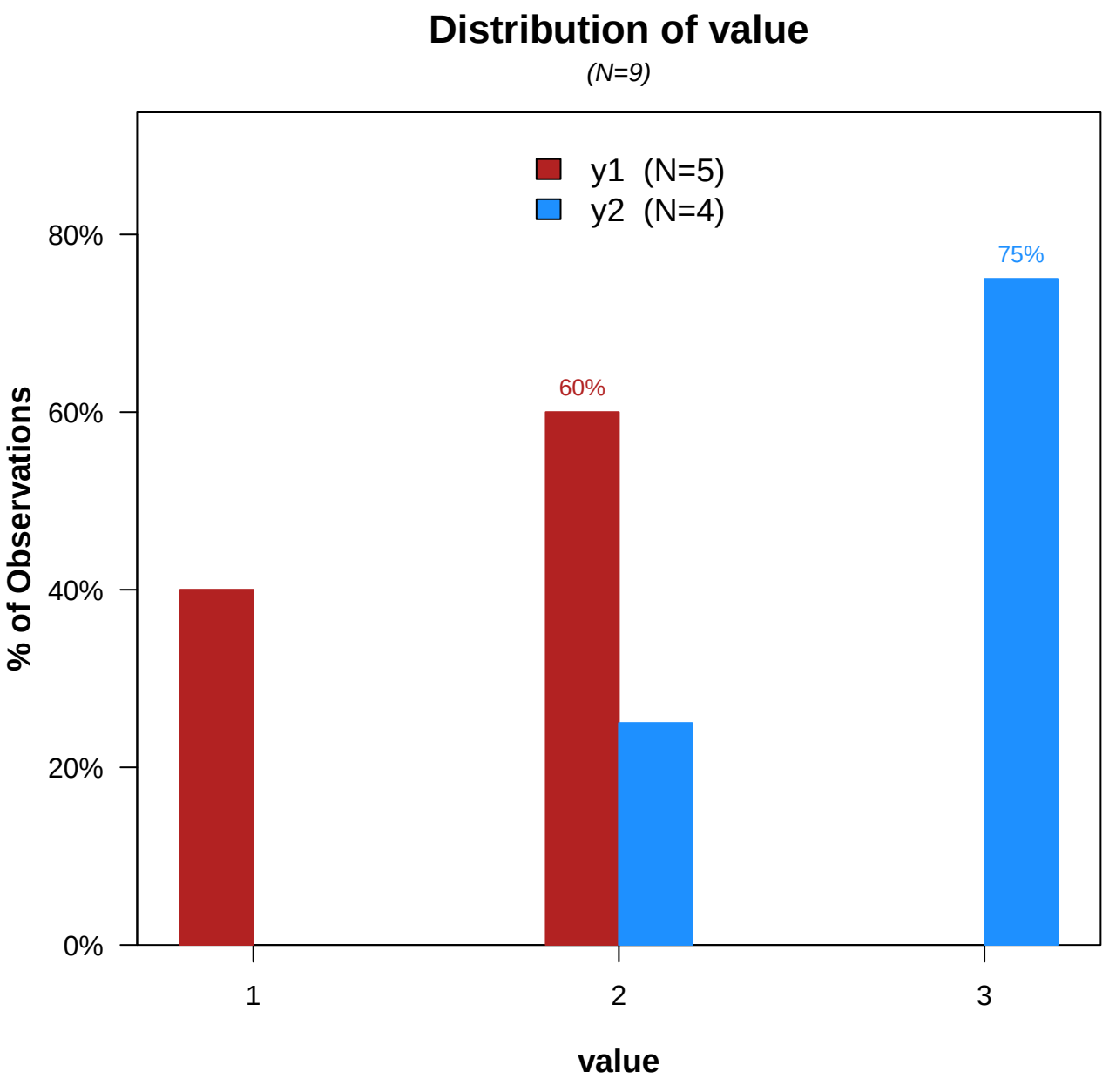

Distribution of x
(N=40)
Frequency
x
0
2
4
6
8
10
5
10
15
20
25
30
35
40

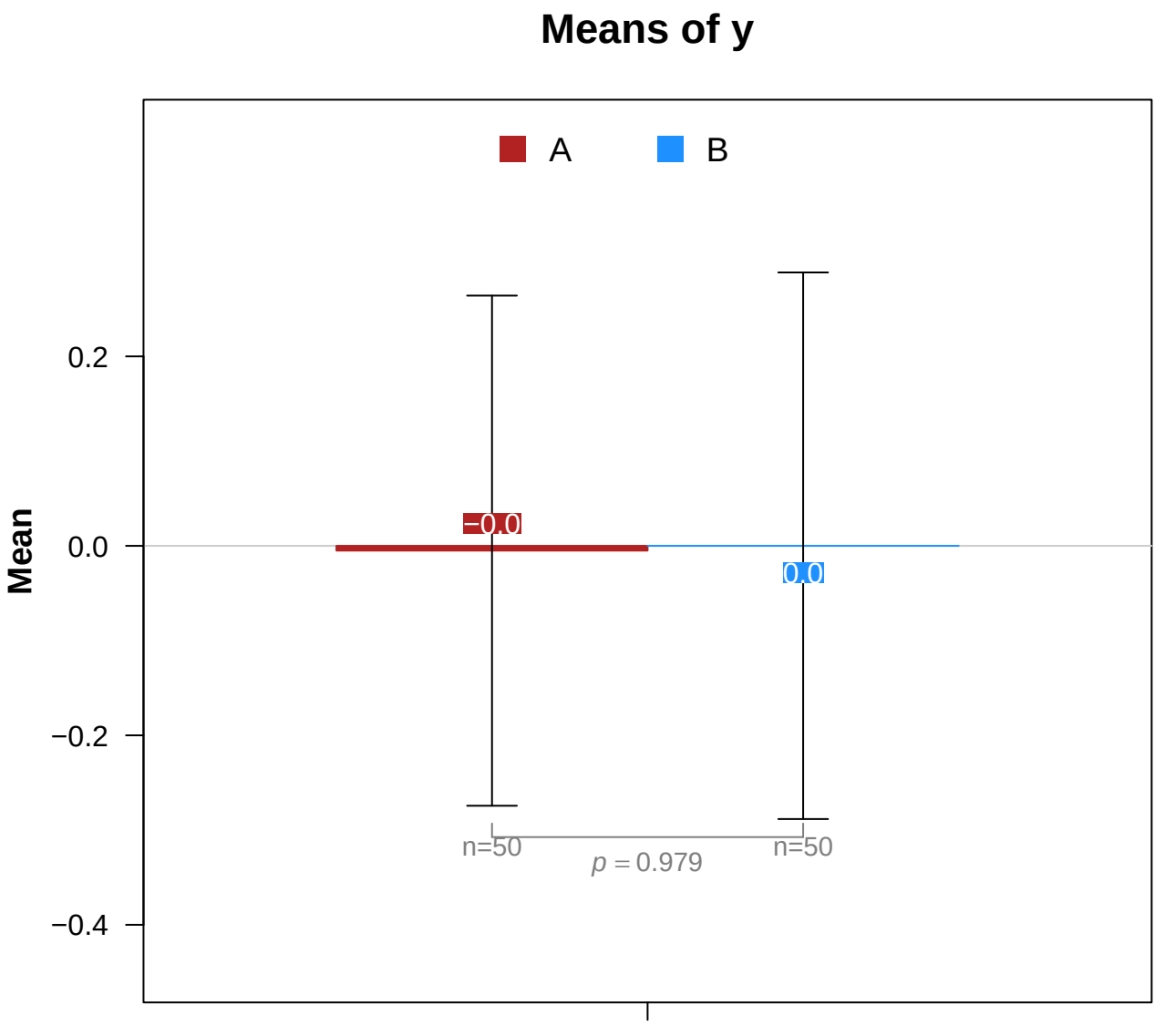

Distribution of value
(N=40)
A (N=20)
B (N=20)
Frequency
value
0
2
4
6
8
10
5
10
15
20
25
30
35
40

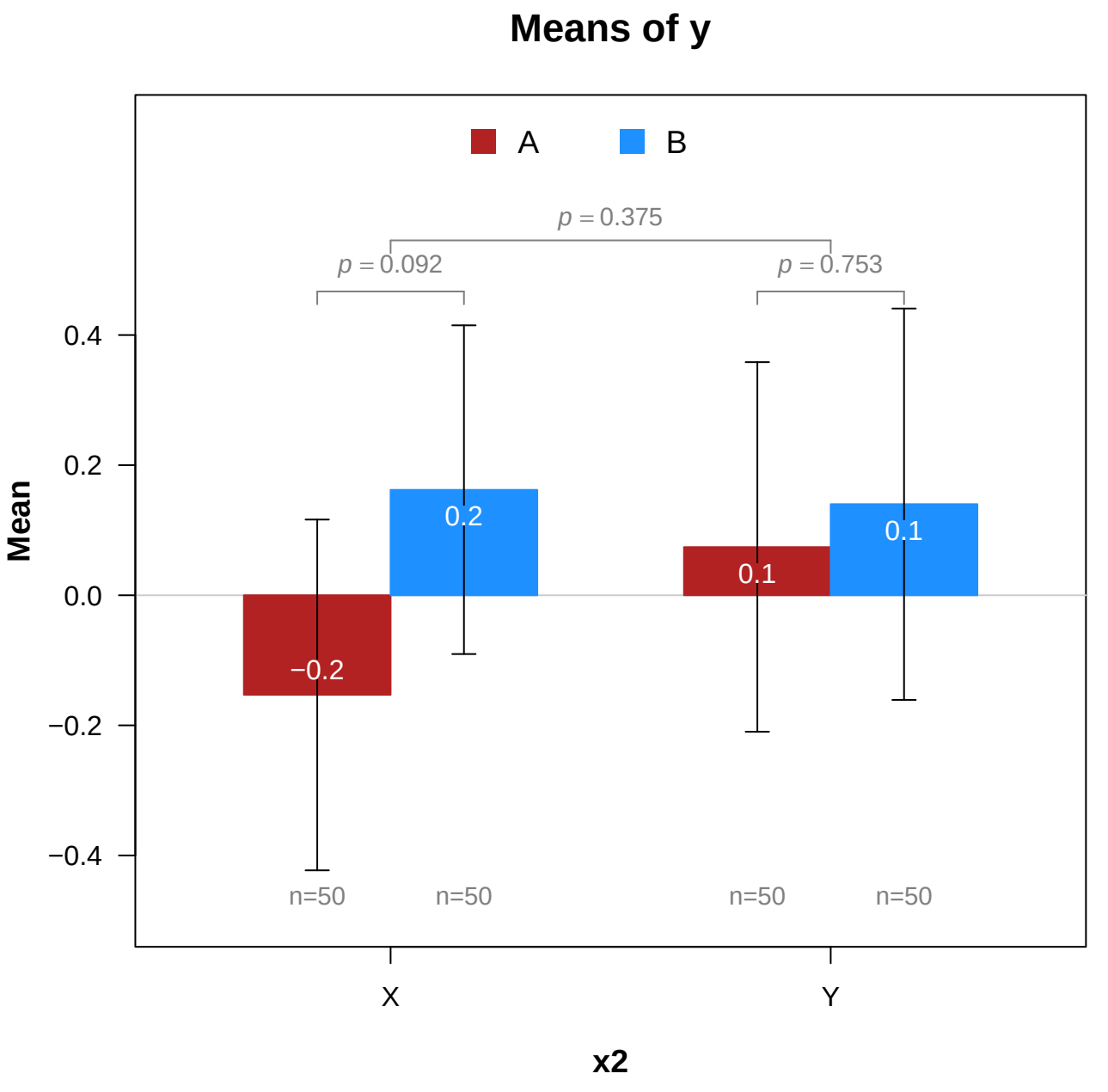

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
Frequency
value
0
1
2
3
1
2
3
4
2
2
3

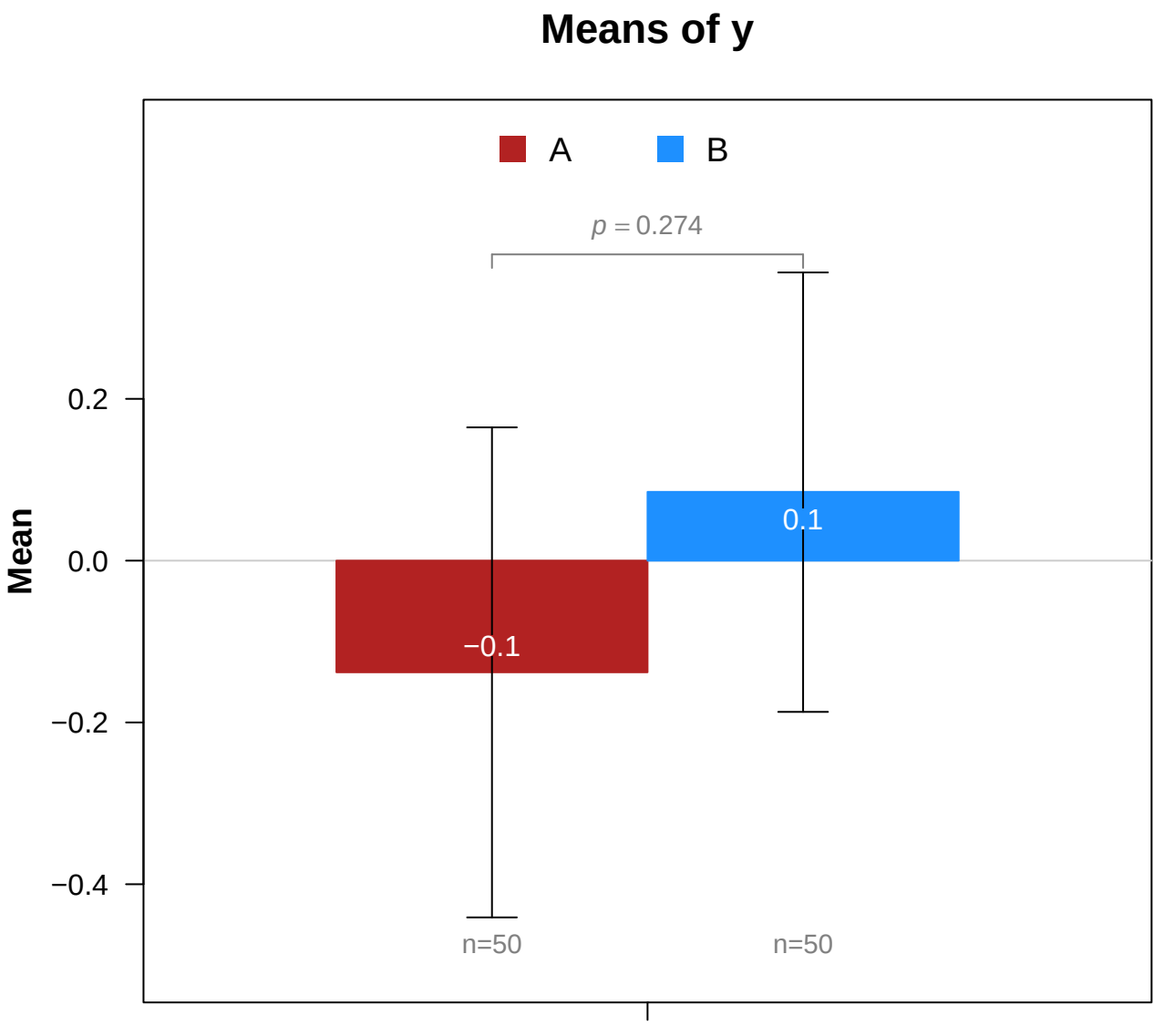

Distribution of value
(N=9)
Frequency
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2
2
1
2
3
4
value

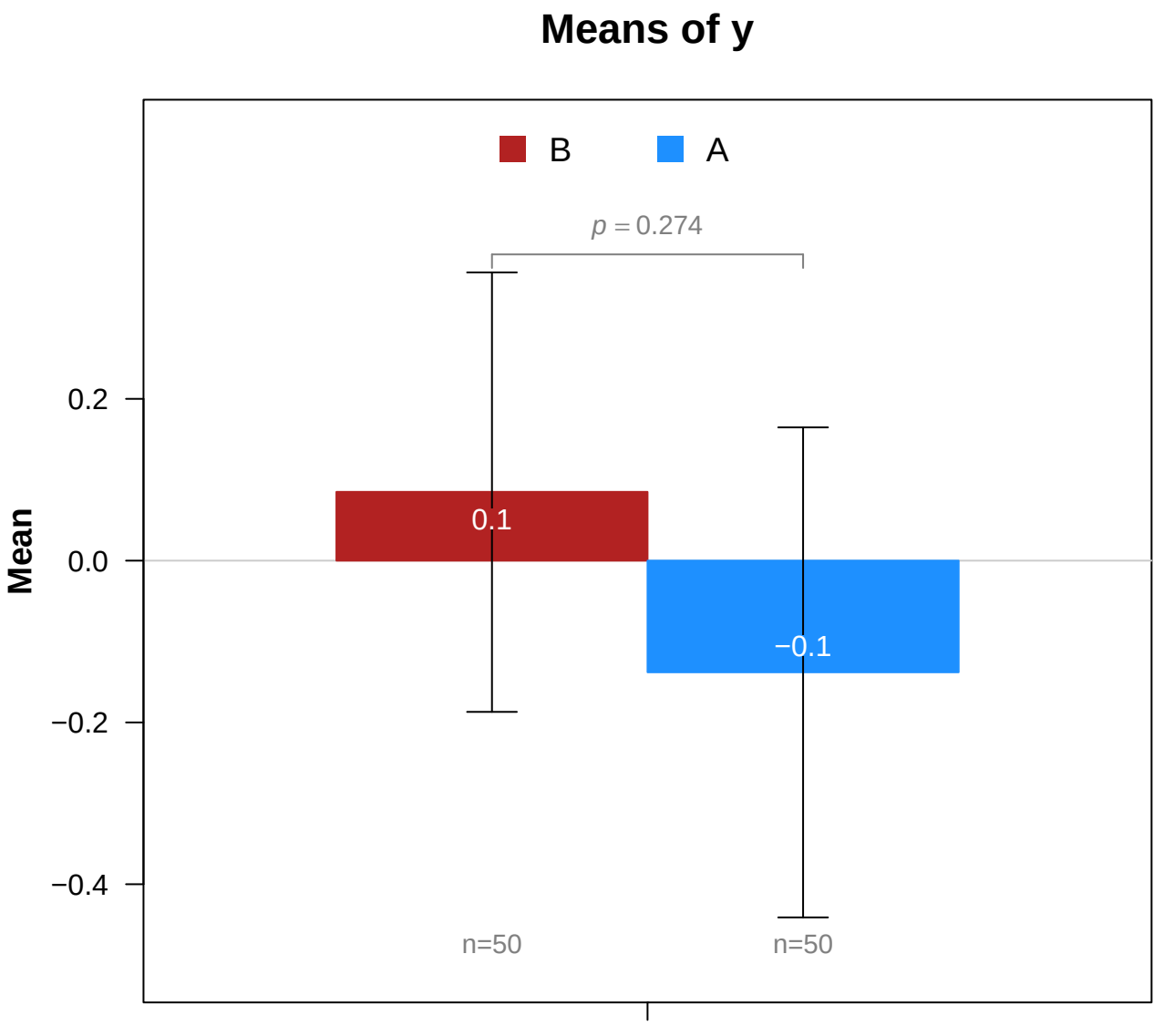

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
3
3
Frequency
0
1
2
3
1
2
3
value

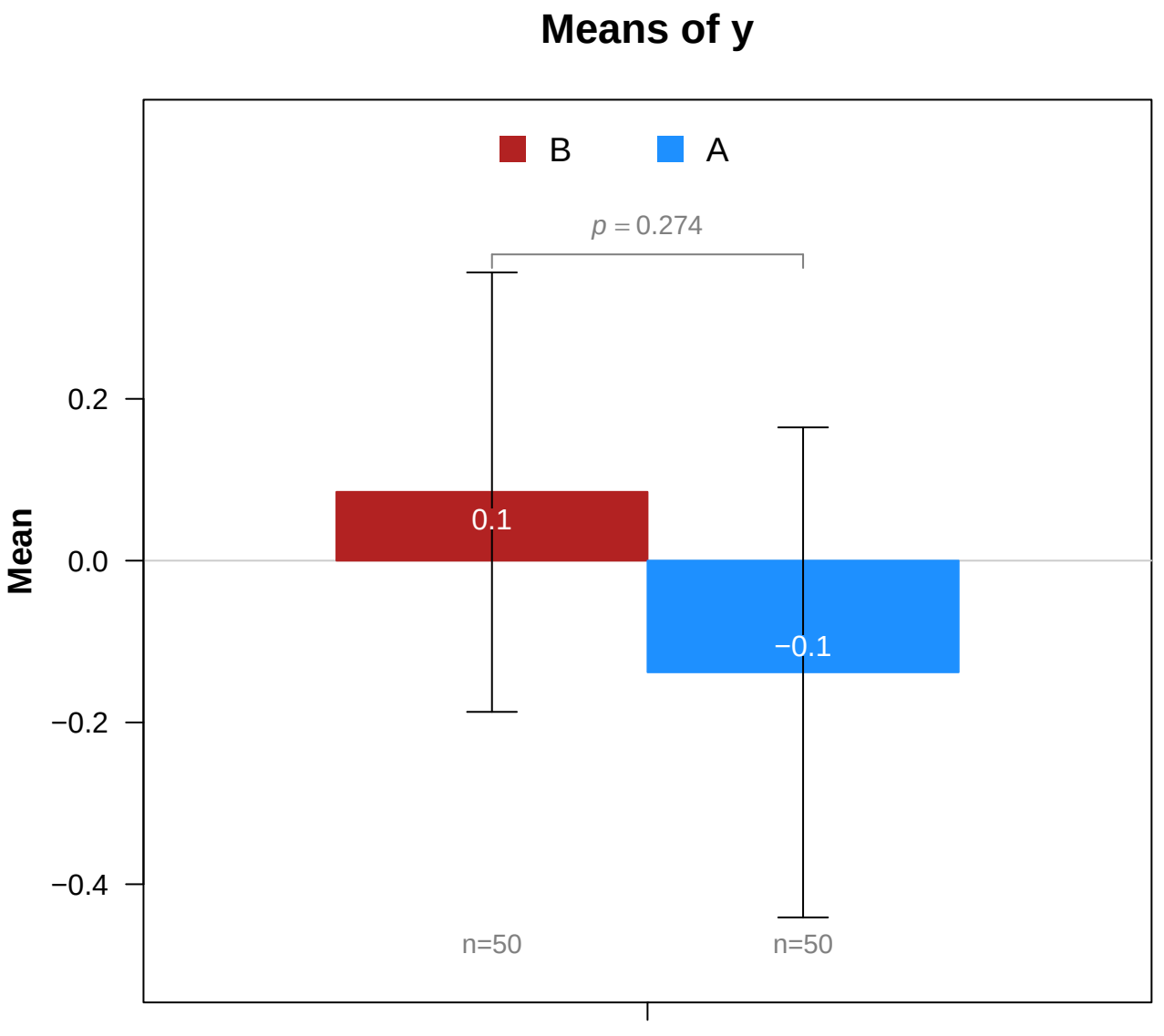

Distribution of value
(N=9)
y1 (N=5)
y2 (N=4)
80%
60%
40%
20%
0%
% of Observations
60%
75%
1
2
3
value

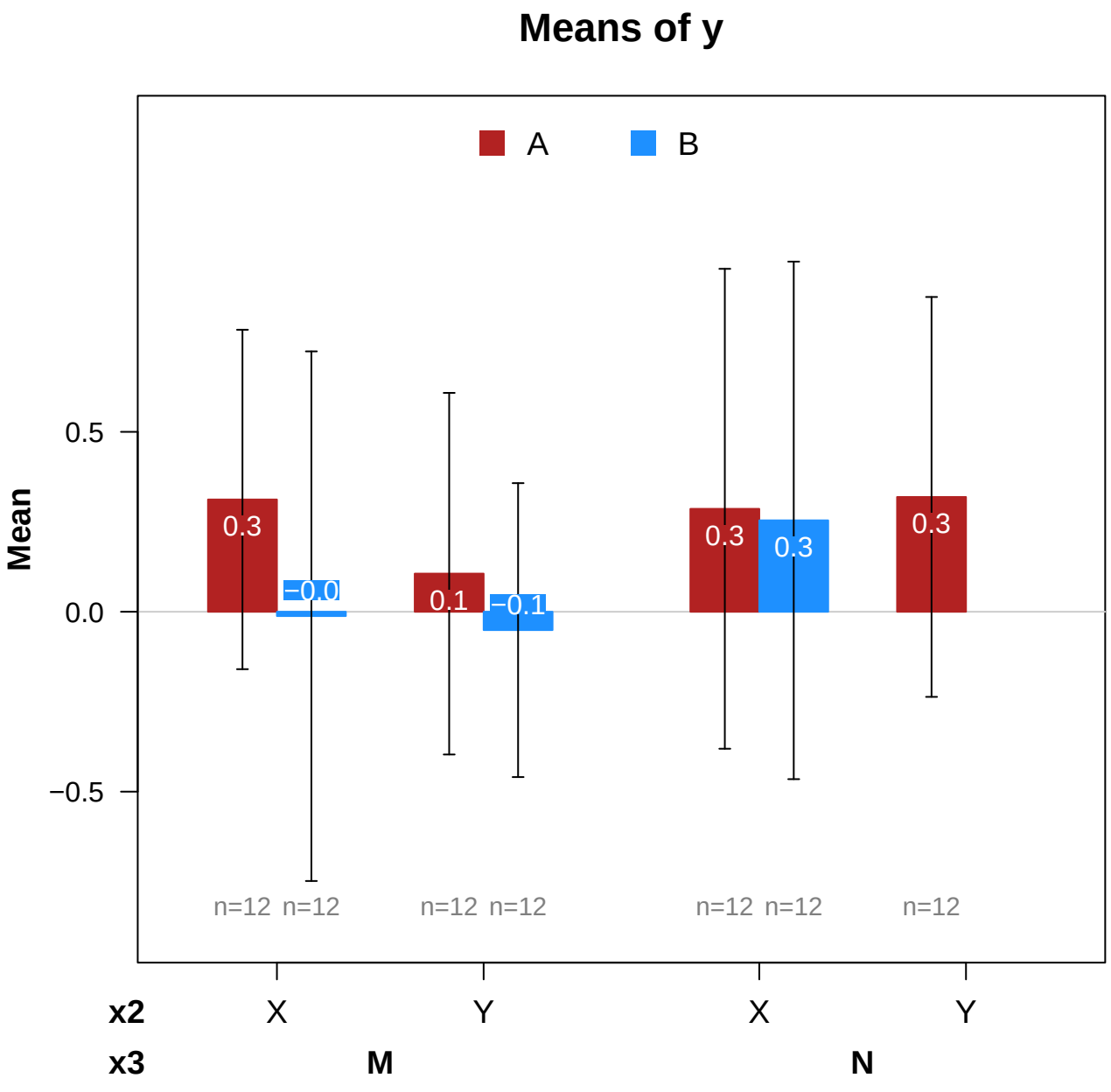

Means of y
A
B
Mean
0.2
0.0
−0.2
−0.4
−0.0
0.0
n=50
n=50
p = 0.979

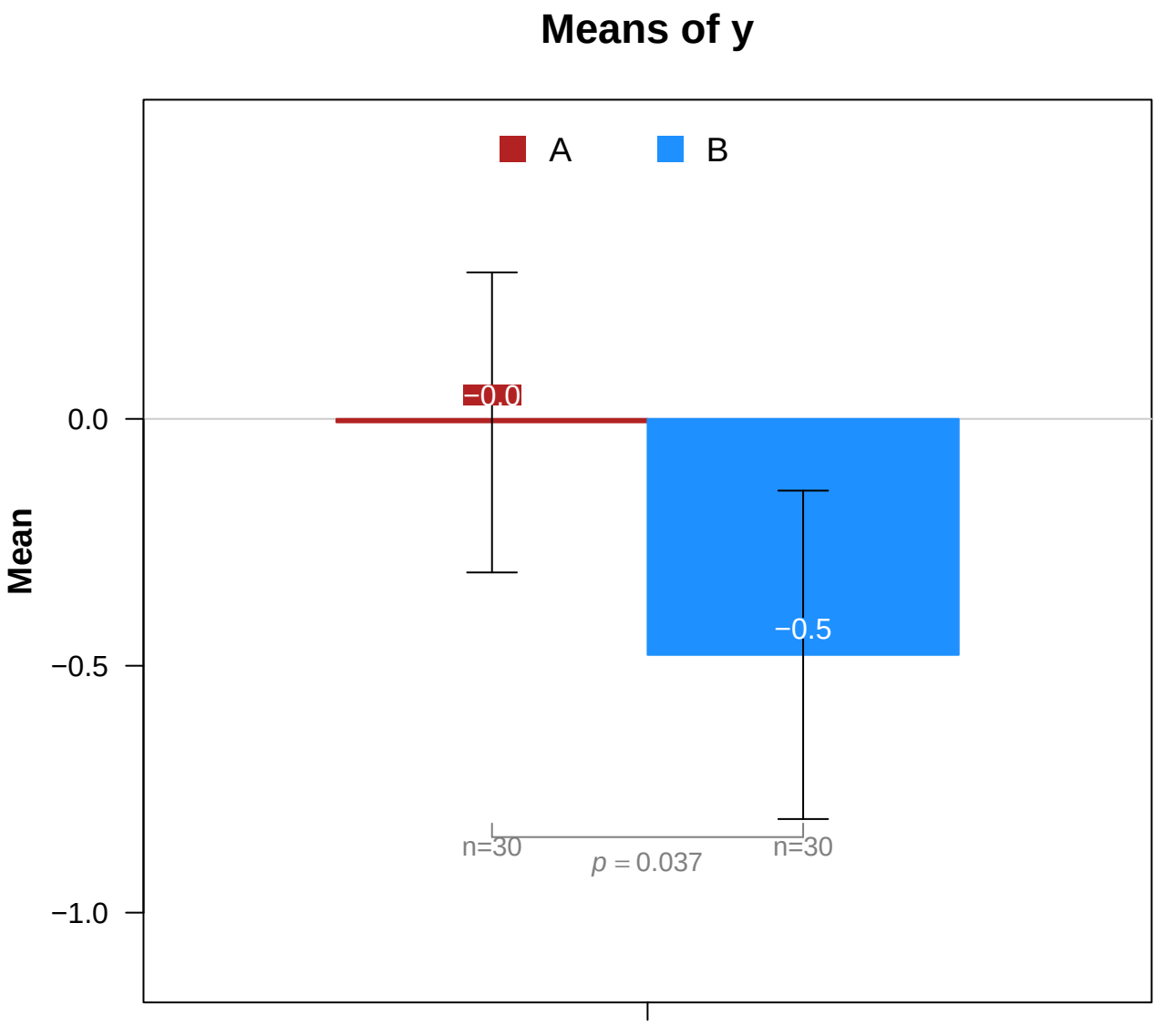

Means of y
A
B
p = 0.375
p = 0.092
p = 0.753
0.4
0.2
0.0
−0.2
−0.4
Mean
0.2
0.1
0.1
−0.2
n=50
n=50
n=50
n=50
X
Y
x2

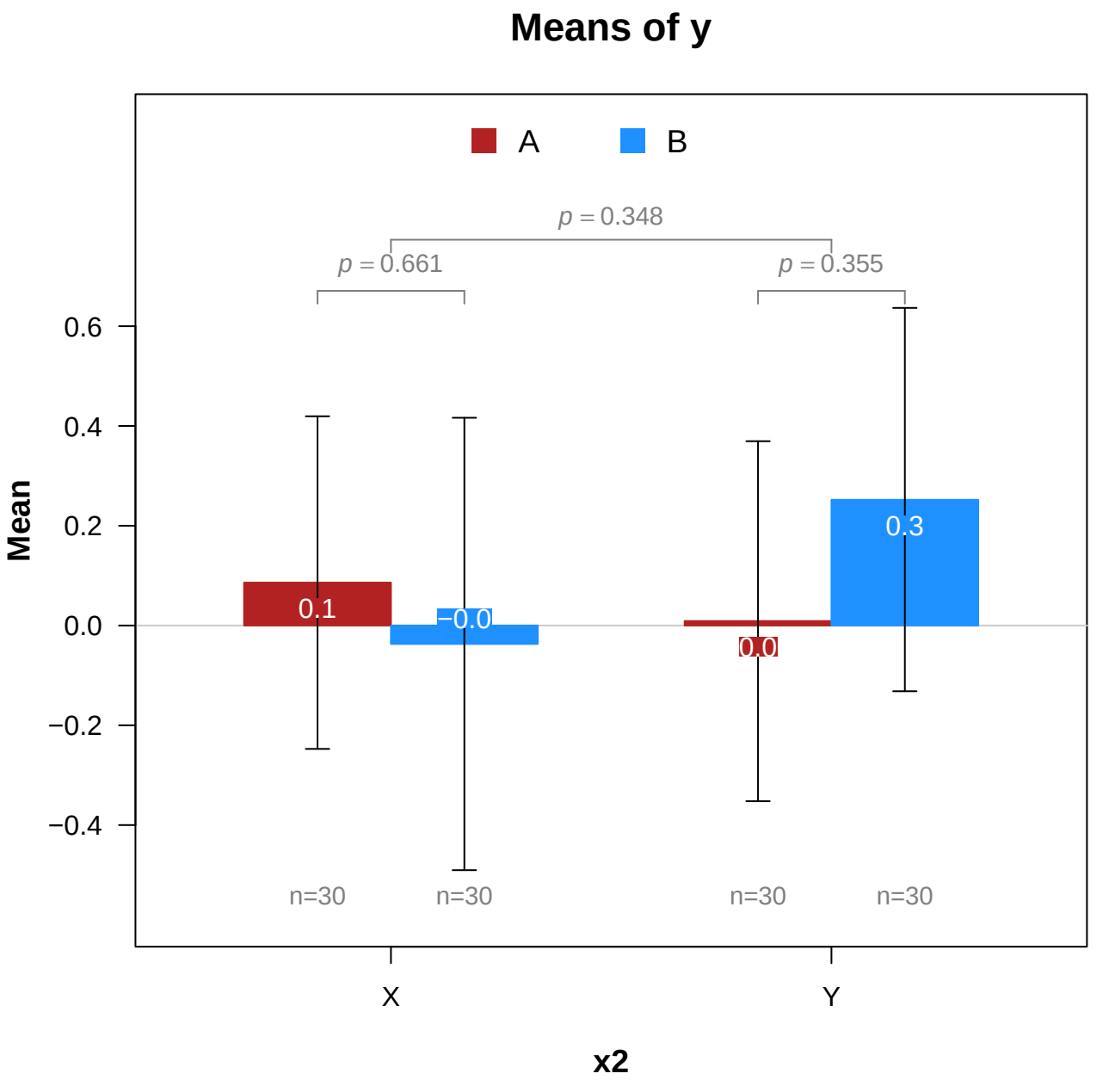

Means of y
A
B
p = 0.274
0.2
0.0
−0.2
−0.4
Mean
0.1
−0.1
n=50
n=50

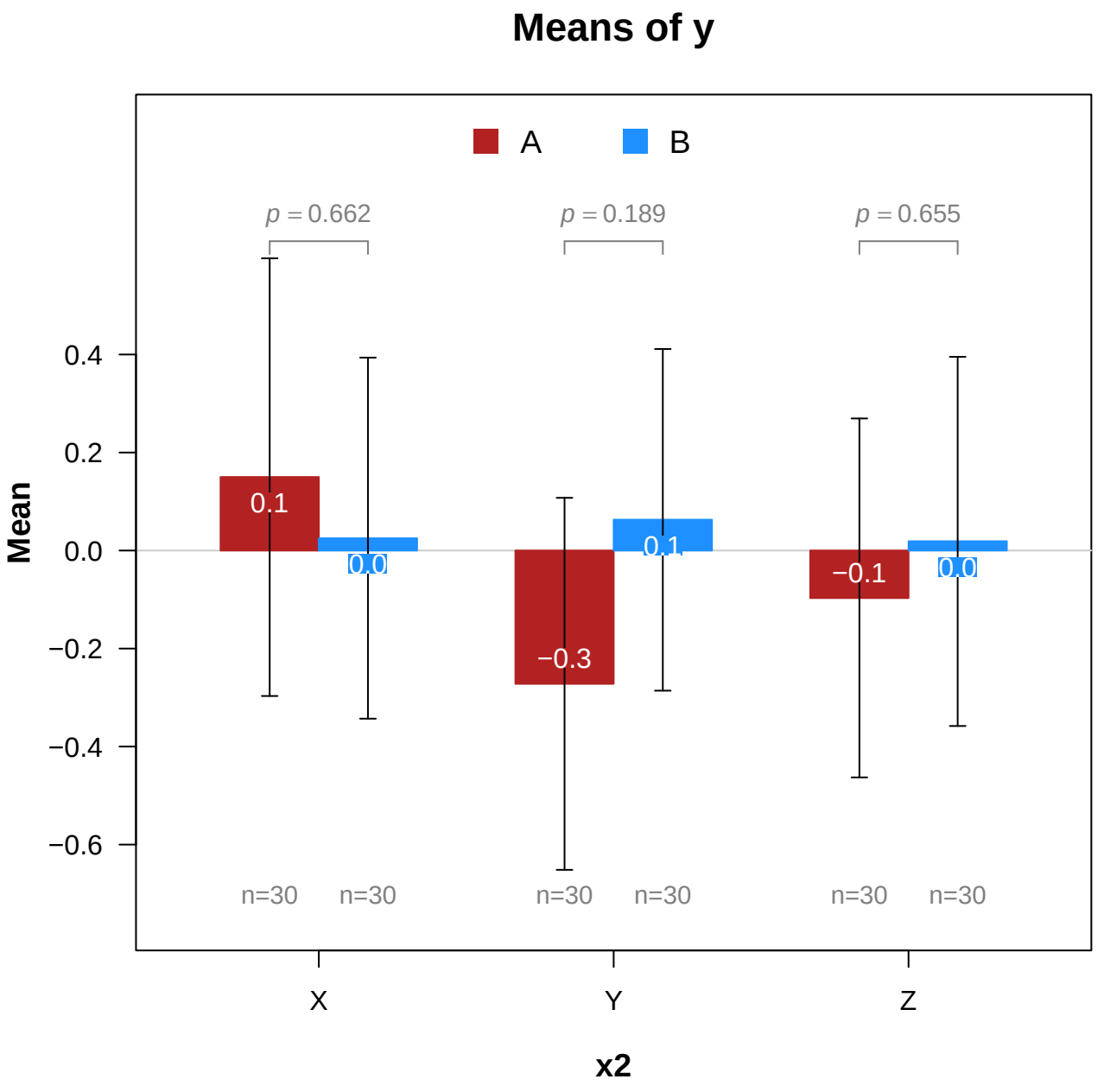

Means of y
B
A
p = 0.274
0.2
0.0
−0.2
−0.4
Mean
0.1
−0.1
n=50
n=50


Means of y
B
A
p = 0.274
0.2
0.0
−0.2
−0.4
Mean
0.1
−0.1
n=50
n=50


Means of y
A
B
Mean
0.5
0.0
−0.5
0.3
−0.0
0.1
−0.1
0.3
0.3
0.3
n=12 n=12
n=12 n=12
n=12 n=12
n=12
x2
X
Y
X
Y
x3
M
N


Means of y
A
B
Mean
0.0
−0.5
−1.0
−0.0
−0.5
n=30
n=30
p = 0.037


Means of y
A
B
p = 0.348
p = 0.661
p = 0.355
0.6
0.4
0.2
0.0
−0.2
−0.4
Mean
0.1
−0.0
0.0
0.3
n=30
n=30
n=30
n=30
X
Y
x2


Means of y
A
B
p = 0.662
p = 0.189
p = 0.655
0.4
0.2
0.0
−0.2
−0.4
−0.6
Mean
0.1
0.0
−0.3
0.1
−0.1
0.0
n=30
n=30
n=30
n=30
n=30
n=30
X
Y
Z
x2